

Birgir Hrafnkelsson *Editor*

Statistical Modeling Using Bayesian Latent Gaussian Models

With Applications in Geophysics and
Environmental Sciences



Springer

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Preface

Bayesian latent Gaussian models are a subclass of Bayesian hierarchical models. They can handle continuous and discrete data with complex structures. They are built with conditional probabilistic models, which help break down the modeling phase into manageable steps. Furthermore, they can be designed to handle computation for models with a large number of parameters. Due to these properties of Bayesian latent Gaussian models, they have been used to analyze datasets of all sizes in the fields of engineering, chemistry, sociology, agriculture, medicine, physics, economics, astronomy, biology, Earth sciences, political science, and psychology. One of the strengths of Bayesian models is their ability to naturally take into account information based on previous knowledge. This is an important feature when analyzing geophysical and environmental data with models that entail processes and parameters that are difficult to estimate. The Bayesian approach makes it possible to represent knowledge from the literature about the nature of the processes and scale of the parameters within the statistical model.

The aim of this book is to make Bayesian latent Gaussian models accessible to researchers, specialists, and graduate students in geophysics, environmental sciences, statistics, and other related fields. The aim is also to demonstrate how these models can be applied to data in geophysics and environmental sciences. The first chapter of this book provides a general background for Bayesian latent Gaussian models. In each of the following six chapters, a case study is given on how to apply these models to geophysical or environmental data. It is assumed that the reader has knowledge of calculus, linear algebra, and the basics of probability and statistics, in particular, the concepts of random variables, probability density functions, and maximum likelihood estimation. The reader will learn about the structure of Bayesian latent Gaussian models, how to construct prior densities for their parameters, how to use their posterior density to learn about their parameters, and how to make predictions with them. Through the case studies, the reader will see how these models are applied to various problems found in geophysics and environmental sciences. In particular, the case studies involve analysis of data from glaciology, hydrology, engineering seismology, meteorology and climatology.

The web page <https://blgm-book.github.io> provides information about this book. Links to code, which correspond to the case studies within each chapter, are available at this web page under the code section. These links are also given within each chapter.

There are several people whom I would like to thank, people that have directly or indirectly been involved in creating this book. First, I would like to thank the corresponding authors of the chapters, namely, Atefe Darzi, Giri Gopalan, Raphaël Huser, Sahar Rahpeyma, and Stefan Siegert, for believing in this project and making sure that their chapters were completed. I would like to thank Haakon Bakka for his contribution as a co-author of the chapter on the Bayesian latent Gaussian models and for the fruitful discussions about the project at its early stage. I am thankful for the contributions of the other authors, namely, Andrew Zammit-Mangion, Arnab Hazra, Axel Örn Jansson, Árni Víðir Jóhannesson, Benedikt Halldórsson, Felicity McCormack, Joshua Lovegrove, Milad Kowsari, Rafael Danél Vias, Sigurður Magnús Garðarson, Sölvi Rögnvaldsson, and Tim Sonnemann. Additional thanks to Rafael for reading over more than half of the chapters at their final stage, and for rigging up the book's web page with me. Dear co-authors, due to your hard work and diligence, we can share this book with people all over the world and be proud of the final outcome.

My thanks go to Eva Hiripi at Springer for believing in this project and for all her help through this process. I want to thank Håvard Rue for constructive conversations on practical issues when starting this book project. I would like to give thanks to my former graduate students at the University of Iceland, namely, Atli Norðmann Sigurðarson, Helgi Sigurðarson, Ólafur Birgir Davíðsson and Óli Páll Geirsson, as well as my former colleague, Egil Ferkingstad, and my current colleagues, Finnur Pálsson and Guðfinna Th. Aðalgeirsdóttir at the University of Iceland. The projects we worked on together paved the road for many of the chapters in this book. Finally, I would like to thank my wife, Helga, for supporting me through this project.

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Birgir Hrafnkelsson

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Bayesian Latent Gaussian Models



Birgir Hrafnkelsson and Haakon Bakka

1 Introduction

We want to understand the wide variety of processes on the Earth. How do the physical and chemical processes shape the Earth minute by minute and decade by decade? How do these processes influence the diverse biological world, and especially, how do they affect humans? To understand these complex processes and interactions, we need a large interdisciplinary effort, with geophysicists, atmospheric scientists, oceanographers, geologists, geographers, chemists, biologists, ecologists, engineers, social scientists, economists, computer scientists, applied mathematicians, statisticians, and more. When we study these processes, we need to make observations and experiments, but we also need precise models to make sense of these observations. Some models, such as Newton's laws, are universal equations, but many processes cannot be pinned down with precise equations. We need statistics! We need tools to handle uncertainty and random variables. We need statistical models that utilize knowledge and an understanding of the underlying physics, chemistry, biology, and sociological and economical dynamics. The process of linking this knowledge and the observed data within a statistical model is important for making the best use of all available information. Analyses based on these statistical models are one of the cornerstones for creating new knowledge of the Earth's physical and environmental processes and are vital for informed decision-making. And this is what this book is about; a large class of statistical models that we believe are extremely useful.

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The class of Bayesian hierarchical models (BHM) is an extensive class of statistical models that includes the Bayesian latent Gaussian models (Bayesian LGMs) as a subclass. Over the past three decades, modeling with BHM has advanced rapidly, and computational methods for BHM have evolved substantially due to extensive research and an increase in computational power. BHM have been applied within various fields: atmospheric sciences, medicine, biology, Earth sciences, genetics, social sciences, engineering, and economics, to name a few. The focus of this book is on the class of Bayesian latent Gaussian models and its application to geophysical and environmental data.

The probabilistic structure of BHM is based on presenting the joint probability density of the observed response and the parameters as a product of conditional probability densities using probability theory. The response is also known as the outcome or the dependent variable. Under the most generic form of BHM, the parameters are split into two groups, the latent parameters, $\mathbf{x}$, and the hyperparameters, $\boldsymbol{\theta}$, and observations of the response are stored in the vector $\mathbf{y}$. The joint density of $(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta})$ is what we refer to as the model, and it can be presented as

$$\pi(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta}) = \pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})\pi(\boldsymbol{\theta}), \quad (1)$$

where $\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})$ is the conditional density of the response given the parameters, $\pi(\mathbf{x}|\boldsymbol{\theta})$ is the conditional density of the latent parameters given the hyperparameters, and $\pi(\boldsymbol{\theta})$ is the marginal density of the hyperparameters. The product $\pi(\mathbf{x}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})$ is referred to as the prior density of the parameters, or simply the prior density. The model in (1) has a hierarchy that comes in three levels where each of the three densities on the right-hand side of (1) defines a level. These three levels are referred to as: the response level, the latent level, and the hyperparameter level. Previously, the response level has been referred to as the data level (e.g., Gelman et al., 2013). We opt for the term response level since at this level the variables that are being modeled are the response variables conditional on the latent model. However, covariates are part of the data, and they are used to define fixed and random effects at the latent level, so using the term data level can be confusing.

The hierarchical structure of BHM makes them easier to build since each component or density can be built conditional on the variables in the levels above, i.e., the response is modeled at the response level conditional on the latent parameter and the hyperparameters, and the latent parameters are modeled at the latent level conditional on the hyperparameters. BHM are flexible and can handle a high degree of complexity, in particular, a large number of factors, which potentially affect the response variable, can be included in the model.

The main advantage of Bayesian statistical models, over the ones that do not rely on the Bayesian approach, is the possibility of adding prior knowledge, based on other data sources or scientific knowledge, into the model. For example, when modeling geophysical or environmental data, it is possible to use physical theory or previous observations to construct informative prior densities. When we construct complex models with many different components, a Bayesian model is always well-defined and theoretically founded given that the prior densities are proper, that is,

they integrate to one. In the case of complex non-Bayesian models, a regularization structure plays a similar role as the prior density for the latent parameters, and regularization parameters play a similar role as the hyperparameters. However, the hyperparameters in a Bayesian model are controlled by their prior density, while the regularization parameters in a non-Bayesian model are usually not constrained in any way. So, the Bayesian models are equipped with their priors to regularize the model parameters in order to make inference stable and to give meaningful results.

Take, for example, a model for time series. If we have one model component for the day-to-day behavior, one for the seasonal behavior, and another one for the yearly behavior, then these components can be very confounded. If we regularize with the priors, we can make sure that the different components are better separated. For example, the priors for the yearly and seasonal components could specify that these components are likely to only change slightly from year to year, but the prior for the daily component could specify that the component changes quickly and is therefore very different from the yearly and seasonal components.

Another important advantage of the Bayesian approach is the generality of the posterior density, which is the main tool for conducting Bayesian statistical inference. There is just a single mathematically correct way of defining a posterior density for a given model, and computation of posterior quantities from the posterior density, such as posterior quantiles, can be conducted with the same approach regardless of the underlying model. On the other hand, non-Bayesian methods often rely on defining estimators and studying their properties and robustness. For a general overview of BHM, see Gelman et al. (2013) and Wakefield (2013).

Bayesian latent Gaussian models will be the focus of this chapter and this book. They are BHM with a Gaussian assumption on the conditional density of $\mathbf{x}$, $\pi(\mathbf{x}|\boldsymbol{\theta})$. The Gaussian assumption at the latent level makes inference and prediction more manageable relative to a non-Gaussian assumption at the latent level, (e.g., Rue et al., 2009; Hrafnkelsson et al., 2021).

One general modeling approach was presented by Berliner (2003). He showed how a mathematical formulation of physical processes can be incorporated within a Bayesian hierarchical model. The geophysical response variable is modeled conditional on the physical processes at the response level, while the physical processes are modeled at the latent level conditional on physical parameters that govern the physical processes. The physical parameters and other statistical parameters found at the response level and at the latent level are assigned prior densities at the hyperparameter level. Models of this type are referred to as physical–statistical models. The terminology of these models as presented by Berliner (2003) and Cressie and Wikle (2011) is different from the terminology used here, namely, instead of using response/latent/hyperparameter levels, they use data/process/parameter levels.

The physical processes at the latent level are modeled by coupling the mathematical structure of the underlying physics with Gaussian processes. In some cases, the modeling of the physical processes is based on structuring Gaussian processes using the partial differential equations or the differential equations that describe the physical processes, (e.g., Royle et al., 1999; Wikle et al., 2001). In other cases,

an output from a numerical model or an explicit mathematical model based on the underlying physics is added to a Gaussian process with the same coordinates. This process is referred to as the error-correcting process (e.g., Gopalan et al., 2019). In Gopalan et al. (2019), a spatio-temporal model was proposed for the surface elevation of a glacier. The physics was described with a partial differential equation that was evaluated with a numerical solver, and the error-correcting process was a spatio-temporal Gaussian process. The above examples involve unknown physical parameters that need to be inferred. When knowledge about the scale of the physical parameters is present, informative prior densities should be constructed for them as these prior densities stabilize the inference for all the parameters.

Outputs from physical models may be such that they give valuable information about the physical process of interest; however, in some cases, the scale of the output is not precisely at the scale of the observations, and scaling is needed to make the best use of the output, see, e.g., Sigurdarson and Hrafnkelsson (2016) and chapter “Improving Numerical Weather Forecasts by Bayesian Hierarchical Modelling”, which is on daily temperature forecast based on an output from a climate model. When there is no model output available for the geophysical or environmental variable of interest, then the statistical model is often based on covariates that are associated with the variable of interest, for example, ground-motion models in engineering seismology applications, see, e.g., Rahpeyma et al. (2018) and chapter “Bayesian Modeling in Engineering Seismology: Ground-Motion Models”. Since geophysical and environmental data commonly have a spatial and temporal reference, the statistical models for these data frequently involve components that are modeled with spatial models (e.g., Cressie, 1993; Rue & Held, 2005; Lindgren et al., 2011), time series models (e.g., Prado et al., 2021), and spatio-temporal models (e.g., Cressie & Wikle, 2011; Wikle et al., 2019). Furthermore, the effect of continuous variables, other than space and time, can also be modeled with Gaussian processes with respect to the coordinates of these continuous variables. Models for discharge rating curves provide a good example of these models, see Hrafnkelsson et al. (2022) and chapter “Bayesian Discharge Rating Curves Based on the Generalized Power Law”.

Environmental and geophysical studies often involve analysis of extremes, e.g., events that involve, extreme precipitation (e.g., Geirsson et al., 2015), floods (e.g., Jóhannesson et al., 2022), earthquakes (e.g., Dutfoy, 2021), droughts (e.g., Alam et al., 2015), and heatwaves (e.g., Raha & Ghosh, 2020). An introduction to Bayesian latent Gaussian models for spatial extremes along with an example on extreme precipitation is given in chapter “Bayesian Latent Gaussian Models for High-Dimensional Spatial Extremes”. The locations of geophysical or environmental events that occur at random across space are naturally modeled with point processes (e.g., Ripley, 1981; Cressie, 1993; Rathbun & Cressie, 1994; Møller et al., 1998). A good example of statistical modeling of a point process is given in Lombardo et al. (2019). They model landslide data as a point process and frame it within a Bayesian latent Gaussian model that allows the intensity of the point process to vary spatially.

1.1 *Structure of This Chapter and the Book*

In this chapter, we introduce Bayesian latent Gaussian models, present their structure, show how their parameters can be inferred, show how to apply them to make predictions, and provide a few examples on how to apply them to real data. These examples are meant to be simple and pedagogical, with limited detail. Here, we focus on specific aspects of Bayesian latent Gaussian models, in increasing complexity. The other chapters of this book provide in-depth statistical analyses of real geophysical and environmental data.

In Sect. 2, the structures of three types of Bayesian latent Gaussian models are presented, and it is demonstrated how their parameters are inferred with the Bayesian approach and how to make predictions with them. The three types of Bayesian latent Gaussian models presented here are: (i) Bayesian Gaussian–Gaussian models (Sect. 2.1); (ii) Bayesian latent Gaussian models with a univariate link function (Sect. 2.2); and (iii) Bayesian latent Gaussian models with a multivariate link function (Sect. 2.3). Section 3 is on the construction of prior densities for the latent parameters and the hyperparameters in Bayesian LGMs. Section 4 provides an example of the application of the Bayesian Gaussian–Gaussian model to experimental data on discharge in an open channel. In Sect. 5, the Bayesian latent Gaussian models with a univariate link function are presented through an analysis of precipitation data. Section 6 is on statistical modeling that calls for applying Bayesian latent Gaussian models with a multivariate link function. Three real-data examples are given; Sect. 6.1 is on observed daily temperature on a spatial grid covering Europe and a corresponding covariate based on an output from a climate model that is used to forecast daily temperature several days ahead; Sect. 6.2 is on extreme precipitation data from Saudi Arabia on a high-dimensional grid; Sect. 6.3 is on monthly precipitation data from Iceland, with focus on the construction of prior densities for the hyperparameters.

The examples in Sects. 6.1 and 6.2 are presented in detail in chapters “Improving Numerical Weather Forecasts by Bayesian Hierarchical Modelling” and “Bayesian Latent Gaussian Models for High-Dimensional Spatial Extremes”, respectively. A review of Bayesian models in glaciology and two examples on the application of these models are given in chapter “A Review of Bayesian Modelling in Glaciology”. Chapter “Bayesian Discharge Rating Curves Based on the Generalized Power Law” involves the modeling of discharge in rivers as a function of water elevation. The predictive function is referred to as a discharge rating curve and is widely used in hydrology. Spatial models for the prediction of peak ground acceleration are the topic of chapter “Bayesian Modeling in Engineering Seismology: Ground-Motion Models”. These models, referred to as ground motion models, are an important tool in engineering seismology and earthquake engineering. The models in chapters “A Review of Bayesian Modelling in Glaciology”, “Bayesian Discharge Rating Curves Based on the Generalized Power Law”, and “Bayesian Modeling in Engineering Seismology: Ground-Motion Models” are Bayesian Gaussian–Gaussian models.

Spatial modeling of the magnitude of earthquakes is the topic of chapter “Bayesian Modelling in Engineering Seismology: Spatial Earthquake Magnitude Model”. The proposed models are Bayesian LGMs that assume either the generalized Pareto distribution or the exponential distribution at the response level. The bibliographic note at the end of this chapter provides further references on BHMs, Bayesian LGMs, and the application of these models to geophysical and environmental data.

The reader can go through this book in various ways. A reader that is primarily interested in Bayesian Gaussian–Gaussian models can, for example, go through Sects. 2.1, 3 and 4, in this chapter and chapters “A Review of Bayesian Modelling in Glaciology”, “Bayesian Discharge Rating Curves Based on the Generalized Power Law”, and “Bayesian Modeling in Engineering Seismology: Ground-Motion Models”. A reader with an interest in Bayesian latent Gaussian models with a univariate link function can take a path through Sects. 2.2, 3, and 5 in this chapter and chapter “Bayesian Modelling in Engineering Seismology: Spatial Earthquake Magnitude Model”. Another reader that is already familiar with Bayesian LGMs with a univariate link function and is keen on learning about Bayesian latent Gaussian models with a multivariate link function can read through Sects. 2.3, 3, and 6 in this chapter and chapters “Improving Numerical Weather Forecasts by Bayesian Hierarchical Modelling” and “Bayesian Latent Gaussian Models for High-Dimensional Spatial Extremes”. For those that are not familiar with BHMs in general, we suggest first learning about Bayesian Gaussian–Gaussian models, then learning about Bayesian LGMs with a univariate link function, and finally, learning about Bayesian LGMs with a multivariate link function.

2 The Class of Bayesian Latent Gaussian Models

In this section, we introduce the class of Bayesian latent Gaussian models. These models are Bayesian hierarchical models with a Gaussian assumption on the latent parameters. Three types of Bayesian LGMs are introduced here; Sect. 2.1: the Bayesian Gaussian–Gaussian models; Sect. 2.2: the Bayesian latent Gaussian models that model a single parameter in the density of an observation y_i at the latent level; Sect. 2.3: the Bayesian latent Gaussian models that model more than one parameter in the density of y_i at the latent level. We add the term Bayesian in front of the names of these models to underline the fact that their parameters are inferred with the Bayesian approach and also to point out that Gaussian–Gaussian models and latent Gaussian models can be inferred using the frequentist approach. In this chapter and in other chapters of the book, these models will also be mentioned without the term Bayesian in front; however, note that all models in the book are Bayesian. These three model types assume that the model at the latent level is linear in terms of the parameters. They also assume conditional independence at the response level, i.e., the observations are independent conditional on the

latent parameters. Other types of Bayesian LGMs exist, e.g., models that assume dependence at the response level.

The Bayesian Gaussian–Gaussian models assume the response follows a Gaussian distribution, and they are an extension of linear regression models. Within both Bayesian and frequentist settings, Gaussian–Gaussian models are also referred to as hierarchical linear models (Gelman et al., 2013), linear mixed models (Jiang & Nguyen, 2021), linear mixed-effects models (Faraway, 2016), multilevel models (Gelman & Hill, 2007), and additive models (Wood, 2017). The latent Gaussian models fall under the class of generalized linear mixed models (e.g. Wakefield, 2013; Jiang & Nguyen, 2021), a class that also includes generalized additive models and generalized additive mixed models (Wood, 2017).

2.1 Bayesian Gaussian–Gaussian Models

When the response variable of interest is continuous and measurements of several factors and other continuous variables that may be associated with the response are available, then a linear regression model is a natural first choice. If the complexity of the data is high due to various effects, such as a spatial effect, a temporal effect, and nonlinear effects of continuous variables, then the number of regression coefficients can become large. These settings often call for an advanced model for the regression coefficients. The Bayesian Gaussian–Gaussian model is well-suited for this type of modeling since it has the same structure as the linear regression model, but it assumes prior distributions for the regression coefficients that restrict their sizes and make use of their structure. To fulfill the Gaussian assumption at the response level, the continuous response variable is sometimes transformed using, e.g., the logarithmic transformation or the Box–Cox transformation (Box & Cox, 1964).

2.1.1 The Structure of Bayesian Gaussian–Gaussian Models

Bayesian Gaussian–Gaussian models are regression models with a Gaussian assumption on the error term, and their linear parameters are assigned Gaussian prior densities. As mentioned above, the Gaussian–Gaussian model can be termed as a linear mixed model, attributed to the model consisting of a linear combination of two types of components. The first type we refer to as fixed effects, and these are coefficients for the covariates of the model. The second type we refer to as random effects, and these can model complex relationships. The linear model can be set forth as

$$y_i = \beta_0 + \sum_{k=1}^K \beta_k v_{i,k} + \sum_{j=1}^J u_j a_{i,j} + \epsilon_i, \quad (2)$$

where y_i is the observed response, β_0 is an intercept, $v_{i,k}$ is the k -th covariate of the i observation, β_k is the coefficient of that covariate, u_j is the j -th random effect and $a_{i,j}$ is the weight of u_j for the i -th observation, and ϵ_i is the error term of the i -th observation. The term ϵ_i is modeled as a mean zero Gaussian random variable with unknown variance σ_ϵ^2 , and it is independent of other error terms. The total number of observations is n . The mean and variance of y_i conditional on $\beta_0, \beta_1, \dots, \beta_K, u_1, \dots, u_J$ are η_i and σ_ϵ^2 , respectively, where $\eta_i = \beta_0 + \sum_{k=1}^K \beta_k v_{i,k} + \sum_{j=1}^J u_j a_{i,j}$, and its conditional probability density is given by

$$\pi(y_i | \eta_i, \sigma_\epsilon^2) = \mathcal{N}(y_i | \eta_i, \sigma_\epsilon^2) \propto \frac{1}{\sigma_\epsilon} \exp\left(-\frac{(y_i - \eta_i)^2}{2\sigma_\epsilon^2}\right),$$

where $\mathcal{N}(y_i | \eta_i, \sigma_\epsilon^2)$ denotes a Gaussian density with mean η_i and variance σ_ϵ^2 and $\propto$ is such that the quantity to the right of it is equal to the quantity to the left of it up to a constant that is independent of y_i .

In the statistics literature, it is common to refer to the β parameters as the fixed effects and the u parameters as the random effects. Within the Bayesian Gaussian–Gaussian model, the β parameters are modeled with a Gaussian distribution with known mean and variance, while the random effects are modeled with a Gaussian distribution with some mean and variance that again are modeled with parameters. These second-level parameters are hyperparameters. They are placed in the vector θ when using the notation of the model given by (1), and their values need to be learned from the data.

The model in (2) can be presented in a vector and matrix notation. All the observations of the response are lined up in the response vector, $\mathbf{y}$, the β parameters are allocated in the vector $\boldsymbol{\beta}$, with β_0 as its first element, and X contains the covariates, with a vector of ones as its first column vector. Furthermore, by lining up the u parameters in the second sum in (2) in the vector $\mathbf{u}$, and allocating the weights of the random effects, $a_{i,j}$, in the matrix A , then the model in (2) can be written as

$$\mathbf{y} = X\boldsymbol{\beta} + A\mathbf{u} + \boldsymbol{\epsilon}, \quad (3)$$

where $\boldsymbol{\epsilon}$ contains the error terms. The conditional probability density of $\mathbf{y}$ conditional on $\boldsymbol{\beta}, \mathbf{u}$, and σ_ϵ^2 is

$$\begin{aligned} \pi(\mathbf{y} | \boldsymbol{\beta}, \mathbf{u}, \sigma_\epsilon^2) &= \mathcal{N}(\mathbf{y} | \boldsymbol{\eta}, \Sigma_\epsilon) \\ &\propto |\Sigma_\epsilon|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{y} - \boldsymbol{\eta})^\top \Sigma_\epsilon^{-1}(\mathbf{y} - \boldsymbol{\eta})\right), \end{aligned}$$

i.e., a multivariate Gaussian density with mean and covariance $\boldsymbol{\eta}$ and Σ_ϵ , where $\boldsymbol{\eta} = X\boldsymbol{\beta} + A\mathbf{u}$, $\Sigma_\epsilon = \sigma_\epsilon^2 I$, and I is an identity matrix of size n .

The prior mean and variance of $\boldsymbol{\beta}$ are given specific values before seeing the data, and they should represent the current knowledge about $\boldsymbol{\beta}$. If there is some information available from past data and/or previous studies about all or some of

the β parameters, then this knowledge is formulated through the prior mean and variance of β . Likewise, if there is limited or non-existing information about all or some of the β parameters, then that is represented through the prior mean and variance, most commonly by setting the mean equal to zero and selecting a large variance for these β parameters. A more detailed discussion about the selection of prior densities for the fixed effects is given in Sect. 3.1.

The selection of prior densities for the random effects is slightly different in nature since the aim is not to predetermine the values of the prior mean and variance but to select the structure of the mean and variance that are functions of the hyperparameters in θ . The random effects, i.e., the u parameters, come in one set for simpler models and in several sets for more complicated models, where each set corresponds to a particular model component, e.g., a temporal component, a spatial component, a spatio-temporal component, an effect that varies nonlinearly with a given covariate, or an independent effect for individuals or items under study. Each component is modeled jointly with a Gaussian density, and in most cases, the mean is set equal to zero, while the covariance structure can correspond to some Gaussian process, or, in the case of independent effects, the covariance matrix is a diagonal matrix with an unknown variance on the diagonal. A temporal component is modeled with a time series covariance matrix, while a spatial component is modeled with a spatial covariance function that can be used to construct a spatial covariance matrix. The inverse of the covariance matrix (i.e., the precision matrix) can also be modeled instead of the covariance matrix itself, and there exist models that represent spatial, temporal, and spatio-temporal components with sparse precision matrices. These sparse precision matrices facilitate faster computation and are very useful for handling high-dimensional spatial and temporal data. Section 3.2 provides examples of Gaussian random effects that have structures that correspond to independence, temporal dependence, and spatial dependence.

The model in (3) can be rewritten in terms of the multiple model components mentioned above. If there are L components, then the vector u in (3) can be decomposed into L vectors, u_l , $l \in \{1, \dots, L\}$, and the corresponding weights of all the observations can be placed in L matrices, A_l , $l \in \{1, \dots, L\}$. Under this decomposition of the vector u , the model in (3) can be written as

$$y = X\beta + \sum_{l=1}^L A_l u_l + \epsilon. \quad (4)$$

The parameter vectors β , $u_1, \dots, u_L$, have prior means, $\mu_\beta, \mu_{u,1}, \dots, \mu_{u,L}$, respectively, and prior precision matrices, $Q_\beta, Q_{u,1}, \dots, Q_{u,L}$, respectively, or prior covariance matrices, $\Sigma_\beta, \Sigma_{u,1}, \dots, \Sigma_{u,L}$, respectively. We will consider both approaches and show that when the precision matrices are sparse, an advantage in posterior computation is achieved, see Sect. 2.1.2. The mean of ϵ is zero, its precision matrix is denoted by Q_ϵ , and its covariance matrix is denoted by Σ_ϵ . Both Q_ϵ and Σ_ϵ are diagonal matrices since the error terms are assumed to be mutually independent.

For a more compact notation, the following matrices and vectors are constructed,

$$\begin{aligned} Z &= (X \ A_1 \ \dots \ A_L), \\ \mathbf{x} &= (\boldsymbol{\beta}^\top, \mathbf{u}_1^\top, \dots, \mathbf{u}_L^\top)^\top, \\ \boldsymbol{\mu}_x &= (\boldsymbol{\mu}_\beta^\top, \boldsymbol{\mu}_{u,1}^\top, \dots, \boldsymbol{\mu}_{u,L}^\top)^\top, \\ Q_x &= \text{bdiag}(Q_\beta, Q_{u,1}, \dots, Q_{u,L}), \\ \Sigma_x &= \text{bdiag}(\Sigma_\beta, \Sigma_{u,1}, \dots, \Sigma_{u,L}), \end{aligned}$$

where $\text{bdiag}(\cdot, \dots, \cdot)$ combines matrices into a block-diagonal matrix.

In the case of the Gaussian–Gaussian model, the distribution of the response, conditional on the latent parameters and the hyperparameters of the model, specified at the response level, is given by

$$\mathbf{y}|\mathbf{x}, \boldsymbol{\theta} \sim \mathcal{N}(Z\mathbf{x}, Q_\epsilon^{-1}), \quad (5)$$

where the right-hand side denotes a Gaussian distribution with mean $Z\mathbf{x}$ and covariance Q_ϵ^{-1} . Note that Q_ϵ may depend on the hyperparameters in $\boldsymbol{\theta}$. Furthermore, it is possible to allow the elements of Z , namely, those of X and A_l , $l \in \{1, \dots, L\}$, to depend on unknown parameters. These unknown parameters would be stored in the hyperparameter vector $\boldsymbol{\theta}$.

The latent parameters are specified at the latent level as

$$\mathbf{x}|\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_x, Q_x^{-1}), \quad (6)$$

and the elements of $\boldsymbol{\mu}_x$ and Q_x may depend on the hyperparameters in $\boldsymbol{\theta}$. The selection of prior densities for the hyperparameters of Bayesian latent Gaussian models, including the Bayesian Gaussian–Gaussian models, is discussed in Sect. 3.3.

2.1.2 Posterior Inference for Gaussian–Gaussian Models

The posterior density of the unknown parameters, $\mathbf{x}$ and $\boldsymbol{\theta}$, stems from Bayes' rule,

$$\begin{aligned} \pi(\mathbf{x}, \boldsymbol{\theta}|\mathbf{y}) &= \frac{\pi(\boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})}{\pi(\mathbf{y})} \\ &\propto \pi(\boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta}) \\ &= \pi(\boldsymbol{\theta})\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_x, Q_x^{-1})\mathcal{N}(\mathbf{y}|Z\mathbf{x}, Q_\epsilon^{-1}). \end{aligned} \quad (7)$$

Posterior samples can be obtained from $\pi(\mathbf{x}, \boldsymbol{\theta}|\mathbf{y})$ by using:

Step 1: Sample $\boldsymbol{\theta}$ from $\pi(\boldsymbol{\theta}|\mathbf{y})$.

Step 2: Sample $\mathbf{x}$ from $\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$.

The density in Step 1 is the marginal posterior density of $\boldsymbol{\theta}$, i.e., it does not depend on $\mathbf{x}$. In the case of Gaussian–Gaussian models, the density $\pi(\boldsymbol{\theta}|\mathbf{y})$ can be written explicitly up to a normalizing constant. Samples can be obtained from the marginal posterior density of $\boldsymbol{\theta}$ by using a sampling technique that can handle probability densities that have an unknown normalizing constant. Examples of sampling techniques that are suited for this task are the Metropolis algorithm (Metropolis & Ulam, 1949; Metropolis et al., 1953), the Metropolis–Hastings algorithm (Hastings, 1970), Metropolis-adjusted Langevin algorithm (MALA) (Roberts & Rosenthal, 1998), and the Hamiltonian Monte Carlo algorithm (Duane et al., 1987; Neal, 1994; Girolami & Calderhead, 2011; Hoffman & Gelman, 2014). We will not go deeply into these algorithms nor other advanced algorithms for posterior sampling. We refer to Liu (2001), Robert and Casella (2004), and Gelman et al. (2013) for further insight. In the Appendix, an algorithm for sampling from the joint posterior density of $(\mathbf{x}, \boldsymbol{\theta})$, which applies the Metropolis algorithm to the marginal posterior density of $\boldsymbol{\theta}$, is given. There it is assumed that the hyperparameters in $\boldsymbol{\theta}$ all have support on the real line, achieved by transforming the original parameters appropriately. When the hyperparameters are transformed to the real line with transformations that lead to a reduction in the skewness of the posterior density of the transformed parameters, e.g., using the logarithmic transformation for variance parameters, then the posterior computation often becomes more efficient.

The density in Step 2 is the posterior density of $\mathbf{x}$ conditional on $\boldsymbol{\theta}$. It is a Gaussian density, and its mean and variance can be represented explicitly. Thus, given $\boldsymbol{\theta}$, samples of $\mathbf{x}$ can be generated directly from this density. The densities in Steps 1 and 2 are represented below in two ways: by using the precision matrix representation (based on Q_ϵ and Q_x) and by using the covariance matrix representation (based on Σ_ϵ and Σ_x).

By using properties of Gaussian densities and the form of the joint Gaussian density of $\mathbf{y}$ and $\mathbf{x}$ conditional on $\boldsymbol{\theta}$ (see the Appendix), it can be shown that the posterior density of $\mathbf{x}$ conditional on $\boldsymbol{\theta}$ in Step 2, in terms of precision matrices, is

$$\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_{x|\mathbf{y}}, Q_{x|\mathbf{y}}^{-1}), \quad (8)$$

where

$$\boldsymbol{\mu}_{x|\mathbf{y}} = Q_{x|\mathbf{y}}^{-1}(Q_x \boldsymbol{\mu}_x + Z^\top Q_\epsilon \mathbf{y}), \quad (9)$$

$$Q_{x|\mathbf{y}} = Q_x + Z^\top Q_\epsilon Z. \quad (10)$$

When the matrices Q_x and Z are sparse, that is, a relatively large fraction of their elements are equal to zero, the computation for $\boldsymbol{\mu}_{x|\mathbf{y}}$ and $Q_{x|\mathbf{y}}$ in (9) and (10) can be reduced substantially. For example, diagonal matrices of sizes 20×20 and 200×200

have 400 and 40,000 elements, respectively, but only 20 and 200 of them are non-zero. Sparse matrices take less storage, and multiplication with zero elements is omitted, hence the reduction in computation time (e.g., Rue & Held, 2005). Recall that Q_ϵ is a diagonal matrix, and thus, it is sparse. The elements of Z that correspond to the A_l matrices are usually sparse since only a few u parameters are used to model each y_i . The matrix Q_x is built from the matrices Q_β , $Q_{u,1}, \dots, Q_{u,L}$. Usually, Q_β is a diagonal matrix, and some of the $Q_{u,l}$ matrices may be diagonal due to an assumption of mutually independent elements in the corresponding u_l vectors. Other u_l vectors correspond to components that are assumed to be dependent, and by modeling their dependence structure with sparse precision matrices, the computation time can be reduced.

The density of $\mathbf{y}$ conditional on $\boldsymbol{\theta}$ only, referred to as the marginal likelihood when conducting inference for $\boldsymbol{\theta}$, is

$$\begin{aligned}\pi(\mathbf{y}|\boldsymbol{\theta}) &= \frac{\pi(\mathbf{y}, \mathbf{x}|\boldsymbol{\theta})}{\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})} \\ &= \frac{\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})}{\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})},\end{aligned}\tag{11}$$

where $\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$, $\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})$, and $\pi(\mathbf{x}|\boldsymbol{\theta})$ are as above. The marginal posterior density in Step 1 is given by

$$\pi(\boldsymbol{\theta}|\mathbf{y}) = \frac{\pi(\boldsymbol{\theta})\pi(\mathbf{y}|\boldsymbol{\theta})}{\pi(\mathbf{y})} \propto \pi(\boldsymbol{\theta})\pi(\mathbf{y}|\boldsymbol{\theta}).$$

Note that even though the right-hand side of (11) is a function of $\mathbf{x}$, it does not depend on $\mathbf{x}$. This is because, by the laws of probability, (11) is the density of $\mathbf{y}$ conditional on $\boldsymbol{\theta}$ only. Hence, any value of $\mathbf{x}$ can be selected to evaluate the right-hand side of (11). Often $\mathbf{x}$ is set equal to the zero vector for convenience.

The marginal likelihood of $\boldsymbol{\theta}$, represented in terms of covariance matrices, is

$$\pi(\mathbf{y}|\boldsymbol{\theta}) = \mathcal{N}(\mathbf{y}|Z\boldsymbol{\mu}_x, Z\Sigma_x Z^\top + \Sigma_\epsilon),$$

and the corresponding marginal posterior density in Step 1, $\pi(\boldsymbol{\theta}|\mathbf{y})$, is proportional to $\pi(\boldsymbol{\theta})\pi(\mathbf{y}|\boldsymbol{\theta})$. The posterior density of $\mathbf{x}$ conditional on $\boldsymbol{\theta}$ is

$$\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_{x|\mathbf{y}}, \Sigma_{x|\mathbf{y}}),$$

where

$$\boldsymbol{\mu}_{x|\mathbf{y}} = \boldsymbol{\mu}_x - \Sigma_x Z^\top (Z\Sigma_x Z^\top + \Sigma_\epsilon)^{-1} (Z\boldsymbol{\mu}_x - \mathbf{y}),$$

$$\Sigma_{x|\mathbf{y}} = \Sigma_x - \Sigma_x Z^\top (Z\Sigma_x Z^\top + \Sigma_\epsilon)^{-1} Z\Sigma_x.$$

If the precision matrices above are sparse, then computation of $\pi(\boldsymbol{\theta}|\mathbf{y})$ in Step 1 will be faster compared to computation based on the version of $\pi(\boldsymbol{\theta}|\mathbf{y})$ that relies on covariance matrices. Further details about the above posterior sampling scheme for $(\mathbf{x}, \boldsymbol{\theta})$ are given in the Appendix.

The advantage of the posterior sampling scheme for $(\mathbf{x}, \boldsymbol{\theta})$, presented above, over a posterior sampling scheme based on a Gibbs sampler with a Metropolis step, or a Metropolis–Hasting step (e.g., Robert & Casella, 2004) for the conditional posterior density of $\boldsymbol{\theta}$, along with an exact draw from the conditional posterior density of $\mathbf{x}$, lies in removing the strong dependence between $\boldsymbol{\theta}$ and $\mathbf{x}$. Filippone et al. (2013) and Filippone and Girolami (2014) found that sampling from the marginal posterior density of the hyperparameters in a Bayesian LGM leads to more effective sampling schemes than sampling from the conditional density of the hyperparameters and latent parameters in Gibbs sampling settings.

2.1.3 Predictions Based on Gaussian–Gaussian Models

Predictions of the response variable are needed for various purposes. We may, for example, be interested in predicting a future value of the response variable or predicting the response variable at an unobserved spatial location. We may want to investigate out-of-sample model performance, e.g., by cross-validation. Furthermore, the model assumptions at the response level can be tested by predicting the response variable under the same setting as the observed data, and a comparison of the predictions and the observations can be made.

The Bayesian tool for making predictions is the posterior predictive distribution (e.g., Gelman et al., 2013). Assume we are interested in making predictions under the Bayesian Gaussian–Gaussian model that are based directly on the random effect $\mathbf{u}$. Let $\tilde{X}$ and $\tilde{A}$ denote matrices that contain the values of the covariates and the weights of the random effects, respectively, for which predictions of the response variable are desired, where $\tilde{\mathbf{y}}$ denotes the predictions. The posterior predictive density of $\tilde{\mathbf{y}}$ is given by

$$\pi(\tilde{\mathbf{y}}|\mathbf{y}) = \int \pi(\tilde{\mathbf{y}}|\tilde{X}, \boldsymbol{\beta}, \tilde{A}, \mathbf{u}, \boldsymbol{\theta})\pi(\boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\beta}d\mathbf{u}d\boldsymbol{\theta}, \quad (12)$$

where the integral is with respect to $(\boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta})$, the density of $\tilde{\mathbf{y}}$ within the integral is Gaussian with mean $\tilde{X}\boldsymbol{\beta} + \tilde{A}\mathbf{u}$ and variance $\sigma_\epsilon^2 I$, and the second density within the integral is the posterior density of $(\boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta})$. Samples can be obtained from the posterior predictive density in (12) by sampling first $(\boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta})$ from the posterior density. Then samples representing predictions of the response variable, $\tilde{\mathbf{y}}$, are obtained by sampling from the model

$$\tilde{\mathbf{y}} = \tilde{X}\boldsymbol{\beta} + \tilde{A}\mathbf{u} + \tilde{\boldsymbol{\epsilon}},$$

where β and u are the posterior samples, and $\tilde{\epsilon}$ is sampled from a Gaussian density with mean zero and variance σ_ϵ^2 , using posterior samples of the variance.

Sampling is slightly more complicated if the predictions are based on random effects other than those in the original model. This is the case if we, for example, extend the spatial or temporal domain. Denote by $\tilde{u}$ samples of the random effects that are not in the original model. In many cases, we can extend the domain already in the fitted model to avoid this problem. In fact, when sparse precision matrices are used to model correlated random effects, then it is computationally easier to include $\tilde{u}$ in the original model and infer it with the model parameters. If the covariance matrices of u and $\tilde{u}$ are based on a well-specified covariance function, then predictions for $\tilde{u}$ can be made after inferring (β, u, θ) in the original model. In particular, say we model the dependence structure with covariance matrices where Σ_{11} and Σ_{22} denote the covariance matrices of u and $\tilde{u}$, respectively, and denote by Σ_{21} the covariance between u and $\tilde{u}$. These three matrices depend on θ . Then the density of $\tilde{u}$ conditional on u , $\pi(\tilde{u}|u, \theta)$, is Gaussian with mean $\Sigma_{21} \Sigma_{11}^{-1} u$ and covariance $\Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{21}^T$.

Here, the posterior predictive density is

$$\pi(\tilde{y}|y) = \int \pi(\tilde{y}|\tilde{X}, \beta, \tilde{A}, \tilde{u}) \pi(\tilde{u}|u, \theta) \pi(\beta, u, \theta|y) d\beta du d\tilde{u} d\theta, \quad (13)$$

and posterior predictive samples are obtained with

$$\tilde{y} = \tilde{X}\beta + \tilde{A}\tilde{u} + \tilde{\epsilon},$$

where samples of $\tilde{u}$ are drawn from $\pi(\tilde{u}|u, \theta)$, samples of $\tilde{\epsilon}$ are drawn as above, and samples of (β, u, θ) are drawn from the posterior density.

Cross-validation can be conducted for the Bayesian Gaussian–Gaussian model by splitting the original dataset into a training dataset and a test dataset and using the training dataset to infer the parameters of the model and then predict the responses of the data points in the test data using the methods described above. The responses in the test dataset are compared to the corresponding predictions from the posterior model, for example, through the mean prediction, prediction intervals, or prediction quantiles. K -fold cross-validation involves splitting the original dataset randomly into K equally large datasets. K is often a number from 5 to 20. Each of these datasets will be used as a test dataset, and K model fits are performed. When the k -th dataset is selected as the test dataset, all the other $K - 1$ datasets are joined to form the training dataset that is used to infer the model parameters. Then the posterior predictions from this model are compared to the values in the test dataset.

A standard way to grade predictive performance on a test dataset is using the log-score, $\log \pi(\tilde{y}|y)$, where $\tilde{y}$ and y are the responses from the test dataset and the training dataset, respectively. This number is approximated by using samples from the posterior density based on the training dataset. In particular, let $\tilde{\eta}_s$ be the s -th posterior sample, and then the log-score for $\tilde{y}$ is approximated

with

$$\log(\pi(\tilde{\mathbf{y}}|\mathbf{y})) \approx \log \left(S^{-1} \sum_{s=1}^S \pi(\tilde{\mathbf{y}}|\tilde{\boldsymbol{\eta}}_s) \right),$$

where S is the number of posterior samples. The expression $\pi(\tilde{\mathbf{y}}|\tilde{\boldsymbol{\eta}}_s)$ is easy to compute in the case of the Gaussian response density. Indeed, this is true for any response density that has an explicit expression.

Liu and Rue (2022) introduced an efficient and accurate method to approximate calculation of leave-group-out cross-validation (LGOCV) in R-INLA and proposed an automatic procedure to construct groups for LGOCV to evaluate the predictive performance when the prediction task is not specified. This method applies to both Gaussian and non-Gaussian specifications at the response level.

2.2 Bayesian LGMs with a Univariate Link Function

In this section, we define the Bayesian latent Gaussian model with a univariate link function. Under this model, the density of the observed response, y_i , is such that only one of its parameters is modeled at the latent level as opposed to two parameters or more. It is an extension of the Gaussian–Gaussian model defined in (2), in particular, the Gaussian assumption at the response level is relaxed. For example, the response variables can be counts, binary variables, or extremes, to name a few types of data that cannot be modeled properly with a Gaussian distribution. Counts can often be modeled with the Poisson distribution, binary variables are usually modeled with the binomial distribution, observed extremes are frequently modeled with the generalized extreme value distribution, and in the case of other types of non-Gaussian responses, other classes of distributions can be applied to model the response variables adequately well.

2.2.1 The Structure of Bayesian LGMs with a Univariate Link Function

In Sect. 2.2 and its subsections, the focus is on Bayesian latent Gaussian models specified such that one of the parameters in the density of the observed response, y_i , denoted by μ_i , usually the location parameter, is transformed with a so-called link function of the form $g(\mu_i) = \eta_i$ (Rue et al., 2009), and η_i is modeled as a linear combination of covariates and random effects. The coefficients of the covariates and the random effects are modeled at the latent level. Alternatively, an LGM can be such that two or more parameters in the density of the observed response are modeled at the latent level; these models will be introduced in Sect. 2.3. The transformation can be the unity transformation as in the case of the Gaussian–Gaussian model, it can be the logarithmic transformation, which is commonly

used, e.g., for the mean in a Poisson density, and in the case of the probability parameter in the binomial model, the logit transformation is a common choice, i.e., $g(\mu_i) = \log(\mu_i) - \log(1 - \mu_i)$.

The linear combination η_i is also referred to as a structured additive predictor. It has the same form as the right-hand side of (2), and covariates and random effects are added to the model through it, that is,

$$\eta_i = \beta_0 + \sum_{k=1}^K \beta_k v_{i,k} + \sum_{j=1}^J u_j a_{i,j} + \epsilon_i, \quad (14)$$

where, as in Sect. 2.1.1, β_0 denotes the intercept, the $v_{i,k}$'s are covariates, the β_k 's are the coefficients of the covariates, the u_j 's are the random effects, and the $a_{i,j}$'s are the corresponding weights of the u_j 's. The total number of observations is n . The parameters β_0 , the β_k 's, and the u_j 's are modeled jointly with a Gaussian prior distribution, and the last terms, the ϵ_i 's, are error terms modeled as independent Gaussian variables. The parameters β_0 , the β_k 's, the u_j 's, and the ϵ_i 's are referred to as the latent parameters, and hence, the name, latent Gaussian model. The variability in η_i , which the covariates and the random effects cannot capture, is modeled with ϵ_i . In some cases, ϵ_i may be very small, and it would be reasonable to set it equal to zero. However, in these cases, it is often better to model ϵ_i with a known small variance to ensure computational stability.

The hierarchical structure of the latent Gaussian model with a single predictor is specified through three levels: namely, the response level, the latent level, and the hyperparameter level. Let $\mathbf{x}$ contain the latent parameters β_0 , the β_k 's, the u_j 's, and the ϵ_i 's. Alternatively, the vector $\mathbf{x}$ can consist of β_0 , the β_k 's, the u_j 's, and the η_i 's as opposed to the ϵ_i 's. In both cases, the prior density of the parameters in $\mathbf{x}$ is Gaussian conditional on the hyperparameters of the latent Gaussian model, which are stored in the vector $\boldsymbol{\theta}$. The prior density of $\mathbf{x}$ conditional on $\boldsymbol{\theta}$ is specified below.

The latent Gaussian model presented here also adopts the conditional independence assumption like the Gaussian–Gaussian model introduced in Sect. 2.1, namely, the observed responses, y_i , are independent of each other conditional on the vector $\mathbf{x}$. The three levels of the latent Gaussian model are as follows.

Response Level The observed responses in $\mathbf{y}$ depend on the latent parameters in $\mathbf{x}$, and potentially on hyperparameters in $\boldsymbol{\theta}$, and have a non-Gaussian density $\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})$. The conditional independence assumption entails that $\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})$ factorizes as $\prod_i \pi(y_i|\mathbf{x}, \boldsymbol{\theta})$, where $\pi(y_i|\mathbf{x}, \boldsymbol{\theta}) = \pi(y_i|\eta_i, \boldsymbol{\theta})$. Most commonly, the linear combination η_i is linked to the mean of y_i , i.e., $\mu_i = E(y_i|\eta_i) = g^{-1}(\eta_i)$, where $g^{-1}(\cdot)$ is the inverse of $g(\cdot)$. Another setup involves η_i being linked to the p quantile, i.e., $\mu_i = Q_p(y_i|\eta_i) = g^{-1}(\eta_i)$, where μ_i is defined in terms of the quantile function Q_p stemming from the density $\pi(y_i|\eta_i, \boldsymbol{\theta})$ and $0 < p < 1$.

Latent Level By the definition of latent Gaussian models, a Gaussian density is assigned to the latent parameters in $\mathbf{x}$. This density might depend on the hyperparameters in $\boldsymbol{\theta}$, i.e., the mean and the covariance matrix of the Gaussian

density may depend on θ . The density of $\mathbf{x}$ is

$$\pi(\mathbf{x}|\theta) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_x(\theta), \Sigma_x(\theta)),$$

i.e., a Gaussian density with mean vector $\boldsymbol{\mu}_x(\theta)$ and covariance matrix $\Sigma_x(\theta)$. This Gaussian prior can be expressed in terms of a precision matrix $Q_x(\theta)$ as opposed to a covariance matrix by replacing $\Sigma_x(\theta)$ with $Q_x^{-1}(\theta)$.

Hyperparameter Level The hyperparameters in θ are modeled with the prior density $\pi(\theta)$. A hyperparameter in θ can be the marginal variance of a random effect, and it can, for example, be a dependence parameter such as correlation, or the range of a spatial random effect or its smoothness. A discussion about prior densities for the hyperparameters is given in Sect. 3.3.

Let $\boldsymbol{\eta}$ denote the vector containing the η_i parameters. The vector version of (14) is

$$\boldsymbol{\eta} = X\boldsymbol{\beta} + A\mathbf{u} + \boldsymbol{\epsilon}, \quad (15)$$

where X , $\boldsymbol{\beta}$, A , $\mathbf{u}$, and $\boldsymbol{\epsilon}$ have the same structure as in Sect. 2.1. The vector $\mathbf{u}$ may consist of L vectors, $\mathbf{u}_l, l \in \{1, \dots, L\}$, and then $A\mathbf{u}$ can be presented as $\sum_{l=1}^L A_l \mathbf{u}_l$, see Sect. 2.1.1.

2.2.2 Posterior Inference for LGMs with a Univariate Link Function Using INLA

Here, posterior inference for the latent Gaussian models specified in Sect. 2.2.1 is considered. It is assumed that $\boldsymbol{\epsilon}$ is either zero or non-zero and that $\mathbf{u}$ consists of L vectors. Let $Z = (X \ A_1 \ \dots \ A_L)$ correspond to $\boldsymbol{\epsilon} = \mathbf{0}$; then $\mathbf{x} = (\boldsymbol{\beta}^\top, \mathbf{u}^\top)^\top$, $\boldsymbol{\eta} = Z\mathbf{x}$, $\boldsymbol{\mu}_x = (\boldsymbol{\mu}_\beta^\top, \boldsymbol{\mu}_{u,1}^\top, \dots, \boldsymbol{\mu}_{u,L}^\top)^\top$ and $Q_x = \text{bdiag}(Q_\beta, Q_{u,1}, \dots, Q_{u,L})$. When $\boldsymbol{\epsilon}$ is not a vector of zeros, then we set $Z = (X \ A_1 \ \dots \ A_L \ I)$, $\mathbf{x} = (\boldsymbol{\beta}^\top, \mathbf{u}^\top, \boldsymbol{\epsilon}^\top)^\top$, $\boldsymbol{\eta} = Z\mathbf{x}$, $\boldsymbol{\mu}_x = (\boldsymbol{\mu}_\beta^\top, \boldsymbol{\mu}_{u,1}^\top, \dots, \boldsymbol{\mu}_{u,L}^\top, \mathbf{0}^\top)^\top$ and $Q_x = \text{bdiag}(Q_\beta, Q_{u,1}, \dots, Q_{u,L}, Q_\epsilon)$.

The posterior density of the unknown parameters $\mathbf{x}$ and θ is

$$\begin{aligned} \pi(\mathbf{x}, \theta | \mathbf{y}) &\propto \pi(\theta) \pi(\mathbf{x} | \theta) \pi(\mathbf{y} | \mathbf{x}, \theta) \\ &= \pi(\theta) \pi(\mathbf{x} | \theta) \prod_i \pi(y_i | \mathbf{x}, \theta). \end{aligned} \quad (16)$$

Several approaches can be applied to obtain summary statistics from this posterior density. The approach that will be presented here is the integrated nested Laplace approximation (INLA) (Rue et al., 2009, 2017).

The central approximation in INLA, often called the Laplace approximation, is a high-dimensional quadratic approximation of the conditional log-posterior,

$$f(\mathbf{x}) = \log \pi(\mathbf{x}|\mathbf{y}, \boldsymbol{\theta}), \quad (17)$$

around some vector $\mathbf{x}^*$. We condition on a fixed $\boldsymbol{\theta}$ for the quadratic approximation. We later discuss how to infer $\mathbf{x}$ by either integrating over $\boldsymbol{\theta}$ or by using a plug-in estimate. Using Bayes' rule, the function f can be written as

$$f(\mathbf{x}) = \log \pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta}) + \log \pi(\mathbf{x}|\boldsymbol{\theta}) + c,$$

for some constant c , where the first term is the log-likelihood of $\mathbf{x}$, and the second term is the log-prior of $\mathbf{x}$.

To build up this quadratic approximation, we start with a Taylor series of the log-likelihood in terms of $\boldsymbol{\eta}$,

$$\log \pi(\mathbf{y}|\boldsymbol{\eta}) \approx a + \mathbf{b}^\top (\boldsymbol{\eta} - \boldsymbol{\eta}^*) - \frac{1}{2} (\boldsymbol{\eta} - \boldsymbol{\eta}^*)^\top C (\boldsymbol{\eta} - \boldsymbol{\eta}^*), \quad (18)$$

around some $\boldsymbol{\eta}^*$, where a is a constant. This is a multivariate approximation; hence, $\mathbf{b}$ is a vector and C is a matrix. When the model fulfills the conditional independence assumption $\pi(\mathbf{y}|\boldsymbol{\eta}) = \prod_i \pi(y_i|\eta_i)$, this part of the approximation can be computed element by element, and C is a diagonal matrix. Using $\boldsymbol{\eta} = Z\mathbf{x}$, we get

$$\log \pi(\mathbf{y}|\mathbf{x}) \approx a + \mathbf{b}^\top Z(\mathbf{x} - \mathbf{x}^*) - \frac{1}{2} (\mathbf{x} - \mathbf{x}^*)^\top Z^\top C Z (\mathbf{x} - \mathbf{x}^*). \quad (19)$$

Since the prior for $\mathbf{x}$ is Gaussian, its log-prior is a quadratic polynomial, namely, $\log \pi(\mathbf{x}|\boldsymbol{\theta}) = c' - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_x)^\top Q_x (\mathbf{x} - \boldsymbol{\mu}_x)$; hence,

$$\begin{aligned} f(\mathbf{x}) &\approx c_0 + \left(\mathbf{b}^\top Z + Q_x \boldsymbol{\mu}_x \right) (\mathbf{x} - \mathbf{x}^*) \\ &\quad - \frac{1}{2} (\mathbf{x} - \mathbf{x}^*)^\top \left(Z^\top C Z + Q_x \right) (\mathbf{x} - \mathbf{x}^*), \end{aligned} \quad (20)$$

where c' and c_0 are some constants.

To find an $\mathbf{x}^*$ where this is a good approximation of f , we initialize $\mathbf{x}^*$ at some vector (e.g., a vector of zeroes) and then iteratively optimize it, see Algorithm 1. At the final $\mathbf{x}^*$, we get an approximate Gaussian density,

$$\pi(\mathbf{x}|\mathbf{y}, \boldsymbol{\theta}) \approx \pi_G(\mathbf{x}|\mathbf{y}, \boldsymbol{\theta}) = \mathcal{N} \left(\mathbf{x} \middle| Q_*^{-1} \left(\mathbf{b}^\top Z + Q_x \boldsymbol{\mu}_x \right), Q_*^{-1} \right), \quad (21)$$

where $Q_* = (Z^\top C Z + Q_x)$. Note that all these variables depend on $\boldsymbol{\theta}$, e.g., $Q_x = Q_x(\boldsymbol{\theta})$, except for Z . Also, all computations here take advantage of sparse matrices,

Require: $(\mu_x, Q_x, Z, L1, L2)$
1: $\mathbf{x}^* = \text{rep}(0, \text{nrow}(Q_x))$
2: $\text{iterdiff} = 1$
3: **while** $\text{iterdiff} > 1\text{E-}6$ **do**
4: $\boldsymbol{\eta}^* = Z\mathbf{x}^*$
5: $\mathbf{b}_* = L1(\boldsymbol{\eta}^*)^\top Z + Q_x \mu_x$
6: $Q_* = -Z^\top L2(\boldsymbol{\eta}^*)Z + Q_x$
7: $\mu_* = Q_*^{-1} \mathbf{b}_*$
8: $\text{iterdiff} = \max_i |\mu_{*i} - x_i^*|$
9: $\mathbf{x}^* = \mu_*$
10: **end while**
Ensure: (μ_*, Q_*)

Algorithm 1: Iteratively computing the Taylor approximation of $f(\mathbf{x})$. The input $L1$ is the function producing the first derivative of the likelihood $\mathcal{L}(\boldsymbol{\eta}; y)$ with respect to $\boldsymbol{\eta}$, and $L2$ is the same for the second derivative. The output are the mean and precision of the Gaussian approximation of the conditional posterior density for $\mathbf{x}|\mathbf{y}, \boldsymbol{\theta}$

and the inverse of C is never computed. Using this approximation, we can follow Eq. (11), and we can use different approaches to estimate $\boldsymbol{\theta}$.

If the likelihood is Gaussian, the approximation is exact, as the Taylor polynomial of the log density is the density itself, and Eq. (21) reduces to Eq. (8) with mean and precision according to (9) and (10), respectively. Most common likelihoods have a shape close to the Gaussian density, and coupled with having a Gaussian prior for $\boldsymbol{\eta}$, the approximation works very well. Let us look at a typical worst-case scenario, where the Taylor approximation of the likelihood is challenging. With a Binomial($N = 4$) likelihood, observing y to be 1, 2, or 3 gives a very good approximation, but observing $y = 0$ or $y = 4$ is more tricky, due to the likelihood being flat for some values of the parameter $\boldsymbol{\eta}$. Figure 1 shows that the likelihood for $\boldsymbol{\eta}$ when $y = 0$ is approximately flat for values of $\boldsymbol{\eta}$ below -6 . This means that any flat (or nearly flat) prior would give a function that is not well-approximated by a quadratic Taylor expansion. However, since we use a prior for $\boldsymbol{\eta}$, this problem is almost always resolved. To illustrate what can happen, we add a $\mathcal{N}(0, 1)$ prior as an example of a prior density and then compute the approximation in Fig. 2. The approximation is very good in this case. Most models have some type of pooling or smoothing, which means that the effect of the prior on a single observation is quite strong, and/or the likelihood is not flat; hence, the quadratic approximation is very good. However, if there are (e.g., geographically) isolated points where the likelihood is flat, the quadratic approximation can be poor.

We discuss two INLA approaches, which we call Simple-INLA and Full-INLA. For both approaches, the starting point is to use

$$\pi(\boldsymbol{\theta}|\mathbf{y}) \propto \pi(\boldsymbol{\theta})\pi(\mathbf{y}|\boldsymbol{\theta}) = \frac{\pi(\boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})}{\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})} = \frac{\pi(\mathbf{x}, \boldsymbol{\theta}, \mathbf{y})}{\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})}, \quad (22)$$

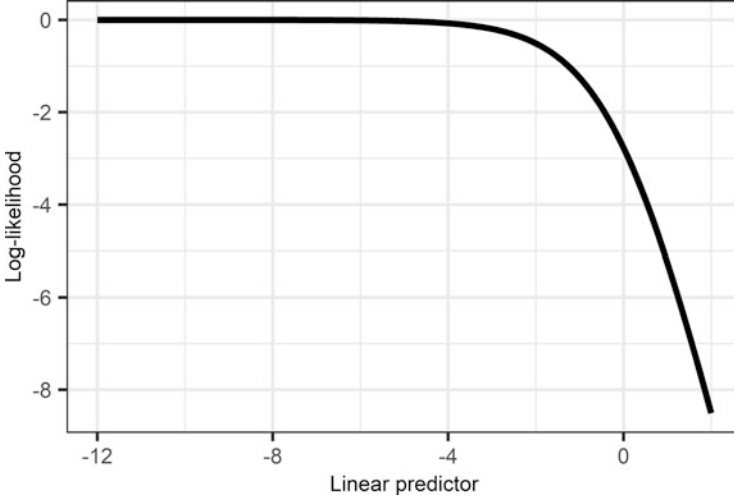


Fig. 1 Log-likelihood of η with a Binomial($N = 4$) likelihood with logit link, conditioned on observing $y = 0$. Any quadratic approximation of this function will not be accurate for all values of η

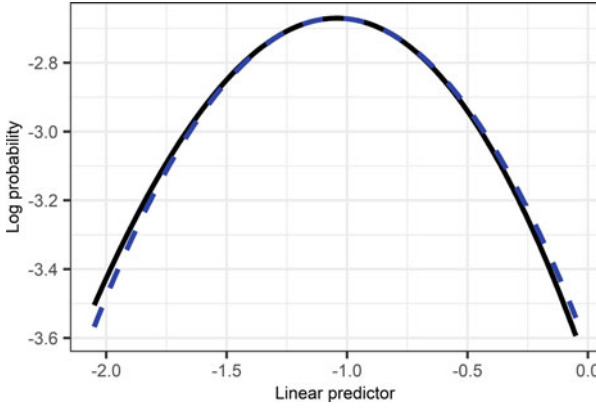


Fig. 2 In black, the log-probability of the posterior $\eta|(y = 0)$ observed with a Binomial($N = 4$) likelihood with logit link, and using a $\mathcal{N}(0, 1)$ prior. In blue, the Laplace approximation, which is a quadratic approximation around $\eta = -1.05$

and approximate the last term with

$$\frac{\pi(\mathbf{x}, \boldsymbol{\theta}, \mathbf{y})}{\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})} \approx \frac{\pi(\mathbf{x} = \mathbf{x}^*, \boldsymbol{\theta}, \mathbf{y})}{\pi_G(\mathbf{x} = \mathbf{x}^*|\boldsymbol{\theta}, \mathbf{y})}, \quad (23)$$

where π_G denotes the quadratic approximation we just discussed. The approximation is evaluated at $\mathbf{x}^*$ because this is where the approximation is the most accurate,

but it could in theory be evaluated anywhere. Simple-INLA finds the maximum posterior for θ in a “good parametrization” (see the R-INLA documentation for the internal parametrizations). This “good parametrization” is chosen such that the posterior mode and mean are usually close. If we use this $\hat{\theta}$ estimate as a plug-in estimate, we can use the quadratic approximation directly,

$$\pi(\mathbf{x}|\mathbf{y}) \approx \pi_G(\mathbf{x}|\mathbf{y}, \theta = \hat{\theta}). \quad (24)$$

However, the posterior density of $\theta|\mathbf{y}$ is then a point mass at $\hat{\theta}$, which is a very poor approximation. This is especially problematic if we want to investigate the uncertainty in $\theta|\mathbf{y}$.

For Full-INLA, we do a numerical integration over θ . The goal of this integration is to compute the marginals

$$\pi(x_i|\mathbf{y}) = \int \pi(x_i|\theta, \mathbf{y})\pi(\theta|\mathbf{y})d\theta. \quad (25)$$

From the quadratic approximation, we can compute the conditional marginals according to Eqs. (4) and (5) in Rue et al. (2009), using

$$\pi(x_i|\theta, \mathbf{y}) \approx \mathcal{N}\left(x_i \middle| \mu_i(\theta), \sigma_i^2(\theta)\right), \quad (26)$$

$$\pi(x_i|\mathbf{y}) \approx \sum_k \pi(x_i|\theta^{(k)}, \mathbf{y})\pi(\theta^{(k)}|\mathbf{y})\Delta_k, \quad (27)$$

where $\theta^{(k)}$ are the grid points in a discretization of θ -space, and Δ_k are the volume weights in the discretization. For details on how to set up this discretization grid, see Rue et al. (2009). Furthermore, Full-INLA performs a second Laplace approximation in order to compute these marginals and has a technique for integrating over parts of the θ -space to get the marginals of θ in a stable way (Rue et al., 2009).

2.2.3 Predictions Based on LGMs with a Univariate Link Function

Predictions of a non-Gaussian response that is modeled with a Bayesian LGM with a univariate link function are based on the assumed distribution of the response and the linear predictor of the LGM. Let us assume that the random effect vector $\mathbf{u}$ contains the elements that are needed for the predictions we want to make. Let $\tilde{\mathbf{y}}$ denote the predicted response, and let $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{A}}$ be matrices that store the corresponding values of the covariates and the weights of the random effects, respectively. The posterior

predictive density is

$$\pi(\tilde{y}|\mathbf{y}) = \int \pi(\tilde{y}|\tilde{\eta})\pi(\tilde{\eta}|\tilde{X}, \boldsymbol{\beta}, \tilde{A}, \mathbf{u}, \boldsymbol{\theta})\pi(\boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\beta}d\mathbf{u}d\boldsymbol{\theta}, \quad (28)$$

where $\pi(\tilde{y}|\tilde{\eta})$ is the non-Gaussian density in the LGM, the parameter of the i -th response variable to be predicted, $\tilde{y}_i$, is $g(\tilde{\eta}_i)$, and $g(\cdot)$ is the univariate link function of the LGM. Furthermore, $\pi(\tilde{\eta}|\tilde{X}, \boldsymbol{\beta}, \tilde{A}, \mathbf{u}, \boldsymbol{\theta})$ is the conditional Gaussian density of $\tilde{\eta}$, with mean $\tilde{X}\boldsymbol{\beta} + \tilde{A}\mathbf{u}$ and variance $\sigma_\epsilon^2 I$, and $\pi(\boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta}|\mathbf{y})$ is the posterior density.

In order to sample from the posterior predictive density in (28), three steps are needed. First, samples are drawn from the posterior density, second, samples are drawn from the linear predictor $\tilde{\eta}$ through the model

$$\tilde{\eta} = \tilde{X}\boldsymbol{\beta} + \tilde{A}\mathbf{u} + \tilde{\epsilon},$$

where $\boldsymbol{\beta}$ and $\mathbf{u}$ are sampled from the posterior density, and $\tilde{\epsilon}$ denotes samples from a Gaussian density with mean zero and variance σ_ϵ^2 , and samples of this variance are drawn from the posterior density. Note that if the LGM assumes $\boldsymbol{\epsilon} = \mathbf{0}$, then $\tilde{\eta}$ is sampled with $\tilde{\epsilon} = \mathbf{0}$. In the third step, samples are drawn from the non-Gaussian density of the LGM with the parameters in $\tilde{\eta}$.

As mentioned in Sect. 2.1.3, the posterior predictive density can be used for various purposes, such as making temporal and spatial predictions, performing cross-validation, and testing the distributional assumptions at the response level.

2.3 Bayesian LGMs with a Multivariate Link Function

Certain types of response variables are such that parameters other than the location parameter require an advanced model, for example, the scale parameter or a set of regression parameters. One example is data with a spatial structure where the observed responses at each site can be modeled with a probability distribution with two or more parameters. If these parameters vary spatially, then a model for each parameter might improve inference and be useful for predictions. Furthermore, if a transformation of each parameter can be modeled adequately well with a linear predictor, then a Bayesian hierarchical model with a Gaussian assumption on the latent parameters could be a sensible model choice. In this case, a latent Gaussian model with a linear predictor for only one of the parameters will not be sufficient because the density of each observation contains two or more parameters that need to be modeled at the latent level. Thus, an extension of Bayesian LGMs with a univariate link function is needed, referred to as Bayesian LGMs with a multivariate link function. These models will be presented here. We will also refer to them here as extended Bayesian LGMs in line with Geirsson et al. (2020).

2.3.1 The Structure of Bayesian LGMs with a Multivariate Link Function

In general, the number of parameters that are modeled at the latent level with a linear predictor can be any number greater than one, and the grouping of the data does not have to be based on spatial grouping, and it could be based on temporal or spatio-temporal categories, or categories that are neither spatial nor temporal. For simplicity, let us consider a model for data that are observed at several geological sites, multiple measurements are taken at each site, conditional independence can be assumed, and each observed response is modeled with a probability density with three parameters. Denote the number of sites by G . Let $y_{i,j}$ denote the j -th observation from site i , and assume the number of observations at site i is n_i . The density of $y_{i,j}$ is denoted by $\pi(y_{i,j}|\mu_i, \sigma_i, \zeta_i)$. The three parameters are transformed jointly with the link function $g(\cdot)$ to a three-dimensional vector such that each of its elements is on the real line, that is,

$$(\eta_{1,i}, \eta_{2,i}, \eta_{3,i}) = g(\mu_i, \sigma_i, \zeta_i).$$

The transformed parameters, $\eta_{1,i}$, $\eta_{2,i}$, and $\eta_{3,i}$, are modeled with linear predictors with the form given in (14), that is,

$$\eta_{m,i} = \beta_{m,0} + \sum_{k=1}^K \beta_{m,k} v_{m,i,k} + \sum_{s=1}^J u_{m,s} a_{m,i,s} + \epsilon_{m,i}, \quad (29)$$

where m refers to the parameter and here $m \in \{1, 2, 3\}$. The vector version of these equations is

$$\begin{aligned} \boldsymbol{\eta}_1 &= X_1 \boldsymbol{\beta}_1 + A_1 \mathbf{u}_1 + \boldsymbol{\epsilon}_1, \\ \boldsymbol{\eta}_2 &= X_2 \boldsymbol{\beta}_2 + A_2 \mathbf{u}_2 + \boldsymbol{\epsilon}_2, \\ \boldsymbol{\eta}_3 &= X_3 \boldsymbol{\beta}_3 + A_3 \mathbf{u}_3 + \boldsymbol{\epsilon}_3, \end{aligned} \quad (30)$$

where X_1 , X_2 , and X_3 are design matrices containing covariates, $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ are the corresponding regression coefficients, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ are random effects, and A_1 , A_2 , and A_3 are their corresponding weight matrices. Furthermore, $\boldsymbol{\epsilon}_1$, $\boldsymbol{\epsilon}_2$, and $\boldsymbol{\epsilon}_3$ are vectors of independent and unstructured mean zero error terms, referred to as model errors. The vectors $\boldsymbol{\beta}_m$, $\mathbf{u}_m$, and $\boldsymbol{\epsilon}_m$ are assigned Gaussian prior densities and assumed to be a priori mutually independent for each $m \in \{1, 2, 3\}$.

The means of the vectors $\boldsymbol{\beta}_1$, $\mathbf{u}_1$, $\boldsymbol{\beta}_2$, $\mathbf{u}_2$, $\boldsymbol{\beta}_3$, and $\mathbf{u}_3$ are denoted by $\boldsymbol{\mu}_{\beta,1}$, $\boldsymbol{\mu}_{u,1}$, $\boldsymbol{\mu}_{\beta,2}$, $\boldsymbol{\mu}_{u,2}$, $\boldsymbol{\mu}_{\beta,3}$, and $\boldsymbol{\mu}_{u,3}$, respectively, and their precision matrices are denoted by $Q_{\beta,1}$, $Q_{u,1}$, $Q_{\beta,2}$, $Q_{u,2}$, $Q_{\beta,3}$, and $Q_{u,3}$, respectively. The means and precision matrices of $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ are assumed to be fixed, while hyperparameters, denoted by $\boldsymbol{\theta}$, govern the means and the precision matrices of the random effects, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$.

2.3.2 Posterior Inference for LGMs with a Multivariate Link Function

In this section, we outline an inference scheme for extended LGMs, referred to as Max-and-Smooth (Hrafnkelsson et al., 2021). An alternative inference scheme for these models, the LGM split sampler (Geirsson et al., 2020), is presented in the Appendix. First, we form the posterior density of the extended latent Gaussian model. The vectors of the model in (30) are combined into three vectors,

$$\boldsymbol{\eta} = (\boldsymbol{\eta}_1^\top, \boldsymbol{\eta}_2^\top, \boldsymbol{\eta}_3^\top)^\top, \quad \mathbf{v} = (\boldsymbol{\beta}_1^\top, \mathbf{u}_1^\top, \boldsymbol{\beta}_2^\top, \mathbf{u}_2^\top, \boldsymbol{\beta}_3^\top, \mathbf{u}_3^\top)^\top, \quad \boldsymbol{\epsilon} = (\boldsymbol{\epsilon}_1^\top, \boldsymbol{\epsilon}_2^\top, \boldsymbol{\epsilon}_3^\top)^\top.$$

Thus, the prior mean and precision of $\mathbf{v}$ are

$$\boldsymbol{\mu}_v = (\boldsymbol{\mu}_{\beta,1}^\top, \boldsymbol{\mu}_{u,1}^\top, \boldsymbol{\mu}_{\beta,2}^\top, \boldsymbol{\mu}_{u,2}^\top, \boldsymbol{\mu}_{\beta,3}^\top, \boldsymbol{\mu}_{u,3}^\top)^\top, \\ Q_v = \text{bdiag}(Q_{\beta,1}, Q_{u,1}, Q_{\beta,2}, Q_{u,2}, Q_{\beta,3}, Q_{u,3}).$$

The precision matrices of $\boldsymbol{\epsilon}_1$, $\boldsymbol{\epsilon}_2$, and $\boldsymbol{\epsilon}_3$ are denoted by $Q_{\epsilon,m}$, $m \in \{1, 2, 3\}$, where $Q_{\epsilon,m} = \sigma_{\epsilon,m}^{-2} I$, I is the identity matrix, and $\sigma_{\epsilon,m}$ are the corresponding standard deviations. The precision matrix of $\boldsymbol{\epsilon}$ is thus given by

$$Q_\epsilon = \text{bdiag}(Q_{\epsilon,1}, Q_{\epsilon,2}, Q_{\epsilon,3}).$$

The matrix Z is based on X_1 , A_1 , X_2 , A_2 , X_3 , and A_3 and given by

$$Z = \begin{pmatrix} X_1 & A_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & X_2 & A_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & X_3 & A_3 \end{pmatrix},$$

where the zeros denote zero matrices. Now (30) can be rewritten as

$$\boldsymbol{\eta} = Z\mathbf{v} + \boldsymbol{\epsilon}.$$

Given the above assumptions, then the prior density of $\mathbf{v}$, $\pi(\mathbf{v}|\boldsymbol{\theta})$, is Gaussian with mean $\boldsymbol{\mu}_v$ and precision Q_v , and the density of $\boldsymbol{\eta}$ conditional on $\mathbf{v}$, $\pi(\boldsymbol{\eta}|\mathbf{v}, \boldsymbol{\theta})$, is Gaussian with mean $Z\mathbf{v}$ and precision Q_ϵ .

The posterior density of the unknown parameters $(\boldsymbol{\eta}, \mathbf{v}, \boldsymbol{\theta})$ is

$$\pi(\boldsymbol{\eta}, \mathbf{v}, \boldsymbol{\theta}|\mathbf{y}) \propto \pi(\boldsymbol{\theta})\pi(\boldsymbol{\eta}, \mathbf{v}|\boldsymbol{\theta})\pi(\mathbf{y}|\boldsymbol{\eta}),$$

where $\mathbf{y}$ is a vector containing the response, $\pi(\mathbf{y}|\boldsymbol{\eta})$ is the response density, $\pi(\boldsymbol{\eta}, \mathbf{v}|\boldsymbol{\theta})$ is the joint Gaussian prior density of $\boldsymbol{\eta}$, and $\mathbf{v}$; in fact, it can be specified with $\pi(\mathbf{v}|\boldsymbol{\theta})\pi(\boldsymbol{\eta}|\mathbf{v}, \boldsymbol{\theta})$, and $\pi(\boldsymbol{\theta})$ is the prior density for the hyperparameters.

Here, we show how Max-and-Smooth (Hrafnkelsson et al., 2021) can be applied to the above posterior density. Max-and-Smooth is a two-step approximate inference

scheme for extended latent Gaussian models. The first step involves approximating the likelihood function with a Gaussian density function (maximization). The second step involves fitting the model from the first step using the Gaussian prior densities specified at the latent level (smoothing). The two steps are equivalent to fitting a Gaussian–Gaussian model. The term Max stands for the first step, i.e., the fit of the Gaussian density to the likelihood is maximized, while the term Smooth refers to the second step, i.e., the information in the approximate likelihood is smoothed with respect to the prior densities of η , $\mathbf{v}$, and θ . Hrafnkelsson et al. (2021) proposed two methods to approximate the likelihood function. The first approximation is based on the mode and the Hessian matrix of the log-likelihood function, i.e., essentially maximum likelihood estimation is applied to the likelihood function. The second approximation is based on normalizing the likelihood function such that it integrates to one, which gives it the properties of a probability density function, and then finding the mean and variance of the normalized likelihood function. Here, the focus will be on the first approximation, but a few details on the second approximation will be given below.

The likelihood function of the extended LGM is denoted by $L(\eta|\mathbf{y})$. In general, the likelihood function is equal to the response density when it is treated as a function of the parameters, and thus, $L(\eta|\mathbf{y}) = \pi(\mathbf{y}|\eta)$. The Gaussian approximation of $L(\eta|\mathbf{y})$ is denoted by $\hat{L}$. Since $\hat{L}$ is normalized and L is not, then $c\hat{L}(\eta|\mathbf{y}) \approx L(\eta|\mathbf{y})$, where

$$\hat{L}(\eta|\mathbf{y}) = \mathcal{N}(\eta|\hat{\eta}, \Sigma_{\eta\mathbf{y}}),$$

and c is a constant that is independent of η . The mean of the Gaussian density, $\hat{\eta}$, is the maximum likelihood estimate for η , i.e., the mode of $L(\eta|\mathbf{y})$, and its variance, $\Sigma_{\eta\mathbf{y}}$, is based on the Hessian matrix of $\log(L(\eta|\mathbf{y}))$, $H_{\eta\mathbf{y}}$, evaluated at $\hat{\eta}$, i.e., $\Sigma_{\eta\mathbf{y}} = (-H_{\eta\mathbf{y}})^{-1}$. The matrix $-H_{\eta\mathbf{y}}$ is referred to as the observed information (e.g., Schervish, 1995).

The extended latent Gaussian model presented here assumes conditional independence at the response level, i.e., the responses are independent conditional on the latent parameters. This means that the response density factorizes as $\pi(\mathbf{y}|\eta) = \prod_{i=1}^G \pi(\mathbf{y}_i|\eta_i)$, where $\mathbf{y}_i$ and η_i are the observed responses and the parameters at site i , respectively. The likelihood function $L(\eta|\mathbf{y})$ factorizes in the same way,

$$L(\eta|\mathbf{y}) = \prod_{i=1}^G L(\eta_i|\mathbf{y}_i),$$

where $L(\eta_i|\mathbf{y}_i) = \pi(\mathbf{y}_i|\eta_i)$ is the likelihood contribution of site i . The Gaussian approximation of $L(\eta|\mathbf{y})$ can now be based on the joint likelihood contributions of the sites, that is, each $L(\eta_i|\mathbf{y}_i)$ is approximated with $c_i \hat{L}(\eta_i|\mathbf{y}_i)$, where

$$\hat{L}(\eta_i|\mathbf{y}_i) = \mathcal{N}(\eta_i|\hat{\eta}_i, \Sigma_{\eta_i\mathbf{y}_i}),$$

c_i is a constant that is independent of η_i , $\hat{\eta}_i$ is the maximum likelihood estimate of η_i , based on $L(\eta_i | y_i)$ only, and $\Sigma_{\eta y_i}$ is the inverse of the negative Hessian matrix of $L(\eta_i | y_i)$ evaluated at $\hat{\eta}_i$, i.e., $\Sigma_{\eta y_i} = (-H_{\eta y_i})^{-1}$. The matrix $-H_{\eta y_i}$ is the observed information corresponding to the parameters at the i -th site, derived from $L(\eta_i | y_i)$. Thus, the approximated full likelihood is given by $\hat{L}(\eta | y) = \prod_{i=1}^G \hat{L}(\eta_i | y_i)$, and the approximated posterior density, $\hat{\pi}(\eta, v, \theta | y)$, can now be presented as

$$\begin{aligned} \hat{\pi}(\eta, v, \theta | y) &\propto \pi(\theta) \pi(\eta, v | \theta) \hat{L}(\eta | y) \\ &\propto \pi(\theta) \pi(\eta, v | \theta) \prod_{i=1}^G \hat{L}(\eta_i | y_i) \\ &\propto \pi(\theta) \pi(\eta, v | \theta) \prod_{i=1}^G \mathcal{N}(\eta_i | \hat{\eta}_i, \Sigma_{\eta y_i}). \end{aligned} \quad (31)$$

The second Gaussian approximation to the posterior density presented in Hrafnkelsson et al. (2021) has the same form as (31); however, $\hat{\eta}_i$ and $\Sigma_{\eta y_i}$ are replaced by the mean and the covariance of the normalized version of the likelihood function $L(\eta_i | y_i)$. When the normalization constant is not finite, then a more adequate model parametrization may be selected. If such a parametrization cannot be found, then the likelihood function can be replaced by an alternative generalized likelihood that consists of the likelihood times an extra prior density for η . For further details, see Hrafnkelsson et al. (2021).

To infer the model parameters through the approximate posterior density in (31), a model for $\hat{\eta}$ is set forth. In this model, the elements of the vector $\hat{\eta}$ are treated as noisy measurements of the latent parameters. Motivated by the approximate posterior density in (31), it is assumed that $\hat{\eta} \sim \mathcal{N}(\eta, Q_{\eta y}^{-1})$, where $Q_{\eta y}$ is known. The numerical values of $Q_{\eta y}$ are evaluated from the likelihood of the already observed responses, in fact, $Q_{\eta y} = -H_{\eta y} = \Sigma_{\eta y}^{-1}$. The model for $\hat{\eta}$ is referred to as the surrogate-model. Its hierarchical structure is

$$\begin{aligned} \pi(\hat{\eta} | \eta, Q_{\eta y}, \theta) &= \mathcal{N}(\hat{\eta} | \eta, Q_{\eta y}^{-1}), \\ \pi(\eta | v, \theta) &= \mathcal{N}(\eta | Zv, Q_{\epsilon}^{-1}), \\ \pi(v | \theta) &= \mathcal{N}(v | \mu_v, Q_v^{-1}), \end{aligned} \quad (32)$$

and the prior density for the hyperparameters is the same as before. The posterior density for the unknown model parameters, η , v , and θ , is

$$\begin{aligned} \pi(\eta, v, \theta | \hat{\eta}) &\propto \pi(\theta) \pi(\eta, v | \theta) \pi(\hat{\eta} | \eta, Q_{\eta y}, \theta) \\ &\propto \pi(\theta) \pi(\eta, v | \theta) \mathcal{N}(\hat{\eta} | \eta, Q_{\eta y}^{-1}) \\ &\propto \pi(\theta) \pi(\eta, v | \theta) \hat{L}(\eta | y). \end{aligned}$$

It turns out that this posterior density is exactly the same as the approximated posterior density in (31). That is, the selected Gaussian approximation for the ML estimates in $\hat{\boldsymbol{\eta}}$ in the surrogate-model and the Gaussian approximations of the likelihood function used in (31) lead to the same posterior density.

The densities in (32) reveal that the surrogate-model for $\hat{\boldsymbol{\eta}}$ is a Gaussian–Gaussian model. This makes the inference for the unknown parameters convenient. Samples from the approximate posterior density can be obtained through the setup presented in Sect. 2.1.2, that is, by using the following two steps:

Step 1: Sample $\boldsymbol{\theta}$ from $\pi(\boldsymbol{\theta}|\hat{\boldsymbol{\eta}})$.

Step 2: Sample $\mathbf{x}$ from $\pi(\mathbf{x}|\boldsymbol{\theta}, \hat{\boldsymbol{\eta}})$.

Here, $\mathbf{x} = (\boldsymbol{\eta}^\top, \mathbf{v}^\top)^\top$, and the above densities are different from those in Sect. 2.1.2; however, the density in Step 2 is Gaussian as before. The marginal posterior density of $\boldsymbol{\theta}$ given $\hat{\boldsymbol{\eta}}$ is

$$\pi(\boldsymbol{\theta}|\hat{\boldsymbol{\eta}}) \propto \pi(\boldsymbol{\theta})\pi(\hat{\boldsymbol{\eta}}|\boldsymbol{\theta}) = \pi(\boldsymbol{\theta}) \frac{\pi(\hat{\boldsymbol{\eta}}|\mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})}{\pi(\mathbf{x}|\hat{\boldsymbol{\eta}}, \boldsymbol{\theta})}, \quad (33)$$

where $\pi(\hat{\boldsymbol{\eta}}|\mathbf{x}, \boldsymbol{\theta})$ is Gaussian with mean $\boldsymbol{\eta}$ and precision matrix $\mathbf{Q}_{\eta y}$, while the mean and precision of the Gaussian density $\pi(\mathbf{x}|\boldsymbol{\theta})$ are

$$\boldsymbol{\mu}_x = \begin{bmatrix} \mathbf{Z}\boldsymbol{\mu}_v \\ \boldsymbol{\mu}_v \end{bmatrix}, \quad \mathbf{Q}_x = \begin{bmatrix} \mathbf{Q}_\epsilon & -\mathbf{Q}_\epsilon \mathbf{Z} \\ -\mathbf{Z}^\top \mathbf{Q}_\epsilon & \mathbf{Q}_v + \mathbf{Z}^\top \mathbf{Q}_\epsilon \mathbf{Z} \end{bmatrix}.$$

The conditional posterior density of $\mathbf{x}$, $\pi(\mathbf{x}|\hat{\boldsymbol{\eta}}, \boldsymbol{\theta})$, is Gaussian with precision

$$\mathbf{Q}_{x|\hat{\boldsymbol{\eta}}} = \mathbf{Q}_x + \mathbf{B}^\top \mathbf{Q}_{\eta y} \mathbf{B} = \begin{bmatrix} \mathbf{Q}_\epsilon + \mathbf{Q}_{\eta y} & -\mathbf{Q}_\epsilon \mathbf{Z} \\ -\mathbf{Z}^\top \mathbf{Q}_\epsilon & \mathbf{Q}_v + \mathbf{Z}^\top \mathbf{Q}_\epsilon \mathbf{Z} \end{bmatrix} \quad (34)$$

and mean

$$\boldsymbol{\mu}_{x|\hat{\boldsymbol{\eta}}} = \mathbf{Q}_{x|\hat{\boldsymbol{\eta}}}^{-1} (\mathbf{Q}_x \boldsymbol{\mu}_x + \mathbf{B}^\top \mathbf{Q}_{\eta y} \hat{\boldsymbol{\eta}}),$$

where $\mathbf{B} = [\mathbf{I}_{p \times p} \mathbf{0}_{p \times q}]$, p is the dimension of $\boldsymbol{\eta}$, and q is the dimension of $\mathbf{v}$.

Since this density is Gaussian, it is straightforward to obtain posterior samples of $\mathbf{x}$. The computation will be more efficient if the precision matrix $\mathbf{Q}_{x|\hat{\boldsymbol{\eta}}}$ is sparse. For $\mathbf{Q}_{x|\hat{\boldsymbol{\eta}}}$ to be sparse, $\mathbf{Q}_v$ and $\mathbf{Z}$ need to be sparse. The precision matrix $\mathbf{Q}_\epsilon$ is diagonal, and $\mathbf{Q}_{\eta y}$ is sparse due to the conditional independence assumption.

If the dimension of $\boldsymbol{\theta}$ is small (≤ 4), then grid sampling can be used to obtain posterior samples of $\boldsymbol{\theta}$. In general, a Metropolis step or other samplers, which are well-suited for densities with non-tractable form, can be applied to $\pi(\boldsymbol{\theta}|\hat{\boldsymbol{\eta}})$.

The computational benefit of the approximations in (31) lies in the fact that the conditional posterior density of $(\boldsymbol{\eta}, \mathbf{v})$ is Gaussian. This is due to $\pi(\boldsymbol{\eta}, \mathbf{v}|\boldsymbol{\theta})$ and

$\hat{L}(\boldsymbol{\eta}|\mathbf{y})$ being Gaussian densities. As a result, posterior samples can be obtained directly from this approximated posterior density. To ensure fast computation when the random effects, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$, are high-dimensional, it is important that their precision matrices, $\mathbf{Q}_{u,1}$, $\mathbf{Q}_{u,2}$, and $\mathbf{Q}_{u,3}$, are sparse. By specifying the random effects with Gaussian Markov random fields (e.g., Rue & Held, 2005), their precision matrices become sparse.

2.3.3 Predictions Based on LGMs with a Multivariate Link Function

The posterior predictive density of an LGM with a multivariate link function is similar to one for an LGM with a univariate link function. The main difference is that the former requires predictions of more than one parameter for each predicted response, while the latter only requires predictions of exactly one parameter for each predicted response.

To simplify the notation, let us assume that each response requires three parameters and that the random effect vectors, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$, contain the elements that are needed for the predictions we desire. Furthermore, $\tilde{\mathbf{y}}$ is the predicted response, $\tilde{X}_1$, $\tilde{X}_2$, and $\tilde{X}_3$ contain the corresponding values of the covariates, and $\tilde{A}_1$, $\tilde{A}_2$, and $\tilde{A}_3$ contain the corresponding weights of the random effects. These matrices are joined in the following matrix:

$$\tilde{Z} = \begin{pmatrix} \tilde{X}_1 & \tilde{A}_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \tilde{X}_2 & \tilde{A}_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \tilde{X}_3 & \tilde{A}_3 \end{pmatrix}.$$

The latent parameters of the predictions are given by $\tilde{\boldsymbol{\eta}} = \tilde{Z}\mathbf{v} + \tilde{\boldsymbol{\epsilon}}$, where $\mathbf{v}$ is based on $\boldsymbol{\beta}_1$, $\mathbf{u}_1$, $\boldsymbol{\beta}_2$, $\mathbf{u}_2$, $\boldsymbol{\beta}_3$, and $\mathbf{u}_3$, as before, and $\tilde{\boldsymbol{\epsilon}}$ has mean zero and precision matrix $\mathbf{Q}_\epsilon$. The posterior predictive density is

$$\pi(\tilde{\mathbf{y}}|\mathbf{y}) = \int \pi(\tilde{\mathbf{y}}|\tilde{\boldsymbol{\eta}})\pi(\tilde{\boldsymbol{\eta}}|\tilde{Z}, \mathbf{v}, \boldsymbol{\theta})\pi(\mathbf{v}, \boldsymbol{\theta}|\mathbf{y})d\mathbf{v}d\boldsymbol{\theta}, \quad (35)$$

where the density $\pi(\tilde{\mathbf{y}}|\tilde{\boldsymbol{\eta}})$ has the same form as the density of the response in the LGM, the i -th response variable to be predicted, $\tilde{y}_i$, requires the parameter vector $g(\tilde{\boldsymbol{\eta}}_{1,i}, \tilde{\boldsymbol{\eta}}_{2,i}, \tilde{\boldsymbol{\eta}}_{3,i})$, and $g(\cdot)$ is the multivariate link function of the LGM. The density $\pi(\tilde{\boldsymbol{\eta}}|\tilde{Z}, \mathbf{v}, \boldsymbol{\theta})$ is the conditional Gaussian density of $\tilde{\boldsymbol{\eta}}$, with mean $\tilde{Z}\mathbf{v}$ and precision $\mathbf{Q}_\epsilon$, and $\pi(\mathbf{v}, \boldsymbol{\theta}|\mathbf{y})$ is the posterior density.

To obtain posterior predictive samples based on (35), three steps are needed. In the first step, samples are drawn from the posterior density. In the second step, samples are drawn from the linear predictor $\tilde{\boldsymbol{\eta}}$ through $\tilde{\boldsymbol{\eta}} = \tilde{Z}\mathbf{v} + \tilde{\boldsymbol{\epsilon}}$, where $\mathbf{v}$ is sampled from the posterior density, and $\tilde{\boldsymbol{\epsilon}}$ are Gaussian samples with mean zero and precision $\mathbf{Q}_\epsilon$. The parameters of $\mathbf{Q}_\epsilon$ are drawn from the posterior density. In

the third step, samples are drawn from the density of the response specified by the LGM with the parameters in $\tilde{\eta}$.

The posterior predictive density is used, for example, to make temporal and spatial predictions, perform cross-validation, and test the distributional assumptions at the response level. For more details, see Sect. 2.1.3.

3 Priors for the Parameters of Bayesian LGMs

In this section, we discuss how to construct prior densities for the fixed effects, the random effects, and the hyperparameters in Bayesian LGMs. Recall that the fixed effects and the random effects are within the latent parameter vector. In Sect. 3.1, we review commonly used prior densities for the fixed effects. A few Gaussian models that are commonly used for the random effects are presented in Sect. 3.2. Prior densities for the hyperparameters in the Gaussian models in Sect. 3.3 are discussed in Sect. 3.3. In Sect. 3.3, we focus mainly on one approach for constructing priors for hyperparameters, namely, the penalized complexity approach of Simpson et al. (2017). Even though we are only focusing on a finite set of models and one approach for constructing priors for hyperparameters, the main goal is to provide examples on how to build prior densities. A general discussion about prior densities within the Bayesian modeling approach is given in the bibliographic note at the end of this chapter. Section 3.3 is quite technical, and it can be omitted when reading through this chapter for the first time. One can simply return to it later.

3.1 Priors for the Fixed Effects

The prior density for the latent parameters in Bayesian LGMs is Gaussian by definition. Therefore, since the fixed effects are part of the latent parameter vector, they are assigned a Gaussian density, and the prior selection becomes a matter of selecting their prior mean, variance, and correlation structure. Independence is often assumed a priori between the fixed effects, that is, the prior correlation is set equal to zero. This is often reasonable since each fixed effect corresponds to a covariate and we may not have any idea about how one fixed effect will interact with a fixed effect corresponding to another covariate. The prior mean and variance of the fixed effects are given fixed values that need to be carefully selected in order to represent the knowledge at hand or the lack thereof. It is common to select weakly informative priors for the fixed effects, that is, priors that contain some information about the parameter of interest and capture roughly the scale of the parameter but without exploiting all available scientific knowledge about it (Gelman et al., 2013). In models where we have only a handful of fixed effects but thousands of observations, it is sensible to use weakly informative priors for the fixed effects because they are usually well-identified by the data since most data points give some information about the fixed effects.

In some cases, a uniform density defined over the real line is selected for some or all of the fixed effects. This uniform prior is improper, that is, its density does not integrate to one. We do not recommend using a uniform prior for the fixed effects, since that may lead to an improper posterior density. If the goal is to represent weak information on a given fixed effect, then our suggestion is to use a Gaussian prior with mean zero and variance that is large relative to the covariate it represents and the role this fixed effect plays in the model. For example, the constant $\beta_{m,0}$ in the linear predictor for the m -th transformed parameter vector, η_m , in an extended LGM, can be assigned a prior standard deviation that is larger than the range that can be expected for this parameter. In general, it can be helpful to scale the covariates (except for the vector of ones) such that they have mean zero and standard deviation equal to one or half. Once the covariates have been scaled, then the same prior standard deviation can be assigned to all the fixed effects corresponding to η_m , and this prior standard deviation is set equal to a value that is large relative to the variability that can be expected from η_m . In practice, weakly informative priors for the fixed effects are often implemented in software as a Gaussian with mean zero and a very large variance, e.g., in INLA, it is 1000 for the covariates, and the constant is assigned zero precision. That is a robust choice for most applications when the covariates are scaled to have standard deviation equal to one or half.

Gelman et al. (2008) proposed a prior density for the regression parameters in a non-hierarchical logistic regression model. They suggested using a t -density with one degree of freedom, also referred to as the Cauchy density, with center at zero and scale parameters equal to 10 for the constant and 2.5 for the other parameters. This setup is based on centering each of the binary covariates to have mean zero and rescaling each numeric covariate to have mean 0 and standard deviation 0.5. This prior density was tested with examples in Gelman et al. (2008), and the results were convincing. A Gaussian prior density with the same center and scale parameters was also tested with the same examples, and it did also perform well. The advantage of the Cauchy prior is due to its robustness as it can work well in cases where the regression parameters are unusually large, in which case the Gaussian density might shrink too much toward zero. Gelman et al. (2008) argued that an increase or a decrease of size 5 for a logit probability is unlikely and that led to selecting scale equal to 2.5. A Gaussian prior density with mean zero and scale 2.5 supports values of a regression parameter between -5 and 5 , but it is not supportive of values far below -5 or high above 5 . Gelman et al. (2008) suggested using their Cauchy prior within a hierarchical logistic model. Likewise, we suggest using the Gaussian version of this prior for fixed effects within LGMs that assume the binomial distribution at the response level.

3.2 Priors for the Random Effects

In this section, we introduce Gaussian models that are commonly used to model random effects within latent Gaussian models. The first model is the simplest one.

It is such that each random effect, u_i , is assumed to be independent of the other random effects, and it follows a Gaussian distribution with mean zero and variance σ_u^2 , that is,

$$u_i \sim \mathcal{N}(0, \sigma_u^2), \quad i \in \{1, \dots, n\}.$$

These random effects are independent and identically distributed (iid), and thus, this model will be referred to as the iid model. If the index i refers to equally spaced time points, then this model is referred to as white noise in the time series literature. The logarithm of the prior density of $\mathbf{u} = (u_1, \dots, u_n)^\top$ conditional on σ_u is

$$\log \pi(\mathbf{u}|\sigma_u) = \text{constant} - \frac{n}{2} \log(\sigma_u^2) - \frac{\sigma_u^{-2}}{2} \mathbf{u}^\top \mathbf{u}.$$

The top panel of Fig. 3 shows an example of a realization (based on a simulation) from this model with $\sigma_u = 1$.

Another common model is the random walk. Let $v_i \sim \mathcal{N}(0, \sigma_v^2)$, $i \in \{1, \dots, n\}$, be mutually independent random variables. Then the random walk process, u_i , can be presented as

$$u_1 = v_1, \quad u_i = u_{i-1} + v_i, \quad i \in \{2, \dots, n\}.$$

The process at step i , conditional on the process at step $i - 1$, has mean u_{i-1} and variance σ_v^2 . The marginal mean of u_i is zero, and its variance is $i\sigma_v^2$. The logarithm of the prior density of $\mathbf{u} = (u_1, \dots, u_n)^\top$ conditional on σ_v is

$$\log \pi(\mathbf{u}|\sigma_v) = \text{constant} - \frac{n}{2} \log(\sigma_v^2) - \frac{\sigma_v^{-2}}{2} u_1^2 - \frac{\sigma_v^{-2}}{2} \mathbf{u}^\top R_n \mathbf{u},$$

where R_n is an $n \times n$ matrix such that

$$R_n = \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 & -1 \\ & & & & & -1 & 1 \end{bmatrix}, \quad (36)$$

(see Rue & Held, 2005, Sect. 3.3.1, p. 95). The middle panel of Fig. 3 shows an example of a realization from the random walk model with $\sigma_v = 1$.

The first-order autoregressive time series model (AR(1)) is similar to the random walk model. Its parameters are ϕ and σ_w . It can be presented as

$$u_1 \sim \mathcal{N}(0, \sigma_w^2(1 - \phi^2)^{-1}), \quad u_i = \phi u_{i-1} + w_i, \quad i \in \{2, \dots, n\},$$

where $w_i \sim \mathcal{N}(0, \sigma_w^2)$, $i \in \{2, \dots, n\}$. Here, we assume that $|\phi| < 1$, which ensures that the time series is stationary. The mean and variance of u_i conditional on u_{i-1} , $i \in \{2, \dots, n\}$, are u_{i-1} and σ_w^2 , respectively. The marginal mean and variance of u_i are zero and $\sigma_w^2(1 - \phi^2)^{-1}$, respectively. The correlation between u_i and u_j is $\phi^{|i-j|}$. The log-prior density of $\mathbf{u} = (u_1, \dots, u_n)^\top$ conditional on ϕ and σ_w is

$$\log \pi(\mathbf{u}|\phi, \sigma_w) = \text{constant} - \frac{n}{2} \log(\sigma_w^2) + \frac{1}{2} \log |Q_u| - \frac{\sigma_w^{-2}}{2} \mathbf{u}^\top Q_u \mathbf{u},$$

where Q_u is an $n \times n$ matrix such that

$$Q_u = \begin{bmatrix} 1 & -\phi & & & & \\ -\phi & 1 + \phi^2 & -\phi & & & \\ & -\phi & 1 + \phi^2 & -\phi & & \\ & & \ddots & \ddots & \ddots & \\ & & & -\phi & 1 + \phi^2 & -\phi \\ & & & & -\phi & 1 + \phi^2 & -\phi \\ & & & & & -\phi & 1 \end{bmatrix}, \quad (37)$$

(see Rue & Held, 2005, Sect. 1, p. 2), and $|Q_u|$ denotes its determinant. The bottom panel of Fig. 3 shows an example of a realization from the AR(1) model with $\phi = 0.8$ and $\sigma_w = 1$.

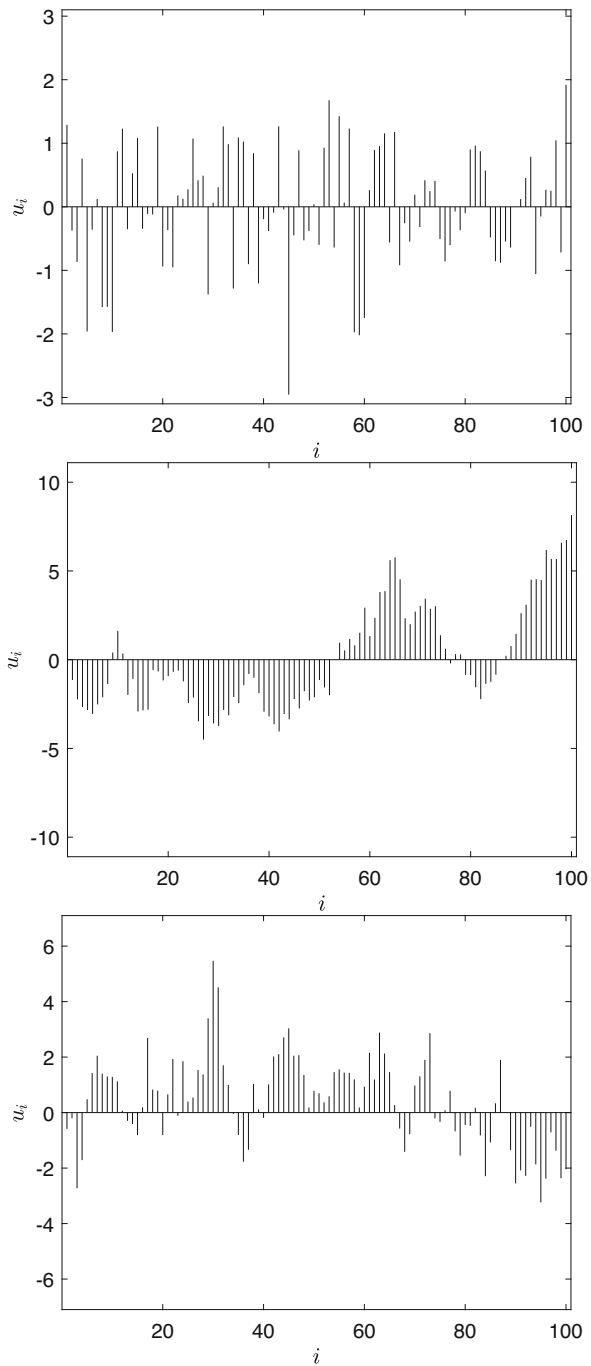
The next model we consider here is a spatial model on a rectangular lattice referred to as a first-order intrinsic Gaussian Markov random field (IGMRF) on a regular lattice (Rue & Held, 2005, Sect. 3.3.2, pp. 104–108). Assume the dimensions of the lattice are K and L , so the total number of grid points is $N = KL$. The distribution of $u_{i,j}$, the element of the field with coordinates (i, j) , conditional on the other elements of the field, $\mathbf{u}_{-(i,j)}$, is Gaussian with mean

$$\mathbb{E}(u_{i,j}|\mathbf{u}_{-(i,j)}) = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}),$$

and variance $\sigma_\kappa^2/4$, provided that (i, j) is not an edge point of the lattice, and σ_κ is the hyperparameter in the model.

Let I_s be an s -dimensional identity matrix, and let R_s be an $s \times s$ matrix as in (36). Furthermore, let Q be such that $Q = I_L \otimes R_K + R_L \otimes I_K$, where $\otimes$ denotes the Kronecker product. The precision matrix of $\mathbf{u}$, the vector containing all the u elements of the random field, is given by $\sigma_\kappa^{-2}Q$. The logarithm of the prior density

Fig. 3 Examples of random effects. Top panel: Independent identically distributed Gaussian random effect with mean zero and variance 1.0. Middle panel: A Gaussian random walk with error variance 1.0. Bottom panel: autoregressive time series of order 1 with a Gaussian error term with variance σ_w^2 equal to 1 and autoregression coefficient ϕ equal to 0.8



of $\mathbf{u}$ conditional on σ_κ is

$$\log \pi(\mathbf{u}|\sigma_\kappa) = \text{constant} - \frac{(N-1)}{2} \log(\sigma_\kappa^2) - \frac{\sigma_\kappa^{-2}}{2} \mathbf{u}^\top \mathbf{Q} \mathbf{u}. \quad (38)$$

Note that the precision matrix $\mathbf{Q}$ is not full rank; instead, its rank is $N-1$ and so the prior density in (38) is improper. However, it can be used to infer $\mathbf{u}$ and σ_κ given that the likelihood function of the LGM provides information about $\mathbf{u}$ and a proper prior density is selected for σ_κ .

Figure 4 shows realizations of first-order intrinsic Gaussian Markov random fields on regular lattices. The lattices have dimensions 10×10 , 20×20 , and 40×40 . The variance of the element in the upper-right corner is 1 conditional on the element in the lower-left corner being equal to zero. The three realizations in Fig. 4 show the spatial patterns this IGMRF can generate. The parameter σ_u is largest in the top panel of Fig. 4 and smallest in its bottom panel. As σ_u decreases, the greater the similarity between a given element and its nearest neighbors.

Gaussian processes, based on the Matérn class of covariance functions (Matérn, 1986), are commonly used to model components in LGMs that are assumed to be continuous functions of the coordinates of continuous variables. Note that Gaussian processes are also referred to as Gaussian random fields. A component of this sort is often used to represent a spatial effect that is defined on a geographical domain where the coordinates are longitude and latitude, but it can also be used to represent a continuous function with respect to another continuous variable or variables.

The Matérn covariance function is defined as

$$C(h) = \sigma^2 \frac{2^{-\nu+1}}{\Gamma(\nu)} \left(\sqrt{8\nu} \frac{h}{\rho} \right)^\nu \mathcal{K}_\nu \left(\sqrt{8\nu} \frac{h}{\rho} \right), \quad h > 0, \quad (39)$$

where $C(h)$ gives the covariance between two elements of a Gaussian random field with d -dimensional coordinates that are h units apart, σ is the marginal standard deviation, ρ is the range parameter, ν is the smoothness parameter, $\mathcal{K}_\nu(\cdot)$ is the modified Bessel function of the second kind of order ν , and $\Gamma(\cdot)$ is the gamma function. This definition scales the range parameter with respect to ν such that ρ is the distance at which the correlation is approximately 0.1 (Lindgren et al., 2011). Note that σ , ρ , and ν are positive parameters.

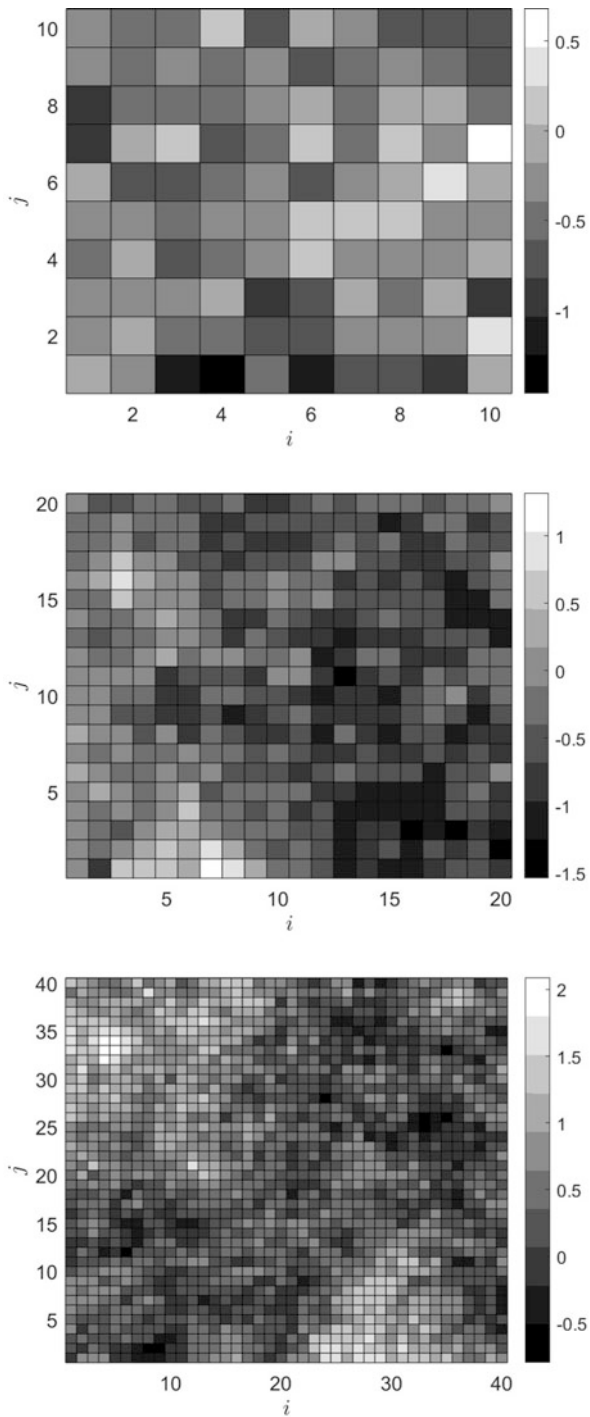
The log-prior density for $\mathbf{u} = (u_1, \dots, u_n)^\top$ conditional on σ , ρ , and ν is

$$\log \pi(\mathbf{u}|\sigma, \rho, \nu) = \text{constant} - \frac{n}{2} \log(\sigma^2) - \frac{1}{2} \log |P| - \frac{\sigma^{-2}}{2} \mathbf{u}^\top P^{-1} \mathbf{u},$$

where P is an $n \times n$ matrix with (i, j) -th element $\rho_{i,j} = \sigma^{-2} C(h_{i,j})$, with $C(\cdot)$ as in (39), and $h_{i,j}$ is the distance between the coordinates of u_i and u_j .

Here, we visualize realizations of Gaussian processes with one- and two-dimensional coordinates that correspond to three special cases within the class of Matérn covariance functions, namely, those that correspond to ν equal to

Fig. 4 Three realizations of first-order intrinsic Gaussian Markov random fields on regular lattices of sizes $K \times K$ with $K \in \{10, 20, 40\}$. $K = 10$ (top panel), $K = 20$ (middle panel), and $K = 40$ (bottom panel). The element $u_{1,1}$ (in the lower-left corner) is set equal to zero. The variance of $u_{K,K}$ (in the upper-right corner), conditional on $u_{1,1} = 0$, is set equal to 1. A consequence of this is that when K increases, σ_u decreases



0.5, 1.5, and 2.5. The first one, $\nu = 0.5$, is such that the covariance function takes the form $C(h) = \sigma^2 \exp(-2h/\rho)$ and is referred to as the exponential covariance function. Its realizations are continuous but not smooth. The second one, $\nu = 1.5$, is such that the covariance function takes the form $C(h) = \sigma^2(1 + \sqrt{12}h/\rho) \exp(-\sqrt{12}h/\rho)$. Gaussian processes corresponding to this covariance function are such that their realizations are continuous and smooth. In particular, they are once mean-square differentiable (Stein, 1999), which can be interpreted such that the limit representing the first derivative of these Gaussian processes is continuous. Finally, when $\nu = 2.5$, the covariance function takes the form $C(h) = \sigma^2(1 + \sqrt{20}h/\rho + 20h^2/(3\rho^2)) \exp(-\sqrt{20}h/\rho)$, and realizations based on it are continuous and smooth. The corresponding Gaussian processes are twice mean-square differentiable (Stein, 1999). The limit representing the second derivative of these Gaussian processes is continuous. Thus, the realizations corresponding to $\nu = 2.5$ are smoother than those corresponding to $\nu = 1.5$.

Figures 5 and 6 show realizations of Gaussian Matérn processes with one- and two-dimensional coordinates, respectively. The top, middle, and bottom panels of Figs. 5 and 6 show realizations with smoothness parameters equal to 0.5, 1.5, and 2.5, respectively, and the increasing degree of smoothness with increasing ν can be seen.

The Matérn covariance function in (39) produces fully populated covariance matrices that lead to high computational cost when the dimension of these matrices is high. The stochastic partial differential equation (SPDE) approach of Lindgren et al. (2011) is based on approximating the precision matrices of Gaussian Matérn processes with sparse matrices. Computation is reduced substantially when the sparse precision matrices of the SPDE approach are applied within LGMs compared to applying the fully populated Matérn covariance matrices.

3.3 Priors for the Hyperparameters

A scale parameter of some sort is always needed to represent Gaussian random effects. This scale parameter is often represented as a conditional variance, an inverse variance (precision), or a marginal standard deviation. Let σ denote the marginal standard deviation or the conditional standard deviation of one of the random effects in an LGMs. The size of σ is related to how much of the variability in the linear predictor is explained by this random effect. A sensibly selected prior for this parameter can help with restricting the parameter space of the random effect it governs. A cautious approach involves assuming a priori that the random effect may not be too large, which means we leave it to the data to inform us that it is large. Through this approach, the data determine which of the random effects have large σ . It is also cautious to assume a priori that σ can be arbitrary close to zero, because, if the data do not suggest that a given random effect is large, then it is shrunk toward zero by the prior.

Fig. 5 Three realizations of one-dimensional Gaussian processes based on the Matérn covariance function with marginal variance equal to 1.0, range parameter equal to 0.6, and smoothness parameters $\nu = 0.5$ (top panel), $\nu = 1.5$ (middle panel), and $\nu = 2.5$ (bottom panel)

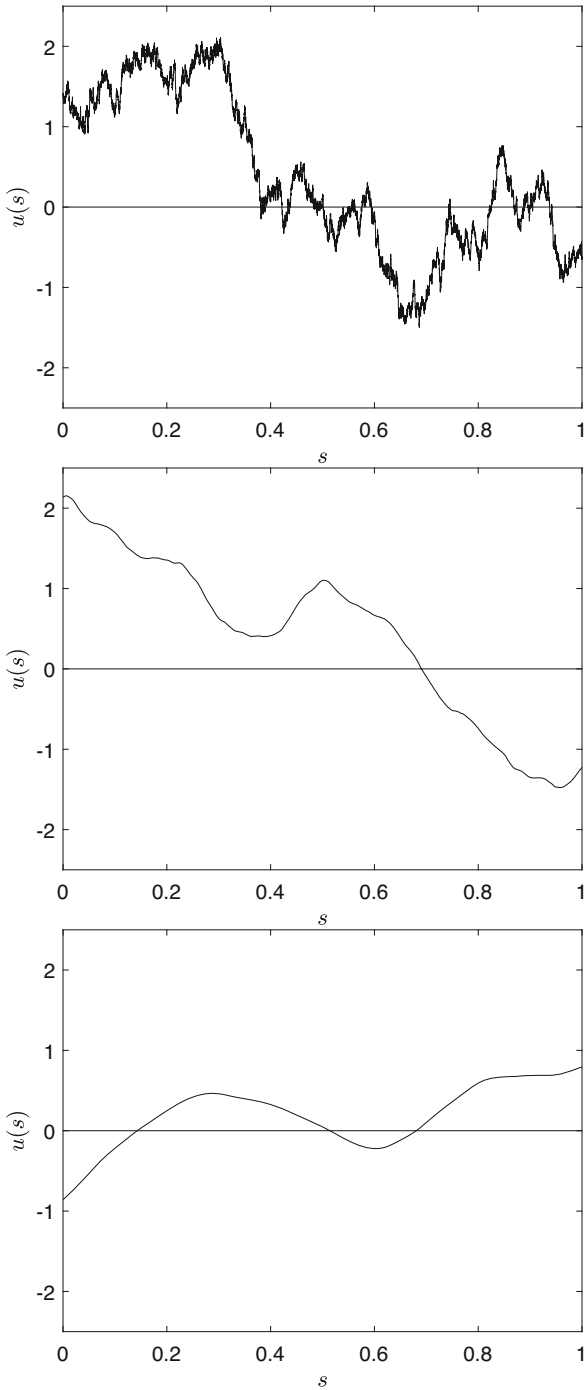
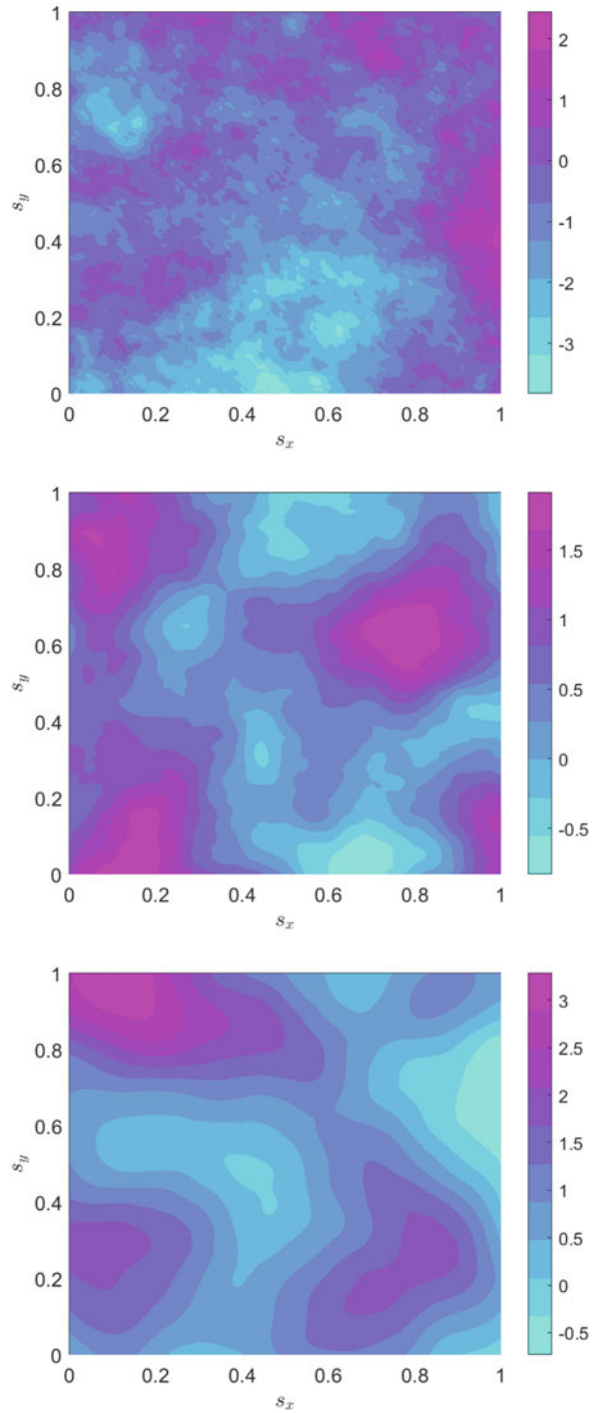


Fig. 6 Three realizations of two-dimensional Gaussian processes based on the Matérn covariance function with marginal variance equal to 1.0, range parameter equal to 0.6, and smoothness parameters $\nu = 0.5$ (top panel), $\nu = 1.5$ (middle panel), and $\nu = 2.5$ (bottom panel)



Gelman (2006) presents this idea in the following way. The prior density for σ has to be larger for values of σ arbitrarily close to zero than the prior density over values further away from zero. He proposed using the half-Cauchy prior density, which is weakly informative when it is properly scaled. When we are not certain about the size of the random effects in our latent Gaussian model, and it is possible that some of them are zero or close to zero in magnitude, then the half-Cauchy density is a suitable choice for the standard deviations of the random effects since that density supports values arbitrarily close to zero. The lognormal and inverse-gamma densities are on the other hand poor prior densities for the standard deviations of the random effects since they are practically zero in some interval containing zero, and thus, push the posterior mass of the standard deviations away from zero.

Simpson et al. (2017) introduce an approach for constructing prior densities for model parameters, referred to as penalized complexity (PC) priors. The PC approach is based on comparing a complex model to a base model where the base model is a special case of the complex model. Some of the components of the complex model are zero in the base model, and typically, the sizes of these components are governed by standard deviation parameters. Large values of the standard deviation parameters are penalized, while values arbitrarily close to zero are supported. In this section, we go through how prior densities for the hyperparameters in Bayesian latent Gaussian models can be constructed using the principles of PC priors.

3.3.1 Penalized Complexity Priors

The penalized complexity (PC) approach of Simpson et al. (2017) is well-suited for the selection of prior densities for the hyperparameters in Bayesian hierarchical models, including the class of Bayesian latent Gaussian models. Simpson et al. (2017) proposed priors that support the shrinkage of random effects within a Bayesian hierarchical model toward a simpler reference model, referred to as the base model. PC priors are based on four underlying principles: (i) Occam's razor: The base model should be preferred unless a more complex model is suggested by the data; (ii) use the Kullback–Leibler divergence (KLD) as a measure of complexity (Kullback & Leibler, 1951), in particular, to measure the distance between the base model, $\mathcal{M}_0$, and the more complex model, $\mathcal{M}_1$, the distance $d_{1,0} = \sqrt{2\text{KLD}(\mathcal{M}_1||\mathcal{M}_0)}$ is used, and the factor 2 gives simpler mathematical derivations; (iii) apply constant-rate penalization, which is natural if there is no additional knowledge suggesting otherwise, this corresponds to an exponential prior on the distance scale $d_{1,0}$; (iv) apply user-defined scaling, that is, the user should use (weak) prior knowledge about the size of the parameter to scale the prior density for $d_{1,0}$.

These four principles can be explained through the following example. Say we are interested in a model of the form

$$y_{i,j} = \beta_0 + u_j + \epsilon_{i,j}, \quad \epsilon_{i,j} \sim \mathcal{N}(0, \sigma_\epsilon^2), \quad (40)$$

where $y_{i,j}$ is the i -th observation within group j and u_j is the effect within the j -th group out of a total of J groups. Within each group, there are n observations. The Gaussian prior density for the u_j 's is the same as the one of the iid model in Sect. 3.2, so the u_j 's have mean zero and unknown standard deviation σ_u and are mutually independent. Here, the base model is the one with no group effect, that is, $u_j = 0$ for all groups, and it can be presented in terms of the more complex model with $\sigma_u = 0$. The specified base model is a suitable reference model.

Denote the density of $\mathbf{u} = (u_1, \dots, u_J)^\top$ under the more complex model by $\pi(\mathbf{u}|\sigma_u)$. Let $\pi(\mathbf{u}|\sigma_{u,0})$ denote the density of $\mathbf{u}$ with standard deviation $\sigma_{u,0}$, where $\sigma_{u,0}$ is arbitrarily close to zero. Thus, this density represents the base model. The distance between these two prior densities according to Principle (ii) is

$$d_{1,0} = \sqrt{2 \int \pi(\mathbf{u}|\sigma_u) \log \frac{\pi(\mathbf{u}|\sigma_u)}{\pi(\mathbf{u}|\sigma_{u,0})} d\mathbf{u}} \approx \sigma_u \sqrt{J \sigma_{u,0}^{-2}},$$

which is derived by assuming $\sigma_{u,0} \ll \sigma_u$ (see Simpson et al., 2017). If we assume that the constant-rate penalization suggested by Principle (iii) is reasonable, then an exponential prior density is assigned to $d_{1,0}$. Set the rate for $d_{1,0}$ equal to $\tilde{\lambda} = \lambda J^{-1/2} \sigma_{u,0}$. Then the prior density for σ_u is exponential with rate λ . Finally, according to Principle (iv), the value of λ is determined by using any available prior information about the scale of σ_u . Typically, an upper bound and the probability of exceeding this upper bound are extracted from this information. If prior knowledge suggests that σ_u exceeds the value U_σ with probability α , then $\lambda = -\log(\alpha) U_\sigma^{-1}$. Alternatively, the user can use simulation to scale the exponential prior density for σ_u by sampling the random effects in $\mathbf{u}$ with various values of σ_u to understand what values of σ_u give plausible random effects.

This exponential prior density for σ_u is such that it supports values arbitrarily close to zero, and the random effect can shrink to zero if that is what the data suggest. When the data suggest the random effect is different from zero, then the prior density provides support for large values of the standard deviation.

3.3.2 PC Priors for Hyperparameters in Common Temporal and Spatial Models

Here, we introduce the PC priors for the hyperparameters in the Gaussian models for random effects presented in Sect. 3.2. The PC prior for the hyperparameter in the iid model, σ_u , was presented in Sect. 3.3.1. It is an exponential density with rate parameter λ . To improve posterior calculation, this parameter is often transformed with the logarithmic transformation. Denote the transformed hyperparameter by ψ_u where $\psi_u = \log(\sigma_u)$. The PC prior density for ψ_u is given by

$$\pi(\psi_u) = \lambda \exp(-\lambda \exp(\psi_u) + \psi_u),$$

where $\psi_u \in \mathbb{R}$.

PC Prior for the Parameter in the Random Walk Model

The parameter of the random walk model, σ_v , measures the size of each step in the random walk. The base model is the one that assumes $\sigma_v = 0$. By applying Principle (iii), it can be shown that the PC prior for σ_v is an exponential density with rate parameter λ . The value of λ is determined by applying Principle (iv). As in the case of the standard deviation in the iid model, λ is selected by controlling the prior mass in the upper tail. An upper bound, U_σ , and corresponding upper tail probability, α , are selected based on weak prior information, resulting in $\lambda = -\log(\alpha)U_\sigma^{-1}$. It may be difficult to get a feel for the standard deviation of each step, while the scale of the n steps that are accumulated in u_n may be a more tangible quantity. In that case, the standard deviation of u_n , $\sqrt{n}\sigma_v$, can be used instead of σ_v to understand what is believed to be a priori large. If $U_{\sigma,n}$ is considered a large value for $\sqrt{n}\sigma_v$ that is a priori exceeded with small probability α , then the rate parameter of the PC prior for σ_v is $\lambda = -\sqrt{n}\log(\alpha)U_{\sigma,n}^{-1}$.

PC Prior for the Parameters in the AR(1) Model

Here, we follow the derivation of PC priors for the parameters in the AR(1) model as presented in Sørbye and Rue (2017), which is based on parametrizing the model in terms of ϕ and the marginal precision of u_i , namely, $\tau = \sigma_w^{-2}(1 - \phi^2)$. The PC prior density is the product of the prior density of τ conditional on ϕ and the marginal prior density of ϕ . The base model, when deriving the PC prior for τ conditional on ϕ , is the one assuming infinite τ , which corresponds to each u_i being zero. The PC prior for τ conditional on ϕ is

$$\pi(\tau|\phi) = \frac{\lambda}{2} \tau^{-3/2} \exp(-\lambda \tau^{-1/2}), \quad \tau > 0,$$

(Sørbye & Rue, 2017; Simpson et al., 2017). The rate λ is found by selecting an upper bound for $1/\sqrt{\tau}$, i.e., the marginal standard deviation of u_i , and specifying a corresponding upper tail probability, α . Denote the upper bound by U_σ , and the rate parameter is given by $\lambda = -\log(\alpha)U_\sigma^{-1}$.

In Sørbye and Rue (2017), two PC priors are considered for ϕ . They stem from base models that assume: (i) no dependency in time ($\phi = 0$); and (ii) no change in time ($\phi = 1$). The first base model results in a PC prior of the form

$$\pi(\phi) = \frac{\lambda_\phi \exp(-\lambda_\phi \sqrt{-\log(1 - \phi^2)})|\phi|}{2(1 - \phi^2)\sqrt{-\log(1 - \phi^2)}}, \quad |\phi| < 1,$$

where λ_ϕ is a rate parameter to be determined from the probability statement $\Pr(|\phi| > U_\phi) = \alpha$, where $0 < U_\phi < 1$ and α is a small probability. The value of λ_ϕ is calculated with $\lambda_\phi = -\log(\alpha)/\sqrt{-\log(1 - U_\phi^2)}$.

The second base model, the one that assumes no change in time, leads to the PC prior

$$\pi(\phi) = \frac{\lambda_\phi \exp(-\lambda_\phi \sqrt{1-\phi})}{2(1 - \exp(-\sqrt{2}\lambda_\phi))\sqrt{1-\phi}}, \quad |\phi| < 1.$$

Here, the value of λ_ϕ can be found by solving

$$\frac{1 - \exp(-\lambda_\phi \sqrt{1-U_\phi})}{1 - \exp(-\sqrt{2}\lambda_\phi)} = \alpha,$$

where the values of U_ϕ and α are such that the statement $\Pr(\phi > U_\phi) = \alpha$ will be meaningful, namely, assume $\alpha > \sqrt{(1-U_\phi)/2}$. If most of the prior probability mass of ϕ is believed to be close to 1, then U_ϕ should be greater than zero and α should be greater than 0.5.

PC Prior for the Parameter in the IGMRF of Order One on a Regular Lattice

The parameter σ_κ in the IGMRF of order one on a regular lattice is such that the standard deviation of the element $u_{i,j}$ conditional on the other elements of the field is equal to $\sigma_\kappa/2$ when (i, j) is not on the edge of the lattice. A natural base model here is $\sigma_\kappa = 0$, which corresponds to the field being zero. By applying the principles of Simpson et al. (2017), we derive an exponential density as the PC prior of σ_κ . The conditional standard deviation, $\sigma_\kappa/2$, might be well-understood in some cases and that understanding could be used to set an upper bound for σ_κ . By setting a corresponding upper tail probability, α , we can determine the rate parameter of this exponential density, namely, $\lambda = -\log(\alpha)U_\sigma^{-1}$, where U_σ is the specified upper bound.

In other cases, the variability of the difference between the elements in the lower-left corner and the upper-right corner might be better understood than σ_κ itself. In that case, assume that the upper bound for the standard deviation of the difference $u_{K,L} - u_{1,1}$, conditional on $u_{1,1} = 0$, is judged to be $U_{\sigma,\text{diff}}$. As before, K and L are the dimensions of the lattice. The standard deviation of $u_{K,L} - u_{1,1}$ is a function of σ_κ , namely, $\sigma_{u,\text{diff}} = c\sigma_\kappa$, where c is a constant derived from the matrix Q in Sect. 3.2 that specifies the IGMRF on a regular lattice.

Let Q^* denote the precision matrix of $\mathbf{u}_{-(1,1)}$ conditional on $u_{1,1} = 0$ when $\sigma_\kappa = 1$. It turns out that Q^* is a $(N-1) \times (N-1)$ matrix, $N = KL$, that can be derived by removing the first row and first column of Q (Rue & Held, 2005). Let $\mathbf{y}$ denote an $(N-1)$ column vector of zeros, except that its last element is one. Let $\boldsymbol{\sigma}_0$ denote an $(N-1)$ vector that is such that $Q^*\boldsymbol{\sigma}_0 = \mathbf{y}$. The square root of the last element of $\boldsymbol{\sigma}_0$ is equal to the standard deviation of $u_{K,L}$ conditional on $u_{1,1} = 0$ when $\sigma_\kappa = 1$. This standard deviation is equal to c in the equation for $\sigma_{u,\text{diff}}$ above.

Now the rate parameter in the exponential PC prior for σ_κ can be calculated with $\lambda = -c \log(\alpha) U_{\sigma, \text{diff}}^{-1}$.

PC Prior for the Parameters in the Matérn Covariance Function

Here, we present the PC prior for the range parameter, ρ , and the marginal standard deviation parameter, σ , in the Matérn covariance function. Fuglstad et al. (2019) proposed a PC prior for (ρ, σ) in Matérn Gaussian random fields, with $d \leq 3$ and fixed ν , that arises from a base model with infinite range and zero marginal standard deviation. This base model corresponds to a zero effect, and models that are close to the base model have an effect that has a small amplitude and is close to being constant with respect to the coordinates of the random field.

The PC prior of (ρ, σ) is based on the PC prior of ρ , $\pi(\rho)$, and the PC prior of σ conditional on ρ , $\pi(\sigma|\rho)$. The PC prior for ρ stems from a base model with infinite range. It is given by

$$\pi(\rho) = \frac{d}{2} \lambda_1 \rho^{-d/2-1} \exp\left(-\lambda_1 \rho^{-d/2}\right). \quad (41)$$

When deriving $\pi(\sigma|\rho)$, the base model is such that $\sigma = 0$, which results in

$$\pi(\sigma|\rho) = \lambda_2 \exp(-\lambda_2 \sigma). \quad (42)$$

The joint PC prior of (ρ, σ) is the product of $\pi(\rho)$ and $\pi(\sigma|\rho)$,

$$\pi(\rho, \sigma) = \frac{d}{2} \lambda_1 \lambda_2 \rho^{-d/2-1} \exp\left(-\lambda_1 \rho^{-d/2} - \lambda_2 \sigma\right). \quad (43)$$

The user defines the parameters λ_1 and λ_2 by specifying the reference values L_ρ and U_σ , and probabilities α_1 and α_2 , which are such that the probability of ρ being less than L_ρ is α_1 , and the probability of σ being greater than U_σ is α_2 . This results in

$$\lambda_1 = -\log(\alpha_1) L_\rho^{d/2}, \quad \lambda_2 = -\log(\alpha_2) U_\sigma^{-1}. \quad (44)$$

The role of the user is to specify what is a relatively large amplitude for the spatial effect or the function and select accordingly a value for U_σ . The user also needs to figure out what is a small value for the range parameter ρ , that is, for what value of L_ρ does the spatial process become too wiggly, and select L_ρ accordingly.

3.3.3 Priors for Multiple Variance Parameters

This section is on the joint PC prior for several variance parameters, where each variance parameter governs a model component in the complex model. Fuglstad et al. (2020) propose joint PC priors for the variance parameters of the random effects in Gaussian–Gaussian models and latent Gaussian models as an alternative to assigning independent prior density to each of the variance parameters. The total variance of the random effects is decomposed in a hierarchical manner along a tree structure to the individual model components. If the modeler is ignorant about this structure, then the Dirichlet prior is appropriate. On the other hand, if the modeler has some knowledge about how to attribute variance to the branches, a PC prior can be applied to achieve a more informed shrinkage.

Expressing Ignorance About the Variance Decomposition

Assume that the linear model at the latent level of our LGM is such that

$$\boldsymbol{\eta} = X\boldsymbol{\beta} + \sum_{l=1}^L A_l \mathbf{u}_l, \quad \mathbf{u}_l \sim \mathcal{N}(\mathbf{0}, \sigma_l^2 \boldsymbol{\Sigma}_l), \quad l \in \{1, \dots, L\}, \quad (45)$$

where the length of $\boldsymbol{\eta}$ is n , the $\boldsymbol{\Sigma}_l$ matrices are correlation matrices, i.e., their diagonal elements are equal to one, and the elements of each A_l are either one or zero, with only one non-zero element per row. Thus, the total variance of the elements in $\sum_{l=1}^L A_l \mathbf{u}_l$ is

$$t = \sigma_1^2 + \dots + \sigma_L^2.$$

We define the weights ω_l as $\omega_l = \sigma_l^2/t$ and $\boldsymbol{\omega} = (\omega_1, \dots, \omega_L)$, so $0 < \omega_l < 1$, $\omega_1 + \dots + \omega_L = 1$, and $\boldsymbol{\omega}$ is defined on a $L - 1$ simplex. The prior density for $(t, \boldsymbol{\omega})$ is

$$\pi(t, \boldsymbol{\omega}) = \pi(t|\boldsymbol{\omega})\pi(\boldsymbol{\omega}).$$

If we are a priori ignorant about the variance decomposition, then we can express our ignorance by assuming that $\boldsymbol{\omega} \sim \text{Dir}(\mathbf{a})$, $\mathbf{a} = a\mathbf{1}$, where $\mathbf{1}$ is an L vector of ones. Here, $\text{Dir}(\mathbf{a})$ denotes a Dirichlet distribution with parameter vector $\mathbf{a}$ (Kotz et al., 2004), and its density is $\pi(\boldsymbol{\omega}) = B(a, \dots, a)^{-1} \prod_{l=1}^L \omega_l^{a-1}$, where $B(\cdot)$ is the multivariate beta function. This prior for $\boldsymbol{\omega}$ is motivated by the work of Bhattacharya et al. (2015). If we set $a = 1$, then $\pi(\boldsymbol{\omega})$ is a uniform density on the $L - 1$ simplex. If $a < 1$, then most of the probability mass of $\pi(\boldsymbol{\omega})$ is on the vertices of the simplex. If $a > 1$, then $\pi(\boldsymbol{\omega})$ has a mode at $\boldsymbol{\omega} = L^{-1}\mathbf{1}$. The marginal densities $\pi(\omega_l)$, $l \in \{1, \dots, L\}$ are beta densities with parameters a and $(L - 1)a$. Fuglstad et al.

(2020) suggest selecting a by specifying ω_0 that satisfies

$$\Pr(\text{logit}(1/4) < \text{logit}(\omega_l) - \text{logit}(\omega_0) < \text{logit}(3/4)) = 1/2, \quad (46)$$

where $\text{logit}(\omega) = \log(\omega) - \log(1 - \omega)$. For example, if $L = 3$ and $\omega_0 = 1/3$, then $a = 0.756$. Likewise, $L = 2$ and $\omega_0 = 1/2$ result in $a = 1$, and the marginal density of ω_1 is the uniform density on $(0, 1)$.

We assume that some knowledge about the total variance is available that can be used to specify a PC prior for t . By assuming a base model with $t = 0$, then the form of the conditional density of t will be

$$\pi(t|\boldsymbol{\omega}) = \lambda \exp\left(-\lambda\sqrt{t}\right) \left(2\sqrt{t}\right)^{-1}, \quad (47)$$

and the fourth principle of PC priors is used to determine the value of λ . Here, the user needs to specify a value for t that reflects a large total variance, say U_t , and set a value α for the probability of exceeding U_t . Then the rate parameter will be $\lambda = -\log(\alpha)(U_t)^{-1/2}$.

PC Priors for a Dual Split

Here, we specify the PC prior for the weights in $\boldsymbol{\omega}$ when there are only two random effects in the linear predictor, i.e., $L = 2$. We refer to this prior as the PC prior for a dual split. This PC prior can be used as a building block in the construction of a PC prior for $\boldsymbol{\omega}$ in a linear predictor with more than two random effects ($L > 2$).

The linear predictor with two random effects is defined as

$$\boldsymbol{\eta} = X\boldsymbol{\beta} + A_1\mathbf{u}_1 + A_2\mathbf{u}_2,$$

where

$$A_l\mathbf{u}_l \sim \mathcal{N}(\mathbf{0}, \sigma_l^2 \tilde{\Sigma}_l), \quad \tilde{\Sigma}_l = A_l \Sigma_l A_l^\top, \quad l \in \{1, 2\},$$

and as above, n is the length of $\boldsymbol{\eta}$, the elements of A_1 and A_2 are either one or zero, with only one non-zero element per row, and the diagonal elements of $\tilde{\Sigma}_1$ and $\tilde{\Sigma}_2$ are one. Thus, the variance of each element of $A_1\mathbf{u}_1 + A_2\mathbf{u}_2$ is $\sigma_1^2 + \sigma_2^2$. Furthermore, let $\omega = \sigma_2^2/(\sigma_1^2 + \sigma_2^2)$ and

$$\tilde{\Sigma}(\omega) = (1 - \omega)\tilde{\Sigma}_1 + \omega\tilde{\Sigma}_2.$$

The user specifies a base model by setting ω equal to some value ω_0 , $0 \leq \omega_0 \leq 1$. The choice of ω_0 depends on how much is known about these two random effects. If the prior knowledge is limited, but yet suggests that $\mathbf{u}_2$ is small relative to $\mathbf{u}_1$, then it is reasonable to set $\omega_0 = 0$, and if the relative sizes of $\mathbf{u}_1$ and $\mathbf{u}_2$ are the other way

around, then set $\omega_0 = 1$. If more knowledge is available a priori, then ω_0 should be given a value between 0 and 1 that reflects this knowledge. For example, if our prior knowledge suggests that $\mathbf{u}_1$ has a standard deviation that is around two times the size of the standard deviation of $\mathbf{u}_2$, then it would be reasonable to set $\omega_0 = 0.2$.

According to Theorem 1 of Fuglstad et al. (2020), the PC prior for a dual split, assuming a base model with $\omega_0 = 0$, is

$$\pi(\omega) = \begin{cases} \frac{\lambda|d'(\omega)|}{1-\exp(-\lambda d(1))} \exp(-\lambda d(\omega)), & 0 < \omega < 1, \tilde{\Sigma}_1 \text{ non-singular,} \\ \frac{\lambda}{2\sqrt{\omega}(1-\exp(-\lambda))} \exp(-\lambda\sqrt{\omega}), & 0 < \omega < 1, \tilde{\Sigma}_1 \text{ singular,} \end{cases} \quad (48)$$

where $d(\omega) = \sqrt{\text{tr}(\tilde{\Sigma}(\omega_0)^{-1} \tilde{\Sigma}(\omega)) - n - \log |\tilde{\Sigma}(\omega_0)^{-1} \tilde{\Sigma}(\omega)|}$, which is defined for $0 \leq \omega_0 \leq 1$, and $\lambda > 0$ is a rate parameter. Fuglstad et al. (2020) suggest setting λ so that the median of ω is equal to 0.25. Note that the PC prior for a dual split, assuming a base model with $\omega_0 = 1$, follows by reversing the roles of $\mathbf{u}_1$ and $\mathbf{u}_2$.

When the base model is such that $0 < \omega_0 < 1$, the PC prior for ω with median equal to ω_0 is

$$\pi(\omega) = \begin{cases} \frac{\lambda|d'(\omega)|}{2[1-\exp(-\lambda d(0))]} \exp(-\lambda d(\omega)), & 0 < \omega < \omega_0, \\ \frac{\lambda|d'(\omega)|}{2[1-\exp(-\lambda d(1))]} \exp(-\lambda d(\omega)), & \omega_0 < \omega < 1. \end{cases}$$

Fuglstad et al. (2020) suggest setting the rate parameter λ so that

$$\Pr(\text{logit}(1/4) < \text{logit}(\omega) - \text{logit}(\omega_0) < \text{logit}(3/4)) = 1/2.$$

Hierarchical Decomposition Priors

The PC prior for a dual split can be applied to construct PC priors for multiple variance parameters associated with the linear predictor $\boldsymbol{\eta}$ in an LGM by defining a tree structure with the L random effects as nodes and the tree consists of a dual split at each step where one starts with all the nodes. For simplicity, take a linear predictor as in (45) with three random effects, A, B, and C. In the first step, one random effect, say C, is split from the other ones, A and B. The weight for C is given a prior that shrinks toward it. In the second step, one of the remaining nodes, say B, is split from the remaining node, that is A, and the weight for B is given a prior that shrinks toward it.

Denote the unknown variances of the three random effects by σ_A^2 , σ_B^2 , and σ_C^2 . One way to specify a tree for these three random effects based on dual splits is as follows. The root node is $T_0 = \{A, B, C\}$. T_0 is split into the nodes $T_1 = \{A, B\}$ and $T_2 = \{C\}$, and T_1 is split into $T_3 = \{A\}$ and $T_4 = \{B\}$. T_1 and T_2 are the descendant nodes of T_0 , stemming from the first split, and T_3 and T_4 are the descendant nodes of T_1 , stemming from the second split. The three hyperparameters, σ_A^2 , σ_B^2 , and σ_C^2 ,

are transformed to $(t, \boldsymbol{\omega}_1, \boldsymbol{\omega}_2)$, where $t = \sigma_A^2 + \sigma_B^2 + \sigma_C^2$, $\boldsymbol{\omega}_1 = t^{-1}(\sigma_A^2 + \sigma_B^2, \sigma_C^2)$, and $\boldsymbol{\omega}_2 = (\sigma_A^2/(\sigma_A^2 + \sigma_B^2), \sigma_B^2/(\sigma_A^2 + \sigma_B^2))$.

An approach referred to as a bottom-up is used to define a prior density for all the S splits, $\{\boldsymbol{\omega}_s\}_{s=1}^S$. Note that $S = 2$ in the example above. In general, the prior density is

$$\pi(\{\boldsymbol{\omega}_s\}_{s=1}^S) = \prod_{s=1}^S \pi(\boldsymbol{\omega}_s | \{\boldsymbol{\omega}_j\}_{j \in D(s)}),$$

where $D(s)$ is the set of descendant splits for split s , $s \in \{1, \dots, S\}$. In the example above, then $\pi(\boldsymbol{\omega}_1, \boldsymbol{\omega}_2) = \pi(\boldsymbol{\omega}_1 | \boldsymbol{\omega}_2) \pi(\boldsymbol{\omega}_2)$.

The hierarchical decomposition (HD) prior of Fuglstad et al. (2020) is defined as the product of the above prior for $\{\boldsymbol{\omega}_s\}_{s=1}^S$ and the conditional prior for t , namely,

$$\pi(t, \{\boldsymbol{\omega}_s\}_{s=1}^S) = \pi(t | \{\boldsymbol{\omega}_s\}_{s=1}^S) \prod_{s=1}^S \pi(\boldsymbol{\omega}_s | \{\boldsymbol{\omega}_j\}_{j \in D(s)}), \quad (49)$$

and in the above example, then $\pi(t, \boldsymbol{\omega}_1, \boldsymbol{\omega}_2) = \pi(t | \boldsymbol{\omega}_1, \boldsymbol{\omega}_2) \pi(\boldsymbol{\omega}_1, \boldsymbol{\omega}_2)$. Fuglstad et al. (2020) suggest simplifying the prior density in (49) by replacing each $\pi(\boldsymbol{\omega}_s | \{\boldsymbol{\omega}_j\}_{j \in D(s)})$ with $\pi(\boldsymbol{\omega}_s | \{\boldsymbol{\omega}_j = \boldsymbol{\omega}_j^0\}_{j \in D(s)})$, where $\boldsymbol{\omega}_j^0$ is a fixed vector corresponding to the base model assumed for split j . The corresponding prior for $(t, \boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_S)$ is

$$\pi(t, \{\boldsymbol{\omega}_s\}_{s=1}^S) = \pi(t | \{\boldsymbol{\omega}_s\}_{s=1}^S) \prod_{s=1}^S \pi(\boldsymbol{\omega}_s | \{\boldsymbol{\omega}_j = \boldsymbol{\omega}_j^0\}_{j \in D(s)}) \quad (50)$$

and is referred to as the HD prior for LGMs.

When the likelihood is non-Gaussian and shrinkage of the total variance of the random effects at the latent level is appropriate, then Fuglstad et al. (2020) suggest using a PC prior for the conditional prior density of t , namely,

$$\pi(t | \{\boldsymbol{\omega}_s\}_{s=1}^S) = \frac{\lambda}{2\sqrt{t}} \exp(-\lambda\sqrt{t}).$$

In the case of the Gaussian–Gaussian model in (4), the variance of the error terms in $\boldsymbol{\epsilon}$, σ_ϵ^2 , needs to be taken into account when specifying the prior density for the hyperparameters. The variance of the sum of the random effects in $\boldsymbol{\eta}$ and $\boldsymbol{\epsilon}$ is equal to $t + \sigma_\epsilon^2$ and denoted by V . Define an additional split, $\boldsymbol{\omega}_\epsilon = (1 - \sigma_\epsilon^2/V, \sigma_\epsilon^2/V)$, and assign to it the PC prior $\pi(\boldsymbol{\omega}_\epsilon | \{\boldsymbol{\omega}_s = \boldsymbol{\omega}_s^0\}_{s=1}^S)$ with base model $\boldsymbol{\omega}_\epsilon^0 = (0, 1)$. When σ_ϵ^2 is expected to be well-identified due to a sufficient amount of data, then it is reasonable to assign a scale-invariant prior to V . In this case, the joint posterior

density is

$$\pi(V, \boldsymbol{\omega}_\epsilon, \{\boldsymbol{\omega}_s\}_{s=1}^S) = \pi(\boldsymbol{\omega}_\epsilon | \{\boldsymbol{\omega}_s = \boldsymbol{\omega}_s^0\}_{s=1}^S) \pi(\{\boldsymbol{\omega}_s\}_{s=1}^S) / V. \quad (51)$$

Alternatively, if there is some knowledge about the variance V that can be utilized, then a PC prior can be assigned to V conditional on $(\boldsymbol{\omega}_\epsilon, \{\boldsymbol{\omega}_s\}_{s=1}^S)$, namely,

$$\pi(V | \boldsymbol{\omega}_\epsilon, \{\boldsymbol{\omega}_s\}_{s=1}^S) = \frac{\lambda_V}{2\sqrt{V}} \exp(-\lambda_V \sqrt{V}).$$

Information about the scale of V is expressed through λ_V . The corresponding joint posterior density of $(V, \boldsymbol{\omega}_\epsilon, \{\boldsymbol{\omega}_s\}_{s=1}^S)$ is like the one in (51) except with $1/V$ replaced by $\pi(V | \boldsymbol{\omega}_\epsilon, \{\boldsymbol{\omega}_s\}_{s=1}^S)$.

4 Application of the Bayesian Gaussian–Gaussian Model—Evaluation of Manning’s Formula

The focus of this section is on Manning’s formula, which gives the mean velocity of a fluid in an open channel. It is an important tool in hydrology and hydraulics since it can be used to compute discharge, found by multiplying the mean velocity with the cross-sectional area. It is based on the geometry of the cross section, the slope of the channel, and the roughness of its surface. This roughness is quantified with Manning’s roughness coefficient that is a widely used measure in hydraulics engineering. Here, we evaluate the adequacy of Manning’s formula for observations of discharge in an open channel with the help of a Bayesian Gaussian–Gaussian model. We show how to build the Bayesian Gaussian–Gaussian model for these data, how to infer the model parameters, and how to interpret the results.

4.1 The Application and Data

Discharge in an open channel, which is driven by gravity, can be calculated through Manning’s formula with

$$Q = \frac{1}{\kappa} R^{2/3} A S^{1/2}, \quad (52)$$

where Q is the discharge, κ is Manning’s roughness coefficient, S is the slope of the channel, A is the cross-sectional area of the flow, R is called the hydraulic radius, which is given by $R = A/P$, where P is the wetted perimeter, that is, the length of the line of contact between the liquid and the channel boundary at the cross section

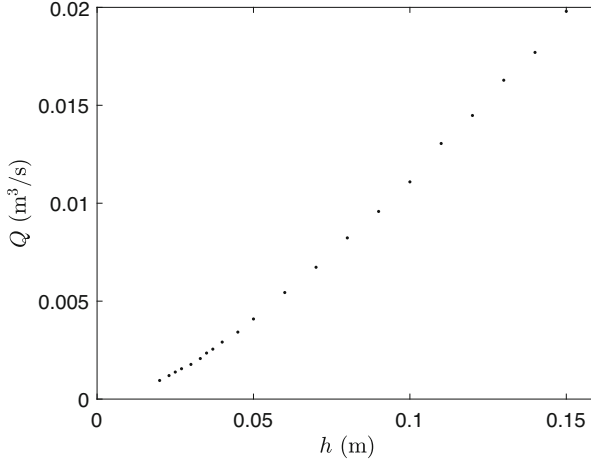


Fig. 7 Data on discharge and water depth (height)

(Chow, 1959). In the case of a rectangular channel with bottom width B and h as the water depth, the area is $A = Bh$ and the wetted perimeter is $P = B + 2h$.

Figure 7 shows measurements of discharge on the y -axis and corresponding water depth on the x -axis in a rectangular channel. These data are from Cheng et al. (2011) and are such that $B = 0.3$ m and $S = 0.586 \times 10^{-3}$.

4.2 Statistical Model

Let Q_i denote the i -th measurement of discharge and other variables with subscript i correspond to the i -th measurement of Q . Assume the total number of measurements is n . A candidate probability model for the measurements is given by

$$Q_i = \frac{1}{\kappa} R_i^{2/3} A_i S^{1/2} e^{\epsilon_i}, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2), \quad i \in \{1, \dots, n\},$$

where ϵ_i is the i -th error term and σ_ϵ^2 is the variance of the error terms. This model can be rewritten as

$$\log(Q_i) = \log\left(\frac{1}{\kappa}\right) + \frac{2}{3} \log(R_i) + \log(A_i) + \frac{1}{2} \log(S) + \epsilon_i.$$

Create a new variable, y_i , such that

$$y_i = \log(Q_i) - \frac{2}{3} \log(R_i) - \log(A_i) - \frac{1}{2} \log(S), \quad (53)$$

and let $\eta = \log(\kappa^{-1})$, which gives

$$y_i = \eta + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2), \quad i \in \{1, \dots, n\}. \quad (54)$$

We explore this model by estimating η and σ_ϵ with a simple Bayesian inference scheme, namely, we assume the prior density is $\pi(\eta, \sigma_\epsilon^2) \propto \sigma_\epsilon^{-2}$. The marginal posterior density for η is a scaled t -density with $n - 1$ degrees of freedom, center $\bar{y}$, and scale $s/\sqrt{n}$, where $\bar{y}$ and s are the sample mean and the sample standard deviation, respectively (Gelman et al., 2013). The posterior predictive density can be used to understand how samples from the proposed model will look. Furthermore, the fit of the proposed model to the observed data can be evaluated by checking how close the observed data fall to the posterior predictive samples. The posterior predictive density of the model in (54) is a scaled t -density with $n - 1$ degrees of freedom, center $\bar{y}$, and scale $s\sqrt{1 + 1/n}$, and its 95% posterior predictive interval is $\bar{y} \pm t_{0.975, n-1} s\sqrt{1 + 1/n}$, where $t_{0.975, n-1}$ is the 0.975 quantile of a standard t -density with $n - 1$ degrees of freedom.

Figure 8 shows the transformed discharge measurements, y_i , calculated with (53), as a function of water depth, along with the posterior mean of η and the 95% posterior predictive interval. Figure 8 reveals that the model $y_i = \eta + \epsilon_i$ does not adequately describe y as a function of water depth since the mean of y is not a constant as a function of water depth.

The model can be improved by adding a nonlinear function to the mean of y , i.e.,

$$E(y_i) = \eta + u(h_i),$$

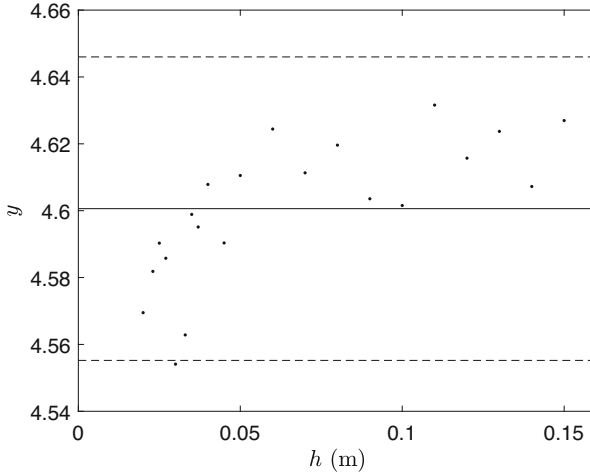


Fig. 8 The transformed discharge measurements, y , versus water depth in meters, h , along with the posterior mean of η (solid line) and the 95% posterior predictive interval for y (dashed lines)

and the model can be presented as $y_i = \eta + u(h_i) + \epsilon_i$, where ϵ_i is the error term of the model. The geometry of the cross section is continuous, and the roughness of the surface is assumed to be constant; thus, we expect $u(h_i)$ to be a continuous function and that at least its first and second derivatives are continuous. Based on these facts, we suggest modeling $u(h_i)$ with a Gaussian process that is twice mean-square differentiable since its realizations mimic a twice differentiable function (Stein, 1999). In particular, we suggest using a mean zero Gaussian process based on the Matérn covariance function (Matérn, 1986) with smoothness parameter $\nu = 2.5$. Its form is $\text{cov}(u(h_i), u(h_j)) = \sigma_u^2 \rho_{i,j}$, where

$$\rho_{i,j} = \left(1 + \frac{\sqrt{20}|h_i - h_j|}{\phi_u} + \frac{20|h_i - h_j|^2}{3\phi_u^2} \right) \exp \left(-\frac{\sqrt{20}|h_i - h_j|}{\phi_u} \right),$$

and σ_u is the marginal standard deviation of u and ϕ_u is the range parameter of u . Here, we opt for a fixed ϕ_u and set it equal to 0.16 m.

The Bayesian approach requires that prior densities are specified for each of the unknown parameters in the model. These prior densities should quantify the uncertainty about these parameters and reflect our knowledge about them before we see the data. Here, the unknown parameters are η , σ_ϵ , σ_u , and $u_i = u(h_i)$, $i \in \{1, \dots, n\}$. A priori knowledge about κ is available in the engineering literature. In the case of the data from Cheng et al. (2011), it is given that the surface of the channel is smooth, which is known to give a value of Manning's roughness coefficient in the interval (0.009, 0.012). These values correspond to values of η in the interval (4.42, 4.71). We make the following assumptions about η : (i) $\eta \in \mathbb{R}$; (ii) the uncertainty about η is quantified by a Gaussian prior density; (iii) this density is such that its mean is the center of the interval (4.42, 4.71), and half of the probability mass is under this interval. This results in a Gaussian prior density for η with mean $\mu_\eta = 4.565$, standard deviation $\sigma_\eta = 0.215$, and a 95% prior interval equal to (4.14, 4.99). In terms of Manning's roughness coefficient, the 95% prior interval for κ is (0.0068, 0.0159).

A penalizing complexity prior density is selected for σ_u (Simpson et al., 2017; Fuglstad et al., 2019), which results in an exponential prior density. We assume that the marginal standard deviation of u can have a magnitude similar to the prior standard deviation of η ; thus, the prior mean of σ_u is set equal to 0.20. We select an exponential prior density for σ_ϵ that is independent of the prior density for σ_u . The measurement error is expected to be relatively small, and that it is likely that $\sigma_\epsilon < 0.10$. The mean of the prior is set equal to 0.10. Let $\mathbf{u}$ denote the vector $(u_1, \dots, u_n)^\top$, and let R_u denote the $n \times n$ correlation matrix with elements $\rho_{i,j}$. The prior density of $\mathbf{u}$ conditional on σ_u is a Gaussian density with mean zero and covariance matrix $\sigma_u^2 R_u$.

4.3 Inference Scheme

Here, we demonstrate how to estimate the unknown parameters of the model proposed in Sect. 4.2 using the Bayesian approach. Let $\mathbf{y} = (y_1, \dots, y_n)^\top$, $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_n)^\top$, $\mathbf{x} = (\eta_1, \mathbf{u}^\top)^\top$, $X = \mathbf{1} = (1, \dots, 1)^\top$, $A = I$, and $Z = (X \ A)$. Our model can be written as

$$\mathbf{y} = X\eta + A\mathbf{u} + \boldsymbol{\epsilon} = Z\mathbf{x} + \boldsymbol{\epsilon},$$

where $\eta \sim \mathcal{N}(\mu_\eta, \sigma_\eta^2)$, $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \sigma_u^2 R_u)$, and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 I)$. The prior distribution of $\mathbf{x}$ is $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_x, \Sigma_x)$, where

$$\boldsymbol{\mu}_x = (\mu_\eta, \mathbf{0}^\top)^\top, \quad \Sigma_x = \begin{pmatrix} \sigma_\eta^2 & \mathbf{0}^\top \\ \mathbf{0} & \sigma_u^2 R_u \end{pmatrix},$$

and the marginal distribution of $\mathbf{y}$ is $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_y, \Sigma_y)$, where

$$\boldsymbol{\mu}_y = \mathbf{1}\mu_\eta, \quad \Sigma_y = \sigma_\eta^2 \mathbf{1}\mathbf{1}^\top + \sigma_u^2 R_u + \sigma_\epsilon^2 I.$$

To sample from the posterior density of $(\sigma_\epsilon, \sigma_u, \eta, \mathbf{u})$, we set up a sampler that is based on sampling first from the marginal posterior density of the hyperparameters, $\boldsymbol{\theta} = (\sigma_\epsilon, \sigma_u)$, $\pi(\boldsymbol{\theta}|\mathbf{y})$, and then from the posterior density of $\mathbf{x}$ conditional on $\boldsymbol{\theta}$, $\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$. That is, (i) draw $\boldsymbol{\theta}$ from

$$\pi(\boldsymbol{\theta}|\mathbf{y}) \propto \pi(\boldsymbol{\theta})\pi(\mathbf{y}|\boldsymbol{\theta}),$$

and (ii) draw $\mathbf{x}$ from

$$\pi(\mathbf{x}|\mathbf{y}, \boldsymbol{\theta}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_{x|\mathbf{y}}, \Sigma_{x|\mathbf{y}}),$$

where

$$\Sigma_{x|\mathbf{y}} = \Sigma_x - \Sigma_x Z^\top (Z \Sigma_x Z^\top + \Sigma_\epsilon)^{-1} Z \Sigma_x, \quad (55)$$

and

$$\boldsymbol{\mu}_{x|\mathbf{y}} = \boldsymbol{\mu}_x + \Sigma_x Z^\top (Z \Sigma_x Z^\top + \Sigma_\epsilon)^{-1} (\mathbf{y} - Z \boldsymbol{\mu}_x).$$

Let $\hat{\boldsymbol{\theta}}$ be the posterior mode of $\pi(\boldsymbol{\theta}|\mathbf{y})$. By setting $\boldsymbol{\theta}$ equal to $\hat{\boldsymbol{\theta}}$, and evaluating $\boldsymbol{\mu}_{x|\mathbf{y}}$ with that value of $\boldsymbol{\theta}$, we obtain the so-called empirical Bayes estimate of $\mathbf{x}$. Furthermore, $\Sigma_{x|\mathbf{y}}$, evaluated with $\hat{\boldsymbol{\theta}}$, gives an estimate of the uncertainty.

4.4 Results

Figure 9 shows the posterior mean of the sum $\eta + u(h)$ as a function of the water depth, h , along with a marginal 95% posterior interval for $\eta + u(h)$ for each h . The 95% posterior predictive interval for the transformed discharge, y , is shown in Fig. 10 as a function of h . The posterior predictive interval for y is wider than the posterior interval for $\eta + u(h)$ since the former takes into account the uncertainty in $\eta + u(h)$ on top of the variability in the error term, ϵ . The intervals represented with dashed lines, in Figs. 9 and 10, show the exact posterior intervals and posterior predictive intervals, while the dotted lines represent these intervals when the uncertainty in the hyperparameters, σ_u and σ_ϵ , is not taken into account. The posterior mode of the marginal posterior density of $(\sigma_u, \sigma_\epsilon)$ was plugged in, and only the uncertainty in $(\eta, \mathbf{u})$ is taken into account. Here, the difference is moderate; however, that may not be the case in general.

The point estimates and 95% credible intervals for the parameters η , κ , σ_u , and σ_ϵ are shown in Table 1. If we assume that η represents Manning's roughness coefficient, κ , then its point estimate for the observed channel is 0.0101, and according to its 95% credible interval, its value most likely lies in the interval (0.0092; 0.0111), which is expected for a smooth channel. In Fig. 9, the estimate of the curve, $\eta + u(h)$, is close to being a constant for values of h above 0.08 m, while for values below 0.08 m the curve increases with h . This indicates that here Manning's formula describes the flow well for water depth above 0.08 m, but it does not fully capture the physics for smaller values of the water depth. Under this assumption, the value of $\exp(-\eta - u(h))$, for some h above 0.08 m, might

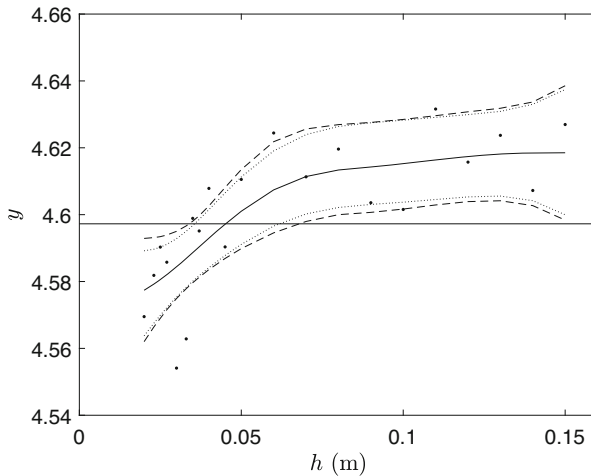


Fig. 9 The posterior mean of $\eta + u(h)$ as a function of water depth (m) (solid line) along with the corresponding 95% exact posterior interval (dashed lines) and the approximated 95% posterior interval (dotted lines)

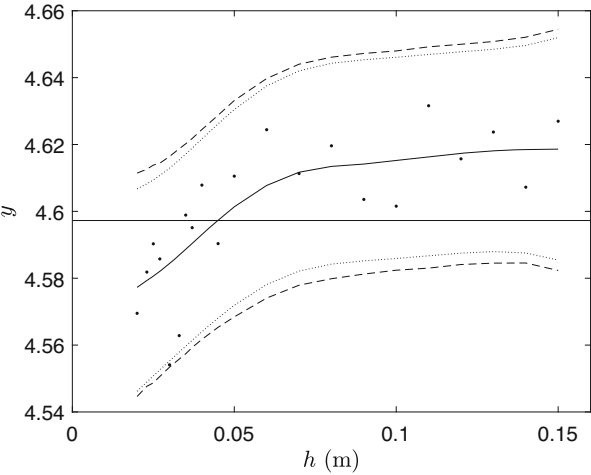


Fig. 10 The 95% posterior predictive interval for y as a function of water depth (m). The solid line represents the posterior prediction, the dashed lines represent the 95% exact posterior predictive interval, and the dotted lines represent the approximated 95% posterior predictive interval

Table 1 Table of point estimates (posterior median) and uncertainty (95% credible intervals) for the parameters η , κ , σ_u , and σ_ϵ

Parameter	Posterior median	95% credible interval
η	4.597	(4.504; 4.685)
κ	0.0101	(0.0092; 0.0111)
σ_u	0.0416	(0.0143; 0.1213)
σ_ϵ	0.0145	(0.0105; 0.0214)

better represent Manning’s roughness coefficient than $\exp(-\eta)$. The parameter σ_ϵ is the standard deviation of the error term in the proposed model and reflects the magnitude of the variability in the observed data not explained with the model $E(y_i) = \eta + u(h_i)$. The point estimate of σ_ϵ is 0.0145, indicating that the unexplained variability has a relatively small magnitude. In particular, here Q_i follows a lognormal distribution, and the ratio of its standard deviation over its mean is $\sqrt{\exp(\sigma_\epsilon^2) - 1} = 0.0145$ using the point estimate of σ_ϵ . This value matches reasonably well the reported discharge measurement error in Cheng et al. (2011) of 0.2-1.3%. The parameter σ_u is the marginal standard deviation of the Gaussian process describing $u(h)$. Its point estimate is 0.0416, and a corresponding 95% marginal interval for $u(h)$ is $(-0.0815; 0.0815)$, but the inferred function $u(h)$ does not stretch this far. Note that the credible interval for σ_u indicates that plausible values for σ_u lie between values that are three times smaller or three times larger than the point estimate. However, the uncertainty about $u(h)$ within the range of the observed data is small relative to the uncertainty in σ_u .

A comparison of Figs. 8 and 10 reveals that the expected value of the transformed variables, y , varies with the water depth, h . The proposed model is such that $E(y_i) =$

$\eta + u(h_i)$, and that means the median of Q_i is

$$Q_{i,\text{median}} = \frac{1}{\kappa} R_i^{2/3} A_i S^{1/2} e^{u(h_i)},$$

which also represents the underlying physical model for discharge as a function of water depth. Based on this formula and the above inference, we can claim that these data show a deviation from Manning’s formula since the term $e^{u(h_i)}$ is different from one. This pattern was also seen in other datasets in Cheng et al. (2011). These results suggest that Manning’s formula does not capture exactly the size of the mean velocity for the smallest values of the water depth even though the roughness of the surface is same. However, for water depth above a certain value (0.08 m in this case), the mean velocity is described adequately well by Manning’s formula.

The code, which was used to perform posterior calculations for the model presented in this section and draw Figs. 7, 8, 9, and 10, can be found at <https://github.com/hrafinkelsson/chapter1blgm>.

5 Application of a Bayesian LGM with a Univariate Link Function—Predicting Chances of Precipitation

In this application, we showcase a non-Gaussian likelihood. We have chosen the binomial likelihood with 4 trials, but other non-Gaussian likelihoods, e.g., Poisson or gamma, can be treated in a similar way. We use the two INLA approaches for inference presented in Sect. 2.2.2. These types of approximations can sometimes be less accurate for approximating the posterior than full MCMC schemes, but they are much faster. In the case of complex models, we may not have time to wait for the MCMC chain to run for long enough, and the approximations presented here may be more accurate than an MCMC algorithm stopped early.

5.1 The Application and Data

Precipitation is a collective term for rain, snow, and other water falling from the clouds. When we discuss or predict weather or climate, precipitation is one of the most interesting components and one that is surprisingly hard to model, e.g., compared to temperature. If we know that there is potential for rain on a given day, it is still hard to predict exactly whether it will rain or not, and how much. We will look at a simplified example on the seasonality of precipitation.

We study precipitation at the location Blindern, which is one of the campuses of the University of Oslo, Norway. In this application, we are interested in studying the general behavior (climate) as opposed to predicting the probability of rain for a given day. In more advanced studies, one could investigate how this seasonality of precipitation changes over decades.

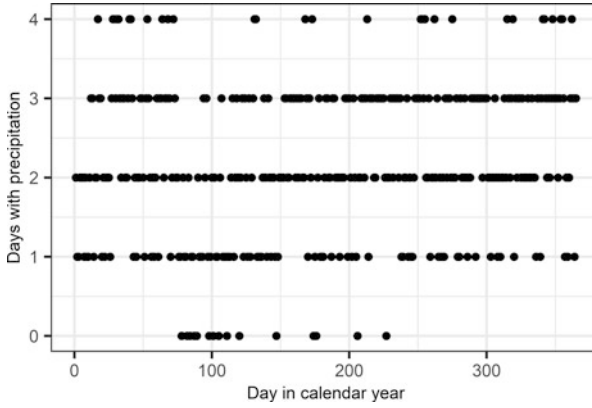


Fig. 11 The number of days with precipitation summed over four years. The x-axis is the day in the year, from 1 January. The maximum number of precipitation days for a given day of the year is four

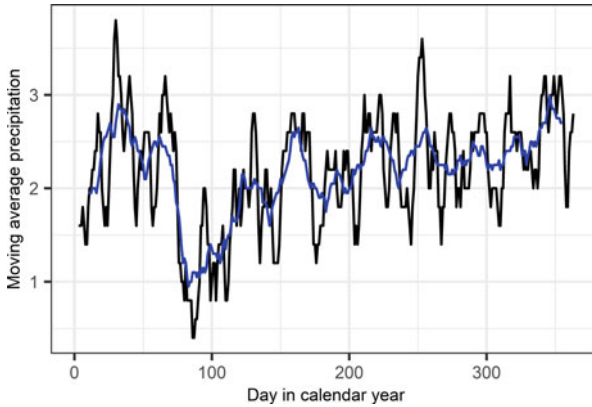


Fig. 12 Moving average of the number of days with precipitation. The black line is a 5-day moving average, and the blue line is a 20-day moving average

Figure 11 provides an exploratory visualization of the precipitation data from Blindern. It shows an aggregation over the four years 2017–2020 of occurrence of precipitation for each day in the year. We deleted the 29th of February year 2020, to make every year have 365 days. It is difficult, if not impossible, to look at this figure and directly get a clear understanding of the seasonal variation. To gain that understanding, we need a model.

To study the seasonality in the data, we show a moving average in Fig. 12. We observe a strong variation throughout the year, including a decrease in precipitation around day 100. The variation looks very different depending on how many days the moving average covers. To get a better understanding of the real seasonality, we

will fit a Bayesian hierarchical model with a smoothing component, in particular, a latent Gaussian model with a univariate link function.

5.2 Statistical Model

The goal of this modeling example is to create a smooth spline across the 365 days. We first set up a likelihood and a link function for the LGM,

$$y_t \sim \text{Binom}(N, p_t), \quad t \in \{1, \dots, 365\},$$

with probability mass function

$$\pi(y_t) = \text{Binom}(y_t|N, p_t) = \frac{N!}{y_t!(N - y_t)!} p_t^{y_t} (1 - p_t)^{1-y_t},$$

and with

$$p_t = \frac{e^{\eta_t}}{1 + e^{\eta_t}},$$

which is the logistic function. In our case, $N = 4$ for all the observations, but this N could have taken different values across the dataset. Furthermore,

$$\eta_t = \beta_0 + u(t),$$

where β_0 is the intercept and $u(t)$ is a spline in time.

Let ρ be a constant in $(0,1)$, and define $r_1 = 1 + 4\rho^2 + \rho^4$, $r_2 = -2\rho - 2\rho^3$, $r_3 = \rho^2$. We model u using the precision matrix

$$Q = \begin{bmatrix} r_1 & r_2 & r_3 & 0 & 0 & \dots & 0 & 0 & r_3 & r_2 \\ r_2 & r_1 & r_2 & r_3 & 0 & \dots & 0 & 0 & 0 & r_3 \\ r_3 & r_2 & r_1 & r_2 & r_3 & \dots & 0 & 0 & 0 & 0 \\ \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\ \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\ \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\ 0 & 0 & 0 & 0 & \dots & r_3 & r_2 & r_1 & r_2 & r_3 \\ r_3 & 0 & 0 & 0 & \dots & 0 & r_3 & r_2 & r_1 & r_2 \\ r_2 & r_3 & 0 & 0 & \dots & 0 & 0 & r_3 & r_2 & r_1 \end{bmatrix}. \quad (56)$$

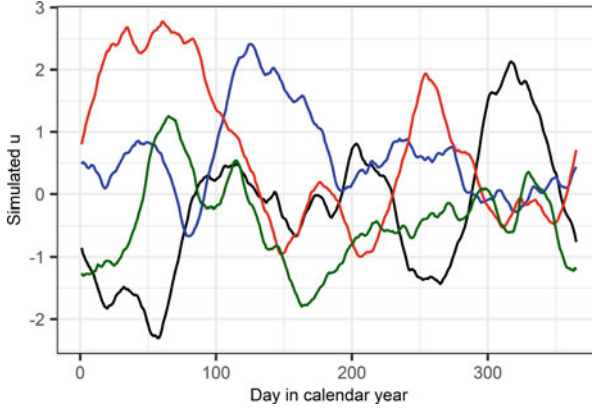


Fig. 13 Simulations from the prior model for u

To add a sensible scaling parameter to this precision matrix, we first find the value of the inverse diagonal (the variance), which turns out to be a constant, and we call it c_u . Then we define $\mathbf{u}$ to be

$$\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \sigma_u^2 c_u^{-1} Q^{-1}), \quad (57)$$

so that the scaling parameter σ_u is the marginal standard deviation.

In this example, we fix $\rho = 0.95$. The reader can use the online code to experiment with other values, see link below. With this value, the correlation after 90 days is 0.056. This means that if t and t' are 90 days apart, then $u(t)$ and $u(t')$ are almost uncorrelated. To further study the prior model of u , we show a few prior simulations in Fig. 13. We note that all these simulations are cyclic, i.e., they have similar values at the start and end of each year.

For the hyperparameter σ_u , we use a penalized complexity prior (PC prior), see Simpson et al. (2017),

$$\sigma_u \sim \text{Exponential}(\lambda),$$

where λ is chosen to be 6.93 so that the prior median for σ_u is 0.1.

5.3 Inference Scheme

The Simple-INLA model refers to the model in Sect. 5.2 inferred with the Simple-INLA approach described in Sect. 2.2.2. Instead of implementing this algorithm ourselves, we use the R package INLA to perform infer-

ence, by adding `control.inla = list(int.strategy="eb", strategy="gaussian")` to the `inla()` call.

The Full-INLA model refers to the model in Sect. 5.2 inferred with the Full-INLA approach described in Sect. 2.2.2. In R-INLA, this is achieved by not adding any `control.inla` argument; this approximation is the default option.

5.4 Results

Figure 14 shows the results of the two models. We note that the mean estimate is almost exactly the same, but that the upper and lower 95% quantiles are slightly different. This is because the posterior marginals are slightly skewed, which is not picked up on by the simpler approximation in the Simple-INLA model (using a plug-in estimate for θ). The figure shows increased precipitation around day 30, decreased precipitation around day 90, and increased precipitation later in the year.

In order to compare the smoothing spline with a non-smoothing model, we now introduce a model we refer to as Single. For each i , we have the observation y_i with probability p_i . On this p_i , we use a Beta(1, 1) prior, which is the conjugate prior to the binomial distribution. This means that we can compute the posterior analytically, and the result is

$$p_i | y_i \sim \text{Beta}(1 + y_i, 5 - y_i).$$

We use the median of the posterior density as the point estimate and represent uncertainty through the 95% equal-tailed credible interval.

We can study specific indices of the estimated probability p , see, e.g., Table 2 for the result at day 101. From this table, we see that, for all models, the point

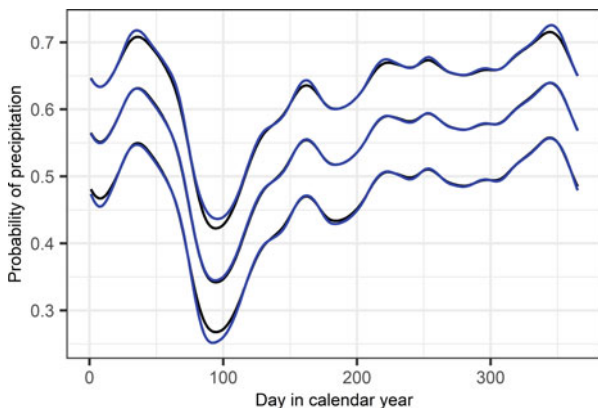


Fig. 14 Probability of precipitation on any given day, with 95% quantiles, from two different models. The Simple-INLA model is shown in black, and the Full-INLA model is shown in blue

Table 2 Table of point estimates (median) and uncertainty (95% credible intervals), for p at day 101. The observed value at day 101 is 0. The Single model is using only day 101 to estimate the probability at that day, with a beta prior. The Simple-INLA and Full-INLA models smooth the response over the year, using the same model but two different approximations

Method	Posterior median	95% credible interval
Single	0.129	(0.005; 0.522)
Simple-INLA	0.287	(0.222; 0.360)
Full-INLA	0.285	(0.213; 0.360)

estimate is shrunk toward 0.5. The Single model has a strong prior, but it shrinks the estimate less than the smoothing models. The smoothing models have weak priors, but the structure of the smoother u shrinks the estimate toward neighboring estimates. The uncertainty is much larger for the Single model because it does not borrow strength across observations. The smoothing spline $u(t)$ on the other hand borrows information from nearby observations to reduce the uncertainty in the parameter p .

This seasonal model can be used to make general statements about the weather. For example, we can look at Fig. 14 and claim that the third and fourth months have substantially less precipitation than an average month. Furthermore, we can use this model to study climate and climate change, by defining one dataset for every decade, compute results similar to Fig. 14, and investigate how the seasonality changes over the decades.

Conclusions on a finer scale, however, must be drawn with caution. Near day 345, there is a local maxima according to the posterior estimate, an increased probability of rain during the last two months relative to the previous three months. The question then is whether that increase is just a random artifact or a significant result. If we pay close attention to the upper and lower quantiles from day 300 to day 365, we see that it is possible to draw a straight horizontal line in between the two quantile lines. This means that the model does not reject the possibility that the precipitation is the same from day 300 to day 365. Hence, we cannot conclude that there has to be a significant precipitation increase at the end of the year, and the model only shows that such an increase would be plausible.

In this section, R-INLA was used to obtain posterior estimates and uncertainty. The code, which was used for these calculations and to draw Figs. 11, 12, 13, and 14, is available at <https://github.com/haakonbakkagit/blgm-book-bakka-code>.

6 Application of Bayesian LGMs with a Multivariate Link Function—Three Examples

Here, we introduce three examples of datasets that can be modeled with Bayesian LGMs with a multivariate link function. The first two examples can be inferred with Max-and-Smooth (Hrafnkelsson et al., 2021). The first dataset is the same as the

one found in chapter “Improving Numerical Weather Forecasts by Bayesian Hierarchical Modelling”. The second dataset is described in detail in chapter “Bayesian Latent Gaussian Models for High-Dimensional Spatial Extremes”. The third dataset is on monthly precipitation, and the model suggested for this dataset is an LGM with a multivariate link function (see Sigurdarson & Hrafnkelsson, 2016). The focus of this example is on the selection of prior densities for the hyperparameters.

6.1 Seasonal Temperature Forecast

This example involves predicting observed daily surface temperature across western Europe by using gridded surface temperature forecasts. The details of this example can be found in chapter “Improving Numerical Weather Forecasts by Bayesian Hierarchical Modelling”. The data are from 2000–2019 ($T = 20$), and the ensemble mean forecasts were calculated from ensembles with 10 members initialized on 29 June in each year. They are run up to 46 days into the future, thus, the temperature on 30 June is forecast at 1 day lead time, 1 July at 2 days lead time, and so on. The latitude–longitude resolution is 1.5 by 1.5 degrees, and the total number of grid points is $N = 1368$.

The predictive model for the observed temperature at grid point s and year t , $y_{s,t}$, is a Gaussian linear model with ensemble mean temperature forecast, $f_{s,t}$, as a covariate. The linear model is

$$y_{s,t} = \alpha_s + \beta_s(f_{s,t} - \bar{f}_s) + \epsilon_{s,t},$$

where $\bar{f}_s$ is the mean of the forecast at each grid point s , and $\epsilon_{s,t}$ is an error term, assumed independent of other error terms. The variance of $\epsilon_{s,t}$ is σ_s^2 , which is transformed to τ_s with the logarithmic transformation. The intercept, α_s , and the slope, β_s , are assumed to be on the real line and are not transformed. Thus, the multivariate link function is $g(\alpha_s, \beta_s, \sigma_s^2) = (\alpha_s, \beta_s, \log(\sigma_s^2)) = (\alpha_s, \beta_s, \tau_s)$.

The parameters $\alpha_1, \dots, \alpha_N$ are stored in $\boldsymbol{\alpha}$, and the vectors $\boldsymbol{\beta}$ and $\boldsymbol{\tau}$ contain $\beta_1, \dots, \beta_N$ and $\tau_1, \dots, \tau_N$, respectively. The vectors $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\tau}$ are modeled at the latent level with

$$\boldsymbol{\alpha} = A\mathbf{u}_\alpha, \quad \boldsymbol{\beta} = A\mathbf{u}_\beta, \quad \boldsymbol{\tau} = A\mathbf{u}_\tau.$$

Here, the fixed effects vectors and the error terms in the general setup in (30) are equal to zero, only the random effects remain, and the A matrix is equal to the identity matrix. The prior density for each of the random effects, $\mathbf{u}_\alpha$, $\mathbf{u}_\beta$, and $\mathbf{u}_\tau$, corresponds to a Gaussian two-dimensional random walk process on a regular grid, also referred to as a first-order intrinsic Gaussian Markov random field on a regular

lattice (see Sect. 3.2 and Rue & Held, 2005, pp. 104–108). The precision matrices of $\mathbf{u}_\alpha$, $\mathbf{u}_\beta$, and $\mathbf{u}_\tau$ are given by $\mathbf{Q}_m = \theta_m \mathbf{Q}_u$, $m \in \{\alpha, \beta, \tau\}$, where $\mathbf{Q}_u$ is the precision matrix of a first-order IGMRF on a regular lattice, and θ_m is a precision parameter.

The ML estimates of the parameters from each grid point, $(\hat{\alpha}_s, \hat{\beta}_s, \hat{\tau}_s)$, are stored in $\hat{\boldsymbol{\eta}}$. The Gaussian approximation of the likelihood function for $(\alpha_s, \beta_s, \tau_s)$ is such that its precision matrix is diagonal, and thus, $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\tau}$ can be inferred separately. For example, the conditional posterior density of $\boldsymbol{\tau}$ is Gaussian with precision matrix $\mathbf{Q}_{\tau|\hat{\boldsymbol{\tau}}} = \theta_\tau \mathbf{Q}_u + (T/2)\mathbf{I}$ and mean

$$\boldsymbol{\mu}_{\tau|\hat{\boldsymbol{\tau}}} = \{(2/T)\theta_\tau \mathbf{Q}_u + \mathbf{I}\}^{-1} \hat{\boldsymbol{\tau}},$$

where $\hat{\boldsymbol{\tau}}$ is the ML estimate of $\boldsymbol{\tau}$. The conditional posterior densities of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are similar. The approximate marginal posterior density of θ_τ is given by

$$\pi(\theta_\tau | \hat{\boldsymbol{\tau}}) \propto \pi(\theta_\tau) \pi(\hat{\boldsymbol{\tau}} | \theta_\tau) = \pi(\theta_\tau) \frac{\pi(\hat{\boldsymbol{\tau}} | \boldsymbol{\tau}, \theta_\tau) \pi(\boldsymbol{\tau} | \theta_\tau)}{\pi(\boldsymbol{\tau} | \hat{\boldsymbol{\tau}}, \theta_\tau)},$$

where $\pi(\hat{\boldsymbol{\tau}} | \boldsymbol{\tau}, \theta_\tau)$ is a Gaussian density with mean $\boldsymbol{\tau}$ and precision $(T/2)\mathbf{I}$. Using cross-validation, it is shown that posterior estimates of the parameters yield better temperature forecasts than the local ML estimates.

6.2 High-dimensional Spatial Extremes

In this example, spatially referenced precipitation data are analyzed with respect to their extremes. Details of these data, and an in-depth analysis of them, can be found in chapter “Bayesian Latent Gaussian Models for High-Dimensional Spatial Extremes”. A high-dimensional dataset on daily satellite-derived precipitation over Saudi Arabia, observed at 2738 grid cells from 2000 to 2019, is used to learn about the current behavior of extreme precipitation in this area. To make better use of the extremes found in this dataset, a statistical model, which relies on the Poisson point process representation of extremes, is set forth. This model uses information from all precipitation observations that exceed their site-wise threshold, which, when properly tuned, makes better use of data than the commonly used block maxima model. Alternatively, a peak-over-threshold model based on the generalized Pareto distribution and a Bernoulli model for the probability of threshold exceedance could have been selected. However, this model can be sensitive to the value of the selected threshold, while the Poisson point process model is less sensitive to it and does not require a Bernoulli model for the probability of threshold exceedance.

Here, the Poisson point process model is assumed to be stationary in time, and thus, the number of parameters at each site is three, i.e., a location parameter, μ_i , a scale parameter, σ_i , and a shape parameter, ξ_i . These three parameters are transformed to the real line using $\psi_i = \log(\mu_i)$, $\tau_i = \log(\sigma_i/\mu_i)$, and $\phi_i = h(\xi_i)$, where h is a function that assumes ξ_i is in the interval $(-0.5, 0.5)$ and

transforms this interval to the real line such that $h(0) = 0$, and for values of ξ_i around zero the function is close to being linear with slope equal to one (see Jóhannesson et al., 2022). So, here the multivariate link function is $g(\mu_i, \sigma_i, \xi_i) = (\log(\mu_i), \log(\sigma_i/\mu_i), h(\xi_i))$.

The transformed parameters across all sites are stored in three vectors, one for each type of parameter, i.e., $\boldsymbol{\psi}$, $\boldsymbol{\tau}$, and $\boldsymbol{\phi}$. The transformed location and scale parameters are modeled with a constant, a spatial component, and an unstructured component, while the transformed shape parameter is modeled with a constant and an unstructured component. The two spatial components are modeled according to the SPDE approach of Lindgren et al. (2011). The model for these three vectors at the latent level is

$$\begin{aligned}\boldsymbol{\psi} &= \mathbf{1}\beta_1 + A\mathbf{u}_1 + \boldsymbol{\epsilon}_1, \\ \boldsymbol{\tau} &= \mathbf{1}\beta_2 + A\mathbf{u}_2 + \boldsymbol{\epsilon}_2, \\ \boldsymbol{\phi} &= \mathbf{1}\beta_3 + \boldsymbol{\epsilon}_3,\end{aligned}$$

where $\mathbf{1}$ is a vector of ones, β_1 , β_2 , and β_3 are intercept parameters, $\mathbf{u}_1$ and $\mathbf{u}_2$ are the SPDE spatial components at the grid points of a triangulated mesh, A is a projection matrix that projects the spatial components from the triangulated mesh to the observational sites, and $\boldsymbol{\epsilon}_1$, $\boldsymbol{\epsilon}_2$, and $\boldsymbol{\epsilon}_3$ are unstructured model components.

Since the model of choice is the extended latent Gaussian model, the prior densities for β_1 , β_2 , β_3 , $\mathbf{u}_1$, $\mathbf{u}_2$, $\boldsymbol{\epsilon}_1$, $\boldsymbol{\epsilon}_2$, and $\boldsymbol{\epsilon}_3$ are Gaussian. Furthermore, at each site i , the generalized likelihood for (ψ_i, τ_i, ϕ_i) , which is supported with an additional prior for ϕ_i , is approximated with a Gaussian density with mean equal to the generalized ML estimates $(\hat{\psi}_i, \hat{\tau}_i, \hat{\phi}_i)$, and precision matrix equal to the observed information matrix at site i . The generalized ML estimates across sites are stored in $\hat{\boldsymbol{\eta}}$. This approximation is used to form the response level of the Gaussian–Gaussian surrogate-model, i.e., the transformed data, $\hat{\boldsymbol{\eta}}$, are assumed to follow a Gaussian density with mean $\boldsymbol{\eta} = (\boldsymbol{\psi}^\top, \boldsymbol{\tau}^\top, \boldsymbol{\phi}^\top)^\top$ and a precision matrix that is equal to the observed information of the joint generalized likelihood.

As a result, the conditional posterior density of $\boldsymbol{\psi}$, $\boldsymbol{\tau}$, $\boldsymbol{\phi}$, β_1 , β_2 , β_3 , $\mathbf{u}_1$, and $\mathbf{u}_2$, all stored in $\mathbf{x}$, is Gaussian with a sparse precision matrix due to the SPDE specification of the spatial components, facilitating fast posterior sampling. Refer to this Gaussian density as $\pi(\mathbf{x}|\hat{\boldsymbol{\eta}}, \boldsymbol{\theta})$. The hyperparameters of the model, $\boldsymbol{\theta}$, can be sampled through an approximation of their marginal posterior density, namely,

$$\pi(\boldsymbol{\theta}|\mathbf{y}) \approx \pi(\boldsymbol{\theta}|\hat{\boldsymbol{\eta}}) \propto \pi(\boldsymbol{\theta}) \frac{\pi(\hat{\boldsymbol{\eta}}|\mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})}{\pi(\mathbf{x}|\hat{\boldsymbol{\eta}}, \boldsymbol{\theta})}.$$

Here, the density $\pi(\boldsymbol{\theta})$ is of a low dimension, $\pi(\hat{\boldsymbol{\eta}}|\mathbf{x}, \boldsymbol{\theta})$ does not depend on $\boldsymbol{\theta}$ and can therefore be excluded in the calculations, the other two densities, $\pi(\mathbf{x}|\boldsymbol{\theta})$ and $\pi(\mathbf{x}|\hat{\boldsymbol{\eta}}, \boldsymbol{\theta})$, have sparse precision matrices, and thus, computation for the above marginal density is feasible even for high-dimensional data.

6.3 Monthly Precipitation

In this example, we explore the latent Gaussian model for monthly precipitation presented in Sigurdarson and Hrafnkelsson (2016). Each month is modeled separately, and for any given month, the observed monthly precipitation in year t at site i , $\tilde{y}_{i,t}$, is transformed to $y_{i,t} = \tilde{y}_{i,t}^{0.4}$. The main focus here is on redefining the prior densities for the hyperparameters using the principles of PC priors. The model for the observed monthly precipitation can be presented as

$$y_{i,t} = \eta_{i,t} + \epsilon_{i,t},$$

where $y_{i,t}$ is Gaussian conditional on $\eta_{i,t}$, with conditional mean

$$\eta_{i,t} = x_{\eta,i} \beta_{\eta} + u_{1,t} + u_{2,i},$$

where $x_{\eta,i}$ is a meteorological covariate associated with the mean of the transformed monthly precipitation (see Sigurdarson & Hrafnkelsson, 2016), β_{η} is the corresponding coefficient, $u_{1,t}$ is a temporal term, and $u_{2,i}$ is a spatial term. The data are observed at J sites over T years. The error term, $\epsilon_{i,t}$, is spatially correlated with other error terms within year t , and its marginal variance is

$$\text{var}(\epsilon_{i,t}) = \sigma_i^2(1 + \kappa E_i),$$

where κ is an unknown parameter and E_i represents the scale at site i , estimated with the median of $y_{i,t}$. Furthermore, σ_i is transformed to $\tau_i = \log(\sigma_i)$, and τ_i is modeled spatially with

$$\tau_i = x_{\tau,i} \beta_{\tau} + u_{\tau,i},$$

where $x_{\tau,i}$ is a meteorological covariate associated with the log-scale parameter, β_{τ} is the corresponding coefficient, and $u_{\tau,i}$ is a mean zero spatial term. The parameters β_{η} , $u_{1,t}$, $u_{2,i}$, β_{τ} , and $u_{\tau,i}$ are assigned Gaussian prior densities. So, the model is a latent Gaussian model with a bivariate link function, $g(\eta_{i,t}, \sigma_i) = (\eta_{i,t}, \log(\sigma_i)) = (\eta_{i,t}, \tau_i)$. The response density is Gaussian; however, the model is not categorized as a Gaussian–Gaussian model since both the location parameter and the scale parameter are modeled at the latent level.

Let $\mathbf{y}$ denote a vector containing the data from one of the twelve months where the first elements of $\mathbf{y}$ correspond to transformed observations from all the observational sites in the first year, followed by observations from the second year and so forth. The model of $\mathbf{y}$ is

$$\mathbf{y} \sim \mathcal{N}(\boldsymbol{\eta}, \mathbf{Q}(\boldsymbol{\tau})(R_y + \kappa E)\mathbf{Q}(\boldsymbol{\tau})), \quad (58)$$

where $\boldsymbol{\eta}$ contains each $\eta_{i,t}$, which corresponds to $y_{i,t}$, and it is modeled with

$$\boldsymbol{\eta} = X_{\boldsymbol{\eta}}\boldsymbol{\beta}_{\boldsymbol{\eta}} + A_1\mathbf{u}_1 + A_2\mathbf{u}_2,$$

where $X_{\boldsymbol{\eta}} = \mathbf{1}_T \otimes (x_{\eta,1}, \dots, x_{\eta,J})^\top$, $\otimes$ denotes the Kronecker product, $\mathbf{u}_1$ and $\mathbf{u}_2$ contain the temporal and spatial effects associated with $\boldsymbol{\eta}$, respectively, $A_1 = I_T \otimes \mathbf{1}_J$, $A_2 = \mathbf{1}_T \otimes I_J$, and I_n and $\mathbf{1}_m$ are an n identity matrix and an m vector of ones, respectively. The matrix R_y is a block-diagonal matrix given by $I_T \otimes R_\psi$, where R_ψ is a Matérn correlation matrix of dimension J with range ψ and smoothness parameter $\nu_y = 1.5$.

Let $\boldsymbol{\tau} = (\tau_1, \dots, \tau_J)^\top$ and $\boldsymbol{\sigma} = (\exp(\tau_1), \dots, \exp(\tau_J))^\top$. The matrix $Q(\boldsymbol{\tau})$ is given by $Q(\boldsymbol{\tau}) = I_T \otimes \text{diag}(\boldsymbol{\sigma})$. The model for $\boldsymbol{\tau}$ is

$$\boldsymbol{\tau} = \mathbf{x}_\tau \boldsymbol{\beta}_\tau + A_3 \mathbf{u}_\tau, \quad (59)$$

where $\mathbf{x}_\tau = (x_{\tau,1}, \dots, x_{\tau,J})^\top$, $A_3 = I_J$, and $\mathbf{u}_\tau$ contains the spatial effects associated with τ . Note that the elements of A_1 , A_2 , and A_3 are either equal to one or zero. The Gaussian prior densities of the random effects, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_\tau$, are

$$\mathbf{u}_1 \sim \mathcal{N}(\mathbf{0}, \sigma_1^2 I_T), \quad \mathbf{u}_2 \sim \mathcal{N}(\mathbf{0}, \sigma_2^2 R_\phi), \quad \mathbf{u}_\tau \sim \mathcal{N}(\mathbf{0}, \sigma_\tau^2 R_\rho),$$

where R_ϕ and R_ρ are Matérn correlation matrices of dimension J with range ϕ and ρ , respectively, both with smoothness parameters equal to 1.5. The parameters σ_1^2 , σ_2^2 , and σ_τ^2 are the marginal variances of $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_\tau$, respectively. Note that this LGM cannot be inferred with the Max-and-Smooth approach of Hrafnkelsson et al. (2021) due to the correlation in R_y . By treating $\boldsymbol{\tau}$ as hyperparameters, then the model parameters could be inferred using the sampling scheme for Gaussian–Gaussian models in the Appendix.

The next step is to update the prior densities of the hyperparameters proposed in Sigurdarson and Hrafnkelsson (2016) with the PC prior densities in Sect. 3.3. Here, a reasonable base model is the one that sets the random effects $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_\tau$ equal to zero. This is equivalent to $\sigma_1^2 = 0$, $\sigma_2^2 = 0$, and $\sigma_\tau^2 = 0$. The range parameters ϕ and ρ also need to be taken into account. The random effect $\mathbf{u}_\tau$ is the only random effect in the model for $\boldsymbol{\tau}$, and thus, we can directly apply the PC prior for the Matérn parameters in Sect. 3.3.2. Here, the variability in the transformed monthly precipitation is modeled on a logarithmic scale, and it is known that the covariate, x_τ , is weakly correlated with the scale. The τ parameter is mainly found in the range between 0.5 and 2.0, and the spatial correlation in τ is expected to be strong for distances less than 20 km. Thus, the PC prior that quantifies our prior beliefs is the one stating that a priori the range parameter, ρ , is less than 20 km with probability 0.10 and the marginal standard deviation, σ_τ , is greater than 0.75 with probability 0.10. Given these quantities, the prior density in (43) can be assigned to (ϕ, σ_τ) using λ_1 and λ_2 calculated according to (44).

We have two options for the construction of a PC prior for the hyperparameters governing $\mathbf{u}_1$ and $\mathbf{u}_2$, that is, σ_1^2 , ϕ , and σ_2^2 . We can either express ignorance about the variance decomposition or we can assign a PC prior for a dual split to it, see Sect. 3.3.3. Both options are considered here. When ignorance is expressed about the variance decomposition, then Fuglstad et al. (2020) suggest a Dirichlet prior for the weights $(\omega_1, \omega_2) = (\sigma_1^2/t, \sigma_2^2/t)$ with parameter $\mathbf{a} = (a, a)$. If we set w_0 in (46) equal to 0.5, then the suggested value of a is 1, and the prior for (ω_1, ω_2) is a Dirichlet density with $\mathbf{a} = (1, 1)$. Thus, the marginal density of ω_2 is a uniform density on the interval $(0, 1)$. Alternatively, we can assign a PC prior for a dual split to ω_2 as suggested in Sect. 3.3.3. Due to the nature of precipitation, we know that the variability between years is likely to be non-zero, while it is possible that the covariate, x_η , can explain much of the spatial variability in the data. Thus, we set the base model such that $\omega_2 = 0$ and assign the PC prior in (48) to ω_2 with λ such that the median is equal to 0.25. Note that here $\tilde{\Sigma} = (1 - \omega_2)\tilde{\Sigma}_1 + \omega_2\tilde{\Sigma}_2$, $\tilde{\Sigma}_1 = A_1 I_T A_1^\top$, and $\tilde{\Sigma}_2 = A_2 R_\phi A_2^\top$. To simplify calculations, we suggest setting ϕ in R_ϕ equal to a preselected value when computing $\tilde{\Sigma}_2$.

Under these two options of priors for ω_2 , the same PC prior is assigned to the variance of $\mathbf{u}_1 + \mathbf{u}_2$, namely, $t = \sigma_1^2 + \sigma_2^2$. When building the PC prior for t , we make use of the fact that the predictive power of the covariate x_η is such that it explains more than half of the variability in the mean of the transformed monthly precipitation. The variability in η is different between months; however, if we take January as an example, then the variance in η is about 9. We use half of that value, 4.5, as the value that t exceeds a priori with probability 0.10. The prior for t has the form given in (47).

A prior density for the range parameter ϕ is needed. Since we have already specified a prior for σ_2 , we opt for using the prior in (41) for ϕ . It is based on the joint PC prior for the range and the marginal standard deviation in the Matérn covariance function given by (43). The dimension of the spatial field is $d = 2$; thus, the prior density for ϕ is $\pi(\phi) = \lambda_\phi \phi^{-2} \exp(-\lambda_\phi \phi^{-1})$, and it shrinks toward $1/\phi = 0$, i.e., a flat spatial field. We expect the spatial dependence in $\mathbf{u}_2$ to be strong for distances under 5 km. Therefore, we select the PC prior for ϕ such that ϕ is less than 5 km with probability 0.10.

Bibliographic Note

The aim of this bibliographic note is to give the reader an idea about the topics that are of interest for those using Bayesian latent Gaussian models, and those developing them. However, the aim is not to give a complete view of everything related to BLGMs, as that would be too extensive. We proceed from the assumption that the reader of this chapter is familiar with the fundamental concepts of Bayesian statistics and has some knowledge of spatial statistics, time series models, and spatio-temporal models.

Accessible text on the fundamental concepts of Bayesian statistics can be found in: Hoff (2009), Gelman et al. (2013), Reich and Ghosh (2019), Albert and Hu (2019), McElreath (2020), and Johnson et al. (2022). Gelman et al. (2020) give an introductory-level tour of linear regression models and generalized linear models from the Bayesian and the classical perspectives. The generalized linear model was introduced by Nelder and Wedderburn (1972), and further details on the generalized linear model can be found, e.g., in McCullagh and Nelder (1989). Hierarchical linear regression models (another term for Gaussian–Gaussian models) are covered in Gelman and Hill (2007). Wakefield (2013) provides a thorough exploration of generalized linear mixed models (which include latent Gaussian models), using both the Bayesian and the frequentist approaches for inference, and covers both theory and applications.

Markov chain simulation of probability densities dates back to the work of Metropolis and Ulam (1949) and Metropolis et al. (1953) on the Metropolis algorithm, which was generalized by Hastings (1970), referred to as the Metropolis–Hastings algorithm. The Gibbs sampler was presented for statistical models by Geman and Geman (1984), and Gelman and Rubin (1992) demonstrated how Metropolis–Hastings steps can be applied within a Gibbs sampler. Posterior simulation for latent Gaussian models dates back to the work of Gelfand et al. (1990) and Zeger and Karim (1991). In Gilks et al. (1996), a variety of examples of Markov chain Monte Carlo sampling schemes are presented. Murray et al. (2010) introduced elliptical slice sampling, which is a robust MCMC sampler for models with Gaussian priors, including Bayesian latent Gaussian models, and is based on slice sampling (Neal, 2003) and the work of Neal (1999).

Knorr-Held and Rue (2002) presented an efficient method to sample from latent Gaussian models by sampling the latent parameters and the hyperparameters jointly. Rue et al. (2009) introduced a new method for fitting latent Gaussian models using integrated nested Laplace approximations (INLA) that does not require posterior sampling but is based on approximating the posterior density numerically. A review of the earlier versions of the R package, R-INLA, is given in Rue et al. (2017). In van Niekerk et al. (2023), a discussion on the latest implementation in R-INLA is given, in particular, a new parallelization scheme has been implemented, which, in the case of large models, leads to speedups of a factor 10 or more (Gaedke-Merzhäuser et al., 2022). Furthermore, a low-rank variational Bayes correction is applied to the posterior mean of the parameters, which is fast, as it maintains the scalability of the Laplace approximation, (van Niekerk & Rue, 2021). The Stan software is based on Hamiltonian Monte Carlo (HMC) for posterior sampling. HMC was introduced by Duane et al. (1987) and presented for statistics problems by Neal (1994). HMC uses ideas from physics for a faster exploration of the posterior density (see, e.g., Neal, 2011; Girolami & Calderhead, 2011; Hoffman & Gelman, 2014). Practical approaches to fitting generalized linear mixed models can be found in Lunn et al. (2012) and Lunn et al. (2000) (using BUGS), Wang et al. (2018) and Gómez-Rubio (2021) (using R-INLA), and Congdon (2019) (using BUGS, JAGS, and Stan). Andreon and Weaver (2015) provide guidance on fitting Bayesian models with JAGS to data from astronomy and physics. Martin et al. (2021) demonstrate

how to do Bayesian modeling and computation in Python. NIMBLE is a statistical software designed for computationally demanding high-dimensional hierarchical models (see de Valpine et al., 2017). It is built in R and can incorporate models written in BUGS.

Priors are an important part of any Bayesian model. In general, the priors should be carefully selected for the problem that they will be applied to. The first step involves gathering the available knowledge on the problem. Once this knowledge has been gathered, the construction of the priors can begin. Below, we discuss methods to formulate this knowledge into prior densities with focus on Bayesian latent Gaussian models.

Priors that are designed to hold as little information as possible are referred to as objective priors (Kass & Wasserman, 1996; Berger, 2006). Jeffreys' priors are an example of objective priors (Jeffreys, 1961). The reference prior framework (Bernardo, 1979; Berger et al., 2009) is an approach to construct objective priors in a rigorous manner. Reference priors have been derived for hyperparameters in some hierarchical models. For example, Berger et al. (2001) proposed objective priors for parameters in Gaussian process that are observed without measurement errors. De Oliveira (2007) extended this work for geostatistical models that assume measurement errors. Other extensions of Berger et al. (2001) can be found in Paulo (2005), Kazianka and Pilz (2012), and Kazianka (2013). Furthermore, Keefe et al. (2019) introduce a reference prior for the parameter σ_u in intrinsic conditional autoregressive models within Gaussian–Gaussian models.

Priors can be constructed by formulating the knowledge of an expert in a probabilistic form. This approach is referred to as eliciting prior densities. It is particularly useful when the data at hand are scarce or the data are such that they provide a limited amount of information about the parameter or parameters of interest. Methods for eliciting prior densities can, for example, be found in Garthwaite et al. (2005), O'Hagan et al. (2006), Oakley (2010), and Daneshkhah and Oakley (2010).

Weakly informative priors are discussed in Sect. 3.1 (see also, e.g., Gelman, 2006; Gelman et al., 2008; Evans & Jang, 2011; Polson & Scott, 2012). In some sense, they fall between the objective priors and the elicited priors. The idea behind weakly informative priors is based on the assumption that we are not completely ignorant about the parameter of interest, while strong prior information about it is not assumed. Most of the probability mass is assigned to a relative wide range of plausible values with respect to the data at hand, while low probability is assigned to non-plausible values. This works well for posterior inference when the data are non-informative about a given parameter, and the worst that can happen is that the posterior density will be nearly the same as the prior density. When the data are informative, the posterior will be dominated by the likelihood.

The PC priors of Simpson et al. (2017) are useful for constructing prior densities for the hyperparameters of Bayesian latent Gaussian models, as discussed in Sect. 3.3. PC priors for hyperparameters, which were not introduced in Sect. 3.3, include priors for the parameters of the autoregressive time series model of order two and higher (Sørbye & Rue, 2017), priors for the degrees of freedom in Bayesian

P-splines (Ventrucchi & Rue, 2016), and priors for hyperparameters of random fields in log-Gaussian Cox point processes (Sørbye et al., 2019).

Statistical models for spatial data are important tools for the fields of geophysics and environmental sciences. In the handbook of Gelfand et al. (2010), a broad view of spatial statistics is given, and the historical development of this branch of statistics is presented in its first chapter (Diggle, 2010). Important contributions from the early days of spatial statistics include Krige (1951), Matérn (1960), Matheron (1971), Besag (1974), and Ripley (1977). These contributions, along with other contributions, were presented in the books of Cliff and Ord (1981), Ripley (1981), and Cressie (1991, 1993). There are three divisions of spatial statistics that are particularly important for statistical modeling within geophysics and environmental sciences, namely, geostatistics, spatial models on lattices, and point processes. It should be noted that there are links between them. These divisions were presented in the contributions mentioned above. Below, we mention a few of the contributions to these divisions.

Some of the key tools in geostatistics are probabilistic models for processes that are defined on continuous coordinates, which are well-suited for point-referenced data. The most commonly used processes are Gaussian processes. They are defined through a Gaussian distribution and a covariance function. The book of Stein (1999) covers, among other topics, a theoretical treatment of Gaussian processes. The mathematical foundations of spatial statistics, including Gaussian processes, are given by van Lieshout (2019), along with examples in R. Somewhat more applied books on the Bayesian approach to geostatistics are those of Diggle and Ribeiro (2007) and Banerjee et al. (2014). In Bivand et al. (2013), it is shown how to use R to handle and analyze spatial data. For a comprehensive background on theory and methods for spatial statistics, see Gaetan and Guyon (2010), Chilès and Delfiner (2012), and Kent and Mardia (2022).

Data observed on regular lattices are common in geophysics and environmental sciences. Gaussian Markov random fields (GMRFs) (Besag, 1974; Besag et al., 1991; Rue & Held, 2005) are well-suited for statistical modeling of data of this type. Rue and Held (2005) cover the theory of Gaussian Markov random fields, how they can be applied within hierarchical models, how to infer the parameters of these models using MCMC, and how to handle computation in a fast and reliable manner. Gaussian Markov random fields are also well-suited for models that handle data on irregular lattices (Cressie, 1993; Rue & Held, 2005; Banerjee et al., 2014). A Gaussian Markov random field is defined through a multivariate Gaussian distribution and its precision matrix. This precision matrix can be built through a conditional model that specifies the conditional distribution of each random element conditional on the other elements in the random vector. In many applications, the conditional distribution of each random element only depends on a few of the other random elements. This leads to a sparse precision matrix, which gives an advantage in terms of computation. See Ferreira et al. (2021) for an example of scalable computations for Gaussian–Gaussian models with intrinsic GMRF random effects.

Lindgren et al. (2011) introduced a method to represent Gaussian processes that are defined through a Matérn covariance function (Matérn, 1986), as Gaussian

Markov random fields with sparse precision matrices. This was done by finding an approximate stochastic weak solution, built on a triangulated grid, to the stochastic partial differential equations that represents the Matérn covariance function. The approximate stochastic weak solution has a sparse precision matrix, which makes computation faster compared to using a fully populated covariance matrix based on the Matérn covariance function. The approach of Lindgren et al. (2011), referred to as the SPDE approach, has been integrated into R-INLA (Bakka et al., 2018). Bolin (2014) introduced how the SPDE approach can be extended to non-Gaussian fields. An overview of the SPDE approach for Gaussian and non-Gaussian fields is given in Lindgren et al. (2022b). Spatial modeling using R-INLA is also presented in Krainski et al. (2018), Blangiardo and Cameletti (2015), and Moraga (2020).

Spatial datasets are often large, and, in these cases, it becomes infeasible to analyze them with traditional Gaussian processes. In Heaton et al. (2019), several methods that are designed to handle large spatial datasets were considered and compared, see references to these methods therein.

Point processes are models for data that have random locations, i.e., their coordinates are random. Point processes are presented in a few of the papers and books mentioned above (Ripley, 1977, 1981; Cressie, 1993; van Lieshout, 2019), see also Møller and Waagepetersen (2004) on inference for spatial point processes. Log Gaussian–Cox processes assume the log intensity of the point process is modeled with a Gaussian process prior. Rathbun and Cressie (1994) were the first to define log Gaussian–Cox processes. Their specification was restricted to intensity measures that are constant within each square and modeled with a Gaussian Markov random field on a regular lattice. Møller et al. (1998) assumed the intensity measures were continuous and modeled them with continuous Gaussian processes. Illian et al. (2012) fitted log Gaussian–Cox processes using INLA assuming a Gaussian Markov random field on a regular lattice similar to Rathbun and Cressie (1994). Simpson et al. (2016) inferred the log intensity measure by assuming a smooth Gaussian process prior for it; in particular, they showed that approximating the Gaussian prior with the SPDE approach of Lindgren et al. (2011), in this context, has good convergence properties compared to a partition of the domain. Examples of applications of the Bayesian latent Gaussian model, which use the log Gaussian–Cox process, are those of Lombardo et al. (2019), who describe the spatial patterns of earthquake-induced landslides, and Diaz-Avalos et al. (2016), who apply this model to wildfires.

Data with temporal reference or spatio-temporal reference are common in geophysics and environmental sciences. There is a vast literature on time series models. The books of West and Harrison (1999), Prado et al. (2021), and Triantafyllopoulos (2021) focus on time series models from the Bayesian perspective. Lindgren et al. (2013) cover topics in time series with emphasis on audience in science and engineering. The following books are on statistical models for spatio-temporal data with focus on geophysics and environmental sciences; Le and Zidek (2006), Cressie and Wikle (2011), Montero et al. (2015), Blangiardo and Cameletti (2015), Krainski et al. (2018), and Wikle et al. (2019). Lindgren et al. (2022a) introduce spatio-temporal models through a diffusion-based family of spatio-temporal

stochastic processes. This class of spatio-temporal models is parametrized such that its covariance functions are either non-separable or separable. Lindgren et al. (2022b) provide a sparse representation based on a finite element approximation for these spatio-temporal models, which is well-suited for statistical inference and has been implemented in the R-INLA software. Xu et al. (2014) propose a computationally efficient Bayesian hierarchical spatio-temporal model designed for large datasets. The spatial dependence is approximated by a GMRF, and a vector autoregressive model is used to handle the temporal correlation. An application to a large precipitation dataset is given. In Salvaña and Genton (2020), an overview of spatio-temporal cross-covariance functions is given, and multivariate spatio-temporal asymmetric nonstationary models are introduced. A class of multiscale spatio-temporal models for multivariate Gaussian data is proposed in Elkhoully and Ferreira (2021). Their approach is based on decomposing the multivariate data and the underlying latent process with a multivariate multiscale decomposition. They apply their approach to the spatio-temporal NCEP/NCAR Reanalysis-I dataset on stratospheric temperatures over North America.

The environmetrics encyclopedia of El-Shaarawi and Piegorsch (2012) covers quantitative methods and their application in the environmental sciences, in fields such as: ecology and environmental biology, public health, atmospheric science, geology, engineering, risk management, and regulatory/governmental policy. The handbook of Gelfand et al. (2019) provides a comprehensive view of statistical methods for the environmental sciences and ecology, covering statistical methodology, ecological processes, environmental exposure, and statistical methods in climate science.

We end this bibliographic note with a few examples of applications of Bayesian latent Gaussian models. Cressie (2018) presents an example of a physical statistical model for atmospheric carbon dioxide, which is inferred using remote sensing data, and discusses the steps from observations to information to knowledge to decision-making, while taking uncertainty into account at each step. Forlani et al. (2020) analyze spatially misaligned air pollution data. In Jóhannesson et al. (2022), flood frequency data are analyzed using a Bayesian LGM with spatial random effects. Opitz et al. (2018) model precipitation intensities with a model that consists of two Bayesian LGMs, both of which use spatial and temporal random effects. The first model is for rate, and it assumes the Bernoulli distribution at the response level, while the other model assumes the generalized Pareto distribution at the response level for the size of threshold exceedances. Bayesian LGMs for spatial data that are observed within a domain that has physical barriers, such as a coastline, are introduced in Bakka et al. (2019). Bowman et al. (2018) propose a hierarchical statistical framework that formally relates projections of future climate to present-day climate and observations and apply it to a future Northern Hemispheric climate projection.

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Appendix

Posterior Computation for the Gaussian–Gaussian Model

The distributional assumptions of the Gaussian–Gaussian model are

$$\begin{aligned}\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta}) &= \mathcal{N}(\mathbf{y}|\mathbf{Z}\mathbf{x}, \mathbf{Q}_\epsilon^{-1}) \\ \pi(\mathbf{x}|\boldsymbol{\theta}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_x, \mathbf{Q}_x^{-1}),\end{aligned}$$

and the joint distribution of $(\mathbf{y}^\top, \mathbf{x}^\top)^\top$ is given by

$$\begin{aligned}\pi(\mathbf{y}, \mathbf{x}|\boldsymbol{\theta}) &= \mathcal{N}\left(\begin{pmatrix} \mathbf{y} \\ \mathbf{x} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{Z}\boldsymbol{\mu}_x \\ \boldsymbol{\mu}_x \end{pmatrix}, \begin{pmatrix} \mathbf{Q}_\epsilon & -\mathbf{Q}_\epsilon \mathbf{Z} \\ -\mathbf{Z}^\top \mathbf{Q}_\epsilon & \mathbf{Q}_x + \mathbf{Z}^\top \mathbf{Q}_\epsilon \mathbf{Z} \end{pmatrix}^{-1}\right) \\ &= \mathcal{N}\left(\begin{pmatrix} \mathbf{y} \\ \mathbf{x} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{Z}\boldsymbol{\mu}_x \\ \boldsymbol{\mu}_x \end{pmatrix}, \begin{pmatrix} \mathbf{Z}\mathbf{Q}_x^{-1}\mathbf{Z}^\top + \mathbf{Q}_\epsilon^{-1} & \mathbf{Z}\mathbf{Q}_x^{-1} \\ \mathbf{Q}_x^{-1}\mathbf{Z}^\top & \mathbf{Q}_x^{-1} \end{pmatrix}\right).\end{aligned}$$

The conditional distribution of $\mathbf{x}$ is

$$\pi(\mathbf{x}|\mathbf{y}, \boldsymbol{\theta}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_{x|\mathbf{y}}, \mathbf{Q}_{x|\mathbf{y}}^{-1}),$$

where

$$\mathbf{Q}_{x|\mathbf{y}} = \mathbf{Q}_x + \mathbf{Z}^\top \mathbf{Q}_\epsilon \mathbf{Z}, \quad \boldsymbol{\mu}_{x|\mathbf{y}} = \mathbf{Q}_{x|\mathbf{y}}^{-1}(\mathbf{Q}_x \boldsymbol{\mu}_x + \mathbf{Z}^\top \mathbf{Q}_\epsilon \mathbf{y}).$$

A computational algorithm for drawing samples from $\pi(\mathbf{x}, \boldsymbol{\theta}|\mathbf{y})$ consists of two steps:

Step 1: Draw $\boldsymbol{\theta}$ from $\pi(\boldsymbol{\theta}|\mathbf{y})$.

Step 2: Draw $\mathbf{x}$ from $\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$.

To sample $\boldsymbol{\theta}$ in Step 1, we use the following relationship:

$$\pi(\boldsymbol{\theta}|\mathbf{y}) \propto \pi(\boldsymbol{\theta})\pi(\mathbf{y}|\boldsymbol{\theta}) = \pi(\boldsymbol{\theta}) \frac{\pi(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x}|\boldsymbol{\theta})}{\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})}.$$

A Metropolis algorithm is used to draw samples from $\pi(\boldsymbol{\theta}|\mathbf{y})$. It is assumed that $\boldsymbol{\theta} \in \mathbb{R}^{\dim(\boldsymbol{\theta})}$. A Gaussian approximation is found for $\pi(\boldsymbol{\theta}|\mathbf{y})$, that is,

$$\pi(\boldsymbol{\theta}|\mathbf{y}) \approx \tilde{\pi}(\boldsymbol{\theta}|\mathbf{y}) = \mathcal{N}(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}}, \mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}^{-1}),$$

$$\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}} = \left(-\frac{\mathrm{d}^2 \log(p(\boldsymbol{\theta}|\mathbf{y}))}{\mathrm{d}\theta_i \mathrm{d}\theta_j} \right)_{i,j} \bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}},$$

where $\hat{\boldsymbol{\theta}}$ is the posterior mode of $\pi(\boldsymbol{\theta}|\mathbf{y})$.

Then θ^* is drawn from

$$J_{\theta,t}(\theta^*|\theta^{t-1}) = \mathcal{N}(\theta^*|\theta^{t-1}, cQ_{\theta|y}^{-1}),$$

where $c = 2.38^2/\dim(\theta)$ (Roberts et al., 1997).

Algorithm 2 shows the calculations that are needed to sample $\mathbf{x}$ from $\pi(\mathbf{x}|\theta, \mathbf{y})$ in Step 2.

Require: $(\mathbf{y}, Z, Q_\epsilon, \mu_x, Q_x)$

- 1: $Q_{x|y} = Q_x + Z^\top Q_\epsilon Z$
- 2: Cholesky decomposition: $Q_{x|y} = R^\top R$.
- 3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_{\dim(\mathbf{x})})$.
- 4: Solve for $\mathbf{u}$: $R\mathbf{u} = \mathbf{z}$.
- 5: $\mathbf{b} = Q_x \mu_x + Z^\top Q_\epsilon \mathbf{y}$
- 6: Solve for $\mathbf{v}$: $R^\top \mathbf{v} = \mathbf{b}$.
- 7: Solve for $\mu_{x|y}$: $R\mu_{x|y} = \mathbf{v}$.
- 8: Compute $\mathbf{x}^* = \mu_{x|y} + \mathbf{u}$.

Ensure: $(\mu_{x|y}, Q_{x|y}, \mathbf{x}^*)$

Algorithm 2: Computing $\mu_{x|y}$ and $Q_{x|y}$, and simulating $\mathbf{x}^*$ from $\mathbf{y}, Z, Q_\epsilon, \mu_x$, and Q_x , assuming $\mathbf{y} = Z\mathbf{x} + \epsilon$, $\mathbf{x} \sim \mathcal{N}(\mu_x, Q_x)$. The outputs are the mean, precision, and a sample of the Gaussian distribution of $\mathbf{x}|y$

The LGM Split Sampler

The LGM split sampler is a posterior sampling scheme for extended LGMs that was introduced by Geirsson et al. (2020). It is a Gibbs sampler with Metropolis–Hastings steps that is structured such that, first, samples are drawn from the conditional posterior of η in one block, referred to as the data-rich block, and then, samples are drawn from the conditional posterior of $(\mathbf{v}, \theta)$ in another block, referred to as the data-poor block.

When sampling from the data-rich block, the mode and the Hessian matrix of the logarithmic transformation of $\pi(\eta|\mathbf{v}, \theta, \mathbf{y})$ are found at each iteration and used to construct a Gaussian approximation of the conditional posterior of η , namely, the mean is set equal to the mode and the precision matrix is set equal to the negative Hessian matrix. This Gaussian approximation is used as the proposal density. The posterior sampling scheme for η conditional on $(\mathbf{v}, \theta)$ within the LGM split sampler is given in Algorithm 2 in Geirsson et al. (2020).

Within the data-poor block, samples of $(\mathbf{v}, \theta)$ are drawn from the posterior distribution of $(\mathbf{v}, \theta)$ conditional on η . The posterior distribution of $(\mathbf{v}, \theta)$ conditional on η does not depend on the response, $\mathbf{y}$, since

$$\pi(\mathbf{v}, \theta|\eta, \mathbf{y}) \propto \pi(\theta)\pi(\mathbf{v}|\theta)\pi(\eta|\mathbf{v}, \theta). \quad (60)$$

Algorithm 1 in Geirsson et al. (2020) shows the steps of the LGM split sampler for obtaining posterior samples of $(\mathbf{v}, \boldsymbol{\theta})$ conditional on $\boldsymbol{\eta}$. Note that the prior density of $\mathbf{v}$ conditional on $\boldsymbol{\theta}$ and the prior density of $\boldsymbol{\eta}$ conditional on $(\mathbf{v}, \boldsymbol{\theta})$ are Gaussian, and $\boldsymbol{\eta}$ depends linearly on $\mathbf{v}$. Algorithm 1 in Geirsson et al. (2020) takes advantage of these facts.

The structure of the right-hand side of (60) is the same as the structure of the Gaussian–Gaussian model. This can be seen by putting $\boldsymbol{\eta}$ and $\mathbf{v}$ into the roles of $\mathbf{y}$ and $\mathbf{x}$ in the Gaussian–Gaussian model, respectively. In fact, Algorithm 1 in Geirsson et al. (2020) and the algorithm for the Gaussian–Gaussian model presented above in the Appendix are very similar. The main difference is that the former samples $(\mathbf{v}, \boldsymbol{\theta})$ jointly, in particular, $\mathbf{v}$ is only sampled if a new proposal for $\boldsymbol{\theta}$ is accepted, while the latter samples $\boldsymbol{\theta}$ first, and then $\mathbf{x}$ is sampled at each iteration, regardless of whether $\boldsymbol{\theta}$ is a new proposal or not.

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A Review of Bayesian Modelling in Glaciology



Giri Gopalan, Andrew Zammit-Mangion, and Felicity McCormack

1 Introduction

Ice sheets and glaciers are large bodies of ice formed from the compaction of accumulated snow, over centuries to millennia. These bodies of ice range in size from the ice sheets of Antarctica and Greenland to comparatively smaller mountain glaciers such as those of Iceland (e.g., Vatnajökull and Langökull; see Fig. 1); about 10% of Earth's surface is covered in ice (Cuffey & Paterson, 2010). Understanding the past, present, and future behaviour of glaciers and ice sheets is of major importance in our warming world because these masses of ice are the largest potential contributors to sea-level rise (Fox-Kemper et al., 2021; Hock et al., 2019).

Glaciers are dynamic and continuously flowing in the direction of steepest ice surface slope due to the force of gravity. This flow can be described using mathematical equations, typically partial differential equations based on Stokes' law or some approximation (e.g., a linearisation) thereof (Blatter, 1995; MacAyeal, 1989; Pattyn, 2003). These equations relate the flow of ice to spatially varying parameters such as ice geometry (surface slope, thickness), ice properties (temperature, microstructure), and the subglacial environment (bed substrate, hydrology). Ice flows through a

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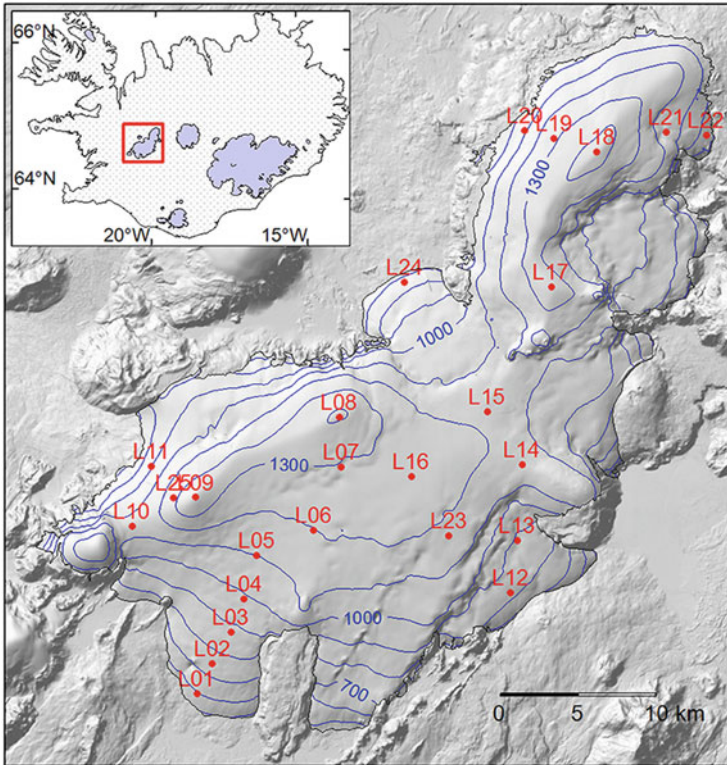


Fig. 1 Surface topographical map of Langjökull from Gopalan et al. (2021), reproduced in accordance with terms of the Creative Commons Attribution License. The inset map illustrates the major Icelandic glaciers, with the largest, Vatnajökull, to the southeast. Blue lines denote elevation contours, with contour labels denoting elevation in metres. The red dots and accompanying labels denote locations at which surface mass balance measurements are recorded twice yearly

combination of ice deformation, basal sliding (as meltwater acts as a lubricant for the ice to slide over the bed), and deformation of the bed.

Statistical modelling has a significant role to play in glaciology for several reasons. First, it can be used to provide probabilistic predictions (i.e., estimates with associated uncertainties) of processes of interest at unobserved locations. This is important, as glaciological data are often sparse and not evenly sampled in a spatial domain of interest: in situ experiments for data collection in a remote place such as Antarctica are time-intensive and very expensive, while it is notoriously difficult to obtain good spatial coverage from Earth-observing satellites at high latitudes. Second, statistical modelling allows one to infer important quantities of interest from indirect observations. This is useful because in cryospheric sciences it is often the case that collected data are only loosely related to the inferential target. For example, the satellite mission Gravity Recovery and Climate Experiment (GRACE) is able to provide a time series of mass change of an area covering part of an ice

sheet. However, this observed mass change does not only include that which is due to ice-sheet processes, but also that which is due to solid-Earth processes such as glacio-isostatic adjustment. Statistical methods can be used to provide probabilistic assessments on what the drivers of the observed mass change are. Third, a statistical model can be used for quickly *emulating* a complex numerical model, which takes a long time to run, and also to account for uncertainty introduced by not running the full numerical model. This is relevant for studies in glaciology, as the only way to reliably project an ice sheet's behaviour into the future is via the use of an ice-sheet model such as the *ice-sheet and Sea-Level System Model* (ISSM; e.g., Larour et al., 2012) or *Elmer/Ice* (e.g., Gagliardini et al., 2013), which requires considerable computing resources. These ice-sheet models need to be equipped with initial and boundary conditions that need to be calibrated from data or other numerical models; statistical emulators thus offer a way forward to make the calibration procedure computationally tractable. These strengths of statistical modelling and inference have led to a vibrant and active area of research in statistical methodology for glaciology and the widespread use of sophisticated, often Bayesian, inferential techniques for answering some of the most urgent questions relating to glaciological processes and their impact on sea-level rise.

Glaciologists have used a variety of techniques to produce estimates of important glacial quantities from data, most notably the class of *control methods* that formulate the estimation procedure as a constrained optimisation problem. There is often no unique solution to an inverse problem given a particular set of measurements, particularly when the number of unknown parameters far exceeds the number of measurements; moreover, solutions to an inverse problem may be very sensitive to small changes in measurements. One way to combat these problems in glaciology is to use regularisation via constrained optimisation, but many sources of uncertainties are not typically accounted for adequately in this way.

Many early works employed control methods for estimating the spatially varying basal friction coefficient—an unobservable spatially varying parameter that informs the basal boundary condition or basal friction law in the momentum equations—by minimising the mismatch between the observed and modelled surface velocity data (MacAyeal, 1993; Morlighem et al., 2010, 2013). Employing such a minimisation yields estimates of glaciological quantities of interest, but not with accompanying full (posterior) probability distributions with which to characterise uncertainty. A contrasting advantage of using Bayesian methodology for such problems is the provision of uncertainties on spatially varying fields (such as the basal friction coefficient) via a posterior probability distribution over all parameters of interest. This comes at a cost, as Bayesian methods require the specification of a complex probability model and generally involve computational challenges, which have been grappled with in recent years.

We now give a few (non-exhaustive) examples of statistical problems in glaciology that we highlight in this chapter:

- Inferring and predicting ice thickness or ice presence/absence
- Inferring subglacial topography (i.e., the geometry of the surface on which a glacier is situated)
- Inferring surface velocity fields
- Inferring basal sliding fields
- Inferring spatial fields of net mass gains or losses (e.g., surface mass balance)
- Inferring the contribution of an ice sheet or glacier to sea-level rise

All of these problems have been addressed on one or more occasions through the use of formal statistical methodology, and the purpose of this review chapter is to highlight some of the works employing such methodology. We will give specific focus to works that are couched in a Bayesian framework, which is particularly useful in glaciology where one needs to combine information from multiple sources in a single model. As we will see, many of these works formulate Bayesian models that are hierarchical, where parameters, geophysical processes, and data processes are modelled separately. An additional advantage of using the Bayesian formalism is that it provides a conditional distribution over unknown quantities given observations; this *posterior distribution* combines information from the data and the *prior distribution* and yields estimates of unknown quantities as well as uncertainty quantification in light of all the available information.

The overarching aim of this review chapter is to provide an accessible introduction of Bayesian and Bayesian hierarchical modelling approaches in glaciology for Earth scientists, as well as a glaciology primer for statisticians. The chapter is structured as follows. In Sect. 2, we begin with a broad summary of literature that has employed Bayesian statistical modelling (not necessarily hierarchical) in a glaciology context, starting from the late 2000s. The overview emphasises the scientific topics being addressed, the statistical modelling employed, and the conclusions or key take-away messages of the reviewed works. Sections 3 and 4 then give a detailed summary of two studies involving Bayesian hierarchical modelling in the context of glaciology (Gopalan, 2019; Zammit-Mangion et al., 2015). These sections give more details on the modelling and inferential strategies employed and clearly demonstrate the role Bayesian statistical methodology has to play in glaciology. Section 5 concludes and discusses current initiatives and research directions at the intersection of Bayesian modelling and glaciology.

2 A Synopsis of Bayesian Modelling and Inference in Glaciology

In the current section, we review work that has applied Bayesian methods to problems in glaciology in the last two decades. Our aim is to provide the reader with a broad overview of what work has been done at the intersection of Bayesian inference and glaciology, with a view to how future work can build on what has been accomplished to date. We do not focus on the technical details but rather

convey the core research goals of the individual works, their Bayesian modelling strategies, and the statistical computing tools they employ. In contrast, Sects. 3 and 4 contain an in-depth treatment of two works that incorporate Bayesian statistics in addressing important questions in glaciology. Section 3 tackles the problem of predicting surface mass balance across an Icelandic mountain glacier, while Sect. 4 details an approach to estimate the Antarctic contribution to sea-level rise. The reader who is interested in these more detailed examples, rather than a broad review, can proceed directly to Sect. 3. The Appendix contains details of the governing ice flow equations, referred to in the chapter, that are often used in glaciology.

In this section, we categorise papers according to the types of models commonly employed in major Bayesian glaciological modelling studies: Bayesian Gaussian–Gaussian models (Sect. 2.1), Bayesian hierarchical models (Sect. 2.2), and models for Bayesian calibration (Sect. 2.3).

2.1 Gaussian–Gaussian Models

A ubiquitous model that appears in much of the glaciology literature employing Bayesian methods and models is the Gaussian–Gaussian model. Assume that the glaciological data of interest are available in a vector $\mathbf{x} \in \mathbb{R}^N$, which in turn is related to a number of physical parameters in the vector $\boldsymbol{\theta} \in \mathbb{R}^M$ via some function $f(\cdot)$. The Gaussian–Gaussian model is defined through the distributions:

$$p(\mathbf{x}|\boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^N |\boldsymbol{\Sigma}_l|}} \exp \left[-\frac{1}{2} (\mathbf{x} - f(\boldsymbol{\theta}))^T \boldsymbol{\Sigma}_l^{-1} (\mathbf{x} - f(\boldsymbol{\theta})) \right] \quad (1)$$

and

$$p(\boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^M |\boldsymbol{\Sigma}_p|}} \exp \left[-\frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}_p^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}) \right]. \quad (2)$$

In (1) and (2), the mean and covariance matrix associated with the distribution $p(\mathbf{x}|\boldsymbol{\theta})$ are given by $f(\boldsymbol{\theta})$ and $\boldsymbol{\Sigma}_l$, respectively. The mean and covariance matrix associated with the prior distribution, $p(\boldsymbol{\theta})$, are given by $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_p$, respectively. We often refer to (1) and (2) as the *data* and *process* model, respectively. Additionally, when (1) is viewed as a function of $\boldsymbol{\theta}$, we refer to it as the *likelihood function*.

While the Gaussian–Gaussian model leads to several computational simplifications, there are two complexities that drive much of the research with these models. The first issue concerns that of inverting and finding the determinant of large covariance matrices, which generally has cubic computational complexity. The second issue concerns the function $f(\cdot)$, which determines the conditional expectation of the data, $\mathbf{x}$. The function $f(\cdot)$ is generally nonlinear, and, therefore, the posterior distribution over $\boldsymbol{\theta}$ is generally non-Gaussian. As we shall see, many

of the papers in the glaciology literature attempt to circumvent these issues with a variety of computational strategies.

One of the earliest approaches employing Bayesian methods with a Gaussian–Gaussian model in a glaciological context is that of Raymond and Gudmundsson (2009), who used a Bayesian approach for inferring basal topography and basal sliding of ice streams from observations of surface topography and velocity and illustrated their methods on simulated data examples. They used Gaussian process priors with squared-exponential kernels for modelling the basal sliding parameters and basal topography (process model) and a physics-based flow forward model, approximated using finite elements, to relate basal conditions to observed surface topography and velocity (data model). A Gauss–Newton iterative optimisation method was then used to compute maximum a posteriori (MAP) estimates of basal sliding and topography. Each iteration required the computation of the gradient of the forward model with respect to the unknown parameters characterising the processes; this was accomplished by using an analytical linear approximation of the forward model. Synthetic tests were used to show that the MAP estimates were often very close to the simulated (true) bedrock velocity and basal sliding profiles. Pralong and Gudmundsson (2011) later applied this methodology to infer basal topography and sliding from observations of the Rutford Ice Stream in West Antarctica.

The work of Raymond and Gudmundsson (2009) was relatively small scale. The increased availability and resolution of remote-sensing instruments, coupled with the need to analyse the behaviour of entire ice sheets, led to work pioneering large-scale tractability for inference with Gaussian–Gaussian models. Developments were especially needed for the analysis of entire ice sheets. The papers of Petra et al. (2014) and Isaac et al. (2015) were among the first to address this issue in a glaciological context. In particular, their focus was on Bayesian inference of large, spatial basal sliding fields at glaciers given surface velocity measurements. In both cases, the process and the observations were linked via a nonlinear function motivated by Stokes’ ice flow equations, which were implemented using Taylor–Hood finite elements. The computational challenge addressed primarily in Petra et al. (2014) was how to efficiently sample from the posterior distribution when the dimensionality of the field is high (as in the case of the Antarctic ice sheets). Their solution was to use a Gaussian (Laplace) approximation (e.g., Gelman et al., 2013, Chapter 13) of the posterior distribution over the basal sliding field, where the mean is set to the MAP estimate, and the precision matrix is set to the Hessian of the negative log of the posterior distribution, evaluated at the MAP estimate. For computational tractability, a low-rank approximation was used for the covariance matrix. Isaac et al. (2015) used a strategy similar to that developed by Petra et al. (2014) to obtain a posterior distribution over the basal sliding field in Antarctica, which was then used to obtain a prediction interval for mass loss in Antarctica.

A Gaussian–Gaussian model was also used by Minchew et al. (2015) for inferring glacier surface velocities, which provide valuable information regarding the dynamics of a glacier. In particular, the goal of Minchew et al. (2015) was to infer velocity fields at Icelandic glaciers (Langjökull and Hofsjökull) solely using repeat-pass interferometric synthetic aperture radar (InSAR) data collected during

early June 2012. In the Gaussian–Gaussian model of Minchew et al. (2015), the function $f(\cdot)$ (see Eq. (1)) is assumed to be a linear function, and so the posterior distribution is Gaussian with an analytically exact mean and covariance matrix (e.g., Tarantola, 2005, Chapter 3). The directions of the inferred horizontal velocities were in the direction of steepest descent, which is consistent with physics models of flow at shallow glaciers (such as those of Iceland). Visualisations of posterior spatial variation in horizontal velocity during the early-melt season (i.e., June) across Langjökull and Hofsjökull revealed magnitudes of horizontal velocities that differ considerably from those obtained from a physical model that ignores basal sliding velocity. Their results suggest that basal sliding is an important factor to consider when determining the velocities of the Icelandic glaciers Langjökull and Hofsjökull.

Brinkerhoff et al. (2016) employed a Gaussian–Gaussian modelling approach to estimate subglacial topography given surface mass balance, surface elevation change, and surface velocity, while employing a mass-conservation model for the rate of change of glacial thickness. Gaussian process priors were used for all processes, employing either an exponential or squared-exponential covariance function. Metropolis–Hastings was used for posterior inference via PyMC (Patil et al., 2010), and the method was applied to Storglaciären, a 3-km-long glacier in northwestern Sweden. The bed topography of Storglaciären is well-known (Brinkerhoff et al., 2016), and the glacier therefore makes an ideal test for bed topography recovery approaches. Brinkerhoff et al. (2016) were thus able to show that the estimate (specifically, the MAP estimate) from their model was reasonably close to the true bed topography. Another glacier was also used as a test case—Jakobshavn Isbræ, on the Greenland ice sheet. While the bed topography is not as well-known for this glacier, the obtained 95% credibility intervals contained bed topography estimates reported elsewhere, although the obtained MAP estimate differed considerably from these other estimates.

2.2 Bayesian Hierarchical Models

The Gaussian–Gaussian model of Sect. 2.1 is a special case of the more general Bayesian hierarchical model (BHM), which is used to denote models that are constructed via the specification of multiple conditional distributions. To the best of our knowledge, the first work that utilised a Bayesian hierarchical model with more than two levels in a glaciological context is that of Berliner et al. (2008). In their paper, the model is couched within a physical–statistical modelling framework (Berliner, 2003; Cressie & Wikle, 2011), and it is used to infer three processes: glacial surface velocity, basal topography, and surface topography, from noisy and incomplete data. The physical–statistical BHM they used is made up of three models: a data model, a process model, and a parameter model:

- The *data model* specifies a statistical model for the data conditional on the physical processes and parameters.

- The *process model* specifies a statistical model for the physical processes of interest conditional on one or more parameters.
- The *parameter model* specifies prior distributions for statistical and/or physical parameters on which inference is made.

In Berliner et al. (2008), the aim was to infer the three latent processes from data of surface elevation, basal topography, and surface velocity. A physical argument was used (van der Veen, 2013) to relate surface velocity (one of the processes of interest) to stress terms that are in turn dependent on glacial thickness (i.e., the difference between surface elevation and the glacier bed elevation, which are additional physical quantities of interest). Although Gaussianity was assumed at the data and process levels, the posterior distributions over the processes were not Gaussian due to nonlinearity in the conditional mean in the data model. The model was used to analyse data from the Northeast Greenland Ice Stream. Posterior distributions over driving stress, basal sliding velocity, and basal topography were sampled from using a Gibbs sampler and importance-sampling Monte Carlo.

A BHM that is considerably higher dimensional is developed by Zammit-Mangion et al. (2015) for assessing the contribution of Antarctica to sea-level rise. Because of the scale of the problem, computational techniques involving sparsity of large precision (i.e., inverse covariance) matrices and a parallel Gibbs sampler were used. A detailed account of this work is given in Sect. 4.

Several recent papers have used BHMs for inferring unknown parameters and predicting spatially varying glacial thickness. The main objective of Guan et al. (2018) was to infer ice thickness and some parameters of glaciological importance at Thwaites Glacier in West Antarctica. The physics model used was a one-dimensional flowline model that relates the net flux of ice to the accumulation of snow and melting of ice. A shallow-ice approximation was used to relate surface velocity to glacial thickness. Surface velocity, slope, ice accumulation, thinning, flow width, and an ice rheological constant are parameters that are input into a glacier dynamics model, which in turn outputs ice thickness along the flowline. Gaussian priors were used for parameters that were treated as unknown, and the likelihood function was Gaussian. A Metropolis–Hastings algorithm was used for posterior inference.

Gopalan et al. (2018) presented a Bayesian hierarchical approach for modelling the time evolution of glacier thickness based on a shallow-ice approximation. As in Berliner et al. (2008), the Bayesian hierarchical model was couched in a physical–statistical framework. The process model used was a novel numerical two-dimensional partial differential equation (PDE) solver for determining glacial thickness through time, as well as an error-correcting process that follows a multivariate (Gaussian) random walk. The model was tested on simulated data based on analytical solutions to the shallow-ice approximation for ice flow (Bueler et al., 2005). Inference of a physical parameter (a rheological constant) and predictions of future glacial thickness across the glacier appear biased, although credibility intervals capture the true (simulated) values.

The model was further developed by Gopalan et al. (2019), with a particular focus on computational efficiency, an issue that must often be addressed in large-scale glaciological applications. Specifically, the use of surrogate process models constructed via first-order emulators (Hooten et al., 2011), parallelisation of an approximation to the log-likelihood, and the use of sparse matrix algebra routines were shown to help alleviate computational difficulties encountered when making inference with large Bayesian hierarchical glaciology models. Additionally, a multivariate random walk assumption in the process model was further developed to include higher-order terms, and it was examined in the context of the shallow-ice approximation numerical solver of Gopalan et al. (2018). This random walk is closely related to the notion of model discrepancy, further discussed in Sect. 2.3.

A related contribution to the literature on BHM in cryospheric science is that of Zhang and Cressie (2020), which addresses the modelling and prediction of sea ice presence or absence. The data model is a Bernoulli distribution with a temporally and spatially varying probability parameter. The process model characterises the log-odds ratios, derived from the time-varying probabilities, as a linear combination of covariates and spatial basis functions. The coefficients of these basis functions evolve through time according to a vector autoregressive process of order one. Sampling from the posterior distribution was done using a Metropolis-within-Gibbs sampler. The approach was used to model Arctic Sea ice absence/presence from 1997–2016 using sea ice extent data from the National Oceanic and Atmospheric Administration (NOAA).

Some of the very recent literature using Bayesian hierarchical models in glaciology has involved non-Gaussian assumptions at either the process, prior, or data levels. For instance, Brinkerhoff et al. (2021) used Bayesian methods for inferring subglacial hydrological and sliding law parameters given a field of time-averaged surface velocity measurements, with application to southwestern Greenland. A physical model was constructed based on an approximate hydrostatic solution to Stokes' equations coupled with a hydrological model. A surrogate model/emulator was used to reduce the computational complexity of running the full model numerically. Specifically, neural networks (fit using *PyTorch*) were chosen to learn the coefficients of basis vectors as a function of input parameters. The basis vectors were derived from a principal component analysis applied to an ensemble of computer simulator runs. The primary benefit of using the neural network emulator was to reduce the time taken to compute an approximate numerical solution to the surface velocity field. The data model used was multivariate normal, and a beta prior distribution was used for the parameters. Posterior computation was achieved with the manifold Metropolis-adjusted Langevin algorithm, a Markov chain Monte Carlo (MCMC) method that efficiently explores complex posterior distributions (Girolami & Calderhead, 2011).

Gopalan et al. (2021) developed a Bayesian approach for inferring a rheological parameter and a basal sliding parameter field at an Icelandic glacier (Langjökull). The model relies on the shallow-ice approximation for relating physical parameters of interest to surface velocities. Two data models were considered—one based on the t -distribution and one based on the Gaussian distribution. A truncated normal

prior was used for the rheological parameter (termed ice softness), whereas a Gaussian process prior was used for the log of the basal sliding field. Sampling from the posterior distribution was done using a Gibbs sampler with an elliptical slice sampling step (Murray et al., 2010) for the basal sliding parameter field. It was found that the inferred rheological parameter was similar to that obtained in other studies and that the inferred spatial variation in sliding velocity and deformation velocity was generally consistent with that of Minchew et al. (2015). Residual analysis suggested that a Gaussian data model yields a worse fit than a t data model, although both data models yielded similar posterior distributions over the rheological parameter. This work also clearly demonstrates the utility of uncertainty quantification when inferring important glaciological parameters via a Bayesian inference approach.

Brinkerhoff et al. (2021) developed a Bayesian approach to jointly estimate a rheological field and a basal sliding parameter field. The approach is similar to that of Gopalan et al. (2021), except that the rheological parameter is also allowed to vary spatially. The scale of the problem addressed is also larger since the methodology is applied to a large marine-terminating glacier in Greenland. To alleviate computational issues, a variational inference scheme (Blei et al., 2017) was used instead of MCMC for evaluation of an approximate posterior distribution. A rank-reduction technique, inspired by Solin and Särkkä (2020), was also used to facilitate computations with large covariance matrices. In contrast to many previous works, Brinkerhoff et al. (2021) used a log-normal distribution for the data model. Moreover, a Gaussian process prior with a squared-exponential kernel was used for both the rheological and basal sliding fields. This paper breaks ground with the use of variational inference instead of MCMC; variational inference has been seeing increased use in recent years in geophysical applications where computational efficiency needs to be addressed.

2.3 *Bayesian Calibration of Physical Models*

Another area of research in glaciology that employs Bayesian methods stems from the computer-model calibration literature (Kennedy & O'Hagan, 2001). Here, the aim is to tune, or calibrate, numerical models from data in a Bayesian framework. An important consideration in Bayesian calibration is the characterisation of numerical-model discrepancy. Briefly, model discrepancy is a model term used to capture the difference between the output of a computer model (usually the output of the computer model using the *best fitting* parameter value) and the true value of the process being modelled. Model discrepancy has been shown to be necessary in order to obtain realistic inferences of numerical-model parameters (Kennedy & O'Hagan, 2001; Brynjarsdóttir & O'Hagan, 2014).

An early contribution applying ideas from Bayesian calibration to ice-sheet models comes from McNeill et al. (2013), in which the authors developed a Gaussian process emulator of an ice-sheet simulator dubbed Glimmer (Rutt et al.,

2009) and introduced a method to quantify the extent to which observational data can constrain input parameters (and, consequently, simulator output). In turn, this method may be used to inform the design of observation strategies for ice sheets, so as to collect data in order to maximise the extent to which parameters are constrained.

Chang et al. (2016) used a model discrepancy term when developing a Bayesian approach to calibrate a numerical model that gives a binary output: ice presence or absence. Their method was used to obtain posterior distributions for a variety of physically relevant parameters (including calving factor, basal sliding coefficient, and asthenospheric relaxation e -folding time), as well as to make projections in ice volume changes, and consequently predictions of sea-level rise, using the PSU3D-ICE model (Pollard & DeConto, 2009) and data on the Amundsen Sea Embayment in West Antarctica. In addition to model discrepancy, Chang et al. (2016) also included an emulator/surrogate model in order to reduce the computational time needed when working with a computationally expensive computer model, much in the spirit of Higdon et al. (2008). Emulators are designed to mimic the output of the computer model but are much less computationally intensive. Inference was made in two steps: First, an emulator for the computer model was constructed using Gaussian processes. Then, the numerical model was calibrated using the emulator and a model discrepancy term, also modelled as a Gaussian process. The work of Chang et al. (2016) and that of McNeill et al. (2013) are among the few that use discrepancy models in a glaciological context.

Lee et al. (2020) revisited the problem of calibrating Antarctic ice-sheet computer models with the goal of improving forecasts of sea-level rise due to ice loss. In contrast to previously summarised approaches at Antarctica, Lee et al. (2020) used a parallelisable particle-sampling algorithm for Bayesian computation, dubbed adaptive particle sampling, for reducing computation times. As in Chang et al. (2016), the PSU3D-ICE model of ice flow was used, though data from the Pliocene era were incorporated into the approach as well.

In this section, we have outlined some major works involving Bayesian statistics and glaciology that fall into three categories: Gaussian–Gaussian models, general Bayesian hierarchical models, and Bayesian calibration models. The list of works reviewed is not exhaustive; we also refer the reader to the following papers that contain core elements of Bayesian statistics and cryospheric science: Klauenberg et al. (2011), Ruckert et al. (2017), Conrad et al. (2018), Edwards et al. (2019), Guan et al. (2019), Irrazaval et al. (2019), Gillet-Chaulet (2020), Rounce et al. (2020), Werder et al. (2020), Babaniyi et al. (2021), Director et al. (2021). Next, in Sects. 3 and 4, we take a detailed look at two case studies that exhibit the utility of Bayesian methods in glaciology.

3 Spatial Prediction of Langjökull Surface Mass Balance

We now illustrate an example of latent Gaussian modelling in glaciology with the problem of prediction of surface mass balance (SMB) over Langjökull, an Icelandic mountain glacier, following Chapter 6 of the PhD thesis of Gopalan (2019). SMB is the temporal rate of change of mass at the surface of a glacier due to factors such as snow accumulation, ice melt, and snow drift. This quantity is usually an integral part of dynamical equations that involve the temporal derivative of glacier thickness, since a negative SMB contributes to a decrease in thickness, while a positive SMB contributes to an increase in thickness (see governing equations in the Appendix). SMB is usually observed at only a few locations across a glacier, and statistical methods are needed to provide probabilistic predictions at spatial locations away from the measurement sites.

Glaciologists at the University of Iceland Institute of Earth Sciences (UI-IES) take measurements of SMB twice a year (late spring and fall) across the surface of Langjökull, usually at all the 25 sites shown in Fig. 1. The measurements taken during late spring are records of winter SMB, whereas the measurements taken during fall are records of summer SMB. The sum of winter and summer SMB yields the net SMB for the year.

Conventionally, linear models are used in glaciology for predicting SMB at unobserved locations; for instance, Pálsson et al. (2012) show that there is a nearly linear relationship between SMB and elevation except at high elevations, possibly due to snow drift at high elevations. Additionally, a model for precipitation (which is directly linked to SMB) is used by Aðalgeirsdóttir et al. (2006) that is linear in the x -coordinate, the y -coordinate, and elevation. This is a reasonable model since analyses of stake mass balance measurements from Hofsjökull in Iceland suggest that linearity is an appropriate assumption to make. These works do not, however, model spatial correlations in the residuals, which in turn can lead to suboptimal predictions and overconfident estimates on spatial aggregations of SMB. This is addressed in the model of Gopalan (2019), which we discuss next.

Gopalan (2019) used two Bayesian hierarchical models (Sect. 2.2) for making spatial predictions of SMB across Langjökull for the summer and winter of every year, respectively, in the recording period 1997–2015. The *data model* they used is

$$Z_{w,t}(\mathbf{s}) = \text{SMB}_{w,t}(\mathbf{s}) + \epsilon_{w,t}(\mathbf{s}), \quad \mathbf{s} \in \mathcal{S}_t, \quad t = 1997, \dots, 2015, \quad (3)$$

where $\mathcal{S}_t \subseteq \{\mathbf{s}_1, \dots, \mathbf{s}_{25}\}$ is the set of locations at which measurements were made in year t , the subscript w refers to winter, and where $\mathbf{s}_1, \dots, \mathbf{s}_{25}$ are the 25 measurement locations containing the x - and y -coordinates from a Lambert projection system (shifted and scaled to lie between 0 to 1). The number of locations in $\mathcal{S}_t$ was greater or equal to 22 in the years 1997–2015 across Langjökull (earlier years tended to have less measurements than more recent years). The terms $\epsilon_{w,t}(\mathbf{s}), \mathbf{s} \in \mathcal{S}_t, t = 1997, \dots, 2015$, are Gaussian white noise terms. The data model used for the summer SMB is analogous to that for winter.

The *process model* model used for SMB is

$$SMB_{w,t}(\mathbf{s}) = \beta_{0,w,t} + \beta_{1,w,t}s_1 + \beta_{2,w,t}s_2 + \beta_{3,w,t}z_t(\mathbf{s}) + U_{w,t}(\mathbf{s}), \quad \mathbf{s} \in D. \quad (4)$$

As before, the subscript w refers to winter, and the spatial domain of interest, D , is the set of x - y coordinates in the glacier Langjökull. The terms $\beta_{0,w,t}, \dots, \beta_{3,w,t}$ are model coefficients corresponding to an intercept, the x -coordinate s_1 , the y -coordinate s_2 , and elevation in metres, $z_t(\cdot)$, in year t , respectively, while $U_{w,t}(\cdot)$, $t = 1997, \dots, 2015$, are independent spatial Gaussian processes. Again, the process model for summer is analogous to that for winter. Gopalan (2019) fit separate SMB models each year so as to account for year-to-year variation in SMB. The latent Gaussian process $U_{w,t}(\cdot)$ has mean 0 with spatial covariance determined by the Matérn kernel:

$$C(\mathbf{s}_a, \mathbf{s}_b) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{8\nu} \|\mathbf{s}_a - \mathbf{s}_b\|}{\rho} \right)^\nu K_\nu \left(\frac{\sqrt{8\nu} \|\mathbf{s}_a - \mathbf{s}_b\|}{\rho} \right), \quad (5)$$

for $\mathbf{s}_a, \mathbf{s}_b \in D$, where σ is the marginal standard deviation, ρ is the spatial range parameter, ν is the smoothness parameter, $K_\nu(\cdot)$ is the modified Bessel function of the second kind of order ν , and $\Gamma(\cdot)$ is the gamma function (e.g., Bakka et al., 2018). A Gaussian process with Matérn covariance function (5) is the solution to the stochastic partial differential equation (SPDE):

$$(\kappa^2 - \Delta)^{\alpha/2} (\tau U_{w,t}(\mathbf{s})) = \mathcal{W}(\mathbf{s}), \quad \mathbf{s} \in D, \quad (6)$$

where recall that here $D \subset \mathbb{R}^2$, Δ is the Laplacian, $\kappa = \sqrt{8\nu}/\rho$, $\alpha = \nu + 1$ (for $d = 2$), $\mathcal{W}(\cdot)$ is a Gaussian white noise spatial process, and where $\tau > 0$ is a function of σ^2 , ρ , and ν (Lindgren & Rue, 2015).

The following specifications were used for the *prior model*. The β parameters were all set to have zero mean prior normal distributions. The precision for the intercept term was set to be very small to induce a non-informative prior; a precision of 0.1 was used for the elevation term; and a precision of 1.0 was used for the x and y terms. Precisions were selected using a leave-one-out cross-validation procedure. These precisions selected were also consistent with the fact noticed that estimated slopes in year-to-year linear regression fits varied more for elevation than for the x - and y -coordinates. Penalised-complexity (PC) priors (Simpson et al., 2017) were used for the range and scale parameters of the Matérn covariance kernel.

In order to fit the SMB models, Gopalan (2019) used the R-INLA software (Rue et al., 2009), which employs an approximate Bayesian inference scheme based on nested Laplace approximations for computing approximate posterior distributions over relevant quantities. Additionally, this software implements the scheme of Lindgren et al. (2011) in order to approximate the solution to the SPDE in (6) as a linear combination of basis functions that are derived from a finite-element

method (FEM). The FEM leads to the use of sparse precision matrices, with which computations can be done efficiently using sparse matrix algebra routines; see, for example, Golub and Van Loan (2012).

Gopalan (2019) fit separate SMB models for summer and winter for each year from 1997 to 2015 and used the fitted models to make predictions for both summer and winter SMB across the glacier at a 100-metre resolution for each year. For illustration, the predictions for summer SMB and winter SMB are shown in Fig. 2 for the year 1997. Accompanying prediction standard deviations, which are quantitative measures of prediction uncertainty, are shown in Fig. 3, while the net SMB is shown in Fig. 4. It is clear that, as expected, melting occurs more in the summer than in winter and that the net SMB tends to be more negative at the perimeter of the glacier, where elevation is lower. On the other hand, the interior of the glacier tends to see little surface mass loss at the higher elevations. This

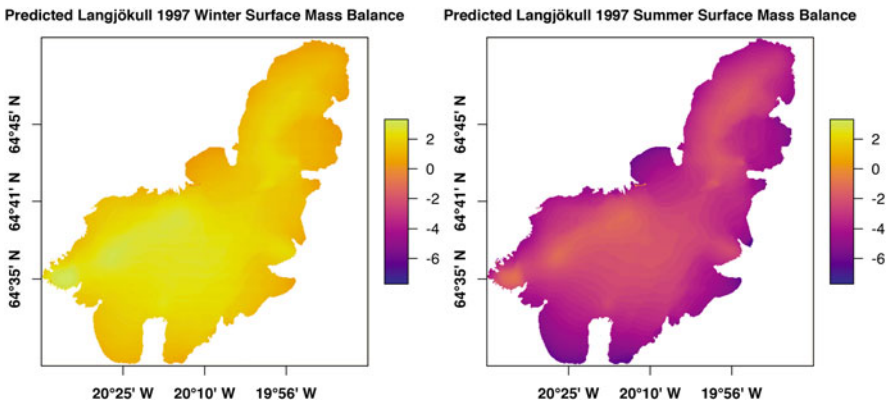


Fig. 2 Summer and winter SMB predictions, in metres per year water equivalent, across Langjökull in the year 1997. Left Panel: Winter SMB predictions. Right panel: Summer SMB predictions

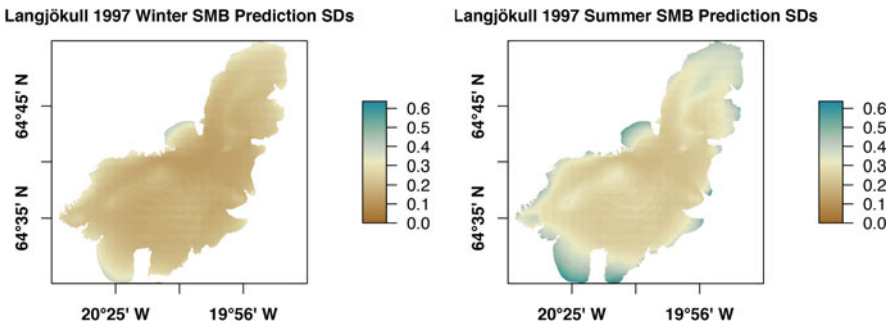


Fig. 3 Prediction standard deviations (SDs) associated with the predictions of Fig. 2, in metres per year water equivalent, across Langjökull in the year 1997. Left Panel: Winter SMB prediction SDs. Right panel: Summer SMB prediction SDs

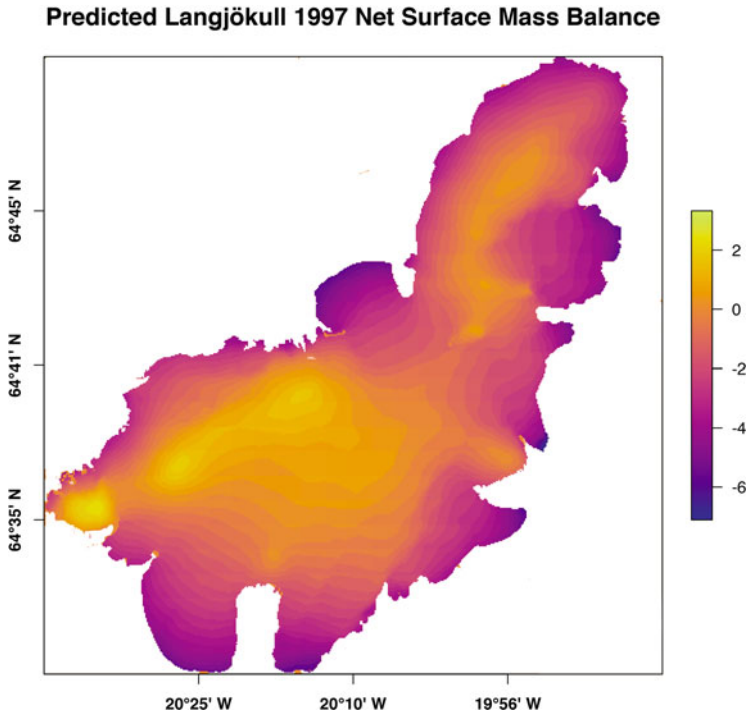


Fig. 4 Net SMB predictions (sum of summer and winter), in metres per year water equivalent, across Langjökull in the year 1997

general pattern is seen also in later years, although in 2015 the amount of mass loss at the perimeter regions is less than in 1997. These results are consistent with what is expected from baseline physical principles. An R script for making mass balance predictions with R-INLA is available at <https://github.com/ggopalan/SMB>.

4 Assessing Antarctica's Contribution to Sea-Level Rise

The Antarctic ice sheet is the world's largest potential contributor to sea-level rise, with enough grounded ice to raise the sea level by approximately 60 m (Morlighem et al., 2020). However, estimating the present day contribution to sea-level rise, let alone the future contribution, is not straightforward: Ice sheets continuously lose mass through basal melting, sublimation, meltwater runoff and flux into the ocean, and gain mass through accumulation of snowfall. When the total mass input matches the total mass output, an ice sheet is said to be in balance and does not contribute to sea-level rise or lowering. On the other hand, mass loss occurs when there is a shift from the balance state.

Deviations from a balance state are detected through changes in ice surface elevation (altimetry) and changes in mass (gravimetry) across time. For example, a detected loss in ice-sheet thickness generally (but not always) corresponds to a loss in ice-sheet mass and, hence, a positive contribution to sea-level rise. However, the problem of estimating ice mass loss/gain from data is ill-posed as there are multiple geophysical processes occurring in Antarctica that all contribute to observed change in ice surface elevation and/or change in mass, the most relevant of which are: (i) glacio-isostatic adjustment (GIA, which is a solid-Earth process), (ii) ice dynamics, (iii) firn compaction, and (iv) a collection of surface processes such as precipitation that affect the mass balance at the surface. Inferring these dynamical processes from observational data is a long-standing problem in glaciology (e.g., Riva et al., 2009; Gunter et al., 2014).

Zammit-Mangion et al. (2014) and Zammit-Mangion et al. (2015) developed a BHM to characterise the multiple processes in Antarctica that contribute to mass balance, which was subsequently used for various studies of the Antarctic ice sheet (e.g., Schoen et al., 2015; Martín-Español et al., 2016; Chuter et al., 2021). In the vein of Berliner (2003) (Sect. 2.2), the BHM is divided into a data model, a process model, and a parameter model. For clarity of exposition, we first describe the process model before proceeding to outline the data and parameter models.

Process Model The process model of the BHM describes the primary quantities of interest, which in this case are the yearly ice surface elevation changes due to the four geophysical processes listed above. Consider a time index $0, 1, \dots, T$ and a spatial domain $D \subset \mathbb{R}^2$. In the BHM, GIA is modelled as a spatial process $Y_{GIA}(\cdot)$ (since it is practically time-invariant on the time scales of interest) with covariance function (5), while ice dynamics are modelled as a spatio-temporal regression of the form

$$Y_{I,t}(\mathbf{s}) = \mathbf{x}'_t \boldsymbol{\beta}(\mathbf{s}) + w_{I,t}(\mathbf{s}); \quad \mathbf{s} \in D, \quad t = 0, 1, \dots, T, \quad (7)$$

where $Y_{I,t}(\mathbf{s})$ is the ice surface elevation that is due to ice dynamics in a time period t (in this case, spanning one year) at location $\mathbf{s}$, $\boldsymbol{\beta}(\mathbf{s})$ are spatially varying weights of covariates $\mathbf{x}'_t = (1, t)$ (chosen in order to model spatially varying linear temporal trends), and $w_{I,t}(\cdot)$ is unexplained variation that is modelled as white noise after discretisation (discussed in more detail below). SMB and firn compaction are modelled as conventional autoregressive-1 processes where the error term is spatially correlated. Specifically, for SMB,

$$Y_{S,t}(\mathbf{s}) = a(\mathbf{s})Y_{S,t-1}(\mathbf{s}) + w_{S,t}(\mathbf{s}); \quad \mathbf{s} \in D, \quad t = 1, \dots, T, \quad (8)$$

where $Y_{S,t}(\mathbf{s})$ is the ice surface elevation change that is due to SMB in a time period t , $a(\mathbf{s})$ is a spatially varying autoregressive parameter, and where each $w_t(\cdot)$ is modelled as an independent spatial process with covariance function (5). The process model for firn compaction, $Y_{F,t}(\cdot)$, is analogous to that for SMB. These modelling choices can be varied as needed; for example, Chuter et al. (2021) instead

modelled ice surface elevation loss due to ice dynamics as an (highly correlated) autoregressive-1 process in the Antarctic Peninsula, where the assumption of a linear or quadratic temporal trend for all given spatial locations is not realistic.

The processes in the BHM are discretised using the same finite-element scheme described in Sect. 3. The triangulation used to construct the elements can vary with the process being modelled; since GIA is smoothly varying, a coarse triangulation suffices for modelling GIA. On the other hand, a fine triangulation is used when modelling ice dynamics, particularly close to the coastline where high horizontal velocities lead to a large variability in ice surface elevation change due to ice dynamics. The finite-element scheme is used to establish a finite-dimensional representation of the four processes being considered for each time period t . Specifically, for each $i \in \{S, F, I\}$, $t = 0, \dots, T$, the finite-element decomposition of $w_{i,t}(\cdot)$ yields the representation $Y_{i,t}(\cdot) = \phi_i(\cdot)' \eta_{i,t}$, where $\phi_i(\cdot)$ are basis functions, and $\eta_{i,t}$ are the coefficients of those basis functions. For GIA, one has that $Y_{GIA}(\cdot) = \phi_{GIA}(\cdot)' \eta_{GIA}$. The process model is completed by specifying multivariate Gaussian distributions over $\eta_{i,t}$, $i \in \{S, F, I\}$, $t = 0, \dots, T$, and η_{GIA} , following the methodology outlined in Lindgren et al. (2011).

Data Model There are several data sets that could be used to help identify the contributors to observed ice surface elevation change in a glacier or ice sheet. GPS data record changes in bedrock elevation and are thus useful for estimating GIA. Satellite altimetry records changes in the ice surface elevation and ice shelf thickness (assuming hydrostatic equilibrium). When combined with ice penetrating radar data—and potentially mass-conservation methods (e.g., Morlighem et al., 2020)—satellite altimetry can yield ice thickness and bed topography estimates for the grounded ice sheet. Satellite gravimeter instrumentation records gravitational anomalies, which in turn give information on changes in ice-sheet mass as well as mantle flow that occurs during GIA.

For the data model, one seeks a mapping between the observed quantity and the geophysical process that, recall, could be either spatio-temporal or, in the case of GIA, spatial only (in which case the observations can be viewed as repeated measurements of a spatial-only process). Assume we have m_t observations at time interval t , where $t = 0, \dots, T$. In this BHM, these mappings take the general form

$$Z_{j,t} = \sum_{i \in \{S, F, I\}} \int_{\Omega_j} f_{i,t}^j(\mathbf{s}) Y_{i,t}(\mathbf{s}) d\mathbf{s} + \int_{\Omega_j} f_{GIA}^j(\mathbf{s}) Y_{GIA}(\mathbf{s}) d\mathbf{s} + v_{j,t}, \quad (9)$$

for $j = 1, \dots, m_t$; $t = 0, \dots, T$, where $Z_{j,t}$ is the j th observation at time t , Ω_j is the observation footprint of the j th observation, $f_{i,t}^j(\cdot)$ and $f_{GIA}^j(\cdot)$ are instrument-and-process-specific functions that, for example, account for volume-to-mass conversions, and where $v_{j,t}$, which is normally distributed with mean zero and variance σ_j^2 , captures both fine-scale process variation and measurement error. For some instruments and processes, $f_{i,t}^j(\cdot) = 0$; for example, $f_{I,t}^j(\cdot) = 0$ if the j th datum is from GPS instrumentation, since GPS instruments only measure

bedrock elevation change, while $f_{F,t}^j(\cdot) = 0$ if the j th datum is from gravimeter instrumentation, since firm compaction is a mass-preserving process. On the other hand, $f_{I,t}^j(\mathbf{s}) = \rho_I(\mathbf{s})$ if the j th datum is from gravimeter instrumentation, where $\rho_I(\mathbf{s})$ is the density of ice at $\mathbf{s}$.

The integrals in (9) are represented as numerical approximations using the finite-element representations of the processes. Specifically, for the first integral in (9), one has that

$$\int_{\Omega_j} f_{i,t}^j(\mathbf{s}) Y_{i,t}(\mathbf{s}) d\mathbf{s} \approx \left(\sum_{l=1}^{n_j^*} f_{i,t}^j(\mathbf{s}_l) \boldsymbol{\phi}_i(\mathbf{s}_l)' \Delta_l \right) \boldsymbol{\eta}_{i,t} \equiv (\mathbf{b}_{i,t}^j)' \boldsymbol{\eta}_{i,t}, \quad (10)$$

where $\mathbf{s}_l, l = 1 \dots, n_j^*$, denote the centroids of a fine gridding of Ω_j , and Δ_l denotes the area of each grid cell. The second integral in (9) is represented in a similar manner. These approximations thus lead to a linear Gaussian relationship between the data and the unknown coefficients $\{\boldsymbol{\eta}_{i,t}\}$ that ultimately represent the processes. Inference proceeds by evaluating the distribution of $\{\boldsymbol{\eta}_{i,t}\}$ conditional on the data $\{Z_{j,t}\}$. When all parameters are assumed known and fixed, this conditional distribution is Gaussian and known. In practice, several parameters also need to be estimated; these are discussed next.

Parameter Model The BHM requires many parameters to be set or estimated from data. It is reasonable for some of the parameters appearing in the process model, such as temporal and spatial length scales (which themselves vary in different regions of the ice sheet), to be derived from numerical models that describe specific physical processes. In Zammit-Mangion et al. (2015), Regional Atmospheric Climate Model (RACMO, Lenaerts et al., 2012) outputs were used to estimate spatio-temporal length scales at various locations on the ice sheet for SMB and firm compaction processes, while the GIA solid-Earth model IJ05R2 (Ivins et al., 2013) was used to obtain a spatial length scale (ρ in (5)) for the GIA process. Correlations between SMB and firm compaction, which are also modelled, were estimated from the firm densification model of Ligtenberg et al. (2011). One of the main attractions of the process models is that, for several of the processes, they allow one to not place any informative prior belief on the process mean value (often simply called the *prior* in geophysical applications), but solely on the second-order properties (i.e., the covariances) such as length scales, which are relatively well-understood.

There are several other parameters in the BHM that need to be estimated online in conjunction with the weights $\{\boldsymbol{\eta}_{i,t}\}$ that define the latent processes. In the BHM of Zammit-Mangion et al. (2015), inference was made over the following parameters that appear in the data and process models:

- Parameters that determine the fine-scale variance components $v_{j,t}, j = 1, \dots, m_t; t = 0, \dots, T$ that were, in turn, allowed to vary with characteristics of the surface topography.

- Parameters that determine the extent of induced spatial smoothing in gravimetry-derived products.
- Multiplicative scaling factors that inflate or deflate the error variances supplied with the data products, when there is evidence that these are too large or too small.
- Parameters used to construct prior distributions for spatially varying weights $\beta(\cdot)$ in (8). The variances of these weights are generally spatially varying; for example, when modelling ice surface elevation change due to ice dynamics, the variance of the spatially varying temporal linear trend was constrained to be a monotonically increasing function of horizontal ice velocity.

Inference with the BHM was done using Metropolis-within-Gibbs MCMC, yielding samples from the posterior distributions over all the unknown processes and those parameters that were estimated concurrently with the processes' weights. The authors found broad agreement between their inferences, which are for the years 2003–2009, and numerical-model outputs and also reported mass balance changes that largely agree, within uncertainty, with those from other approaches that make heavy use of numerical-model output in their analyses. For illustration, Fig. 5 is an adaptation of Figure 8 of Zammit-Mangion et al. (2015) and depicts the inferred ice surface elevation change in Antarctica that is occurring due to ice dynamics in the years 2004 and 2009.

We conclude this section with a brief discussion on the benefits and drawbacks of employing this fully Bayesian approach for modelling the Antarctic contribution to sea-level rise. The advantages are several: First, the posterior inferences are predominantly data-driven and largely agnostic to the underlying physical drivers. They therefore can also be used to validate output from solid-Earth or ice-sheet models. Second, uncertainties are provided on all geophysical quantities, which are interpretable both intra-process and inter-process. For example, a strong anti-correlation between the ice surface elevation contribution from SMB and ice dynamics in the posterior distribution is indicative of the difficulty in separating out the contributors of the observed elevation change; this uncertainty is allowed to vary spatially as well as temporally. Third, it provides a principled way with which to integrate expert knowledge, via the prior distributions placed over the processes and the parameters, with observational data, which can be heterogeneous and have differing support. The limitations are the following: First, it is computationally intensive to make inference with the BHM, and further increasing the spatial and temporal resolution could result in computational difficulties. Second, useful inferences heavily rely on the provision of high-quality data; unrecognised biases or strongly correlated errors in the data products used will adversely affect the predictions and will not be acknowledged in the returned prediction (uncertainty) intervals. Finally, to be used correctly, the framework requires both statistical expertise and glaciological expertise and is thus only useful for teams with experts in both disciplines. Software that automatically performs this type of analysis is lacking; this is an important angle of statistics for cryospheric science that requires

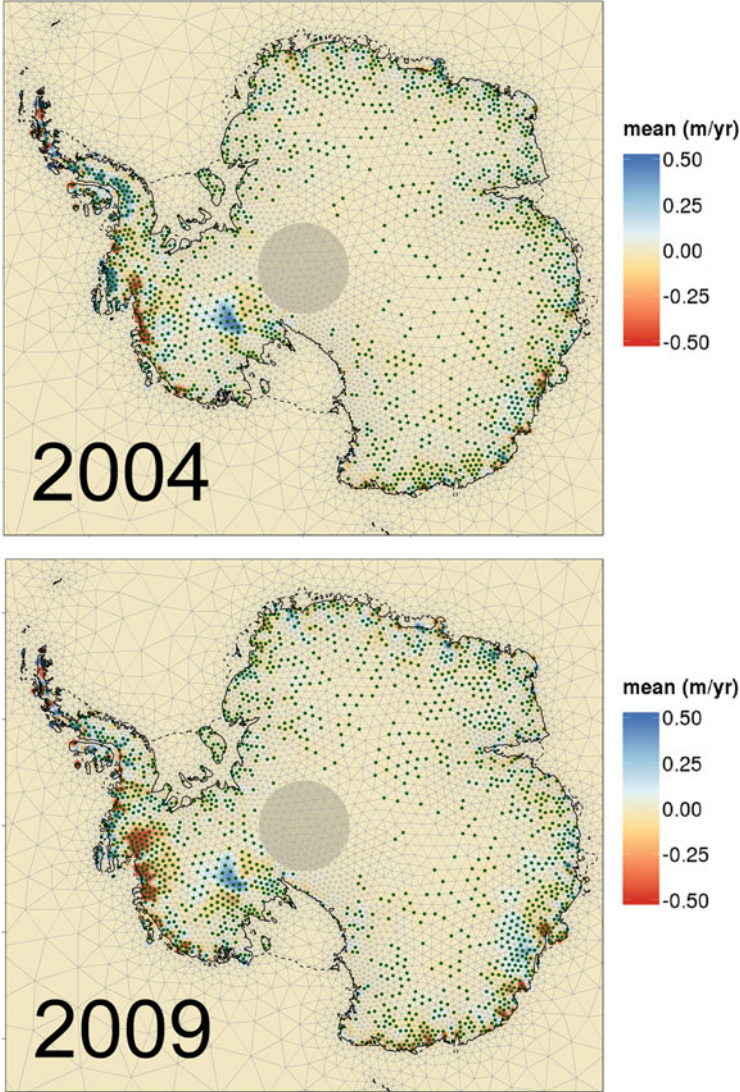


Fig. 5 Posterior mean rate of change of ice surface elevation due to ice dynamics for the years 2004 and 2009. Ice surface elevation changes over ice shelves (enclosed between the dashed and solid lines, which are the coastline and grounding line, respectively) are omitted as these do not contribute to sea-level change. The triangulation (grey lines) is that used to establish the finite-dimensional representation for $Y_{l,t}(\cdot)$ (see *Process model* in Sect. 4). Stipples (green dots) denote areas where the posterior mean is significantly different from zero (in this case, where the ratio of the posterior standard deviation to the absolute posterior mean is less than one). (Figure adapted from Zammit-Mangion et al., 2015)

further development. Code for reproducing the results in Zammit-Mangion et al. (2015) is available from <https://sites.google.com/view/rates-antarctica/home>.

5 Conclusions and Future Directions

Collecting data from mountain glaciers and ice sheets is a laborious and time-intensive process. For instance, the Antarctic Ice Sheet is the most remote place on Earth and experiences some of the harshest conditions in the world. Advances in remote sensing have led to extended spatial coverage of ice surface fields (e.g., elevation and velocity) and ice mass change over recent decades, facilitating advances in our understanding of ice dynamic processes and evolution. However, some data—including bed topography that is crucial in predicting Antarctica’s contribution to future sea-level rise—remain sparsely and unevenly sampled and come at a premium. In such circumstances, the BHM is a powerful framework to maximise the utility of available data, by: (i) allowing one to easily integrate data from multiple sources and with differing support, (ii) allowing one to incorporate prior information and effectively exclude implausible estimates a priori, (iii) allowing one to incorporate dynamical models or simulations into a statistical model, and (iv) providing a means to quantify uncertainty of the inferential target or forecast, potentially informing scientists where to invest next for expensive data collection.

Bayesian statistics is a mature field of research, but further investigation and development of several techniques in the field of cryospheric sciences are warranted. For example, the vast majority of the chapter discussed models that are based on underlying Gaussian assumptions. While Gaussianity may be a reasonable assumption for the process (i.e., the latent) model, Gaussianity is often an unrealistic assumption for the data model, particularly in situations where the measurement-error distribution exhibits skewness and heavy tails. When Gaussianity is a valid assumption, there are computational challenges involved that may be addressed using advances in numerics and computational statistics. The reviewed literature has showcased some of these techniques, including the use of surrogate models, the use of sparse matrix algebraic routines, and the use of parallelisable inference schemes. It is clear that developments in Bayesian computation will also benefit the inferential aspects of cryospheric science.

Despite the benefits of Bayesian reasoning and analysis, there is a paucity of Bayesian methods used by the cryospheric science community. For example, out of the 1744 articles published in the flagship journal *The Cryosphere* between the years 2013 and 2021, only 16 articles contain the word Bayes, or derivatives thereof, in the title, abstract, or keywords, and two of these are co-authored by authors of this chapter. However, recent years have seen a sustained increase in investment in the use of Bayesian methods for the cryospheric sciences. For example, the *GlobalMass*

project¹ that started in the year 2016 is a five-year European Union project where the BHM outlined in Sect. 4 is applied to other ice sheets and glaciers, as well as other processes that contribute to sea-level rise, while *Securing Antarctica's Environmental Future*² that started in the year 2021 is a seven-year Australian Research Council initiative that will see, among other things, Bayesian methods being implemented for data fusion and statistical downscaling to model and generate probabilistic forecasts of changes in environmental conditions and biodiversity in Antarctica. This increased use of Bayesian methods and investment is expected to be sustained for many years to come as data availability and the awareness of the importance of uncertainty quantification increases over time.

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Appendix: Governing Equations

For ease of exposition, in this appendix, we omit notation that establishes dependence of a variable or parameter on space and time. All variables and parameters should be assumed to be a function of space and time unless otherwise indicated.

Modelling ice-sheet flow relies on the classical laws of conservation of momentum, mass, and energy. For incompressible ice flow, conservation of momentum is described by the full Stokes equations:

$$\nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g} = \mathbf{0}, \quad (11)$$

$$\text{Tr}(\dot{\boldsymbol{\epsilon}}) = 0, \quad (12)$$

where $\nabla \cdot \boldsymbol{\sigma}$ is the divergence vector of the stress tensor $\boldsymbol{\sigma}$, ρ is the constant ice density, $\mathbf{g}$ is the constant gravitational acceleration, $\dot{\boldsymbol{\epsilon}}$ is the strain rate tensor, and Tr is the trace operator. The stress and strain rates are related by the material constitutive relation:

$$\boldsymbol{\sigma}' = 2\eta\dot{\boldsymbol{\epsilon}}, \quad (13)$$

¹ <https://www.globalmass.eu>.

² <https://arcsaef.com/>.

where $\sigma' = \sigma + p\mathbf{I}$ is the deviatoric stress tensor, p is the pressure, $\mathbf{I}$ is the identity matrix, and η is the viscosity. Boundary conditions for the mechanical model typically assume a stress-free ice surface and the specification of a friction or sliding law at the ice–bedrock interface.

A number of simplifications to the full Stokes equations exist, including the three-dimensional model from Blatter (1995) and Pattyn (2003), the two-dimensional shallow-shelf approximation (MacAyeal, 1989), and the two-dimensional shallow-ice approximation (Hutter, 1983).

Conservation of mass is described by the mass transport equation:

$$\frac{\partial H}{\partial t} + \nabla \cdot H\mathbf{v} = M_s + M_b, \quad (14)$$

where $\mathbf{v}$ is the horizontal velocity, H is the ice thickness, M_s is the surface mass balance, and M_b is the basal mass balance. For regional models of ice flow, the thickness is prescribed at the inflow boundaries, and a free-flux boundary condition is typically applied at the outflow boundary.

Finally, conservation of energy is described by the following equation:

$$\frac{\partial T}{\partial t} = (\mathbf{w} - \mathbf{v}) \cdot \nabla T + \frac{k}{\rho c} \Delta T + \frac{\Phi}{\rho c}, \quad (15)$$

where $\mathbf{v}$ is the horizontal velocity, $\mathbf{w}$ is the vertical velocity, T is the ice temperature, k is the constant thermal conductivity, c is the constant heat capacity, and Φ is the heat production term. The boundary conditions typically constitute a Dirichlet boundary condition at the ice surface, a relation for the geothermal and frictional heating at the base of the ice sheet, and a relation for the heat transfer at the ice–ocean interface.

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Bayesian Discharge Rating Curves Based on the Generalized Power Law



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1 Introduction

Discharge rating curves are one of the fundamental tools in hydrology. They are used to transform measurements of water elevation in a stream or river to discharge. Quantification of discharge in rivers and its associated uncertainty is essential for infrastructure design (e.g., Patil et al., 2021), flood frequency analysis (e.g., Steinbakk et al., 2016), hydroelectric power generation (e.g., Handayani et al., 2019), and climate science (e.g., Meis et al., 2021). It is much more expensive and complex to directly measure discharge than water elevation. The latter can be measured at a relatively high frequency, and the data can be gathered automatically (e.g., Mosley & McKerchar, 1993). Constructing a discharge rating curve requires measurements of discharge over a range of water elevation values to represent the relationship between these two variables as well as possible. To ensure good rating curve estimates for a variety of different rivers, the mathematical form of the rating curve needs to be relevant and flexible enough to capture the variability in channel size and cross-sectional shape.

Statistical discharge rating curves date back to the power-law relationship proposed by Venetis (1970). It has the form

$$Q = a(h - c)^b, \quad (1)$$

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for $h \geq c$, where Q is the discharge, h is the water elevation, c is the water elevation of zero discharge, b is the exponent of the power law, and a represents the discharge when the water depth is 1 m (that is, when $h - c = 1$ m). The measurement system used here is the International System of Units (SI). Note that water elevation is also referred to as stage and water level in the literature on discharge rating curves. The relationship in (1) is motivated by the mean velocity formulas of Manning and Chézy (Chow, 1959). These formulas are based on the assumption that open-channel flow is approximately uniform, that is, both the frictional resistance due to the riverbed roughness and the channel slope in the flow direction are constant. Both formulas can be presented as

$$\bar{v} = k_1 (A/P)^x,$$

where k_1 and x are constants, A is the cross-sectional area, and P is the wetted perimeter, i.e., the circumference of the cross section excluding the free surface. According to Manning, $x = 2/3$ and $k_1 = n^{-1} S^{1/2}$, where n is Manning's roughness coefficient and S is the slope of the channel. While, according to Chézy, $x = 1/2$ and $k_1 = C S^{1/2}$, where C is Chézy's constant, a measure of frictional resistance. The hydraulic radius, R , is defined as the ratio between the cross-sectional area and the wetted perimeter, i.e., $R = A/P$, which suggests that the discharge can be presented as $k_1 A R^x$. The terms A and R represent the geometry of the river's cross section, and Venetis (1970) argued that empirical evidence often suggests that the geometry can be represented with a power law in terms of the water depth, $d = h - c$, namely, with $A R^x = k_2 d^b$, where k_2 is a constant. Under these assumptions, the discharge is given by

$$Q = k_1 A R^x = k_1 k_2 d^b = a (h - c)^b,$$

where $a = k_1 k_2$. This power-law formula is appropriate under the assumptions of an approximately uniform flow and the cross-sectional geometry being such that the term $A R^x$ can be represented adequately well as a function of water depth.

The power law in (1) is still used in practice today. It is very often a sufficient model, showing that the arguments of Venetis (1970) still hold. When the power law is insufficient, the common practice has been to use segmented rating curves (Petersen-Øverleir & Reitan, 2005; Reitan & Petersen-Øverleir, 2009). They are based on a power-law rating curve within each segment of water elevation values (most commonly, two or three segments are used), and the continuity of the rating curve is usually ensured where the segments meet. The segmented rating curves can provide a better fit than the power-law rating curve. Estimation of segmented rating curves requires either selecting or estimating segmentation points. Selecting the segmentation points may not always give the optimal fit, and estimating these points can be numerically challenging (Reitan & Petersen-Øverleir, 2009). Furthermore, the segmented rating curves do not have a continuous first derivative where the segments meet, which may be a questionable model assumption in the case of natural channels unless there is a weir structure just

downstream of the measuring site, in which case the assumption might be reasonable.

Hrafnkelsson et al. (2022) proposed an alternative rating curve, which is this chapter's main topic. It is an extension of the power-law rating curve, referred to as the generalized power-law rating curve. It does not require segmentation; thus, segmentation points are not selected nor estimated. However, it is more flexible than the power-law rating curve and has a continuous first derivative. Its form is

$$Q(h) = a(h - c)^{f(h)}, \quad (2)$$

for $h \geq c$, where a is a constant, c is the water elevation of zero discharge, and $f(h)$ is referred to as the power-law exponent. As in the power-law model, a is the discharge when the water depth is 1 m.

The formula in (2) is flexible enough to mimic the discharge formulas of Manning and Chézy, namely $Q(h) = k_1 A(h) R(h)^x = k_1 A(h)^{x+1} P(h)^{-x}$. In Hrafnkelsson et al. (2022), it was shown that $f(h)$ is a function of the cross-sectional area and the wetted perimeter. To simplify the algebra, c was set equal to zero (i.e., water elevation and water depth take the same value), which gave

$$f(h) = \frac{(x + 1) \log \left\{ \frac{A(h)}{A(1)} \right\} - x \log \left\{ \frac{P(h)}{P(1)} \right\}}{\log(h)}, \quad (3)$$

where $h > 0$ and $h \neq 1$, and the constant a , in terms of the Manning and Chézy formulas, is given by

$$a = k_1 \frac{A(1)^{x+1}}{P(1)^x},$$

where, as above, under Manning, $k_1 = n^{-1} S^{1/2}$ and $x = 2/3$, and under Chézy, $k_1 = C S^{1/2}$ and $x = 1/2$. Thus, a is a function of the physical parameters and the geometry at $h = 1$ m, while the power-law exponent, $f(h)$, depends only on the geometry. It is assumed that the geometry of the cross section is such that it forms a single area for all values of the water elevation, i.e., there cannot be two or more disjoint areas for any value of the water elevation. Furthermore, it is assumed that the wetted perimeter, P , increases continuously as a function of water elevation. This means that the cross section is assumed to be such that P does not jump for some value of the water elevation; however, P can increase considerably within a narrow interval of water elevation values.

Let us consider a rectangular cross section with width ϕ_0 . Its cross-sectional area and wetted perimeter are $A(h) = \phi_0 h$ and $P(h) = \phi_0 + 2h$, respectively, when the

water elevation is h , assuming $c = 0$. In this case, the power-law exponent is

$$f(h) = (x + 1) - \frac{x \{\log(\phi_0 + 2h) - \log(\phi_0 + 2)\}}{\log(h)},$$

with an upper limit equal to $x + 1$ at $h = 0$, where $x + 1 = 1.67$ when $x = 2/3$, and a lower limit equal to 1, which is obtained when h approaches infinity. At $h = 1$, the power-law exponent takes the value $(x + 1) - 2x/(2 + \phi_0)$. In the case of a triangular cross section, the power-law exponent is equal to $2 + x$, or 2.67 when $x = 2/3$; thus, it is constant with respect to the water elevation. A cross section that has the shape of a parabola (i.e., its width is proportional to $h^{1/2}$) has a power-law exponent with an upper bound equal to $1.5 + x$ at $h = 0$ and a lower bound equal to $1.5 + 0.5x$ that is reached when h approaches infinity. When $x = 2/3$, the upper and lower bounds are 2.17 and 1.83, respectively. Figure 1 shows the shapes of rectangular, parabolic, and triangular cross sections and their corresponding power-law exponents.

Cross sections found in natural open channels are likely to be close to one of these three cross-sectional shapes. These three types of cross sections have power-law exponents that take values in the interval from 1 to 2.67 (using $x = 2/3$), indicating that natural channels have power-law exponents in this range. Moreover, Hrafnkelsson et al. (2022) derived the power-law exponents for three simulated geometries that mimic natural cross sections by using (3) and found that they were all in the range between 1 and 2.67. This fact about the power-law exponent is used when building a Bayesian statistical model for rating curves based on the generalized power law; see Sect. 3.

2 Data

Two datasets are analyzed in Sect. 5. The data are comprised of paired observations of discharge and water elevation from two stream gauging stations in Sweden: Kallstorp and Melby. These datasets were selected to demonstrate that the flexibility of the generalized power-law rating curve is sometimes needed and that the power-law rating curve is sufficient in other cases. The data, which were gathered by the Swedish Meteorological and Hydrological Institute, are such that the discharge is in cubic meters per second (m^3/s) and the water elevation is in meters (m). The water elevation measurements do not represent the actual depth of the water stream; rather, they show the water elevation relative to some local benchmarks. The raw data from both stations are shown in Fig. 2, along with the logarithmic transformation of the observed discharge, $\log(Q_i)$, versus the logarithmic transformation of the water depth, $\log(h_i - \hat{c})$, where $\hat{c}$ is the posterior median of the c parameter based on the data, inferred by the `gplm` model in the R package `bdrcc`; see details in Sect. 5.

These two datasets highlight the rigidity of the power-law model for some datasets where the assumption of a linear relationship between $\log(Q)$ and $\log(h - c)$

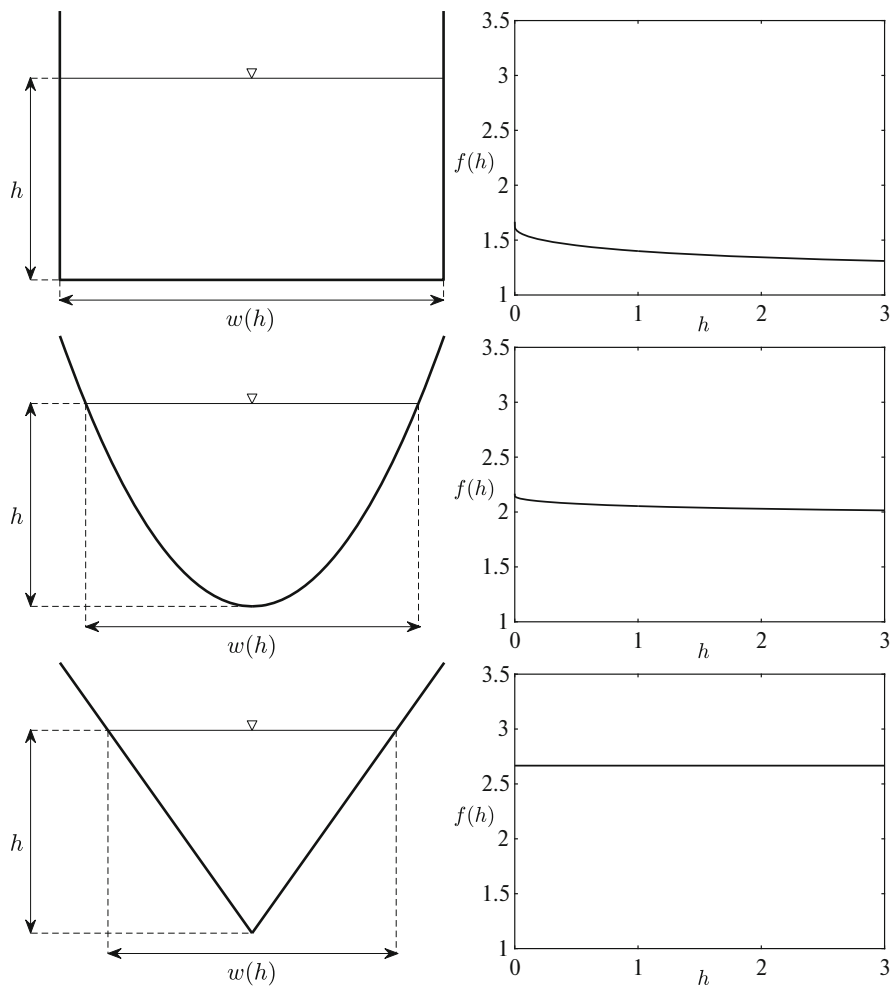


Fig. 1 Three cross sections and their corresponding power-law exponent when $x = 2/3$. Top panel: a rectangular cross section with width $w(h) = \phi_0$. Middle panel: a parabolic cross section with width $w(h) = \phi_{1/2}h^{1/2}$. Bottom panel: a triangular cross section with width $w(h) = \phi_1h$. Here, we select $\phi_0 = \phi_{1/2} = \phi_1 = 3$ m to illustrate examples of cross sections

does not hold. A log-linear assumption does appear reasonable for the transformed data from the Melby station. However, the transformed data from the Kallstorp station are such that the transformed observations do not line up linearly, and thus, a log-linear model will not fit the data well over all the water elevation values.

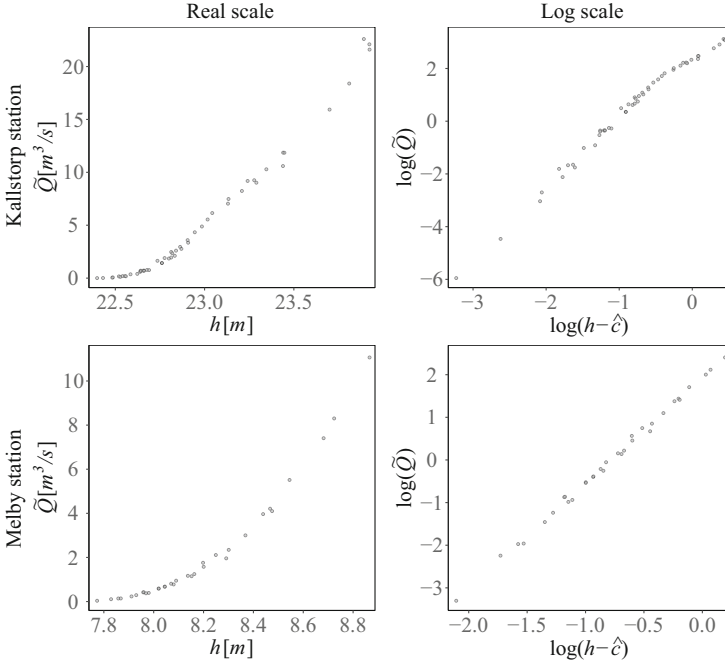


Fig. 2 Scatter plots showing the data on a real scale (left column) and after a log–log transformation of the power-law relationship (right column) for stream gauging stations Kallstorp (top row) and Melby (bottom row). The estimate for the c parameter, $\hat{c}$, is the posterior median of c , inferred by the `gplm` model in the R package `bdr`

3 Statistical Models

The generalized power law given in Sect. 1 is well-suited for discharge rating curve estimation. It is presented as a function of water elevation according to the formulas of Manning and Chézy and is applicable to a wide class of cross-sectional geometries. Hrafnkelsson et al. (2022) proposed a statistical model for rating curve estimation based on the generalized power law in (2). It is given by

$$\tilde{Q}_i = a(h_i - c)^{f(h_i)} e^{\epsilon_i}, \quad (4)$$

where $\tilde{Q}_i$ is the i -th discharge observation, h_i is the corresponding water elevation, a , c , and $f(\cdot)$ are as in (2), and ϵ_i is an error term that represents the discrepancy between the observed discharge and the generalized power law. The power-law exponent is modeled as a constant b and a function $\beta(h)$, that is, $f(h) = b + \beta(h)$. By applying the logarithmic transformation to both sides of (4), the statistical model

can be presented as

$$y_i = a_0 + \{b + \beta(h_i)\} \log(h_i - c) + \epsilon_i, \quad (5)$$

where $y_i = \log(\tilde{Q}_i)$ and $a_0 = \log(a)$. Hrafnkelsson et al. (2022) proposed that the error term in (5), ϵ_i , follows a Gaussian distribution with mean zero and variance that is a function of the water elevation. The exponent of the variance was modeled with B-spline basis functions (Wasserman, 2006),

$$\sigma_\epsilon^2(h) = \exp \left\{ \sum_{k=1}^K \eta_k B_k(h) \right\} = \prod_{k=1}^K \exp(\eta_k B_k(h)), \quad h_{\min} \leq h \leq h_{\max}, \quad (6)$$

where $h_{\min}$ and $h_{\max}$ are the smallest and largest water elevation observations in the paired dataset $(h_1, \tilde{Q}_1), \dots, (h_n, \tilde{Q}_n)$, n being the number of pairs. The parameters $\eta_1, \dots, \eta_K$ are unknown, and K is the number of basis functions. As a compromise between flexibility and the number of parameters, K is here set equal to 6. Note that $\sigma_\epsilon^2(h_{\min}) = \exp(\eta_1)$, $\sigma_\epsilon^2(h_{\max}) = \exp(\eta_K)$, and if $\eta_1 = \dots = \eta_K$, then the variance is constant with respect to water elevation. The interior knots are equally spaced on the interval $[h_{\min}, h_{\max}]$, while the additional knots are set equal to the endpoints of the interval. The error variance outside of the interval $[h_{\min}, h_{\max}]$ is defined as $\sigma_\epsilon^2(h) = \exp(\eta_1)$ for $h < h_{\min}$, and as $\sigma_\epsilon^2(h) = \exp(\eta_K)$ for $h > h_{\max}$. This specification is needed for discharge predictions when the water elevation value is not in $[h_{\min}, h_{\max}]$.

Hrafnkelsson et al. (2022) also set forth a power-law model with an error variance that can vary with the water elevation. That model is a special case of the model specified by (5) and (6), with $\beta(h_i) = 0$. These two models can be simplified by assuming that the variance of the error terms on the logarithmic scale is constant with respect to the water elevation. The two models with a constant variance on the logarithmic scale were not discussed specifically in Hrafnkelsson et al. (2022). Note that the power-law model with a constant error variance on the logarithmic scale is the model introduced by Venetis (1970). The generalized power-law model with a constant error variance on the logarithmic scale is useful when the median of the discharge observations, as a function of water elevation, requires the generalized power law, but the variance on the logarithmic scale can be modeled adequately well with a constant.

The R package `bdrc` was made available on the Comprehensive R Archive Network (CRAN) package archive in 2021. The package has functions to fit discharge rating curves based on these four statistical models. A Shiny app that allows users to fit rating curves with the `bdrc` package in an interactive graphical user interface is also available. An introduction to `bdrc` is given in Sect. 5, along with a link to the Shiny app.

The generalized power-law model with variance that can vary with water elevation is a flexible model that requires prior densities that constrain the parameter space in a sensible way. The prior densities are selected such that the model can

reduce to the power-law model with constant variance. This means that the prior densities of the statistical model are such that both the power-law exponent and the variance can reduce to a constant with respect to the water elevation. Thus, the generalized power-law model with varying variance can reduce to the other three statistical models for rating curve estimation that are based on the power law or the generalized power law. The principles of the penalized complexity (Simpson et al., 2017) were applied in Hrafnkelsson et al. (2022) to select prior densities for $\beta(h_1), \dots, \beta(h_n), \eta_1, \dots, \eta_K$ and their associated hyperparameters. The prior densities of these parameters and other parameters in the model are specified below according to Hrafnkelsson et al. (2022).

The parameter a represents the discharge when the water depth is 1 m. This parameter is given a prior density that supports large and small rivers. Namely, a Gaussian prior density with mean 3.0 and standard deviation 3.0 is assigned to $a_0 = \log(a)$. The parameter b is set equal to 1.835, which is the value at the center of the interval [1.0, 2.67] (see Introduction). Fixing b in this way ensures a more stable inference for the sum $b + \beta(h)$. A Gaussian process with mean zero is selected as a model for $\beta(h)$. Hrafnkelsson et al. (2022) assume that the function $f(h)$ is twice differentiable within the statistical model. This assumption leads to a continuous and smooth power-law exponent, and it is met by opting for a two times mean-square differentiable Gaussian process, in particular, one with a Matérn covariance function with smoothness parameter equal to 2.5 (Matérn, 1986). The prior density of the vector $\mathbf{u} = (\beta(h_1), \dots, \beta(h_n))^T$ is such that $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \Sigma_u)$, where $\mathbf{0}$ is a vector of zeros, and the (i, j) -th element of Σ_u is the covariance between $\beta(h_i)$ and $\beta(h_j)$, given by

$$\{\Sigma_u\}_{i,j} = \sigma_\beta^2 \left(1 + \frac{\sqrt{5}v_{i,j}}{\phi_\beta} + \frac{5v_{i,j}^2}{3\phi_\beta^2} \right) \exp \left(-\frac{\sqrt{5}v_{i,j}}{\phi_\beta} \right), \quad (7)$$

where $v_{i,j} = |h_i - h_j|$ is the distance between water elevations h_i and h_j .

The parameter c is such that its value is lower than the smallest water elevation in the paired dataset $(h_1, \tilde{Q}_1), \dots, (h_n, \tilde{Q}_n)$. If an even smaller value of observed water elevation is found in some other data source for the river of interest, then that value can be used as an upper bound for c . Let $h'_{\min}$ denote the smallest known water elevation of the river. If the data are such that $h'_{\min}$ is much greater than a plausible value of c , then the Bayesian inference scheme for the model parameters of the generalized power-law models can become unstable and produce questionable estimates of the rating curve for water elevation values below $h'_{\min}$. In such cases, the value of $h'_{\min}$ is updated within the `bdrc` package to compensate for this lack of information about c . More specifically, a power-law model with a constant variance is fitted to the data, and the value of $h'_{\min}$ is set equal to the 0.975 quantile of the marginal posterior density of c . After updating the value of $h'_{\min}$, the quantity $h'_{\min} - c$ is assigned an exponential prior density with rate parameter equal to 2. When $h'_{\min}$ is much greater than a plausible value of c , the fitted rating curve below $h'_{\min}$ should be interpreted with caution.

The parameters of the Matérn covariance function, σ_β and ϕ_β , are assigned the PC prior density in Fuglstad et al. (2019); see also Sect. 3.3.2 in chapter “Bayesian Latent Gaussian Models”. For details on the selection of the values of the parameters of the penalized complexity (PC) prior density, see Hrafnkelsson et al. (2022). When the PC approach is applied to σ_β and ϕ_β within the statistical model based on the generalized power law, the corresponding base model is the one with $\beta(h) = 0$, or $f(h) = b$. However, note that this PC prior shrinks toward a constant $\beta(h)$ (that is, toward a constant power-law exponent) by shrinking one over the range parameter ϕ_β toward zero. This PC prior also shrinks σ_β toward zero, which translates to shrinking $\beta(h)$ toward zero. Here, the fact that the PC prior shrinks toward a constant power-law exponent is more important than the fact that it shrinks toward $f(h) = b$ since a constant power-law exponent corresponds to the power-law model, while $f(h) = b$ corresponds to a power-law model with a power-law exponent equal to exactly 1.835. The value of b is somewhat arbitrary and another value close to 1.835 could have been selected.

An exponential prior density is assigned to $\exp(0.5\eta_1)$, and a random walk prior density with standard deviation σ_η is assigned to $\eta_2, \dots, \eta_K$ conditional on η_1 and σ_η (Hrafnkelsson et al., 2022). The standard deviation σ_η is given a PC prior, in particular, an exponential density; see Simpson et al. (2017) and Sect. 3.3.2 in chapter “Bayesian Latent Gaussian Models”. The base model of this PC prior has a constant error variance with respect to the water elevation, that is, $\eta_1 = \eta_2 = \dots = \eta_K$, which corresponds to $\sigma_\eta = 0$. The details on the selection of the value of the rate parameter of the exponential prior density for σ_η can be seen in Hrafnkelsson et al. (2022).

4 Posterior Inference

The parameters of the model in Sect. 3 are inferred according to the posterior sampling scheme for Gaussian–Gaussian models presented in Sect. 2.1.2 in chapter “Bayesian Latent Gaussian Models”. That posterior sampling scheme works well when each hyperparameter is transformed to the real line. The hyperparameters are transformed to the $K + 4$ -dimensional vector $\boldsymbol{\theta}$ according to $\theta_1 = \log(h'_{\min} - c)$, $\theta_2 = \log(\sigma_\beta)$, $\theta_3 = \log(\phi_\beta)$, $\theta_4 = \log(\sigma_\eta)$, $\theta_5 = \eta_1$, and $\theta_6, \dots, \theta_{K+4}$ are such that $\eta_k = \theta_5 + \sum_{m=2}^k \sigma_\eta \theta_{m+4}$, $k \in \{2, \dots, K\}$. The parameters $\theta_6, \dots, \theta_{K+4}$ are assumed to be a priori independent and given Gaussian prior densities with mean zero and variance one. The above parameterization is such that $\boldsymbol{\theta} \in \mathbb{R}^{K+4}$.

A vector and matrix presentation of the model in Sect. 3 is as follows. Let $\mathbf{y} = (y_1, \dots, y_n)^\top$ denote the response vector, $\boldsymbol{\beta} = (a_0, b)^\top$ contains the fixed effects, X is an $n \times 2$ matrix where the i -th row is given by $(1, \log(h_i - c))$, $\mathbf{u} = (\beta(h_1), \dots, \beta(h_n))^\top$ contains the random effects, and the weight matrix for the random effects is $A = \text{diag}(\log(h_1 - c), \dots, \log(h_n - c))$. Note that X and A are functions of c , and c is transformed to the hyperparameter θ_1 .

The variance of the error term ϵ_i , $\sigma_\epsilon^2(h_i)$, is given by (6). Define the covariance matrix $\Sigma_\epsilon = \text{diag}(\sigma_\epsilon^2(h_1), \dots, \sigma_\epsilon^2(h_n))$, and note that it depends on the hyperparameters $\theta_5, \dots, \theta_{K+4}$. Furthermore, define $Z = (X \ A)$, $\mathbf{x} = (a_0, b, \mathbf{u}^\top)^\top$, $\boldsymbol{\mu}_x = (\mu_a, \mu_b, \mathbf{0}^\top)^\top$, $\Sigma_x = \text{bdiag}(\sigma_a^2, \sigma_b^2, \Sigma_u)$, where bdiag denotes a block-diagonal matrix, $\mu_a = 3$, $\mu_b = 1.835$, $\sigma_a = 3$, and $\sigma_b = 0$. The statistical model can be presented as

$$\mathbf{y}|\mathbf{x}, \boldsymbol{\theta} \sim \mathcal{N}(Z\mathbf{x}, \Sigma_\epsilon), \quad \mathbf{x}|\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_x, \Sigma_x), \quad (8)$$

and inferred with the posterior sampling scheme in Sect. 2.1.2 in chapter “Bayesian Latent Gaussian Model” for Gaussian–Gaussian models that use the covariance representation.

The implementation of this posterior sampling scheme in the R package `bdr` is such that σ_b is set equal to 0.01 (as opposed to zero), then the vector $\mathbf{x}$ is sampled from its conditional density, and finally, $\mathbf{x}$ is adjusted according to the constraint $x_2 = 1.835$ using conditioning by Kriging (e.g., Rue & Held, 2005, Section 2.3).

Predictions of unobserved discharge $\tilde{Q}_{\text{pred}}$ for a specified water elevation h_{pred} can be made through the posterior predictive distribution. If the water elevation h_{pred} is equal to some h_i in the paired dataset, then samples are drawn from the posterior predictive distribution of $\tilde{Q}_{\text{pred}}$ by first sampling $\mathbf{x}^{(l)}$ and $\boldsymbol{\theta}^{(l)}$ from the posterior density and then drawing samples from

$$\tilde{Q}_{\text{pred}} \sim \text{LN}\left(a_0^{(l)} + \left\{b + \beta(h_i)^{(l)}\right\} \log\left(h_i - c^{(l)}\right), \sigma_\epsilon^2(h_i)^{(l)}\right),$$

where $\text{LN}(\mu, \sigma^2)$ denotes a lognormal distribution with parameters μ and σ^2 , and $\beta(h_i) = x_{i+2}$. When the water elevation h_{pred} is not found in the paired dataset, posterior predictive samples are drawn from

$$\tilde{Q}_{\text{pred}} \sim \text{LN}\left(a_0^{(l)} + \left\{b + \beta(h_{\text{pred}})^{(l)}\right\} \log\left(h_{\text{pred}} - c^{(l)}\right), \sigma_\epsilon^2(h_{\text{pred}})^{(l)}\right),$$

where $\beta(h_{\text{pred}})^{(l)}$ is drawn from the conditional Gaussian density

$$\pi\left(\beta(h_{\text{pred}}) \middle| \mathbf{u}^{(l)}, \theta_2^{(l)}, \theta_3^{(l)}\right) = \mathcal{N}\left(\beta(h_{\text{pred}}) \middle| \boldsymbol{\gamma}^\top \Sigma_u^{-1} \mathbf{u}^{(l)}, \sigma_\beta^2 - \boldsymbol{\gamma}^\top \Sigma_u \boldsymbol{\gamma}\right),$$

where $\boldsymbol{\gamma} = \text{cov}(\mathbf{u}, \beta(h_{\text{pred}}))$ is the covariance between $\beta(h_{\text{pred}})$ and the elements of $\mathbf{u} = (\beta(h_1), \dots, \beta(h_n))^\top$, evaluated with the Matérn covariance function in (7), and within the l -th iteration $\boldsymbol{\gamma}$, Σ_u , and σ_β^2 are evaluated with $\theta_2^{(l)}$ and $\theta_3^{(l)}$.

5 Results and Software

The R package `bdrc` applies the statistical methods described in the previous sections. It is freely available on the CRAN package archive. For documentation, tutorials, and more details, see the homepage of the package: <https://sor16.github.io/bdrc/>. We also created a Shiny app suitable for users who want to fit rating curves with the models from `bdrc` in a graphical user interface: <https://bdrc.shinyapps.io/bdrc/>. From within R, the package can be installed by running the following code:

```
install.packages("bdrc")
library(bdrc)
```

The four statistical models presented in Sect. 3 for discharge rating curve estimation are included in the `bdrc` package. The least complex of these four models is `plm0`. This model assumes that the median of the discharge observations follows the power law and that the variance of the measurement errors is constant on the logarithmic scale. There are only two mandatory input arguments when running the model: `formula` and `data`. The `formula` is of the form $y \sim x$, where y is the discharge in cubic meters per second (m^3/s) and x is the water elevation in meters (m). The `data` argument must be a `data.frame` with column names `x` and `y`. In our case, the data from the Kallstorp and Melby stations are such that the discharge and water elevation measurements are in columns named `Q` and `W`, respectively. Note that these two datasets are included in the `bdrc` package. Before applying the models from the `bdrc` package to these data, we set the seed of R's random number generator so that the results can be reproduced. The `plm0` model is fitted to the Kallstorp dataset by running the following code:

```
set.seed(1)
kallstorp.plm0 <- plm0(formula=Q~W, data=kallstorp)
```

The model returns an object of class `plm0` that can be input directly into the `plot` function to visualize the resulting rating curve. If one sets `type="panel"`, then the `plot` function creates a panel plot with four figures:

```
plot(kallstorp.plm0, type="panel")
```

The results are shown in Fig. 3. The fitted power-law rating curve does not adequately describe the underlying relationship between discharge and water elevation. Even though the data points all lie within the 95% prediction interval, the median of the posterior predictive distribution (the estimated rating curve) does not appear to be a good predictor for future discharge measurements. The residuals show a clear nonlinear pattern as a function of water elevation, implying that the assumption of mean zero Gaussian measurement errors on the logarithmic scale does not hold. Due

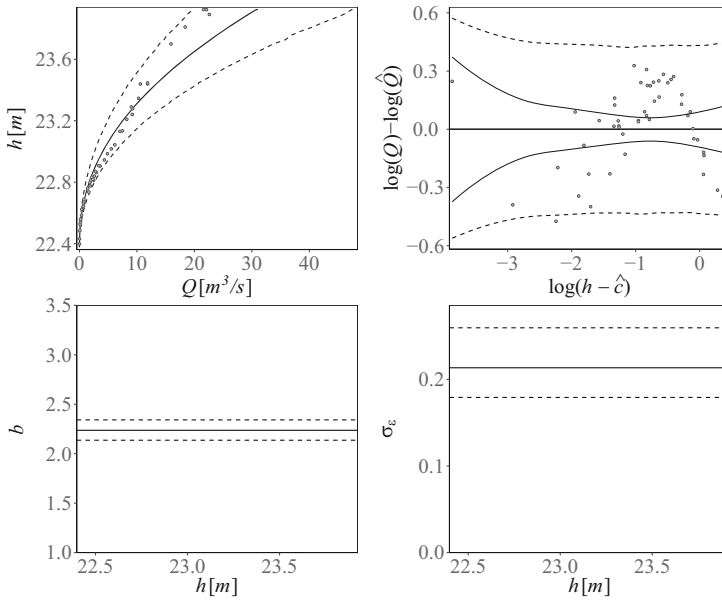


Fig. 3 A panel plot showing the results from the `plm0` model when applied to the data from the Kallstorp station. The upper-left graph shows the estimated rating curve (solid line) and 95% prediction intervals (dashed lines). The residuals are shown in the upper-right graph, along with 95% prediction intervals (dashed lines) and 95% posterior intervals for the log-rating curve minus the log-estimate (solid lines). The lower-left graph shows the estimate of the constant power-law exponent b (solid line), along with 95% posterior intervals (dashed lines). The lower-right graph shows the estimated constant standard deviation of the error term on the logarithmic scale σ_ϵ (solid line), along with 95% posterior intervals (dashed lines)

to the assumptions of the `plm0` model, the estimates of the power-law exponent, b , and the standard deviation of the error terms on the logarithmic scale, σ_ϵ , are constant across all water elevation values. In conclusion, the results based on `plm0` imply that there is not a linear relationship between $\log(Q)$ and $\log(h - c)$, and a more complex model should be considered.

In our search for a better fitting rating curve for the Kallstorp dataset, the `gplm` model in the `bdrbc` package is applied. This model assumes that the median of the discharge observations is given by the generalized power law and that the variance of the measurement errors on the logarithmic scale can vary with water elevation. Running the following code fits the `gplm` model to the Kallstorp dataset and creates a panel plot:

```
set.seed(1)
kallstorp.gplm <- gplm(formula=Q~W, data=kallstorp)
plot(kallstorp.gplm, type="panel")
```

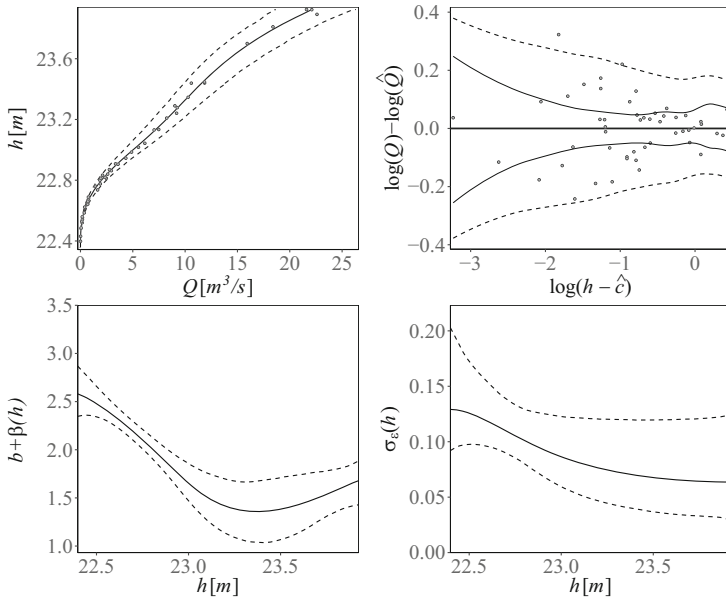


Fig. 4 A panel plot showing the results from the `gp1m` model when applied to the data from the Kallstorp station. The upper-left graph shows the estimated rating curve (solid line) and 95% prediction intervals (dashed lines). The residuals are shown in the upper-right graph, along with 95% prediction intervals (dashed lines) and 95% posterior intervals for the log-rating curve minus the log-estimate (solid lines). The lower-left graph shows the estimate of the power-law exponent $b + \beta(h)$ (solid line), along with 95% posterior intervals (dashed lines). The lower-right graph shows the estimate of the standard deviation of the error term on the logarithmic scale $\sigma_\epsilon(h)$ (solid line), along with 95% posterior intervals (dashed lines)

The results are shown in Fig. 4. The estimated rating curve fits the data much better than the power-law model with a constant variance. The residuals are randomly scattered above and below the rating curve estimate. There appears to be some heteroscedasticity in the residuals, but the `gp1m` model takes this into account by modeling the standard deviation of the error terms as a function of water elevation. The plot of the standard deviation, $\sigma_\epsilon(h)$, shows the extent of this heteroscedasticity. However, the main strength of this model lies in allowing the power-law exponent, $f(h)$, to vary with water elevation. The graph clearly shows that the power-law exponent is not constant over the water elevation values. Jointly, the four graphs in Fig. 4 underscore the need for the full complexity of the generalized power-law model with a standard deviation that can vary with water elevation to obtain a good fit to the Kallstorp data.

Summary statistics of the posterior distributions of the main parameters from both models for the Kallstorp dataset are presented in Table 1. The parameters of the `p1m0` model are a , b , c , and σ_ϵ . Similarly, the `gp1m` model parameters are a , c , $b + \beta(h_{\min})$, and $\sigma_\epsilon(h_{\min})$, where the power-law exponent and the standard deviation

Table 1 Posterior summary statistics for the main parameters in the `plm0` and `gplm` models when applied to the data from the Kallstorp station

Model	Parameter	Posterior median	Posterior std.	95% posterior interval
plm0	a	11.636	0.511	(10.661; 12.663)
	b	2.236	0.052	(2.137; 2.339)
	c	22.377	0.003	(22.371; 22.382)
	σ_ϵ	0.214	0.020	(0.179; 0.259)
gplm	a	10.444	0.337	(9.832; 11.187)
	$b + \beta(h_{\min})$	2.583	0.133	(2.338; 2.865)
	c	22.358	0.008	(22.341; 22.371)
	$\sigma_\epsilon(h_{\min})$	0.126	0.028	(0.092; 0.200)

are evaluated at the smallest water elevation in the dataset, $h_{\min}$. The estimates of the `plm0` model parameters should be interpreted with precaution since the model does not provide an adequate fit to the data. Nevertheless, the estimates of a and c are similar for both models, while, not surprisingly, the estimate of σ_ϵ for `plm0` is much larger than the estimate of $\sigma_\epsilon(h_{\min})$ for `gplm`. The `gplm` model parameters are estimated with a fair amount of certainty; for example, the width of the posterior interval for c is only 3 cm, and the size of the posterior standard deviation for a is only 3% of its posterior median.

An important feature of the relatively complex `gplm` model is that it is constructed such that it mimics the simple `plm0` model in cases where the latter model gives a sufficiently good description of the relationship between discharge and water elevation. To demonstrate this feature, the `gplm` model is fitted to the data from the Melby station, and the results are presented in Fig. 5. The rating curve and residuals indicate a good fit to these data. Despite the flexibility of the `gplm` model, its power-law exponent and error variance both shrink toward a constant as a function of water elevation. These data suggest that the power-law model with a constant variance is suitable, so, in this case, `gplm` essentially reduces to `plm0`.

Table 2 further illustrates the similarity between the fit of the two models in the case of the Melby data. The posterior medians of the parameters are very similar between the two models, and the 95% posterior intervals are consistently wider under the generalized power-law model by a factor of around 1.2. Unsurprisingly, the uncertainty in the parameters of the generalized power-law model is slightly greater than that of the power-law model, since the former is a more flexible model.

To allow for a comparison of the predictive performance of the models in the `bdrc` package, the widely applicable information criterion (WAIC) (Watanabe, 2010; Gelman et al., 2013) is calculated and stored in the model objects. WAIC uses the computed logarithm of the pointwise predictive density, $\widehat{\text{lppd}}$, a measure of the model’s fit, and a penalization term, $\widehat{p}_{\text{WAIC}}$, to correct for the model complexity, referred to as the computed effective number of parameters. It is defined as

$$\text{WAIC} = -2\widehat{\text{lppd}} + 2\widehat{p}_{\text{WAIC}}.$$

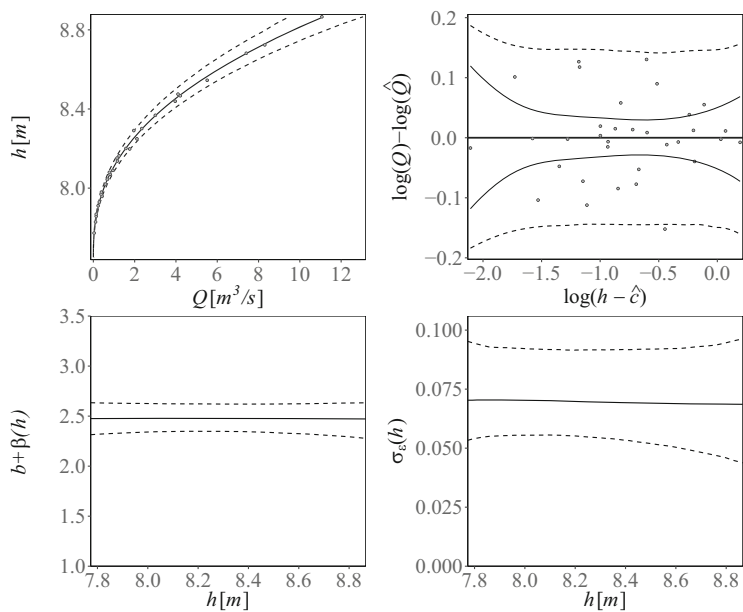


Fig. 5 A panel plot showing the results from the gp1m model when applied to the data from the Melby station. The upper-left graph shows the estimated rating curve (solid line) and 95% prediction intervals (dashed lines). The residuals are shown in the upper-right graph, along with 95% prediction intervals (dashed lines) and 95% posterior intervals for the log-rating curve minus the log-estimate (solid lines). The lower-left graph shows the estimate of the power-law exponent $b + \beta(h)$ (solid line), along with 95% posterior intervals (dashed lines). The lower-right graph shows the estimate of the standard deviation of the error term on the logarithmic scale $\sigma_{\epsilon}(h)$ (solid line), along with 95% posterior intervals (dashed lines)

Table 2 Posterior summary statistics for the main parameters in the plm0 and gp1m models when applied to the data from the Melby station

Model	Parameter	Posterior median	Posterior std.	95% posterior interval
plm0	a	6.875	0.169	(6.547; 7.211)
	b	2.481	0.067	(2.354; 2.620)
	c	7.651	0.010	(7.630; 7.668)
	σ_{ϵ}	0.069	0.009	(0.055; 0.091)
gp1m	a	6.888	0.208	(6.502; 7.334)
	$b + \beta(h_{\min})$	2.475	0.079	(2.311; 2.631)
	c	7.651	0.011	(7.629; 7.673)
	$\sigma_{\epsilon}(h_{\min})$	0.070	0.011	(0.054; 0.095)

The computed logarithm of the pointwise predictive density is given by

$$\widehat{\text{lppd}} = \sum_{i=1}^n \log \left(\frac{1}{L} \sum_{l=1}^L p(y_i | \boldsymbol{\psi}^{(l)}) \right),$$

where $\boldsymbol{\psi} = (\boldsymbol{\theta}, \mathbf{x})$, $p(y_i | \boldsymbol{\psi})$ is the response density of the i -th observation, and $\boldsymbol{\psi}^{(l)}$ is the l -th posterior sample. The computed effective number of parameters, $\hat{p}_{\text{WAIC}}$, is evaluated with

$$\hat{p}_{\text{WAIC}} = \sum_{i=1}^n \left(\frac{1}{L-1} \sum_{l=1}^L (\log p(y_i | \boldsymbol{\psi}^{(l)}) - \bar{p}_i)^2 \right),$$

where $\bar{p}_i$ is the sample average of $\log p(y_i | \boldsymbol{\psi}^{(l)})$ over the posterior samples.

The computed effective number of parameters is usually smaller than the actual number of parameters in the model. As more of the complexity of the model is being put to work to achieve a good fit to the data, the computed effective number of parameters increases. Defined in this way, WAIC quantifies the trade-off between the model's fit and complexity, where a smaller WAIC value is preferable to a larger one.

In the `bdrc` package, the WAIC value, along with the summary statistics of the estimated parameters, can be obtained by simply inserting a model object into the `summary` function:

```
summary(kallstorp.plm0)
```

```
> Formula:
> Q ~ W
> Latent parameters:
>   lower-2.5% median-50% upper-97.5%
> a  10.66      11.64      12.66
> b   2.14       2.24       2.34
>
> Hyperparameters:
>           lower-2.5% median-50% upper-97.5%
> c           22.371      22.377      22.382
> sigma_eps   0.179       0.214       0.259
>
> WAIC: 48.39526
```

These model comparison statistics for the case study at hand are extracted from the `plm0` and `gplm` model objects and presented in Table 3. Furthermore, the `plm` and `gplm0` models from the `bdrc` package are also applied to the Kallstorp and Melby data for comparison in Table 3. The `plm` model assumes that the median of the discharge observations follows the power-law rating curve and that the variance

Table 3 Model comparison based on WAIC. The table shows $\widehat{lppd}$, $\hat{p}_{WAIC}$, WAIC, and $\Delta WAIC$ for the data from the Kallstorp and Melby stations when comparing the four models in the `bdr` package. The values in the table have been rounded to the first decimal place

Station	Model	$\widehat{lppd}$	$\hat{p}_{WAIC}$	WAIC	$\Delta WAIC$
Kallstorp	plm0	−19.3	4.9	48.4	81.0
	plm	−18.3	5.7	48.0	80.6
	gplm0	17.4	6.5	−21.8	10.8
	gplm	24.4	8.1	−32.6	
Melby	plm0	42.8	3.2	−79.2	−0.1
	plm	42.9	3.3	−79.3	−0.3
	gplm0	42.8	3.3	−78.8	0.2
	gplm	42.9	3.4	−79.0	

can vary with the water elevation, whereas for the `gplm0` model, the median is made to follow the generalized power-law rating curve, and the error variance is assumed to be constant on the logarithmic scale.

In the case of the Kallstorp dataset, the measure of fit, $\widehat{lppd}$, is highest for the `gplm` model, indicating a better fit to the data than the other models. The increased complexity of the generalized power-law models is being put to work, as can be seen in the computed effective number of parameters, $\hat{p}_{WAIC}$, which is greater than $\hat{p}_{WAIC}$ of the power-law models. The full complexity of the `gplm` model is needed here, as indicated by the relatively large difference in the WAIC values of the models, presented as $\Delta WAIC = WAIC_s - WAIC_{gplm}$, where $s \in \{plm0, plm, gplm0\}$.

In the case of the Melby dataset, the fitted rating curve of the `gplm` model (see Fig. 5) is practically identical to the fitted rating curves of the other three models (results not shown). The measure of fit, presented in Table 3, does not become smaller by choosing a more complex model. Interestingly, since the data suggest that the power-law model with a constant error variance is an adequate model, the computed effective number of parameters is practically the same for all the models. This results in nearly identical WAIC values, as a difference in WAIC of 0.3 or less reflects that two models are practically equivalent in terms of their prediction performance.

6 Summary

The generalized power-law rating curve is based on the physical relationship between discharge and water elevation according to the formulas of Manning and Chézy. It mimics the effect of the cross section for a wide class of geometries, giving it flexibility beyond that of the power-law rating curve, which is, in fact, a special case of the generalized power-law rating curve.

The Bayesian approach is used to infer the parameters of the model, and Bayesian hierarchical modeling is used to structure a statistical model for the paired observations of discharge and water elevation. In some cases, these data are such that the variance of the measurement errors in the discharge observations is not constant

on the logarithmic scale as a function of water elevation. Thus, it is assumed within the statistical model that this variance can vary with water elevation. The statistical model has an inherent flexibility that comes through the Bayesian hierarchical modeling. Namely, it reduces to the power-law model if the data provide evidence in that direction, and similarly, the variance of the measurement errors can reduce to a constant as a function of the water elevation.

We opt for selecting a parsimonious model with the help of WAIC when deciding whether to use the generalized power-law rating curve with a varying variance or one of the reduced versions of that model.

The R package `bdr` is designed to infer four statistical models that are based on the power law and the generalized power law in a user-friendly manner. It robustly runs the Markov chain Monte Carlo scheme presented in Sect. 4 without any tuning to infer the model parameters. It contains plotting methods to create plots like the ones in Figs. 3, 4 and 5. Finally, it includes functions that aid users with model checking and model selection by providing visualization of posterior inference diagnostics, along with using WAIC to compare the prediction performance of the models.

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Bayesian Modeling in Engineering Seismology: Ground-Motion Models



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1 Introduction

In any seismic region with human presence, the estimation of *seismic risk* (i.e., the probability of suffering losses due to strong earthquake ground shaking, referred to as strong motion) is key to the optimized mitigation of the adverse earthquake effects on people's lives and the man-made environment. A reliable risk estimation, in turn, relies on the assessment of *seismic hazard*, e.g., the estimation of the ground-motion levels that a particular ground-shaking parameter is expected to exceed over time period (the output of probabilistic seismic hazard assessment, PSHA). PSHA is the international standard practice for seismic risk management worldwide and is used as the foundation for structural building codes for earthquake-resistant design (e.g., Eurocode 8). A reliable PSHA, however, requires the state-of-the-art specification of its three key elements: (1) the locations of earthquake sources, (2) their seismic activity and maximum magnitudes, and (3) the ground-motion models (GMMs) that describe the Earth shaking at any given location. The field of engineering seismology deals with the study of seismology with focus on its practical applications that lead to improved, comprehensive, and systematic assessment of the seismic hazard (see Lee et al., 2003). That in turn involves utilizing our knowledge of the three main factors that control the seismic ground motion.

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Namely, the earthquake source radiation of seismic waves; the path effects on the seismic waves as they propagate through Earth's crust; and the effects of localized geology on the seismic waves at an engineering site of interest. Such knowledge lends itself particularly to a key tool of engineering seismologists, namely, the above-mentioned *ground-motion model* (also known as ground-motion prediction equation, or attenuation relationship) that combines in a single mathematical model how seismic ground-shaking scales with the strength of the earthquake source (i.e., magnitude), the propagation path (i.e., distance from the site to the source), and conditions at the site of interest (i.e., potential soil amplification). Traditionally, for the fast and efficient hazard assessment, the GMMs are empirical equations that, given key independent parameters of the above factors, predict the amplitudes of various ground-motion parameters of engineering interest. Those parameters are primarily the peak horizontal acceleration of ground-shaking (PGA) and the pseudo-acceleration spectral response (PSA) of a simple oscillator of natural resonance period T , the latter modeling the maximum dynamic response of a simple structure during seismic ground shaking. The free parameters of the empirical GMMs are determined by fitting the model to a dataset of the ground-motion parameters of interest derived from recordings of earthquake ground motions in the seismic region. In this chapter, we show the application of the Bayesian statistical framework to GMMs, using Icelandic strong-motion data as an example.

2 Ground-Motion Models

Empirical GMMs have been employed for decades in the prediction of earthquake strong motion, i.e., the seismic ground motion that is large enough to have effects on the man-made environment. It was reported as a function of magnitude and source-to-site distance, and its parameters were inferred by using strong-motion data (Esteva & Rosenblueth, 1964). Since then, many researchers have concentrated on improving the estimation of the ground-motion variability by using non-parametric techniques, implementing a more specific definition of site characterization, including additional independent parameters to reduce the ground-motion variability, and appropriately considering the epistemic uncertainty, i.e., the uncertainty due to limited amount of data.

In general, empirical GMMs represent simplified mathematical–physical models of the characteristics of peak intensity parameters of earthquake strong motion and are developed from recorded time histories of seismic ground motion. The most common mathematical form of a ground-motion model is

$$Y = f(\mathbf{X}, \boldsymbol{\theta}) + \Delta, \quad (1)$$

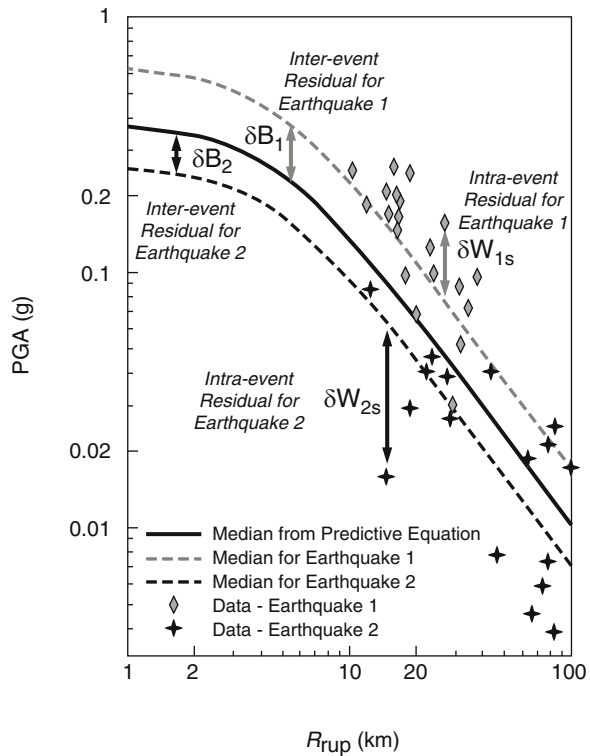
where (usually, the logarithm of) Y is the observed earthquake strong-motion intensity measure; $f(\mathbf{X}, \boldsymbol{\theta})$ represents the GMM; $\mathbf{X}$ is a vector of key independent parameters such as earthquake magnitude, source-to-site distance, and site condi-

tions; θ is a vector of model coefficients; and Δ is a random variable representing the model residuals, i.e., $Y_{es} - f(X, \theta)$, the difference between the observations and the GMM predictions conditioned on X . Further, for a given earthquake event e and site s , Δ can be decomposed into an inter-event residual (also known as between-event residual), δB_e , and an intra-event residual (also known as within-event residual), δW_{es} , and (1) can be rewritten as

$$Y_{es} = f(X, \theta) + \delta B_e + \delta W_{es}, \quad (2)$$

where the components δB_e and δW_{es} are independent, zero-mean Gaussian random variables with standard deviations τ and ϕ , respectively (Al Atik et al., 2010). The term δB_e denotes the average shift of the observed ground motion from an individual earthquake e from the population median predicted by the GMM, as schematically shown in Fig. 1. The inter-event residuals thus capture the effects of earthquake-source-specific variations that affect all its ground motions. Then, the intra-event residuals, δW_{es} , are the extent to the individual observations s deviate from the ensemble median of earthquake e . The intra-event residual thus captures the additional effects of variations due to, e.g., source–site geometry variations such as azimuthal variations in radiation strength, propagation path variations, and

Fig. 1 Schematic illustration of the median GMM prediction of the attenuation of PGA with distance from the earthquake rupture (solid black line) for a given magnitude, along with a hypothetical dataset from two earthquakes (diamonds and stars) of the same magnitude. The deviation of the model from the observations is referred to as the model residual. The inter-event terms δB_e for earthquakes $e = 1$ and 2, respectively, capture the deviation of the model from the medians of each dataset. Then, the intra-event terms δW_{es} capture the deviation of each data point s from the median of that earthquake e (Strasser et al., 2009)



localized site effects (see e.g., Strasser et al., 2009). The total standard deviation of the ground-motion model, i.e., the standard deviation of Δ , referred to as σ (sigma), where $\sigma = \sqrt{\tau^2 + \phi^2}$; here, τ and ϕ denote the inter-event and intra-event standard deviations, respectively. We note that, in essence, the long-term objective of engineering seismology has been, and still is, the improvement of $f(X, \theta)$ that leads to a reduction of σ for a given dataset Y_{es} . With this objective in mind, we find the Bayesian framework an especially effective tool.

3 Methods

3.1 Regression Analysis

Classically, GMMs have been largely developed based on the regression analysis (Draper & Smith, 1981), and the quality of the model is evaluated using its goodness-of-fit to the observations. Regression analysis is a statistical process that relates a set of input variables to an output variable. Although this scheme is conceptually simple, it is a well-known and popular method for investigating mathematical relationships among variables.

Implementing an appropriate regression method is important to constrain the GMM's coefficients. Donovan (1973) found that the correlation between magnitude and source-to-site distance can lead to trade-offs between the derived coefficients. However, in some cases, the scaling of seismic ground motion with distance could not be appropriately determined due to a lack of understanding or information about the correlation between observations at different sites for a given event (Campbell, 1981; Joyner & Boore, 1993). As a result, weighted nonlinear least-square regression methods, such as the two-stage maximum-likelihood and the random-effect methods, were introduced to improve the regression analysis (Draper & Smith, 1981; Campbell, 1981; Joyner & Boore, 1981; Brillinger & Preisler, 1985; Abrahamson & Youngs, 1992; Joyner & Boore, 1993; Campbell, 1993; Boore et al., 1993). In such an approach, the distance-dependent coefficients are derived first, using individual amplitude-scaling factors for each earthquake. In the second stage, the magnitude-dependent coefficients are derived by fitting a curve to these amplitude-scaling factors.

Although several researchers confirmed the usefulness of the two-stage regression method (e.g., Abrahamson & Litehiser, 1989; Fukushima et al., 1995; Molas & Yamazaki, 1995), some preferred to apply a one-stage method. Ambraseys and Bommer (1991) showed that using a two-stage method leads to information loss since more than half of the earthquakes in their set of records were only recorded by one instrument and are therefore excluded from the calculation of the magnitude dependence in the second stage. Moreover, Spudich et al. (1999) revealed that the two-stage methods underestimate τ for sets of records like theirs with many singly recorded earthquakes. Caillot and Bard (1993) stated that the two-stage

method might be misleading because it does not reduce the variance for some spectral methods. There have also been found significant changes in predictions of GMMs calibrated with one- and two-stage methods, respectively (Douglas, 2003).

3.2 *Bayesian Inference*

Developing reliable GMMs for regions with sparse earthquake strong-motion data is challenging. The lack of data leads to restrictions on the functional form of GMMs. In other words, some important terms and their parameters (e.g., ones related to magnitude saturation, magnitude–distance-dependent scaling, and inelastic attenuation terms) cannot be fully constrained when calibrated to sparse datasets. This can result in GMMs with median predictions that are associated with large uncertainty, which in turn has significant adverse influence on the probabilistic seismic hazard assessment, especially at low probability levels (Atkinson, 2006). As a result, many researchers have made efforts to develop better empirical GMMs, with the hope of decreasing their standard deviation. However, the use of more complicated functional forms of GMMs over the last few decades (Abrahamson et al., 2008; Abrahamson & Shedlock, 1997; Douglas & Gehl, 2008; Strasser et al., 2009) has not led to the expected decrease in the total variability, despite the vast increase in recorded data after the turn of the century (Douglas, 2010). Unavoidably therefore, improving GMMs involves capturing better the physics of what controls earthquake strong motion. One way is to apply a rigorous regression that quantifies the reliability of the inferred parameters and that of the model residuals, which ideally should be partitioned into physics-based terms such as *source*, *path*, and *site* terms, the very least (e.g., Aki & Richards, 1980; Lay & Wallace, 1995). Furthermore, such a partition allows us to inspect in detail the behavior of the residuals and potentially identifying relationships between the residuals and other independent variables (e.g., magnitude, distance, source depth, source–site geometry, surface geology, radiation pattern, and rupture complexities that affect near-fault motions). Modeling such residual behavior leads to a reduction in residual variability, thus improving the GMM and the PSHA.

To that end, the Bayesian statistical framework is especially useful. Bayesian statistical modeling presents a well-defined scheme based on Bayes' theorem that allows us to update our prior knowledge about some unknown parameters in light of new observations and make inferences about them (Gelman & Rubin, 1992; Diggle et al., 1998; Berger, 2013; Gelman et al., 2013; Congdon, 2014). The Bayesian methodology principally differs from the classical frequentist methods in that all of the unknown parameters in the underlying probability model are considered random variables, in contrast to classical statistics, where parameters are treated as unknown constants. The Bayesian methodology is sometimes referred to as Bayesian inversion, and

its output is a probability density of the unknown model parameters. This probability density is referred to as the posterior density and can be used to obtain parameter point estimates (e.g., the mode of the posterior density) and quantify the parameters' uncertainty (e.g., the marginal posterior standard deviations).

A Bayesian model consists of a probabilistic model for the observations, and it requires specifying prior knowledge about the model parameters through a prior density. The Bayesian statistical approach can be particularly useful to overcome this problem as it allows taking prior information about the model parameters into account and adding it to the information in the likelihood function that stems from the observed data. Moreover, knowledge obtained from the posterior distributions of the inferred GMM parameters and a physics-based interpretation of the behavior of the residuals from a seismological point of view can help to improve the GMMs (Cotton et al., 2018; Rahpeyma et al., 2022). Finally, the Bayesian methodology can also be quite effective over time as the GMMs can be systematically updated each time new observations become available (Rahpeyma et al., 2018, 2019; Kowsari et al., 2020).

The inference for the model parameter vector θ is based on the data y that contain information about θ through the sampling distribution, denoted by $\pi(y|\theta)$, also known as the likelihood function. The prior density, denoted by $\pi(\theta)$, describes assumptions about θ probabilistically. The posterior density, denoted by $\pi(\theta|y)$, represents the knowledge about θ after observing the data and can be thought of as an updated prior density. It is given by

$$\pi(\theta|y) = \frac{\pi(\theta)\pi(y|\theta)}{\pi(y)}, \quad (3)$$

where $\pi(y)$ represents the marginal density function of the data, y , which is independent of the parameters θ and is given by

$$\pi(y) = \int \pi(\theta)\pi(y|\theta)d\theta. \quad (4)$$

Since the integral in (4) cannot be evaluated analytically in most cases, a numerical approximation method is most often used instead that uses the fact the density in (3) can be defined up to a constant, namely

$$\pi(\theta|y) \propto \pi(\theta)\pi(y|\theta). \quad (5)$$

In general, computing these properties requires optimizing and integrating the posterior probability density, which must be carried out numerically for nonlinear problems (Molnar & Cassidy, 2006).

4 Applications

In this section, we illustrate the application of the Bayesian approach in engineering seismology through three examples involving the development and selection of GMMs for assessing seismic hazards.

First, we present a three-level Bayesian hierarchical model (BHM) and show how one computes the posterior distributions of the model parameters over multiple levels using Markov chain Monte Carlo (MCMC) simulations. In this example, a dataset of *aftershock* strong motions is used. Aftershocks are the intense sequence of earthquakes that occur immediately on and around the causative fault on which a *mainshock* took place, the largest aftershocks being usually about 1.5 magnitude units smaller than the mainshock (e.g., Utsu & Ogata, 1995). The aftershocks were recorded on a small array of recording sites in a town in South Iceland located very close to the fault. The strong-motion scaling is modeled using a simple GMM form since the main emphasis is on partitioning the residuals into different terms and evaluating their respective contributions to the overall variation of the model. That allows us to determine the key sources of uncertainties in the model (source vs. path vs. site effects) along with mapping spatially the station terms (i.e., the localized site effects). We then present a Bayesian random-effect model that uses MCMC simulations to recalibrate various GMMs of different functional forms to a dataset of mainshock ground motions recorded on a regional network of recording sites across Southwest Iceland. In this example, we show how using informative priors can help address certain limitations of the dataset and produce improved GMMs that do not suffer from those limitations. Finally, we present a new Bayesian data-driven GMM selection method based on the deviance information criterion (DIC). Given a regional ground-motion dataset, we show how to select the most optimal GMMs to use in that region's seismic hazard assessment.

4.1 Site Effect Characterization Using a Bayesian Hierarchical Model for Array Strong Ground Motions

In 2007, the first Icelandic small-aperture strong-motion array (ICEARRAY I) was deployed in the town of Hveragerði in the western part of the South Iceland Seismic Zone (SISZ) in Southwest Iceland. It consists of 12 strong-motion stations in an area of around 1.23 km^2 with interstation distances ranging from 50 to 1900 m (Halldórsson & Sigbjörnsson, 2009). On 29 May 2008 at 15:45 local time, the $M_w 6.3$ Ölfus earthquake occurred in the western part of the SISZ (Sigbjörnsson et al., 2009). As shown in Fig. 2, the ICEARRAY I strong-motion stations recorded the mainshock and more than 1700 of its aftershocks over several months. The aftershock distribution delineates two parallel and near-vertical north-south striking earthquake faults approximately 4.5 km apart. The analysis of the

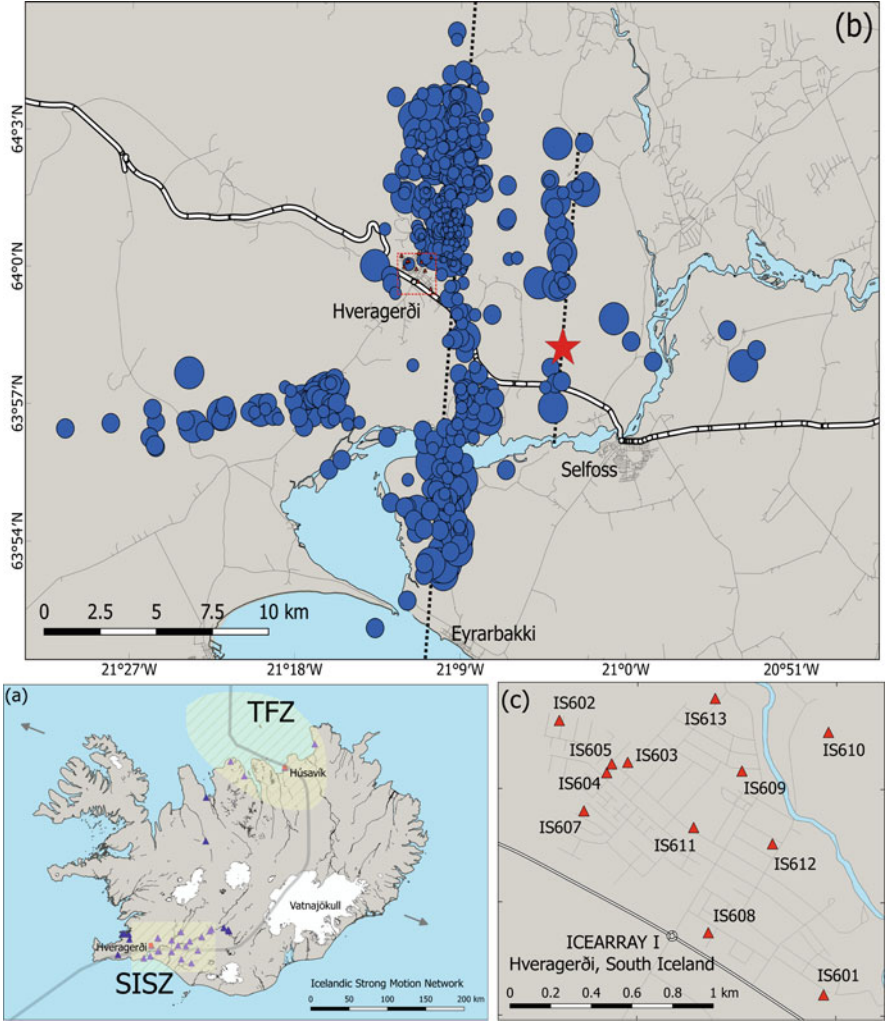


Fig. 2 (a) Map of Iceland showing the approximate centerline of the present-day tectonic margin between the North American and Eurasian plates, respectively (gray line), and the two transform zones in the country, the South Iceland Seismic Zone (SISZ) and the Tjörnes Fracture Zone (TFZ), marked by hatched areas. The red rectangles show the locations of the towns of Hveragerði and Húsavík. The town of Hveragerði is shown in more detail in figures (b) and (c). (b) The aftershock distribution (blue circles) from the 29 May 2008 Ölfus earthquake in Southwest Iceland outlines the two causative earthquake faults (dotted lines), and the epicenter of the mainshock is shown as a red star. (c) The locations of the ICEARRAY I stations (red triangles) within the town of Hveragerði are presented along with the station ID codes

large aftershock dataset showed considerable variation in the distribution of strong-motion amplitudes between stations, both in the recordings of the same earthquake and in the recordings of different earthquakes. No clear systematic trends could

be identified using conventional methods of analysis, and therefore, the Bayesian hierarchical modeling was employed.

4.1.1 Bayesian Hierarchical Modeling Framework

As presented in (2), the classical model for the prediction of a ground-motion parameter (e.g., PGA, PGV, PSA at different periods) depends on earthquake source, path, and site parameters (Al Atik et al., 2010; Abrahamson & Youngs, 1992). In this chapter, considering the notation recommended in Al Atik et al. (2010), the model in (2) can be rewritten as

$$Y_{es} = \mu_{es}(M_e, R_{es}, D_e) + \delta B_e + \delta W_{es}, \quad e = 1, \dots, N, \quad s = 1, \dots, Q, \quad (6)$$

where Y_{es} is the strong-motion parameter of interest in the logarithmic unit (base 10) for the event (earthquake) e recorded at recording site (station) s that follows a Gaussian distribution. Here, N and Q are the total number of events and stations, respectively. The predictive model, μ_{es} , provides the median ground motion in terms of independent seismic variables (moment magnitude M_e , source-to-site epicentral distance R_{es} , and depth of event D_e) for event e at station s . Although GMMs come in a variety of different functional forms that depend on the desired inputs, due to the small magnitudes of the aftershocks and the short source-to-site distances, we used the following simple linear model:

$$\log_{10} \mu_{es} = \beta_1 + \beta_2 M_e + \beta_3 \log_{10}(R_{es}) + \beta_4 D_e, \quad (7)$$

where the GMM coefficient vector $\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3, \beta_4)$ consists of the parameters that capture the characteristics of the seismic region and the geological structure. In (6), δB_e , the inter-event residual (i.e., the event term), represents the average shift, corresponding to an individual earthquake, of the observed ground motions from the corresponding median estimates of the ground-motion model. Let $\delta \mathbf{B}$ denote the vector $(\delta B_1, \dots, \delta B_N)$. In the BHM formulation, the event terms are assumed to be independent zero-mean Gaussian random variables with variance τ^2 . On the other hand, the intra-event residual δW_{es} represents the difference between an individual observation at station s from the earthquake-specific median prediction (Strasser et al., 2009; Al Atik et al., 2010). Using the BHM model, the intra-event residual can be further divided into three terms as follows:

$$\delta W_{es} = \delta S_2 S_s + \delta W S_{es} + \delta R_{es}. \quad (8)$$

The station term vector (i.e., the vector of inter-station residual), $\delta S_2 S_s$, represents the average of the intra-event residuals at each station, and it is used to scale the prediction of the GMM to a site-specific prediction. The inter-event residuals are modeled spatially with a zero-mean Gaussian distribution with a covariance structure governed by an exponential function from the Matérn family with variance

ϕ_{S2S}^2 . Let $\delta S2S$ denote the vector $(\delta S2S_1, \dots, \delta S2S_Q)$. The event-station residual, $\delta W S_{es}$, captures the record-to-record variability and can be investigated for other repeatable effects. In the BHM formulation, $\delta W S_{es}$ are assumed to be spatially correlated variables from a zero-mean Gaussian distribution with a covariance structure governed by a covariance function from the Matérn family with variance ϕ_{SS}^2 . Finally, δR_{es} , the error term (or unexplained term), accounts for effects that are not modeled by the other terms. The unexplained terms are assumed to be independent and are modeled with a mean-zero Gaussian distribution with variance ϕ_R^2 .

The total variability of a GMM can be separated into two main parts: (i) inter-event variability and (ii) intra-event variability. The intra-event variability can further be divided into inter-station variability (i.e., station-to-station variability), event-station variability (i.e., variability between stations within an event), and another unexplained variability (e.g., measurement and model error). This ability to separate the variability is of great importance for many engineering applications. The total variability of the model in (6) can therefore effectively be written as the sum of the four independent variance terms, namely

$$\sigma_T^2 = \tau^2 + \phi_{S2S}^2 + \phi_{SS}^2 + \phi_R^2. \quad (9)$$

The inter-event variance, τ^2 , quantifies the variation between events relative to the average ground-motion level predicted by the GMM. The inter-station variance, ϕ_{S2S}^2 , quantifies the ground-motion variability between stations, primarily a manifestation of the localized variations such as differences in the geological profiles beneath the stations. The event-station variance, ϕ_{SS}^2 , can be defined as a measure of the spatial variability in the ground-motion amplitudes between stations within an event after taking into account the event and station terms. The purpose of this term is to quantify the remaining variations not already captured by the GMM or the event and station terms. Lastly, the unexplained term variance, ϕ_R^2 , quantifies the variability in the measurement errors and other deviations not accounted for by other terms in the model.

4.1.2 The Hierarchical Formulation

The BHM represents a flexible probabilistic framework for multi-level modeling of earthquake ground-motion parameters of interest (e.g., PGA, PGV, and PSA), in which a collection of random variables can be decomposed into a series of conditional models (Rahpeyma et al., 2018). In this chapter, we set up the BHM in three levels as follows:

1. Data level
2. Latent level
3. Hyperparameter level

At the data level, the conditional density of the observed data is given by a function of the model parameters defined at the latent and hyperparameter levels. Furthermore, the prior probability distributions for the latent and hyperparameters are given at the latent and hyperparameter levels, respectively. The purpose of these prior distributions is to regularize or constrain the model parameters in a sensible way to improve their inference. To hierarchically formulate (6), we define $\mathbf{y} = (Y_{11}, Y_{12}, \dots, Y_{NQ})$ at the first level as a vector containing all the observations, Y_{es} , ordered by station number for all the events. The vector of latent parameters is defined as $\boldsymbol{\eta} = (\boldsymbol{\beta}, \boldsymbol{\delta B}, \boldsymbol{\delta S2S})$ at the second level, and the vector of hyperparameters $\boldsymbol{\theta} = (\tau, \phi_{S2S}, \phi_{SS}, \phi_R, \Delta_{SS})$ at the third level. A detailed mathematical formulation of the proposed BHM will now be given.

1. Data Level

At the first level, the conditional density of the observed data, $\mathbf{y}$, is given by

$$\begin{aligned} \pi(\mathbf{y}|\boldsymbol{\beta}, \boldsymbol{\delta B}, \boldsymbol{\delta S2S}) &= \mathcal{N}(\mathbf{y}|\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}_1\boldsymbol{\delta B} + \mathbf{Z}_2\boldsymbol{\delta S2S}, \boldsymbol{\Sigma}_y) \\ &= \mathcal{N}(\mathbf{y}|\mathbf{K}\boldsymbol{\eta}, \boldsymbol{\Sigma}_y), \end{aligned} \quad (10)$$

where $\mathbf{X}$ is a $(NQ) \times p$ -dimensional matrix containing the linear predictors for all combinations of stations and events. The matrices $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are index matrices, as they have exactly one non-zero entry per row and can be defined as

$$\mathbf{Z}_1 = \mathbf{I}_N \otimes \mathbf{1}_Q, \quad \mathbf{Z}_2 = \mathbf{1}_N \otimes \mathbf{I}_Q, \quad (11)$$

where $\otimes$ refers to the Kronecker product, $\mathbf{1}_n$ is an n -element vector of ones, and $\mathbf{I}_n$ is an $n \times n$ identity matrix. The integers N , Q , and p are the total number of events, stations, and parameters, respectively. The vector $\boldsymbol{\eta}$ contains the latent parameter vectors $\boldsymbol{\beta}$, $\boldsymbol{\delta B}$, and $\boldsymbol{\delta S2S}$. The matrix $\mathbf{K}$ contains the linear predictors and index matrices, $\mathbf{K} = [\mathbf{X} \ \mathbf{Z}_1 \ \mathbf{Z}_2]$. And lastly, $\boldsymbol{\Sigma}_y$ is an $(NQ) \times (NQ)$ -dimensional block-diagonal matrix composed as

$$\boldsymbol{\Sigma}_y = \phi_R^2 \mathbf{I}_{NQ} + \mathbf{I}_N \otimes \boldsymbol{\Sigma}_{SS} = \begin{pmatrix} \phi_R^2 \mathbf{I}_Q + \boldsymbol{\Sigma}_{SS} & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \phi_R^2 \mathbf{I}_Q + \boldsymbol{\Sigma}_{SS} \end{pmatrix}, \quad (12)$$

where

$$\{\boldsymbol{\Sigma}_{SS}\}_{ij} = \left\{ \phi_{SS}^2 \left(1 + \frac{d_{ij}}{\Delta_{SS}} \right) \exp \left(-\frac{d_{ij}}{\Delta_{SS}} \right) \right\}_{ij}, \quad (13)$$

where d_{ij} is the inter-station distance between stations i and j , and the covariance matrix, $\{\Sigma_{SS}\}_{ij}$, is based on a Matérn covariance function with a standard deviation ϕ_{SS} , smoothness parameter $\nu_{SS}=1.5$, and range parameter, Δ_{SS} . It should be highlighted that rows and columns are removed from all the matrices and vectors according to missing data points.

2. Latent Level

The vector η consists of the parameters β , δB , and $\delta S2S$ at the latent level of the BHM. The prior distribution of η follows a Gaussian distribution with mean μ_η and covariance matrix Σ_η , where

$$\eta = \begin{pmatrix} \beta \\ \delta B \\ \delta S2S \end{pmatrix}, \quad \mu_\eta = \begin{pmatrix} \mu_\beta \\ \mathbf{0}_N \\ \mathbf{0}_Q \end{pmatrix}, \quad \Sigma_\eta = \begin{pmatrix} \Sigma_\beta & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \tau^2 I_N & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Sigma_{S2S} \end{pmatrix}, \quad (14)$$

where $\mathbf{0}_D$ is a column vector of zeros of length D , $\mathbf{0}$ is a matrix of zeros, $\mu_\beta = \mathbf{0}_p$, and $\Sigma_\beta = 100^2 I_p$. The covariance matrix Σ_{S2S} is based on the Matérn covariance function with standard deviation ϕ_{S2S} , smoothness parameter $\nu_{S2S} = 0.5$, and range parameter Δ_{S2S} and defined as

$$\{\Sigma_{S2S}\}_{ij} = \left\{ \phi_{S2S}^2 \exp\left(-\frac{d_{ij}}{\Delta_{S2S}}\right) \right\}_{ij}. \quad (15)$$

It should be noted that the sensitivity analysis revealed that the parameter Δ_{S2S} is difficult to infer. Therefore, we fix $\Delta_{S2S} = 0.06$ km.

3. Hyperparameter Level

The vector $\theta = (\tau, \phi_{S2S}, \phi_{SS}, \phi_R, \Delta_{SS})$ contains the hyperparameters. The hyperparameters are assumed to be independent and exponentially distributed as follows:

$$\begin{aligned} \tau &\sim \text{Expon}(\lambda_\tau), & \phi_{S2S} &\sim \text{Expon}(\lambda_{\phi_{S2S}}), & \phi_{SS} &\sim \text{Expon}(\lambda_{\phi_{SS}}), \\ \phi_R &\sim \text{Expon}(\lambda_{\phi_R}), & \Delta_{SS} &\sim \text{Expon}(\lambda_{\Delta_{SS}}). \end{aligned} \quad (16)$$

In order to sample only positive values of the hyperparameters and enhance the efficiency of the sampling process, a logarithmic transformation is used for the

hyperparameters, i.e., $\psi_i = \log(\theta_i)$. More precisely,

$$\begin{aligned}\psi_1 &= \log(\tau), & \psi_2 &= \log(\phi_{S2S}), & \psi_3 &= \log(\phi_{SS}), \\ \psi_4 &= \log(\phi_R), & \psi_5 &= \log(\Delta_{SS}),\end{aligned}\tag{17}$$

and the marginal prior densities are given by

$$\pi(\psi_i | \lambda_i) = \lambda_i \exp(-\lambda_i \exp(\psi_i) + \psi_i).\tag{18}$$

4.1.3 Posterior Inference

The joint posterior density can be presented as

$$\pi(\boldsymbol{\eta}, \boldsymbol{\psi} | \mathbf{y}) \propto \pi(\mathbf{y} | \boldsymbol{\eta}, \boldsymbol{\psi}) \pi(\boldsymbol{\eta} | \boldsymbol{\psi}) \pi(\boldsymbol{\psi}),\tag{19}$$

where

$$\pi(\boldsymbol{\eta} | \boldsymbol{\psi}) = \pi(\boldsymbol{\beta}) \pi(\boldsymbol{\delta} \mathbf{B} | \psi_1) \pi(\boldsymbol{\delta} \mathbf{S} \mathbf{2} \mathbf{S} | \psi_2, \Delta_{S2S}), \pi(\boldsymbol{\psi}) = \prod_{k=1}^5 \pi(\psi_k).$$

The posterior distribution of $\boldsymbol{\eta}$ conditional on $\boldsymbol{\theta}$ can be shown to be a normal density, namely

$$\begin{aligned}\pi(\boldsymbol{\eta} | \mathbf{y}, \boldsymbol{\psi}) &\propto \pi(\mathbf{y} | \boldsymbol{\eta}, \boldsymbol{\psi}) \pi(\boldsymbol{\eta} | \boldsymbol{\psi}) \\ &\propto \mathcal{N}(\mathbf{y} | \mathbf{K} \boldsymbol{\eta}, \boldsymbol{\Sigma}_y) \mathcal{N}(\boldsymbol{\eta} | \boldsymbol{\mu}_\eta, \boldsymbol{\Sigma}_\eta) \\ &\propto \exp\left(-\frac{1}{2}(\mathbf{y} - \mathbf{K} \boldsymbol{\eta})^\top \boldsymbol{\Sigma}_y^{-1}(\mathbf{y} - \mathbf{K} \boldsymbol{\eta})\right) \\ &\quad \times \exp\left(-\frac{1}{2}(\boldsymbol{\eta} - \boldsymbol{\mu}_\eta)^\top \boldsymbol{\Sigma}_\eta^{-1}(\boldsymbol{\eta} - \boldsymbol{\mu}_\eta)\right) \\ &\propto \mathcal{N}(\boldsymbol{\eta} | \boldsymbol{\mu}_{\eta, \text{post}}, \boldsymbol{\Sigma}_{\eta, \text{post}}),\end{aligned}\tag{20}$$

where

$$\begin{aligned}\boldsymbol{\mu}_{\eta, \text{post}} &= \left(\boldsymbol{\Sigma}_\eta^{-1} + \mathbf{K}^\top \boldsymbol{\Sigma}_y^{-1} \mathbf{K}\right)^{-1} (\boldsymbol{\Sigma}_\eta^{-1} \boldsymbol{\mu}_\eta + \mathbf{K}^\top \boldsymbol{\Sigma}_y^{-1} \mathbf{y}), \\ \boldsymbol{\Sigma}_{\eta, \text{post}} &= \left(\boldsymbol{\Sigma}_\eta^{-1} + \mathbf{K}^\top \boldsymbol{\Sigma}_y^{-1} \mathbf{K}\right)^{-1}.\end{aligned}$$

The marginal posterior distribution of the transformed hyperparameters $\boldsymbol{\psi}$ can be written up to a normalizing constant as

$$\pi(\boldsymbol{\psi}|\mathbf{y}) \propto \pi(\mathbf{y}|\boldsymbol{\psi})\pi(\boldsymbol{\psi}). \quad (21)$$

Consequently, $\boldsymbol{\Sigma}_\eta$ and $\boldsymbol{\Sigma}_y$ depend now on $\boldsymbol{\psi}$, and we have

$$\boldsymbol{\Sigma}_\eta(\boldsymbol{\psi}) = \begin{pmatrix} \boldsymbol{\Sigma}_\beta & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \exp(2\psi_1)\mathbf{I}_N & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \exp(2\psi_2) \left\{ \exp\left(-\frac{d_{ij}}{\Delta_{S2S}}\right) \right\}_{ij} \end{pmatrix}, \quad (22)$$

$$\begin{aligned} \boldsymbol{\Sigma}_y(\boldsymbol{\psi}) &= \exp(2\psi_4)\mathbf{I}_{NQ} \\ &+ \mathbf{I}_N \otimes \left\{ \exp(2\psi_3) \exp\left(1 + \frac{d_{ij}}{\exp(\psi_5)}\right) \exp\left(-\frac{d_{ij}}{\exp(\psi_5)}\right) \right\}_{ij}. \end{aligned} \quad (23)$$

The mean and covariance of the Gaussian density $\pi(\mathbf{y}|\boldsymbol{\psi})$ are

$$\mathbf{E}(\mathbf{y}) = \mathbf{K}\boldsymbol{\mu}_\eta, \quad (24)$$

$$\text{cov}(\mathbf{y}) = \mathbf{K}\boldsymbol{\Sigma}_\eta\mathbf{K}^\top + \boldsymbol{\Sigma}_y, \quad (25)$$

and thus,

$$\pi(\mathbf{y}|\boldsymbol{\psi}) = \mathcal{N}(\mathbf{y}|\mathbf{K}\boldsymbol{\mu}_\eta, \mathbf{K}\boldsymbol{\Sigma}_\eta\mathbf{K}^\top + \boldsymbol{\Sigma}_y). \quad (26)$$

The marginal posterior density of the hyperparameters can therefore be presented as

$$\begin{aligned} \pi(\boldsymbol{\psi}|\mathbf{y}) &\propto \mathcal{N}(\mathbf{y}|\mathbf{K}\boldsymbol{\mu}_\eta, \mathbf{K}\boldsymbol{\Sigma}_\eta\mathbf{K}^\top + \boldsymbol{\Sigma}_y) \prod_{k=1}^5 \exp(\psi_k|\lambda_k) \\ &\propto \left| \mathbf{K}\boldsymbol{\Sigma}_\eta\mathbf{K}^\top + \boldsymbol{\Sigma}_y \right|^{-\frac{1}{2}} \\ &\times \exp\left(-\frac{1}{2}(\mathbf{y} - \mathbf{K}\boldsymbol{\mu}_\eta)^\top (\mathbf{K}\boldsymbol{\Sigma}_\eta\mathbf{K}^\top + \boldsymbol{\Sigma}_y)^{-1} (\mathbf{y} - \mathbf{K}\boldsymbol{\mu}_\eta)\right) \\ &\times \prod_{k=1}^5 \lambda_k \exp(-\lambda_k \exp(\psi_k) + \psi_k). \end{aligned} \quad (27)$$

Then, the logarithmic transformation of $\pi(\boldsymbol{\psi}|\mathbf{y})$ is

$$\begin{aligned} \log(\pi(\boldsymbol{\psi}|\mathbf{y})) = & C - \frac{1}{2} \log \left| \mathbf{K} \boldsymbol{\Sigma}_\eta \mathbf{K}^\top + \boldsymbol{\Sigma}_y \right| \\ & - \frac{1}{2} \left[(\mathbf{y} - \mathbf{K} \boldsymbol{\mu}_\eta)^\top (\mathbf{K} \boldsymbol{\Sigma}_\eta \mathbf{K}^\top + \boldsymbol{\Sigma}_y)^{-1} (\mathbf{y} - \mathbf{K} \boldsymbol{\mu}_\eta) \right] \\ & + \sum_{k=1}^5 \{ \log(\lambda_k) - \lambda_k \exp(\psi_k) + \psi_k \}. \end{aligned} \quad (28)$$

4.1.4 Posterior Sampling

Markov chain Monte Carlo (MCMC) simulation algorithm with embedded Metropolis steps is used to sample from the posterior density of the proposed BHM through two main steps:

1. The transformed hyperparameters, $\boldsymbol{\psi}$, are sampled jointly from the marginal posterior density $\pi(\boldsymbol{\psi}|\mathbf{y})$ using the Metropolis step of Roberts et al. (1997).
2. The latent parameters, $\boldsymbol{\eta}$, are sampled jointly from the conditional posterior density $\pi(\boldsymbol{\eta}|\mathbf{y}, \boldsymbol{\psi})$.

In step 1 of this algorithm (for the k -th iteration), random samples $\boldsymbol{\psi}^*$ are drawn from a multivariate Gaussian proposal density centered at the previous draw $\boldsymbol{\psi}^{k-1}$ with a covariance matrix $c(-\mathbf{H})^{-1}$, where $\mathbf{H}$ is the Hessian matrix of $\log \pi(\boldsymbol{\psi}|\mathbf{y})$ evaluated at the posterior mode, $\boldsymbol{\psi}_0$, and $c = 2.38^2/\dim(\boldsymbol{\psi})$. The Hessian matrix can be defined as a square matrix of second-order partial derivatives of a scalar-valued function and can be presented here as

$$\mathbf{H} = \nabla^2 \log \pi(\boldsymbol{\psi}|\mathbf{y})|_{\boldsymbol{\psi}=\boldsymbol{\psi}_0}, \quad (29)$$

where ∇^2 is the second derivative operator for a multivariable function. Consequently, the resulting proposal density is

$$q(\boldsymbol{\psi}^*|\boldsymbol{\psi}^{k-1}) = \mathcal{N}(\boldsymbol{\psi}^*|\boldsymbol{\psi}^{k-1}, c(-\mathbf{H})^{-1}). \quad (30)$$

The scaling parameter c can be shown to yield optimal acceptance rates in a particular large dimensional scenario (Roberts et al., 1997). In step 2, samples of $\boldsymbol{\eta}$ are drawn directly from the Gaussian density in (20).

The following steps summarize the sampling of the marginal posterior density of ψ :

1. Initialize the MCMC process with estimates of parameters.
Set $\Delta_{S2S} = 0.06$. Evaluate the mode, ψ_0 , of the marginal posterior density of ψ , and the Hessian matrix, H , at $\psi = \psi_0$. Generate the initial value for ψ from a Gaussian density with mean ψ_0 and covariance $c(-H)^{-1}$.
2. At iteration k , sample a proposal value ψ^* from a Gaussian density with mean ψ^{k-1} and covariance $c(-H)^{-1}$.
3. Calculate

$$r_k = \min \left\{ 1, \frac{\pi(\psi^*|y)}{\pi(\psi^{k-1}|y)} \right\}. \quad (31)$$

4. Sample u_k from a uniform density on $[0, 1]$. Accept or reject the proposed values of (ψ, η) according to

$$(\psi^k, \eta^k) = \begin{cases} (\psi^{k-1}, \eta^{k-1}), & \text{if } r_k \leq u_k, \\ (\psi^*, \eta^*), & \text{if } r_k > u_k, \end{cases} \quad (32)$$

where η^* is the vector of latent parameters at iteration k (sampled only if $r_k > u_k$). The proposal value of η^* can be sampled from the conditional posterior density of $\eta = (\beta, \delta B, \delta S2S)$, conditioned on $\psi = \psi^*$.

5. Repeat steps 2–4 until an adequate amount of posterior samples has been obtained.
6. Calculate posterior summaries for ψ and η using their posterior samples.

4.1.5 Bayesian Convergence Diagnostics

Although the MCMC algorithms largely ensure the convergence of the simulation process to the target density, it is necessary to check the convergence of the simulated sequences once the simulation algorithm has been implemented and the simulated samples have been drawn. In general, fast convergence and low dependence between successive samples lead to higher-quality MCMC chains. There are several diagnostic techniques to evaluate the quality of the MCMC chains. The convergence diagnostics tools normally used for assessing the computational efficiency of MCMC chains are the following.

Trace Plots

A trace plot shows the value of a single parameter in the MCMC chain plotted as a function of the iteration number. Visual inspection of the trace plots makes it

possible to identify if and where the MCMC chain gets stuck at the same value for many consecutive iterations. If the MCMC chain does get stuck, that indicates low computational efficiency.

Gelman–Rubin Statistic ($\hat{R}$)

Gelman and Rubin (1992) proposed a metric for assessing the convergence of iterative MCMC simulations. The $\hat{R}$ is evaluated from the multiple simulated MCMC chains, which have different initial values and have been simulated independently of each other. The algorithm for calculating the $\hat{R}$ is thoroughly outlined in Brooks and Gelman (1998). The $\hat{R}$ can be interpreted as follows. Values close to 1.00 suggest that the MCMC simulations are close to the target distribution. In most practical cases, values below 1.05 are acceptable. However, values above 1.10 are typically said to indicate that the simulations have not converged to the target density. Gelman–Rubin plots can be used as a visual tool for assessing the rate of convergence of the given MCMC chains.

Autocorrelation (AC)

The dependence between successive samples of the Markov chain is evaluated with the autocorrelation, which is estimated with the sample correlation. The j -th lag autocorrelation, ρ_j , is defined as the correlation between every j successive draws. The j -th lag autocorrelation of an MCMC chain $\{\theta_k\}_{k=1}^K$ can be estimated by

$$\hat{\rho}_j = \frac{\sum_{k=1}^{K-j} (\theta_k - \bar{\theta}) (\theta_{k+j} - \bar{\theta})}{\sum_{k=1}^K (\theta_k - \bar{\theta})^2}, \quad (33)$$

where $\bar{\theta} = K^{-1} \sum_{k=1}^K \theta_k$. How the j -th lag autocorrelation decreases as a function of lag j yields insight into the computational efficiency of the MCMC sampler. The j -th lag autocorrelation decreases rapidly if the MCMC algorithm is computationally efficient. However, high j -th lag autocorrelation for relatively high values of j indicates poor computational efficiency. Autocorrelation plots, which show the j -th lag autocorrelation as a function of lag j , are a useful visual diagnostic tool for assessing the behavior of the autocorrelation.

4.1.6 Results

The BHM model was applied for an analysis of the ICEARRAY I PGA data. Histograms of the posterior samples of the station terms for PGA recorded by ICEARRAY I stations are illustrated in Fig. 2. The figure shows that seismic motions were recorded over a relatively small area (i.e., aperture ~ 1.9 km) compared to the

source–station distance (i.e., hypocentral distance 1.8–17.8 km). Nonetheless, the station terms vary considerably across the uniform site condition within Hveragerði. Trace plots of the MCMC samples from the marginal posterior densities of the station terms are presented in Fig. 4, along with the corresponding Gelman–Rubin and autocorrelation plots.

The elements of $\delta S2S$ meet the convergence diagnostics criteria, i.e., the Gelman–Rubin diagnostic falls below <1.1 , and the degree of autocorrelation is less than 0.10 at a lag of 50. Furthermore, their marginal posterior densities are well-determined, nearly symmetric, and unimodal, with similar standard deviations but different means. The same is true for other model parameters, i.e., they are well-determined, unimodal, and nearly symmetric. All the hyperparameters meet the convergence diagnostics criteria. In addition, the overall acceptance rate for model parameters is approximately 42%, which is within the recommended range.

Contrary to the classical regression analysis designed to separate the observed ground-motion parameters and determine point estimates of the unknown model parameters, the BHM presents the unknown model parameters in terms of the posterior probability density. This is highly important and practical as the posterior distribution describes the uncertainty of the model parameters given the observed data. Figures 3 and 4 strongly indicate that the station terms cannot be considered equal since the marginal posterior densities of the stations with the smallest and largest posterior means have their masses almost entirely below zero and above zero, respectively.

For comparison, the posterior distribution of the station terms for longer periods (i.e., $T = 0.1$ – 3.0 s) is presented in Fig. 5. The station terms are well constrained, as is evident from their narrow posterior intervals that are associated with small posterior standard deviations. However, there is a noticeable increase in the individual posterior widths (i.e., station term's posterior standard deviation) and their ensemble variations (i.e., their medians vary greatly) in the period 0.1–0.3 s, where the posterior standard deviations are around twice as large as in the other periods. This increase in the posterior variability of the individual station terms thus translates to an increase in the station-to-station standard deviation, ϕ_{S2S} . This observation is peculiar; therefore, further analysis of the residual behavior of the station terms requires a physics-based interpretation. According to logs of shallow boreholes in Hveragerði, most of the town is located on a young and competent lava-rock layer, which is underlain by a layer of softer sedimentary deposits (Rahpeyma et al., 2016, 2017). Namely, station IS609 is located on older bedrock, while the remaining stations are located on a post-glacial lava-rock layer that lies on top of a sedimentary layer. A geologic profile of this sort is a typical layering encountered primarily in geologically younger parts of Iceland in the vicinity of the volcanic zones (Rahpeyma et al., 2021). In Hveragerði, the uppermost lava layer is relatively young basaltic lava of varying thickness (ranging typically from 10 to 30 meters based on borehole data). The sedimentary layer (similar in thickness variability) is then found to lie on top of an older lava layer. No information about the thickness of the lower lava layer exists; however, it has a similar origin and flowed from the same volcanic fissure, and based on the spatial extent of the lava, it is likely

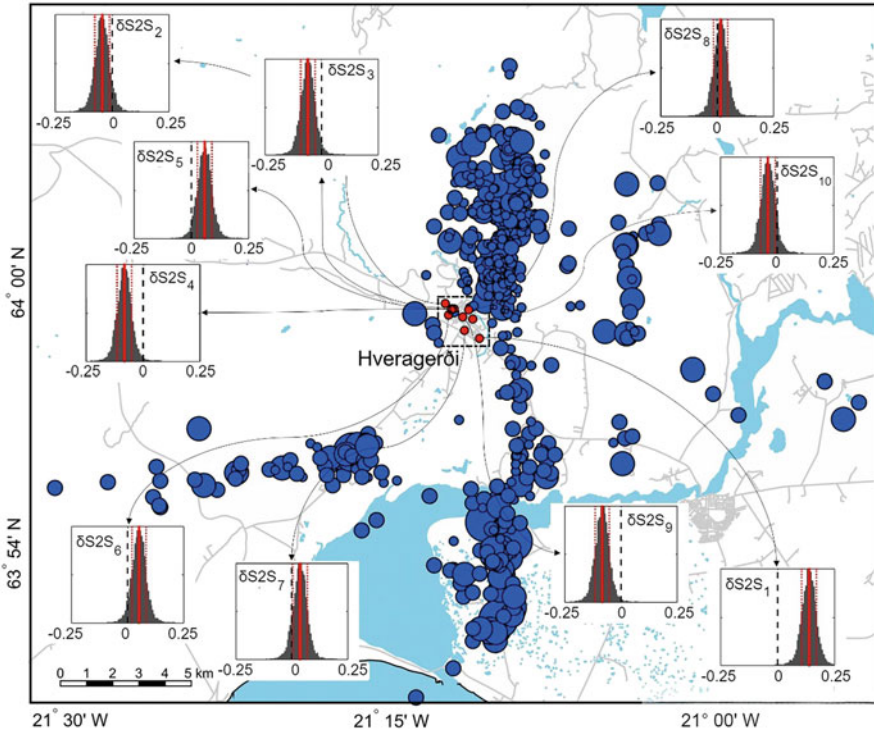


Fig. 3 The marginal posterior densities of the station terms for ICEARRAY I (estimated with histograms) using aftershock recordings of the M_w 6.3 Ólfus earthquake in the SISZ in 2008. The relative differences in the mean values of the station terms (the dashed black line is the average) show that site effects in Hveragerði differ significantly over short distances

of similar thickness as the top lava layer. Therefore, the soil structure underneath Hveragerði appears to be characterized by a dense surficial lava-rock layer, on top of a looser sedimentary layer that, in turn, lies on top of an older dense lava layer. Therefore, there is reversal in the density and the shear wave velocity of the materials underneath the town. Such reversals lead to the resonance of horizontal ground shaking at the surface of the soil structure when subjected to the propagating seismic waves.

It is of note in this context that Rahpeyma et al. (2016) implemented the horizontal-to-vertical spectral ratio (HVSr) method (Nakamura, 1989) to reveal the presence of resonance frequencies in the seismic ground motions of the same dataset. Figure 6 shows the HVSr over the frequency range of interest to this study (0.5–20 Hz), along with the geometric mean HVSr and the associated $\pm 1\sigma$ (Rahpeyma et al., 2016). The results clearly show the presence of resonance frequencies and that they are evident to different degrees at the stations. Some stations exhibit bimodal amplification curves (one between 3–4 Hz and the other at 8–9 Hz), with one mode being more dominant and of relatively larger amplitude than

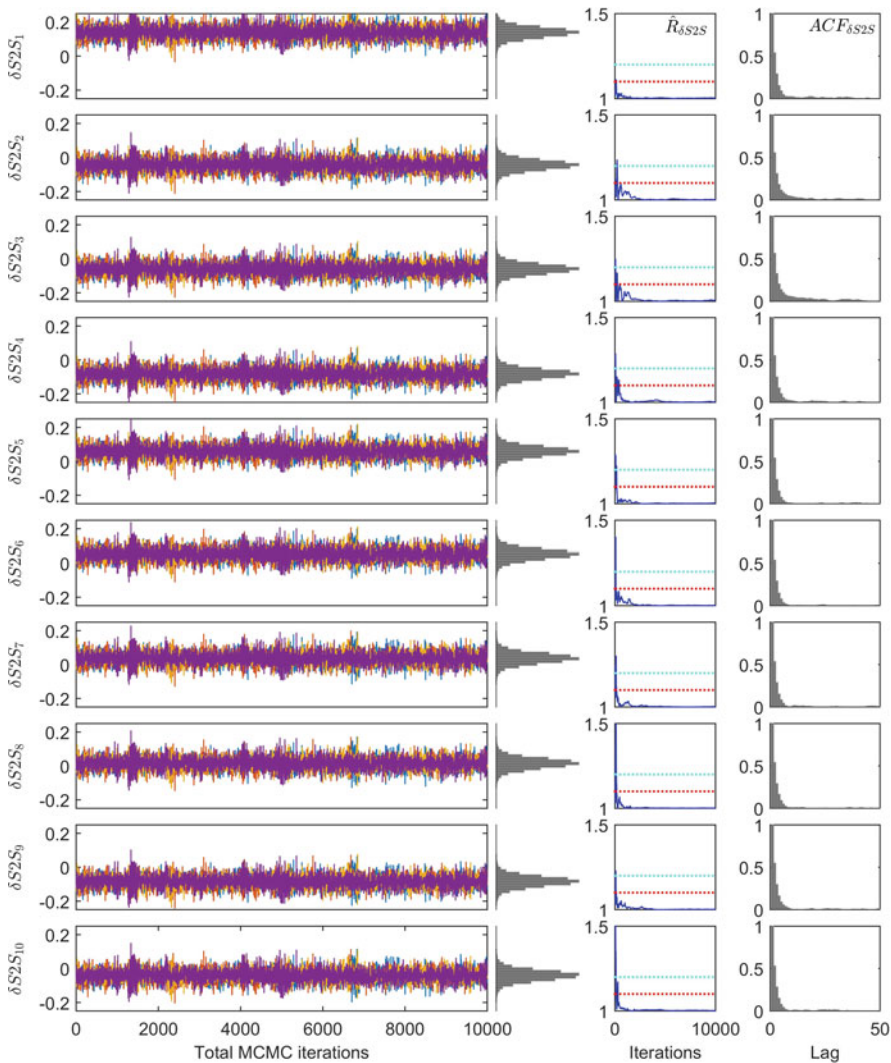


Fig. 4 Trace plots of the posterior samples from the MCMC simulations (and the corresponding histograms) of the station terms, $\delta S2S$, for PGA. The convergence diagnostics, e.g., the Gelman–Rubin convergence statistics ($\hat{R}_{\delta S2S}$) and the autocorrelation function plots ($ACF_{\delta S2S}$), are also shown

the other (e.g., IS605, IS604, and IS608). Other stations have a single narrow-band peak of relatively low amplitudes (e.g., IS602) or amplification curves of relatively high amplification over a wide frequency range (e.g., IS601 and IS603), or even very low and uniform amplification curves across the frequency range (IS609, IS611, and IS612). Therefore, these results reveal the considerable variability in the site response to seismic ground excitation across the town. Such differences would

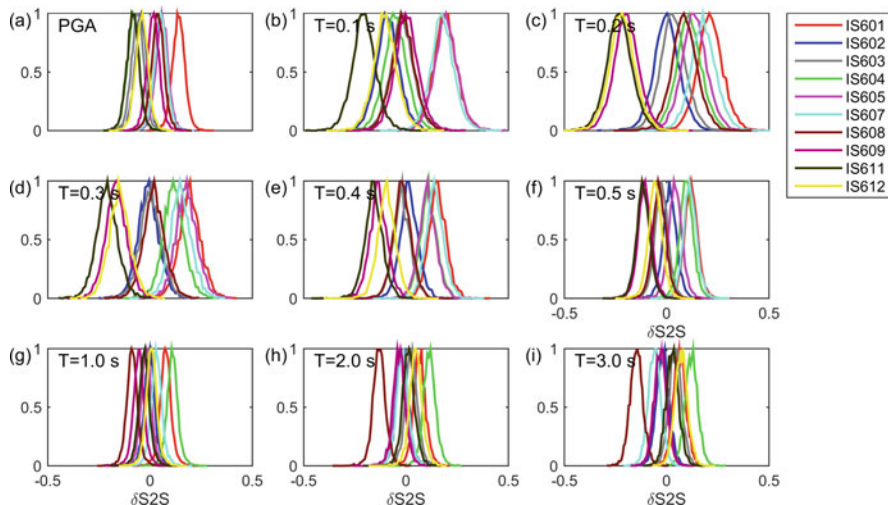


Fig. 5 Posterior distribution of station terms, $\delta S2S$, for ICEARRAY I at different periods (a) PGA, (b) $T = 0.1$ s, (c) $T = 0.2$ s, (d) $T = 0.3$ s, (e) $T = 0.4$ s, (f) $T = 0.5$ s, (g) $T = 1.0$ s, (h) $T = 2.0$ s, and (i) $T = 3.0$ s

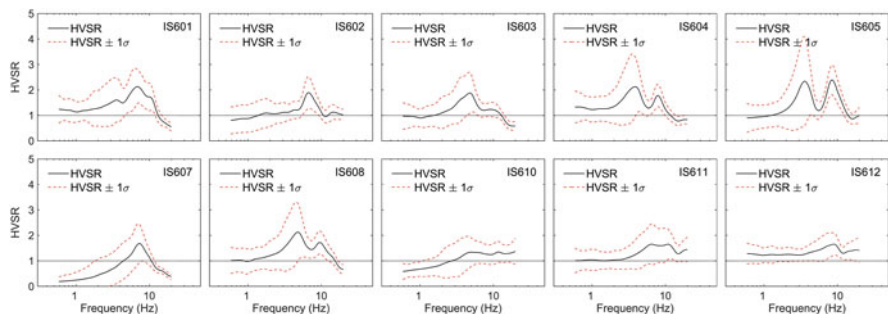


Fig. 6 Mean HVSR (solid black line) and mean $\pm 1\sigma$ HVSR (dashed red lines) from earthquake data from the ICEARRAY I strong-motion stations located on lava rock in Hveragerði, South Iceland

manifest as increased station terms variability around those frequencies. Comparing Figs. 5 and 6, it is clear that the site-specific characteristics shown in Fig. 6 play a significant role in the resonant frequency bandwidth of the region's subsoil structure of the region. The inter-station variability near the resonance frequencies shows larger values and associated uncertainties due to the resonance phenomenon that are observed in some stations, such as IS604, IS605, and IS608. Therefore, constructing a GMM in the Bayesian hierarchical framework, in particular, the specification of the residuals into different terms (most notably the station terms), allows us to fully quantify the relative frequency-dependent site amplification of the ICEARRAY I stations and attain posterior information at each discrete frequency.

The localized site amplification is one of the three key elements required for the specification of the seismic hazard. Its specification at the local level in a man-made environment can make the hazard assessment more accurate, reliable, and useful, e.g., for urban planning.

4.1.7 Supplementary MATLAB Code

A MATLAB code with synthetic data is provided as supplementary material for this section, see <https://github.com/hjordislb/BHM2020>.

4.2 *Bayesian Inference of Empirical GMMs Based on Informative Priors*

Empirical GMMs are calibrated by fitting equations of certain functional forms to ground-motion data using various statistical techniques. The more plentiful the dataset is, the better constrained the GMM predictions are along with their overall uncertainties for the dataset in question. The level of model uncertainty (as well as model bias, if any) allows us to evaluate the confidence that we have in the predicted ground-motion values.

The quality and quantity of recorded data have increased dramatically over the last couple of decades, facilitating the GMM calibration along with various regression methods (Douglas, 2018). However, datasets are generally unbalanced in terms of earthquake magnitudes, since larger earthquakes are much less common than smaller ones. As a result, data availability is much lower at larger magnitudes. The same can be said for data availability as a function of distance from the earthquake source, as near-fault recordings (closer than ~ 20 km from the earthquake fault of moderate to strong earthquakes) are generally much rarer than those in the far-field region. In addition, in regions where data are nonexistent (e.g., at larger magnitudes or in the near-fault region), the GMM parameters that control the scaling of ground motion with earthquake magnitude are essentially unconstrained. This is the case in the seismic region of Southwest Iceland where the largest recorded earthquake magnitude is $M_w 6.5$, but earthquakes of ~ 7 are known to happen (Einarsson, 2014). Thus, the scaling of earthquake ground motion is essentially unconstrained at larger magnitudes along with their magnitude–distance-dependent saturation of strong motions from large earthquakes, regardless of what functional form of the GMM is employed. This is a critical issue, as larger earthquakes produce larger ground shaking that can cause greater damage. Further compounding the issue, GMMs developed from more plentiful datasets from other seismic regions in the world cannot be used as they have been shown to be strongly biased against Icelandic strong-motion data (Ólafsson & Sigbjörnsson, 2002, 2006; Kowsari et al., 2019b, 2020). Namely, if the model residuals, which are a measure of how the GMM

predictions differ from the data, show a systematic deviation from a zero-mean and symmetric distribution, then the model is considered to be biased and, as a result, associated with very low confidence in the validity of its predictions. This was the case for well-known GMMs developed for other seismic regions but had been used or recommended for seismic hazard studies in Iceland. To address this issue, Kowsari et al. (2019b), Kowsari et al. (2020) used Bayesian inference to recalibrate several of these GMMs as they had functional forms that satisfied the criteria of such GMMs for use in hazard assessment; see Cotton et al. (2006), Bommer et al. (2010). They proposed a random-effect statistical model that partitions the model variability into inter-event and intra-event components, and the GMM parametric inference is carried out in the Bayesian framework using a Markov Chain Monte Carlo algorithm.

In this section, the recalibration framework is presented, focusing on the necessary selection of informative priors for those key parameters that would essentially be unconstrained by the Icelandic dataset.

4.2.1 Bayesian Random Effects

For GMMs, the random-effect regression presented by Brillinger and Preisler (1985, 1984) was first used to separate the inter-event and the intra-event components of ground-motion variability. This one-stage maximum-likelihood approach was then improved by Abrahamson and Youngs (1992) through a computational algorithm, considering that, in the case of a nonlinear model, the regression coefficients could be estimated incorrectly if the coefficients in the nonlinear regression analysis model did not start with good initial estimates. A typical GMM with the partitioned error into inter-event and intra-event terms can take the following form:

$$\begin{aligned} \log(y_{ij}) &= f(M_i, r_{ij}, \boldsymbol{\theta}) + \eta_i + \varepsilon_{ij}, \\ \eta_i &\sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \phi^2), \end{aligned} \quad (34)$$

where y_{ij} is a ground-motion intensity measure (here, PGA and PSA at different periods) for the j -th recording from the i -th event; M , r , and $\boldsymbol{\theta}$ are magnitude, distance, and a vector of model parameters, respectively; and f is a nonlinear function of these variables and parameters. The η_i and ε_{ij} are the inter-event and intra-event variations, respectively, which are assumed to follow mean-zero Gaussian distributions with standard deviations τ and ϕ , respectively.

Establishing earthquake GMMs for Southwest Iceland is challenging due to a relatively small strong-motion dataset and a lack of data from earthquakes greater than $M_w 6.5$. The parametric description of the dataset is shown in Fig. 7, with data recorded on rock accounting for the majority of the 83 records from six strike-slip earthquakes (mainshocks) that occurred in the SISZ between 1987 and 2008.

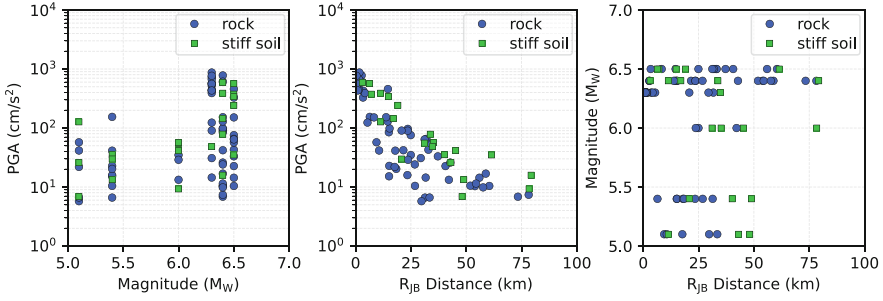


Fig. 7 The parametric characterization of the Icelandic strong-motion dataset in terms of recorded peak ground acceleration (PGA) values versus the independent variables magnitude (left), distance (middle), and magnitude–distance (right). The site conditions are indicated by circles (rock) and squares (stiff soil)

It has been known for a long time that the earthquake strong-motion attenuation characteristics in Iceland are different from those found in other regions of Europe. Consequently, the selected GMMs appear to underestimate the Icelandic ground motions in the near-fault region and overestimate them in the far field (Ólafsson & Sigbjörnsson, 2002, 2004; Ornthammarath et al., 2011). One of the GMMs that shows a strong bias against the Icelandic dataset is the model proposed by Akkar and Bommer (2010, AB10), which can be written as

$$\log(y_{ij}) = C_1 + C_2 M + C_3 M^2 + (C_4 + C_5 M) \log \sqrt{C_6^2 + r^2} + C_7 S + \eta_i + \varepsilon_{ij}, \quad (35)$$

where C_{1-7} are the regression coefficients, with the last coefficient added to model site conditions (S is a flag that takes the value of 0 for rock sites and 1 otherwise). As the functional form shows, it is nonlinear in terms of magnitude scaling and has a magnitude–distance-dependent term. As a result, the model is flexible, but due to the limitations of the dataset, some parameters will be more unconstrained than others in the recalibration of the model due to the lack of data above $M_W 6.5$.

As presented in (5), the posterior distribution is proportional to the product of the prior density of the parameters, $\pi(\theta)$, and the likelihood function, $\pi(\mathbf{y}|\theta)$. An important feature of the Bayesian method is that it allows for prior knowledge of model parameters to be included in the form of an informative prior probability distribution, which, when carefully chosen, improves the inference for the model parameters. We assume that the prior distribution density of the model parameters, $\pi(\theta)$, is a product of uniform densities, i.e., a priori, each model parameter is assumed to be uniform on the interval from $\theta_{i_{\min}}$ to $\theta_{i_{\max}}$, or

$$\pi(\theta_i) = \begin{cases} (\theta_{i_{\max}} - \theta_{i_{\min}})^{-1}, & \text{for } \theta_{i_{\min}} \leq \theta_i \leq \theta_{i_{\max}}, \\ 0, & \text{otherwise,} \end{cases} \quad (36)$$

where $\pi(\boldsymbol{\theta}) = \prod_{i=1}^K \pi(\theta_i)$, and K is the number of parameters in $\boldsymbol{\theta}$. On the other hand, the logarithmic ground-motion residual is generally assumed and observed to follow a Gaussian distribution that has proven to be a reasonable assumption (Kowsari et al., 2019b; Brooks & Gelman, 1998). Therefore, the log-likelihood of the observation is given by

$$\begin{aligned} \log(\pi(\mathbf{y}|\boldsymbol{\theta})) = & -\frac{N}{2} \log(2\pi) - \frac{1}{2} \log |\mathbf{C}| \\ & - \frac{1}{2} (\mathbf{y}^{\text{obs}} - \mathbf{y}^{\text{sim}})^{\top} \mathbf{C}^{-1} (\mathbf{y}^{\text{obs}} - \mathbf{y}^{\text{sim}}), \end{aligned} \quad (37)$$

where $\mathbf{y}^{\text{obs}}$ is a vector containing the observations, $\log(y_{ij})$, and $\mathbf{y}^{\text{sim}}$ is a vector containing the corresponding elements of the model, $f(M_i, r_{ij}, \boldsymbol{\theta})$. Due to the random-effect model in (34), the data covariance matrix $\mathbf{C}$ in the likelihood function in (37) is not a diagonal matrix with ϕ^2 on the diagonal, but a block-diagonal matrix composed as

$$\mathbf{C} = \phi^2 \mathbf{I}_N + \tau^2 \oplus_{i=1}^M \mathbf{J}_{n_i}, \quad (38)$$

where $\oplus_{i=1}^M$ is the matrix direct sum (creating a block-diagonal matrix) of square matrices of ones, each matrix denoted by $\mathbf{J}_{n_i}$ whose size, n_i , corresponds to the number of records for event i ; thus, $\mathbf{C}$ is presented as

$$\mathbf{C} = \begin{pmatrix} \phi^2 \mathbf{I}_{n_1} + \tau^2 \mathbf{J}_{n_1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \phi^2 \mathbf{I}_{n_2} + \tau^2 \mathbf{J}_{n_2} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \dots & \phi^2 \mathbf{I}_{n_M} + \tau^2 \mathbf{J}_{n_M} \end{pmatrix}. \quad (39)$$

4.2.2 Results

The Bayesian random-effect model has been applied to recalibrate various different GMMs to the Icelandic strong-motion dataset, including the AB10 model (Kowsari et al., 2020). However, as previously mentioned, some parameters (e.g., parameters for magnitude-dependent scaling) cannot be sufficiently constrained for regions with limited data at higher magnitudes. One of the advantages of the Bayesian approach is that it allows prior information on the GMM parameters to be combined with the likelihood of the observed data. However, defining the prior distribution in Bayesian analysis is a challenging task. Typically, if the data alone hold enough information about a variable, a noninformative prior is used that exerts minimal influence on the inference. Otherwise, informative priors (i.e., known or suspected information on a given parameter) can be used, which influences the posterior inference and increases its precision. In other words, using an informative prior density that incorporates experience and judgment into the model can be especially effective for regions

with a small amount of data (Wang & Takada, 2009). Therefore, the inference using noninformative vs. informative priors for the recalibration of the GMMs is contrasted in what follows. For more details on the functional form and the posterior distribution of the model parameters of the selected GMMs, see Kowsari et al. (2020).

The MCMC algorithm forms the backbone of modern Bayesian posterior inference and is used to simulate samples from the posterior density of the GMM parameters, often using several chains of several thousand simulated samples each. To better represent the final stationary distribution and mostly remove the starting values' influence, a certain number of the first iterations are discarded, known as the burn-in. A primary concern in applying such a method is determining that it has in fact converged, i.e., after an initial burn-in period. Therefore, the Gelman–Rubin statistic (Gelman & Rubin, 1992) is used to assess the convergence of iterative MCMC simulations. Good convergence of successive simulations is confirmed by the Gelman–Rubin statistic in all the recalibrated GMMs (see Kowsari et al., 2019a, 2020).

The recalibrated GMMs are shown in Fig. 8, where noninformative priors are used for five GMMs of different functional forms (Y1–Y5, with Y1 being the AB10). As can be seen, near-fault and far-field predictions are very well constrained to fit the Icelandic data where data are available (i.e., $M_w \leq 6.5$). For larger magnitudes ($M_w > 6.5$), where there is a lack of data, the recalibrated GMMs are associated with poorly constrained coefficients and, therefore, with a large uncertainty. Thus, the noninformative inference provides unconstrained models for ground-motion predictions outside the data range. As a result, there is low confidence in using such noninformative GMMs at larger magnitudes, particularly in the near-fault region. In the GMM given by (35), the parameters C_3 and C_5 control the higher-order magnitude scaling and magnitude–distance-dependent scaling, respectively. These are the parameters that are essentially unconstrained due to the lack of data at larger magnitudes. However, the original calibration of AB10 to a much larger European dataset that included multiple earthquakes larger than the maximum magnitude in the Icelandic dataset did not suffer from such limitations. Therefore, these two parameters essentially embody the salient characteristics of earthquake ground-motion scaling at larger magnitudes and its magnitude–distance-dependent characteristics. Namely, the seismic ground motions from shallow crustal interplate earthquakes are found to saturate at larger magnitudes, primarily in the near-fault region but also in the far field. In other words, while ground-motion scaling is self-similar at lower magnitudes, it becomes non-self-similar at larger magnitudes (Abrahamson & Shedlock, 1997; Halldórsson & Papageorgiou, 2005; Abrahamson et al., 2008; Archuleta & Crempien, 2015; Dalguer, 2015). Properly accounting for such scaling is critical in the ground-motion prediction for seismic hazard purposes. However, Bayes' theorem allows us to formally incorporate prior information to influence the inference of model parameters from incomplete datasets. Thus, by generating a prior distribution for each unconstrained parameter (e.g., C_3 and C_5 in AB10) based on its previously estimated median value and variation from other datasets, we can allow the inference from the Icelandic dataset

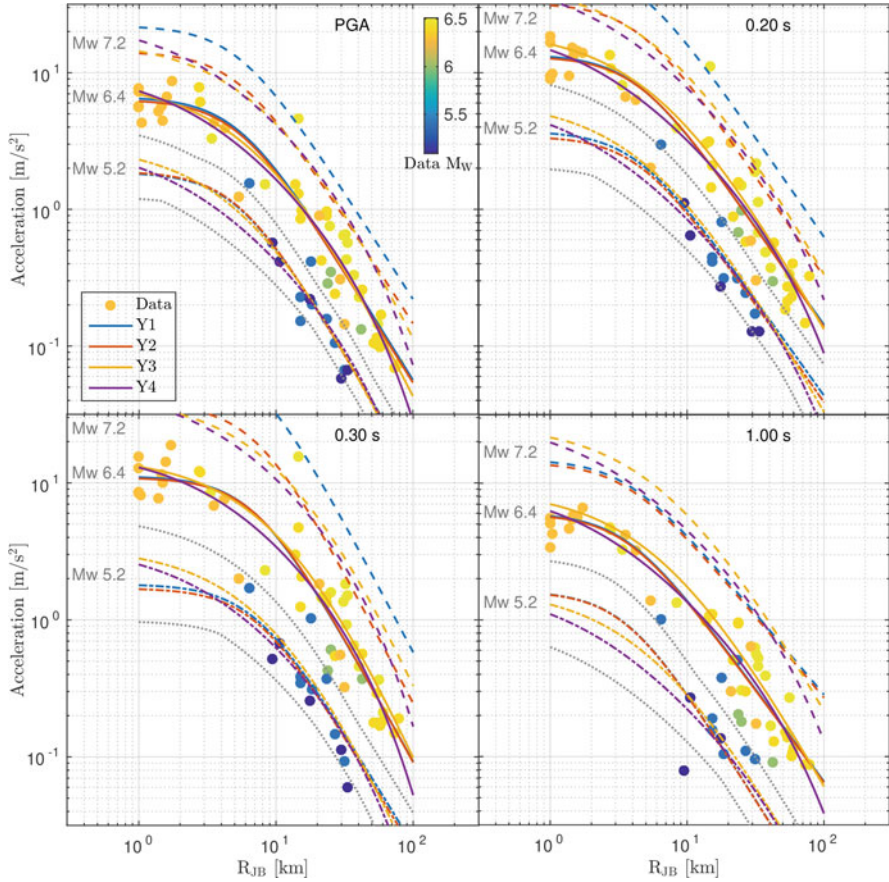


Fig. 8 The distance attenuation of PGA and PSA at various periods (on rock) from the recalibrated ground-motion models when noninformative priors are used for the model parameters. The data points are shown as color-coded circles by magnitude. The dotted lines represent the range of model uncertainties ($\pm 1\sigma$) around the median predictions for $M_w 5.2$. Distance R_{JB} is given after (Joyner & Boore, 1981), which is the shortest distance to the surface projection of the fault plane

to be influenced by this information. In other words, prior to the inference, we can propose a range of values that these two parameters are expected to take. This does not directly affect the other parameters, and they will be more strongly influenced by the characteristics of the Icelandic data. Figure 9 shows the recalibrated GMM predictions based on such informative priors. The plot shows the predictions for M_w 5.2 (dash-dotted line), 6.4 (solid line), and 7.2 (dashed line) for PGA and PSA at periods $T = 0.2, 0.3$, and 1.0 s. The M_w 6.4 is the weighted average magnitude of the three largest earthquakes that account for the majority of the data (i.e., the two 2000 and the 2008 earthquakes with M_w of 6.5, 6.4, and 6.3, respectively). The legend indicates which coefficients in each GMM were subjected to informative

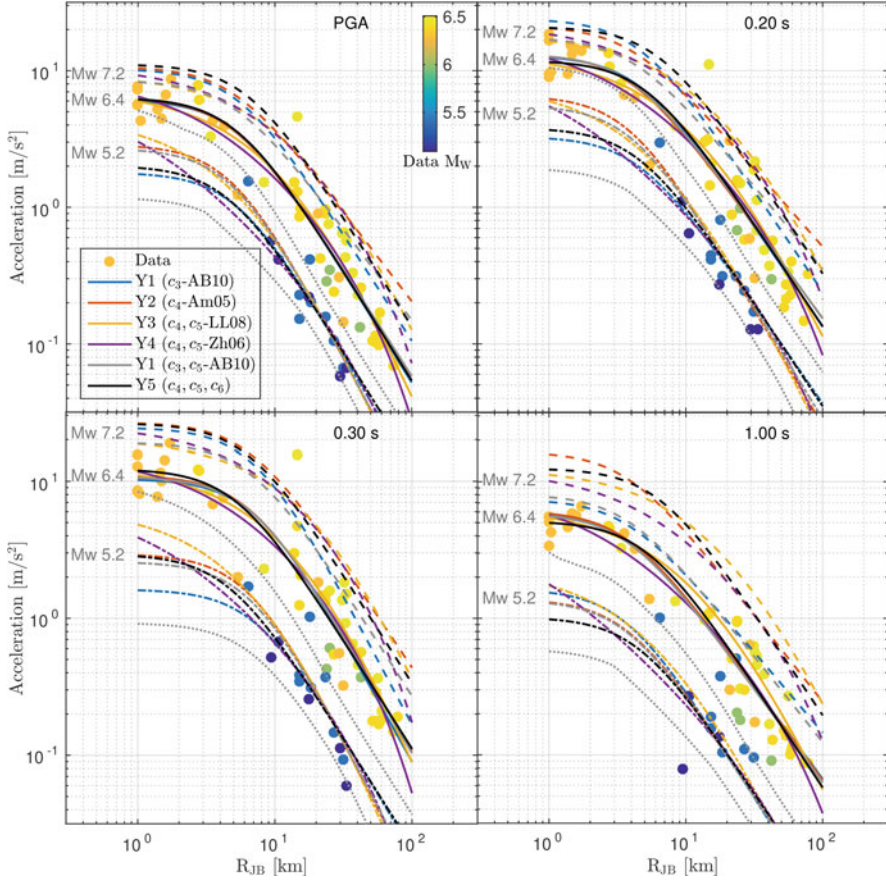


Fig. 9 The attenuation of the recalibrated ground-motion models when an informative prior density is used for PGA and PSA at various periods on rock. The observed data are shown as color-coded diamonds by magnitude. The dotted lines represent the range of model uncertainties ($\pm 1\sigma$) around the median predictions for $M_w 5.2$. The model parameters that are constrained with the help of informative priors are shown for each recalibrated GMM in the legend

priors. For example, two versions of AB10 are shown, first with an informative prior on only one of the unconstrained parameters (C_3 -AB10) and then on both (C_3, C_5 -AB10). Just from a visual comparison of the results presented in Figs. 8 and 9, the effects of the informative priors are now clearly seen on the predicted ground-motion amplitudes that the Icelandic dataset cannot constrain. Namely, the GMMs are seen to be equally well constrained in the near-fault and far-field amplitudes in our data range; but more importantly, all the recalibrated models now are better constrained at large earthquake magnitudes (i.e., they exhibit the desired non-self-similar ground-motion scaling). In other words, the GMMs now predict much more realistic values than when using noninformative priors, which are not only consistent

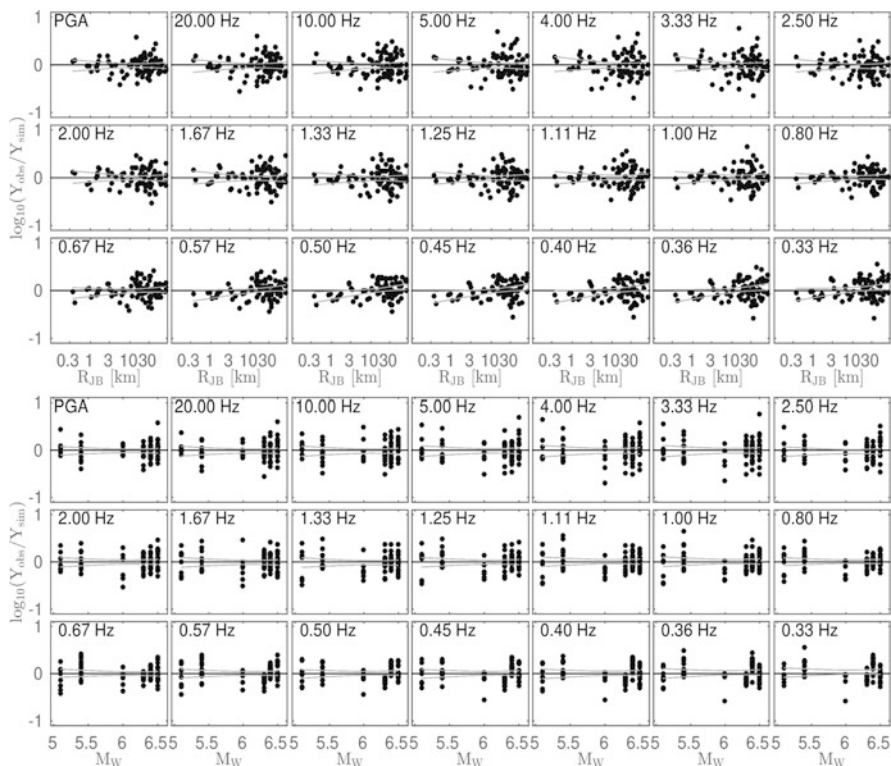


Fig. 10 PSA residuals of Y1 (C_3 , C_5 -AB10) using median model parameter estimates along with the 95% confidence intervals for a least-squares line fit versus the logarithm of distance (top) and magnitude (bottom)

with the behavior of all GMMs for shallow crustal earthquakes in interplate regions where larger-magnitude data are plentiful but also consistent with earthquake-source physics and numerical modeling of near-fault ground motions.

As a measure of the quality of fit of the GMMs to the data, the residuals of model Y1 (C_3 , C_5 -AB10) at 21 discrete oscillator frequencies from 0.33 to 20 Hz and PGA are presented in Fig. 10, plotted versus log-distance and magnitude, respectively. Their distribution shows that there are no significant residual outliers in this dataset. Moreover, there appear to be no clear systematic trends in the overall distribution of residuals, as indicated by the 95% confidence intervals for the individual fitted regression lines, all being effectively horizontal. For the sake of space, only Y1 (C_3 , C_5 -AB10) is shown as an example; however, the same behavior is also seen for the other GMMs with informative priors.

In this section, we have shown how the Bayesian statistical methodology combined the salient characteristics of the Icelandic data with prior parametric information from the original GMMs based on richer datasets. This addresses the two major difficulties of modeling ground motions with the Icelandic data. First,

the problem of local GMMs being fitted to the limited Icelandic dataset and having unacceptable functional forms for seismic hazard assessment. Second, the problem of the GMMs having acceptable functional forms developed from data in other seismic regions that are both strongly biased against the Iceland dataset and have parameters that the Icelandic dataset cannot constrain. The ability of the Bayesian framework to allow a formal way of introducing prior parametric distributions is an elegant way of letting our knowledge influence a robust statistical inference. As a result, the recalibrated GMMs are unbiased with respect to the Icelandic data, exhibit realistic predictions at larger magnitudes due to the flexibility of their functional forms combined with informative priors for key parameters, and satisfy the conditions on GMMs' functional forms for use in hazard assessment. They can therefore be used with confidence in future earthquake hazard assessments in Iceland.

4.2.3 Supplementary MATLAB Code

A MATLAB code with synthetic data is provided as a supplementary material for this section, see <https://github.com/tsonne/bayes-empirical-gmm>.

4.3 *Ground-Motion Model Selection Using the Deviance Information Criterion*

In the previous section, multiple GMMs of various functional forms were recalibrated to the same dataset. As the results show, their ground-motion predictions tend to converge in the range where most data are available and diverge elsewhere. All the models are unbiased and have essentially the same total uncertainty. Then, the question becomes which model is most appropriate to use? Selecting the appropriate GMM is challenging, particularly for regions where indigenous GMMs do not exist (Delavaud et al., 2012). To overcome this problem, data-driven methods can be used to select the best GMM for application in probabilistic seismic hazard analysis (PSHA), reducing subjectivity by quantitatively guiding the selection process. To that end, Scherbaum et al. (2004) proposed a probability-based approach known as the likelihood (LH) method that calculates the normalized residuals for a set of observed and estimated ground-motion data. A few years later, Scherbaum et al. (2009) suggested an information-theoretic approach called the log-likelihood (LLH) method that overcomes several shortcomings of the LH method. The LLH method is less sensitive to the sample size and does not require any ad hoc assumptions regarding classification boundaries (Delavaud et al., 2009). These likelihood-based approaches inspired (Kale & Akkar, 2013) to propose the Euclidean distance-based ranking (EDR) method, which uses the Euclidean distance to account for both variability in ground motions and the trend between the observed and predicted

data. Stewart et al. (2015) presented a GMM selection procedure consisting of expert reviews of several information sources, including the evaluation of multi-dimensional ground-motion trends, functional forms, and a review of published quantitative tests for GMM performance against independent data. Furthermore, Mak et al. (2017) represented the effects of data correlation and score variability on the evaluation of GMMs.

Recently, Kowsari et al. (2019a) proposed a data-driven method using the deviance information criterion (DIC) of Spiegelhalter et al. (2002), a criterion for Bayesian statistical models, to select the most suitable earthquake GMM for application in PSHA. The DIC was used within Bayesian statistical models and was adjusted for posterior inference based on MCMC algorithms. The standard deviation of the GMM (known as sigma) is an important parameter for seismic hazard assessment and plays an essential role in data-driven selection methods. The main advantage of the procedure proposed by Kowsari et al. (2019a) was to introduce the posterior sigma as the key quantity for objectively ranking different candidate models against a given ground-motion dataset. The following subsections describe the deviance information criterion and its application to GMMs. Then, we apply the DIC method to rank several candidate GMMs for Iceland and compare the results to other ranking methods.

4.3.1 Deviance Information Criterion

The DIC is one of the model selection methods that are particularly useful if the posterior distributions of the models are obtained by MCMC simulations. The Kullback–Leibler divergence measures the difference between two probability distributions of the same variable (Kullback & Leibler, 1951; Kullback, 1997). The deviance $D(\mathbf{y}, \boldsymbol{\theta})$ has an important role in statistical model comparison because of its connection to the Kullback–Leibler information measure. It is defined as

$$D(\mathbf{y}, \boldsymbol{\theta}) = -2 \log p(\mathbf{y}|\boldsymbol{\theta}), \quad (40)$$

where p is the probability density function of the observed data $\mathbf{y}$ given the unknown parameters $\boldsymbol{\theta}$. In the limit of large sample sizes, the model with the lowest expected deviance (i.e., the lowest Kullback–Leibler information) will have the highest posterior probability. Therefore, it seems reasonable to estimate the expected deviance as a measure of overall model fit (Gelman et al., 2013). A summary of $D(\mathbf{y}, \boldsymbol{\theta})$ that relies on a single estimated value for $\boldsymbol{\theta}$ is given by

$$D_{\hat{\boldsymbol{\theta}}}(\mathbf{y}) = D(\mathbf{y}, \hat{\boldsymbol{\theta}}(\mathbf{y})), \quad (41)$$

where $\hat{\boldsymbol{\theta}}$ is a point estimate for $\boldsymbol{\theta}$, usually the posterior mean. Another summary is the posterior mean of $D(\mathbf{y}, \boldsymbol{\theta})$, that is,

$$D_{\text{avg}}(\mathbf{y}) = E \{D(\mathbf{y}, \boldsymbol{\theta}) | \mathbf{y}\}, \quad (42)$$

which can be estimated with

$$\hat{D}_{\text{avg}}(\mathbf{y}) = \frac{1}{L} \sum_{l=1}^L D(\mathbf{y}, \boldsymbol{\theta}^l), \quad (43)$$

where $\boldsymbol{\theta}^l$ is the l -th draw from the posterior density of $\boldsymbol{\theta}$, $p(\boldsymbol{\theta} | \mathbf{y})$. The estimated average discrepancy in (43) is a better summary of the model parameter error than the discrepancy based on the point estimate since it averages over the range of possible parameter values (Gelman et al., 2013). Moreover, the difference $p_D = \hat{D}_{\text{avg}}(\mathbf{y}) - D_{\hat{\theta}}(\mathbf{y})$ has been used as a measure of the effective number of parameters and the complexity of the model. In a normal linear model with no constraints on the parameters, p_D is equal to the number of parameters. The deviance and the effective number of parameters can be used to measure the model's predictive accuracy, that is, how well the model performs in terms of out-of-sample predictions. For this purpose, Spiegelhalter et al. (2002) proposed the deviance information criterion, DIC, given by

$$\text{DIC} = 2\hat{D}_{\text{avg}}(\mathbf{y}) - D_{\hat{\theta}}(\mathbf{y}) = \hat{D}_{\text{avg}}(\mathbf{y}) + p_D = D_{\hat{\theta}}(\mathbf{y}) + 2p_D. \quad (44)$$

4.3.2 Application to GMM Selection for Southwest Iceland

The standard deviation (sigma) of GMMs represents the variability of ground motions in the region of origin. In some regions, due to a lack of data, developing a GMM exclusively from local earthquake strong motions is challenging as the GMMs are generally associated with large uncertainties. In such cases, GMMs based on worldwide data or from other regions with similar tectonic or geological settings are usually used for seismic hazard assessment if they are found to be relatively unbiased (contrary to those discussed in the previous section). In addition, the selected GMMs must adequately capture the variability of the ground motions in the region under study. In other words, using GMMs from other regions is effectively equivalent to applying the aleatory uncertainty from other regions to the region under study, thereby potentially introducing unrealistic uncertainties into the hazard assessment (Kowsari et al., 2019b,a). We argue that it is physically more reasonable to assume that the standard deviation is unknown. Then the posterior mean of the standard deviation will be determined by the ground-motion observations of the region where the GMM model selection is carried out. For the model comparison below, we will consider both cases; i.e., each model will have one version where the standard deviation is assumed known and another where it is assumed unknown, their DIC values denoted by DIC^1 and DIC^2 , respectively. The standard deviation in the former case is previously determined from data in other regions (hereafter referred to as prior sigma) and adopted for use in the region of

interest, while sigma in the latter case (hereafter referred to as posterior sigma) is effectively obtained through the misfit between the predicted ground motions (i.e., the median) and the local ground-motion observations of the region. These two models are applied in practice; thus, comparing them is relevant. Given that a sufficient amount of data is available, we expected that the model based on the regional data would fit better. However, it is useful to see the scale of the difference between the regional and adopted models, and DIC is well suited for measuring this difference.

Assuming that the logarithm of the ground-motion parameter in question follows a Gaussian distribution, then

$$p(y|\sigma^2, \boldsymbol{\beta}) = \prod_{i=1}^N \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(y_i - \mu(\boldsymbol{\beta})_i)^2}{2\sigma^2}\right), \quad (45)$$

where N is the number of observations, $\mathbf{y}$ is the vector of the logarithm of the observed ground motions, $\mu(\boldsymbol{\beta})$ is the mean value predicted by the GMM, and σ is the standard deviation of the GMM.

In the first case of the adopted model, the originally adopted parameters are used (i.e., $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$) so there are no unknown parameters (i.e., $p_D = 0$). Therefore, the deviance of the Gaussian model is given by

$$\begin{aligned} \text{DIC}^1 &= D_{\hat{\boldsymbol{\theta}}}(\mathbf{y}) = D(\mathbf{y}|\hat{\boldsymbol{\theta}}) = -2 \log p(\mathbf{y}|\hat{\boldsymbol{\theta}}) \\ &= -2 \log \left\{ \prod_{i=1}^N N(y_i|\hat{\boldsymbol{\beta}}, \hat{\sigma}^2) \right\} \\ &= N \log(2\pi) + N \log(\hat{\sigma}^2) + \hat{\sigma}^{-2} \sum_{i=1}^N (y_i - \mu(\hat{\boldsymbol{\beta}})_i)^2, \end{aligned} \quad (46)$$

where the sigma in (46), $\hat{\sigma}$, is the originally adopted sigma, i.e., the prior sigma. In the second case, however, we assume that sigma is unknown. We assign a scaled inverse chi-squared distribution to σ^2 , which is the conjugate prior distribution for σ^2 in the Gaussian model, namely

$$p(\sigma^2) \propto (\sigma^2)^{-\vartheta/2-1} \exp\left(-\frac{\vartheta s^2}{2\sigma^2}\right), \quad (47)$$

where ϑ is the degrees-of-freedom parameter, and s^2 is the scaling parameter. Therefore, the posterior distribution of σ^2 conditional on $\boldsymbol{\beta}$ is given by

$$p(\sigma^2|\mathbf{y}, \boldsymbol{\beta}) \propto p(\sigma^2)p(\mathbf{y}|\boldsymbol{\beta}, \sigma^2)$$

$$\propto \text{Inv-}\chi^2 \left(\sigma^2 \mid N+\vartheta, \frac{1}{N+\vartheta} \left\{ \vartheta s^2 + \sum_{i=1}^N (y_i - \mu(\boldsymbol{\beta})_i)^2 \right\} \right), \quad (48)$$

a scaled inverse chi-squared distribution with $N + \vartheta$ degrees of freedom, and scaling parameter $(N + \vartheta)^{-1} \left\{ \vartheta s^2 + \sum_{i=1}^N (y_i - \mu(\boldsymbol{\beta})_i)^2 \right\}$. Given what has been presented above, a model comparison can now be made using formulas (40)–(44). Thus, the DIC of the Gaussian model is given by

$$\begin{aligned} \text{DIC}^2 &= 2\hat{D}_{\text{avg}} - D_{\hat{\boldsymbol{\theta}}}(\mathbf{y}) \\ &= 2L^{-1} \sum_{l=1}^L D(\mathbf{y}|\boldsymbol{\theta}_l) - D(\mathbf{y}|\bar{\boldsymbol{\theta}}) \\ &= \frac{2}{L} \sum_{l=1}^L \left(N \log(2\pi) + N \log(\sigma_l^2) + \sigma_l^{-2} \sum_{i=1}^N (y_i - \mu(\boldsymbol{\beta}_l)_i)^2 \right) \\ &\quad - N \log(2\pi) - N \log(\bar{\sigma}^2) - \bar{\sigma}^{-2} \sum_{i=1}^N (y_i - \mu(\bar{\boldsymbol{\beta}})_i)^2, \end{aligned} \quad (49)$$

where L is the number of samples, σ_l^2 and $\boldsymbol{\beta}_l$ are the l -th samples of σ^2 and $\boldsymbol{\beta}$ from their posterior distribution using an MCMC method, $\bar{\sigma}^2$ and $\bar{\boldsymbol{\beta}}$ are their posterior means, and $\bar{\boldsymbol{\theta}} = (\bar{\boldsymbol{\beta}}, \bar{\sigma}^2)$.

Based on the above formulation, we explore several empirical GMMs in three categories; local models calibrated to the earthquake strong-motion data from South Iceland; regional models based on European and Middle Eastern data (primarily those that had been proposed in the previous hazard studies as being suitable models for the seismotectonic environment of Iceland); and lastly, the NGA-West2 models

Table 1 Description of the earthquake ground-motion models (GMM) used in this study for South Iceland

GMM	Mw range	R range	PSA periods(s), PGA (and PGV)	Main regions
Kea20	4.0–8.3	0–100	0.05–5.0, PGA	Iceland
AB10	5.0–7.6	0–100	0.05–3.0, PGA, PGV	Europe and Middle East
Am05	5.0–7.6	0–100	0.05–2.5, PGA	Europe and Middle East
Kea16	4.0–8.0	0–300	0.01–4.0, PGA, PGV	Europe and Middle East
Zh06	5.0–8.3	0–300	0.05–5.0, PGA	Japan
ASK14	3.0–8.5	0–300	0.01–10, PGA, PGV	Worldwide
BSSA14	3.0–8.5	0–400	0.01–10, PGA, PGV	Worldwide
CB14	3.3–8.5	0–300	0.01–10, PGA, PGV	Worldwide
CY14	3.0–8.0	0–300	0.01–10, PGA, PGV	Worldwide
I14	4.5–7.9	0–175	0.01–10, PGA, PGV	Worldwide

that were calibrated to worldwide data. Table 1 gives an overview of the models. We omit other GMMs that fall into these categories but fail the criteria proposed by Cotton et al. (2006) and Bommer et al. (2010) for the minimum requirement of the functional form for GMMs to be applied in PSHA. The final GMMs are the local ones developed by Kowsari et al. (2020), named Kea20-1 to Kea20-6; the regional models Aea05 (Ambraseys et al., 2005), Zh06 (Zhao et al., 2006), AB10 (Akkar & Bommer, 2010), Kea16 (Kotha et al., 2016); and the NGA-West2 GMMs including ASK14 (Abrahamson et al., 2014), BSSA14 (Boore et al., 2014), CY14 (Chiou & Youngs, 2014), CB14 (Campbell & Bozorgnia, 2014), I14 (Idriss, 2014). They will be ranked according to the proposed DIC method. For comparison, they are also ranked according to other previously proposed data-driven ranking methods (i.e., the LLH and EDR methods mentioned above). Their performance is tested on the Icelandic earthquake strong-motion data that were shown in Fig. 7. The PGA and PSA at periods of interest in earthquake engineering (i.e., $T = 0.3, 1.0, 2.0$ s) have been chosen as the intensity measures.

A comparison of the ranking results of the fifteen GMMs considered for the SISZ is shown in Fig. 11. Each of the fifteen models is applied to PGA and PSA at different periods. The models are compared with respect to their EDR, LLH, DIC^1 , and DIC^2 scores. As detailed above, the DIC scores obtained by using the prior and posterior sigma of the candidate models. The prior and posterior sigmas are then listed in Table 2.

As expected, the best model, according to DIC, is the GMM that is developed based on the Icelandic data at most periods. The DIC method has not only been shown to optimize the selection of GMMs for the given region in an unbiased way through Bayesian statistics, but it also solves the problem associated with the previous data-driven selection methods. The results show that when the observed data are congregated relatively far from the median estimations of the two models, LLH favors GMMs with larger sigma. For example, LLH shows a better performance of Zh06 over BSSA14 (at $T = 0.3$ s) because of its larger sigma. On the other hand, when the median of two models is approximately the same, the EDR favors models with smaller sigma. Thus, EDR favors CB14 over CY14 (at PGA) because of its smaller sigma regardless of the bias and true variability. In these cases, the DIC^2 (with the posterior sigma) favors the opposite because it considers both the model bias and the deviation from observations, which is representative of the aleatory variability of the region under study. Furthermore, the DIC is well suited for model selection since it penalizes model complexity. Therefore, the DIC method is a useful and objective method for comparing the performances of different GMMs on a given dataset and has potentially important applications for PSHA.

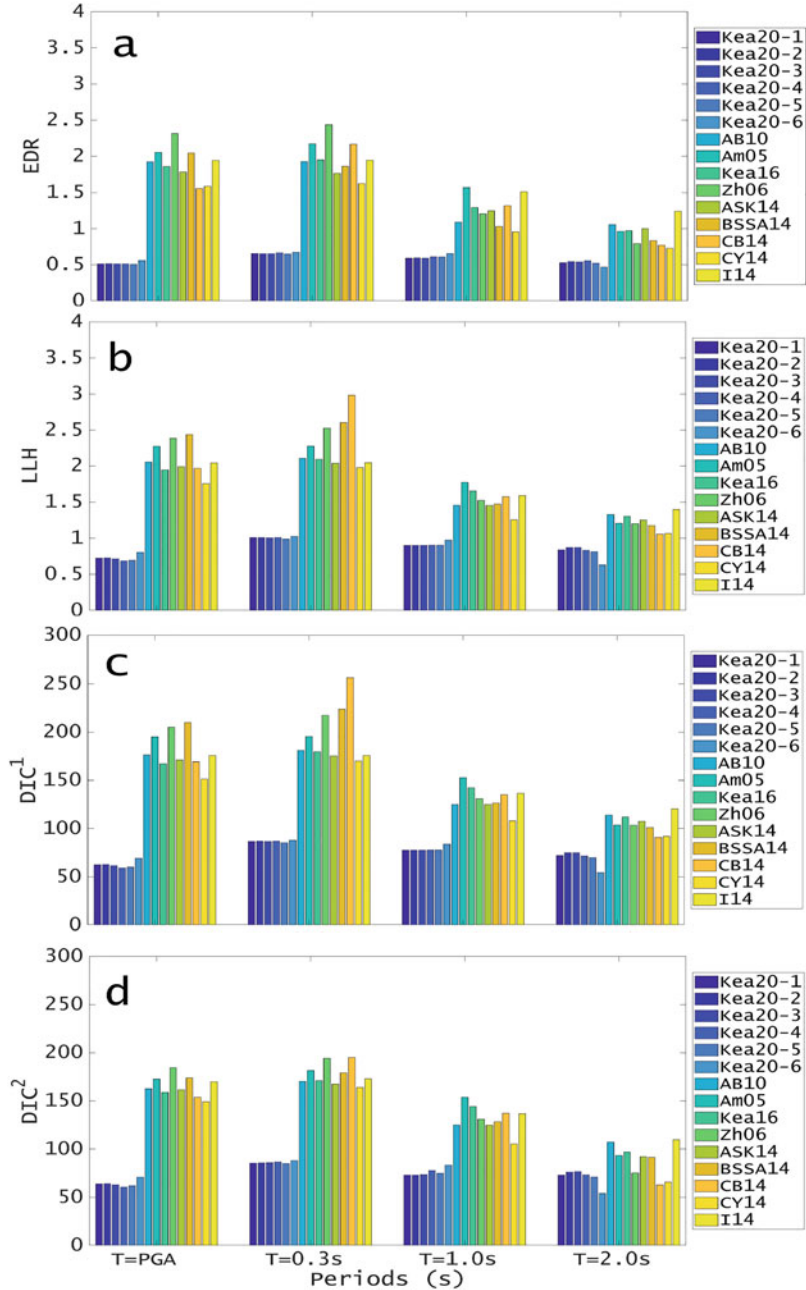


Fig. 11 The scores of (a) EDR, (b) LLH, (c) DIC^1 with the prior sigma, and (d) DIC^2 with the posterior sigma for the candidate GMMs in South Iceland when applied to data on PGA and PSA at different periods. The smaller scores imply a better representation of the observed ground motions by the predictive model (Kowsari et al., 2019a)

Table 2 The scores of the EDR, LLH, DIC¹ with the prior sigma, and DIC² with the posterior sigma, along with the prior and posterior sigma (natural logarithmic scale) of the candidate GMMs for Southwest Iceland at different periods

GMMs		Kea201	Kea202	Kea203	Kea204	Kea205	Kea206	AB10	Am05	Kea16	Zh06	ASK14	BSSA14	CB14	CY14	II14
PGA	EDR	0.51	0.52	0.51	0.51	0.51	0.56	1.92	2.05	1.86	2.32	1.78	2.05	1.55	1.59	1.94
	LLH	0.73	0.73	0.71	0.69	0.70	0.80	2.05	2.27	1.94	2.39	1.99	2.44	1.97	1.76	2.04
	DIC ¹	62.5	62.6	61.4	59.0	60.0	69.2	176.4	195.2	167.1	205.4	171.2	209.8	169.1	151.1	175.6
	DIC ²	64.0	64.2	62.9	60.4	61.9	71.0	162.7	172.5	158.6	184.4	161.5	173.7	153.5	149.2	169.8
	Sig _{org}	0.42	0.42	0.4	0.40	0.40	0.41	0.64	0.65	0.6	60.72	0.66	0.61	0.59	0.67	0.74
	Sig _{pos}	0.41	0.41	0.40	0.40	0.40	0.43	0.90	0.97	0.87	1.07	0.89	0.98	0.83	0.80	0.95
PSA (T = 0.3 s)	EDR	0.65	0.65	0.65	0.67	0.65	0.67	1.92	2.17	1.95	2.44	1.76	1.86	2.17	1.62	1.95
	LLH	1.01	1.01	1.01	1.01	0.99	1.02	2.11	2.27	2.09	2.53	2.04	2.60	2.98	1.98	2.05
	DIC ¹	86.8	86.9	86.6	87.0	85.3	88.0	180.9	195.4	179.7	217.5	175.3	223.9	256.5	170.0	175.8
	DIC ²	85.5	85.6	85.9	86.6	84.9	88.1	170.3	181.4	171.0	194.2	167.4	179.1	194.8	164.0	172.7
	Sig _{org}	0.56	0.56	0.56	0.55	0.55	0.55	0.71	0.75	0.73	0.77	0.71	0.61	0.64	0.71	0.80
	Sig _{pos}	0.48	0.48	0.48	0.49	0.48	0.49	0.95	1.04	0.96	1.16	0.93	1.02	1.16	0.91	0.97
PSA (T = 1 s)	EDR	0.59	0.60	0.59	0.61	0.61	0.65	1.09	1.57	1.29	1.20	1.25	1.03	1.32	0.96	1.51
	LLH	0.90	0.90	0.90	0.90	0.90	0.97	1.46	1.77	1.66	1.52	1.45	1.47	1.57	1.26	1.59
	DIC ¹	77.5	77.5	77.4	77.7	77.8	83.7	125.1	152.6	142.3	130.9	125.0	126.5	135.1	108.0	136.5
	DIC ²	73.1	73.0	73.5	77.7	74.9	83.2	124.9	153.6	144.2	131.2	124.7	128.4	137.0	105.4	136.7
	Sig _{org}	0.55	0.55	0.54	0.51	0.54	0.54	0.75	0.75	0.79	0.78	0.75	0.69	0.72	0.68	0.81
	Sig _{pos}	0.44	0.44	0.44	0.45	0.44	0.47	0.66	0.83	0.77	0.70	0.66	0.68	0.73	0.57	0.73
PSA (T = 2 s)	EDR	0.53	0.55	0.54	0.56	0.52	0.47	1.05	0.96	0.97	0.79	1.00	0.83	0.77	0.73	1.24
	LLH	0.84	0.87	0.87	0.83	0.81	0.63	1.33	1.21	1.30	1.20	1.25	1.18	1.06	1.07	1.40
	DIC ¹	72.1	74.9	74.8	71.7	70.0	54.3	114.0	103.8	111.9	103.3	107.6	101.1	90.8	91.7	120.3
	DIC ²	73.2	76.0	76.6	73.3	71.2	53.9	107.3	93.5	96.9	75.2	92.2	91.4	62.9	65.8	109.7
	Sig _{org}	0.47	0.48	0.46	0.45	0.46	0.42	0.76	0.72	0.79	0.79	0.77	0.70	0.71	0.71	0.82
	Sig _{pos}	0.44	0.45	0.45	0.44	0.43	0.38	0.58	0.52	0.53	0.45	0.51	0.51	0.40	0.41	0.59

5 Supplementary MATLAB Code

A MATLAB code with synthetic data is provided as a supplementary material for this section, see <https://github.com/tsonne/gmm-ranking-dic>.

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Bayesian Modelling in Engineering Seismology: Spatial Earthquake Magnitude Model



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1 Introduction

The previous chapter on Bayesian modelling in engineering seismology, chapter “Bayesian Modeling in Engineering Seismology: Ground-Motion Models”, focused on ground motion models, which constitute one of the three key elements of estimating seismic hazards. The other two are the spatial characterisation of earthquake sources in a seismic region and their seismic activity, respectively. In the conventional engineering approach to seismic hazard assessment, these two are derived from the historical and instrumental earthquake catalogue of the region but are generally uncoupled in the formulation of their spatial versus temporal characteristics. Namely, the spatial distribution of earthquakes is estimated from the approximate locations of earthquake epicentres and serves the key role in confining the earthquake source’s spatial extent. On the other hand, provided the catalogue is of long enough duration, the relative distribution of earthquake magnitudes over the catalogue’s time duration quantifies the region’s seismic activity rate over the catalogue’s magnitude range and the maximum earthquake magnitude believed to be possible in the region. Currently, the time-independent probabilistic seismic hazard assessment (PSHA) is the most critical tool that is utilised to quantify the earthquake hazard in a given seismic region. It provides the necessary results required in seismic risk assessment and the earthquake-resistant design of structures, which are critical preparedness efforts for the mitigation and management of seismic risk (Jordan et al., 2011). Long-term (time-independent) PSHA implicitly assumes

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ergodicity in the spatiotemporal distribution of seismicity in the seismic region of interest. In other words, the seismic activity is assumed to remain constant over time and space as defined by the historical earthquake catalogues (and, if available, additional geological evidence). For a very long historical earthquake catalogue, this may be a reasonable assumption. Most earthquake catalogues, however, are very short compared to the timescale that geological processes operate by, the former being decades of instrumental catalogues and a couple to a few hundred years in the case of most historical accounts of large and devastating earthquakes. The historical catalogues, moreover, unavoidably suffer from considerable and highly time-dependent uncertainties in earthquake magnitudes, dates, and locations. The most recent decades of an earthquake catalogue are generally more accurate and have relatively low values of the magnitude of completeness (e.g., around M_w 5.0). In contrast, the oldest part of the catalogue (over hundreds of years) consists of events for which locations and magnitudes are very uncertain, and with a relatively large value of completeness magnitude since only those earthquakes that were the most notable historical events were reported. However, for modelling purposes in conventional PSHA, the temporal characteristics of the seismicity are based on the assumption that the earthquake occurrence is homogeneous over the time period, and the temporal seismicity rates are estimated from a simple statistical analysis of the catalogue. Then, the spatial characteristics are generally modelled by assigning confined spatial regions as area sources inside which the earthquakes are modelled as randomly located point sources, in particular, when the locations of the causative faults are relatively unknown (see, e.g., Lee et al., 2003, and references therein).

In general, there are two approaches to characterise the spatial variation of seismic activity across a seismic region. The first approach is the conventional seismogenic source zonation, i.e., defining a set of non-overlapping seismic source zones based on a series of information such as seismogenic characteristics, geological structures, and seismicity. A uniform seismicity is assumed over each seismic source zone. Afterwards, the seismic activity parameters (such as a -value and b -value in the Gutenberg–Richter model (Gutenberg & Richter, 1944); see Sect. 3.1 for detailed information) are determined for each zone using the recorded earthquake catalogue accommodated in each seismogenic zone. It is advised to employ the zonation models developed by relevant experts (e.g., seismologists or geophysicists) and recognised by the scientific community. However, one of the main drawbacks of this approach is the sharp change in activity rate at the boundaries of the source zones. Several approaches have been proposed to smooth the drastic jumps from one zone to another (see Bender, 1983; Veneciano & Pais, 1986). Another drawback of this method, commonly used in PSHA, is that the non-uniform geographical distribution of seismicity corresponding to each seismic zone is not considered. This is due to two main reasons. First, the lack of complete data across a broad area of interest due to a small number of seismic recording stations. The second reason is the absence of a seismotectonic model for the computation of critical seismic activity parameters while manifesting their spatial variation over a given region (e.g., Bayat et al., 2022).

The second method for the determination of seismic activity rate is a zoneless approach (Frankel, 1995; Fix & Hodges, 1951; Silverman, 1986). Contrary to the traditional zonation model, in this method, the seismic activity rates are attained disregarding any source zones over which the seismic parameters are assumed constant. This approach allows the expression of the seismic catalogue without biasing the shape of the activity rate function. The dependence on location has a continuous variation, while the dependence on magnitude does not necessarily follow the Gutenberg–Richter relation. Such approaches employ non-parametric density estimation (e.g., basic histogram) in order to detect the density function from which a given sample derives without specifying a priori a particular shape for the density function, e.g., a Gaussian or gamma distribution. For instance, Frankel (1995) used this approach to determine the a -value spatially by relying on a smooth function that decays with distance from the grid centre across the cells covering the region of interest, whereas, for the b -value, they employed the source zonation models, which do not vary spatially. Furthermore, Frankel (1995) assumes that earthquake occurrence above a certain reference magnitude follows a Poisson process with intensity that varies spatially but is constant in time. However, the model for this intensity is such that if the reference magnitude increases, then the intensity will decrease proportionally the same across space since the b -value does not vary over space.

2 ICEL-NMAR Earthquake Catalogue

Recently, Jónasson et al. (2021) compiled a harmonised earthquake catalogue, the ICEL-NMAR, of shallow crustal earthquakes in Iceland and on the northern Mid-Atlantic Ridge (NMAR) by merging local estimates of epicentral locations and moment magnitude estimates from teleseismic catalogues. The earthquake catalogue is available online at <http://data.mendeley.com>. The on-land (ICEL) part of the catalogue contains 1281 events with $M_w \geq 4.0$ in the region 62° – 68° N and 12° – 26° W (see Fig. 1). The estimated completeness magnitude ranges from M_w 5.5 for the oldest part to M_w 4.5 for the most recent catalogue of events.

Figure 1 shows Iceland and the ICEL-NMAR catalogue of events with $M_w \geq 4.0$ for the period 1904–2019. The major transform zones are located in Southwest Iceland and North Iceland, as illustrated by red rectangles. Earthquakes are classified by their moment magnitudes M_w (see legend), and recording stations of the National seismic network (SIL) are shown by red triangles. The larger scale map then shows earthquakes in Southwest Iceland, with $M_w \geq 6.0$ labelled by yellow stars. The largest historical earthquake of $M_w \sim 7$ occurred in the easternmost part of the South Iceland seismic zone (SISZ) on 6 May 1912 (Bjarnason et al., 1993; Bellou et al., 2005; Jónasson et al., 2021). Since then, multiple moderate-to-strong earthquakes have taken place in Southwest Iceland, such as the M_w 6.36 1929 and M_w 6.12 1968 earthquakes in the Reykjanes peninsula oblique rift (RPOR), and the M_w 5.5 1998, M_w 6.52 17 June 2000, M_w 6.44 21 June 2000, and M_w 6.3 29 May

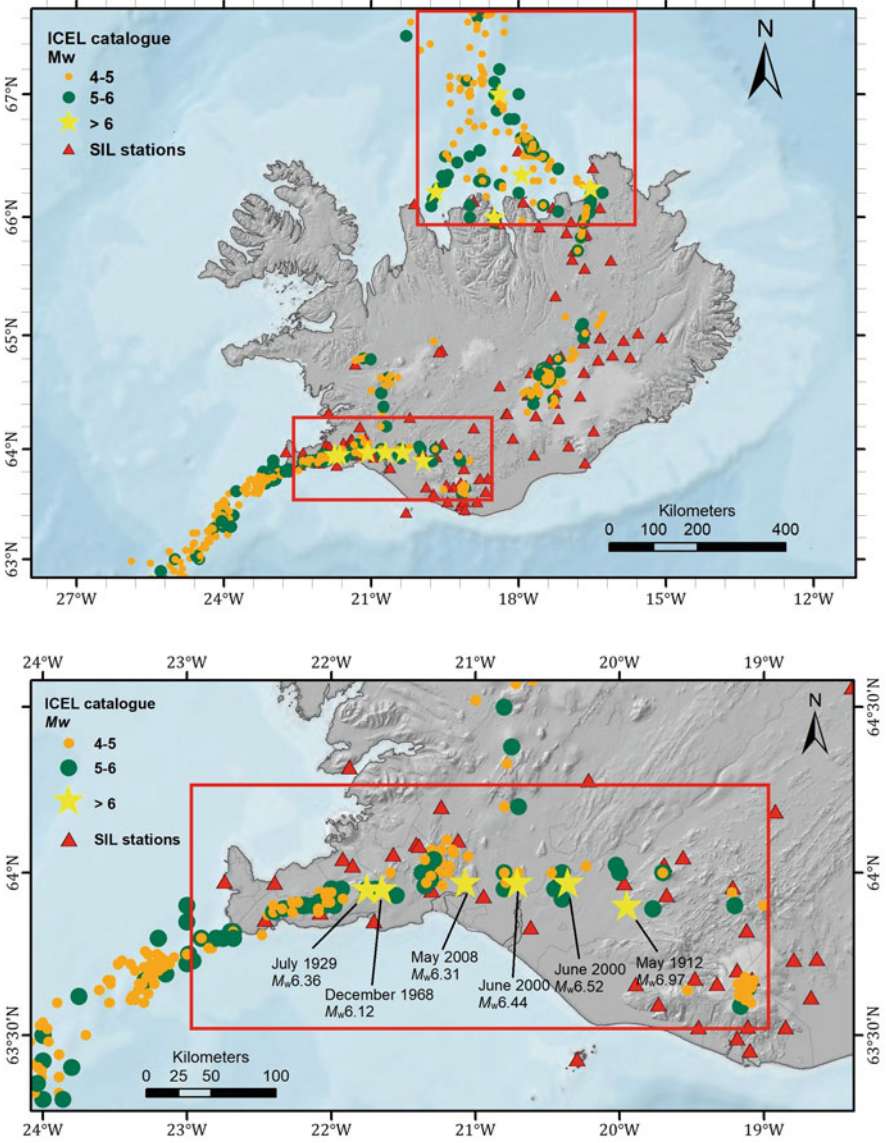


Fig. 1 The geographical distribution of $M_w \geq 4.0$ earthquakes from 1904 to 2019 (ICEL-NMAR, Jónasson et al., 2021). The largest earthquakes in Iceland take place in the two transform zones, in North and Southwest Iceland, respectively (rectangles), as shown by the spatial distribution of M_w (see legend). On the bottom panel, $M_w \geq 6.0$ earthquakes in Southwest Iceland are labelled on the map. For reference, seismic stations of the National seismic network (SIL) are shown as red triangles

2008 earthquakes in the SISZ (Einarsson, 2014). Thus, in addition to the capability of earthquakes of magnitude ~ 7 , the SISZ and RPOR regions are considered to be capable of producing earthquakes in the magnitude range 6–6.5 with return periods of a few decades (Einarsson, 1991, 2010, 2014; Sigmundsson et al., 1995; Sigmundsson, 2006; Einarsson et al., 2020; Steigerwald et al., 2020). In this book chapter, we analyse earthquake parametric data from the ICEL-NMAR catalogue that occurred from 1904 to 2019 in Southwest Iceland. For the sake of simplicity, we designate the area as defined by longitude 19.0° west, longitude 23.0° west, latitude 63.59° north, and 64.07° north, and consider events larger than magnitude 4.5, as it is generally considered the minimum magnitude used in PSHA.

3 Statistical Models for Earthquake Magnitudes

In this section, we investigate potential statistical models for earthquake magnitudes above a given threshold. We first examine the exponential distribution and the generalised Pareto distribution with constant parameters attributed to the spatial domain of interest. Both of these models have been used in the literature. We explore and compare these two models using the ICEL-NMAR catalogue. Based on this exploration and comparison, we propose statistical models that assume the distribution of earthquake magnitudes, above a given threshold, varies spatially in Southwest Iceland.

3.1 The Gutenberg–Richter Model

The two most frequently used models to describe the magnitude–frequency distribution of earthquakes in a seismic region are the Gutenberg–Richter model (Gutenberg & Richter, 1944) and the characteristic earthquake model (Schwartz & Coppersmith, 1984). For earthquakes of small to moderate magnitudes, both models are based on the general Gutenberg–Richter relationship. The latter model, however, relies on its maximum earthquake magnitude, and its recurrence rate is determined from geological analyses of the active faults identified across the region of interest. For regions with diffuse seismicity or non-identified faults, the former is used. In this chapter, we focus on the Gutenberg–Richter relationship for analysis of the frequency–magnitude distribution of seismicity,

$$\log_{10} N(M) = a - bM, \quad (1)$$

where $N(M)$ is the expected cumulative number of events of magnitude equal to or greater than magnitude M occurring within the region of interest throughout the observation period, and the parameters a and b are empirical constants known as the seismicity parameters. The parameter a , referred to as the a -value, is a measure of

the absolute seismicity rate and is equal to the logarithm of the expected number of events with $M > 0$, that is, $a = \log_{10} N(0)$. The a -values can be highly variable as they depend on the size of the region under study and the length of the time period over which earthquakes are observed. For that reason, it is generally normalised to the annual or centennial seismicity rate for a seismic region of a given spatial extent.

The parameter b , referred to as the b -value, is the slope of the Gutenberg–Richter recurrence curve and therefore a measure of the relative cumulative rate of earthquakes by magnitude, thought to be a seismotectonic characteristic. A larger b -value indicates that smaller events occur more frequently relative to larger events. In contrast, a smaller b -value indicates that larger earthquakes happen more often, although their cumulative number is always lower than that of smaller events (Frohlich & Davis, 1993; Kagan, 1999; Utsu, 1999; Godano & Pingue, 2000; Marzocchi & Sandri, 2003; Schorlemmer et al., 2005).

Richter (1958) fitted the linear relation in (1), using least-square regression, to the cumulative number of earthquakes in Southern California over a specific time period. Several limitations of this simple approach have been addressed and improved over the past years. One of the fundamental approaches for assessing the seismicity parameters was proposed by Aki (1965), who estimated the b -value using a maximum likelihood procedure. However, the proposed procedure was only applicable to complete earthquake catalogues. Several researchers attempted to address this weakness in the case where the level of completeness of seismic events varies over time. The most well-known and frequently used method was proposed by Weichert (1980), who developed a maximum likelihood procedure for events divided into different magnitude bins, which correspond to a specific time period. Kijko and Sellevoll (1989, 1992) introduced a maximum likelihood procedure that can be applied to earthquake catalogues encompassing incomplete (historically large events) and complete parts (instrumental catalogue with time-varying completeness level). This approach does not account for uncertainties associated with the applied earthquake occurrence models. To address such problems, Kijko and Smit (2012) extended the procedure of Aki (1965) and addressed the consideration of using multiple seismic event catalogues with different levels of completeness. Alternatively, some studies utilised Bayesian parameter estimation techniques (e.g., Tsapanos et al., 2001; Tsapanos, 2003; Tsapanos & Christova, 2003; Yadav et al., 2013). Moreover, Kijko et al. (2016) introduced compound distribution of gamma for b -value and activity rate parameter, while both are assumed random variables.

Another fundamental modification in the use of the general Gutenberg–Richter relationship in (1) is to account for the completeness of a catalogue that may differ for different magnitude levels. This is commonly done by restricting the events to a particular lower-bound magnitude, generally referred to as the magnitude of completeness and denoted by M_c . Thus, the estimation of seismicity parameters in some methodologies depends on the completeness of a seismic event catalogue, irrespective of the fitting procedure (see the methods in Aki, 1965; Utsu, 1965; Molchan et al., 1970; Rosenblueth & Ordaz, 1987; Weichert, 1980; Kijko & Sellevoll, 1992; Kijko & Smit, 2012). [For an instrumental earthquake catalogue, the magnitude of completeness is the minimum magnitude for which the seismic

network is capable of recording all the events in a space–time volume (Wiemer & Wyss, 2000; Woessner & Wiemer, 2005; Kijko & Smit, 2017).] Several procedures have been proposed for the estimation of M_c from an earthquake catalogue, such as the maximum curvature technique, goodness-of-fit test, methods based on the stability of b -value, and the entire magnitude range method (Ogata & Katsura, 1993; Wiemer & Wyss, 2000; Cao & Gao, 2002; Woessner & Wiemer, 2005; Mignan & Woessner, 2012; Ebrahimian et al., 2014). There are other methods for the determination of seismicity parameters that do not require the M_c estimates in advance and, therefore, exclude the challenges associated with the M_c calculation. For further information, please refer to Lee and Brillinger (1979) and Kijko and Smit (2017).

One of the reasons for the above-mentioned research efforts is that PSHA requires a reliable magnitude–frequency distribution of seismicity. Most recently, Douglas and Danciu (2020) presented a nomogram for simple PSHA that captures the influence of different inputs for different return periods assuming the 5% critical damping. They observed a relatively minor influence of the b -value on PGA but a large influence on long-period spectral accelerations. Moreover, the expected ground motion for a given return period increases as the activity rate (a -value) increases. Usually, the PSHA results are more sensitive to b -value estimates than a -value estimates because, in spite of the b -value, acquiring the a -value is not challenging.

3.2 The Generalised Pareto Distribution

It is common practice to resort to extreme value theory (e.g., Coles, 2001; Ismail-Zadeh et al., 2014; Bousquet & Bernardara, 2021) for probabilistic models when modelling the size of a natural quantity above a carefully selected threshold. Dutfoy (2021) selected the generalised Pareto distribution as a model for the magnitude of earthquakes above a prespecified threshold. Here, we opt for the same model for the magnitude of earthquakes in Southwest Iceland.

Given that the random variable Y follows a generalised Pareto distribution with threshold μ , the probability of Y exceeding the value $y > \mu$ is

$$P(Y > y) = \left(1 + \frac{\xi(y - \mu)}{\sigma}\right)^{-1/\xi},$$

where ξ is the shape parameter and σ is the scale parameter. The shape parameter is defined on the real line, and the scale parameter is positive. The shape parameter affects the probability density function such that $\xi > 0$ corresponds to heavier tails than that of the exponential distribution. The larger the value of ξ is, the heavier the tail, while, if $\xi < 0$, the tail is lighter, and the random variable Y has an upper bound $\mu - \sigma/\xi$.

The special case $\xi = 0$ corresponds to a shifted exponential distribution with zero probability mass below μ . The exponential distribution is the underlying assump-

tion in the Gutenberg–Richter relationship. When the exponential distribution is the appropriate assumption for the magnitude of earthquakes above μ , then the Gutenberg–Richter relationship for the number of earthquakes with $M > \mu$ can be presented in terms of the shifted exponential distribution, that is,

$$\log_{10} N(M) = a - b(M - \mu), \quad M > \mu, \quad (2)$$

where $N(M)$ is the expected cumulative number of earthquakes that have magnitude above M , a is such that 10^a is the expected number of earthquakes greater than μ for the region and time period of interest, and b is the slope of the line. Since the shifted exponential distribution is a special case of the generalised Pareto distribution, the expected cumulative number of earthquakes, $N(M)$ with $M > \mu$, can also be presented in terms of the generalised Pareto distribution (see Dutfey, 2021) with

$$\log_{10} N(M) = a - \sigma^{-1} \left(\frac{\sigma}{\xi} \right) \log_{10} \left(1 + \frac{\xi(M - \mu)}{\sigma} \right), \quad (3)$$

where the bounds for M are such that, if $\xi > 0$, then the bounds are $M > \mu$, while, if $\xi < 0$, they are $\mu < M < \mu - \sigma/\xi$, and lastly, $N(M) = 0$ when M is equal to or greater than the upper bound $\mu - \sigma/\xi$. As in (2), 10^a is the expected number of earthquakes such that $M > \mu$ within the region and period of interest, and σ^{-1} plays a similar role as b in (2). Here, $\sigma^{-1}(\log(10))^{-1}$ is equal to the absolute value of the first derivative with respect to M of the curve in (3) at $M = \mu$. However, the absolute value of the first derivative increases with M if $\xi < 0$, while it decreases with M if $\xi > 0$.

3.3 Comparison of Two Models

To get a better understanding of the relationships given by (2) and (3), and to see how well they perform in practice, we applied them to earthquake data with magnitudes greater than M_w 4.5 in the ICEL-NMAR catalogue that occurred in Southwest Iceland during the period from 1904 to 2019; see Sect. 2. The parameters of the two models were estimated using maximum likelihood estimation. The fitted relationships in (2) and (3) were compared to observed magnitudes in the catalogue to visually inspect whether the shifted exponential distribution or the generalised Pareto distribution is appropriate.

As can be seen in Fig. 2, one plot is drawn for each of the two models. These two plots consist of points representing pairs of the observed earthquake magnitudes (the x-axis) and the observed number of earthquakes above this magnitude (the y-axis, shown on a logarithmic scale). Furthermore, for each of the models in (2) and (3), the expected number of earthquakes above magnitude M , $N(M)$, is added to the corresponding plot. If the magnitudes of the earthquakes in the region follow an exponential distribution with a constant b parameter, then the observed points will

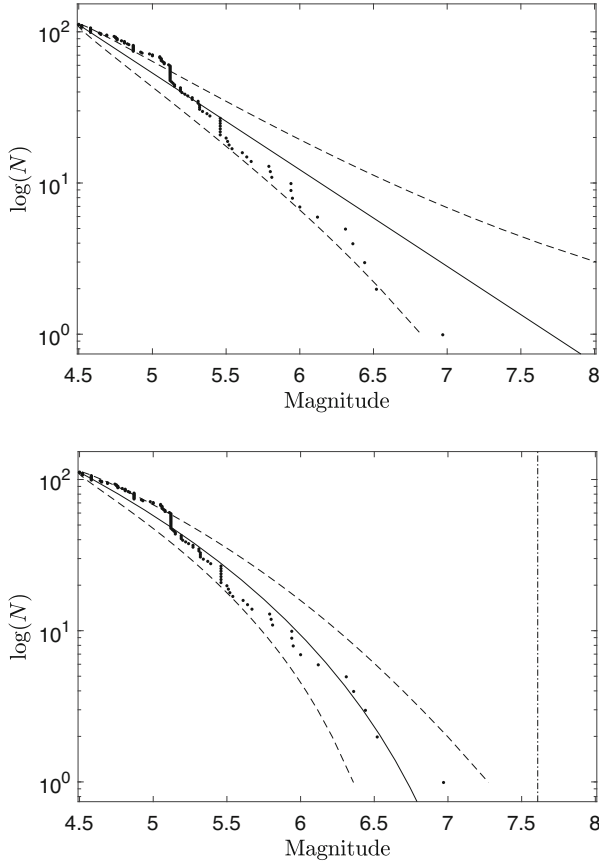


Fig. 2 The top panel shows the Gutenberg–Richter for Southwest Iceland based on the ICEL-NMAR dataset when assuming the shifted exponential distribution. The bottom panel shows the cumulative frequency of earthquakes above a given magnitude (y-axis) as a function of earthquake magnitude (x-axis) when assuming the generalised Pareto distribution (solid line), and the vertical dash-dotted line shows the upper bound of the magnitude. The points represent the observed earthquakes, and the dashed lines represent 95% horizontal bounds for the ordered data according to (4) and (5)

fall close to the curve given by the model in (2), which is a straight line when $N(M)$ is on a log scale. On the other hand, if the magnitudes of the earthquakes follow a generalised Pareto distribution with constant parameters, then the observed points will fall close to the curve given by the model in (3). Marginal prediction intervals are computed for each of the observed magnitudes by using the marginal distribution of the i -th order statistic in a sample of size n from the uniform distribution (Casella & Berger, 2002). These intervals provide a measure on how close the observed points are to the line in the exponential model or the curve in the generalised Pareto model. The i -th order statistic from the uniform distribution, denoted by $v_{(i)}$, follows

a beta distribution with parameters i and $n - i + 1$. The distribution of the i -th order statistic of the magnitudes, $M_{(i)}$, is found through either the quantile function of the shifted exponential distribution, that is, $M_{(i)} = \mu - \sigma \log(1 - v_{(i)})$, or the quantile function of the generalised Pareto distribution, that is, $M_{(i)} = \mu - \sigma \xi^{-1}((1 - v_{(i)})^{-\xi} - 1)$.

The 95% prediction intervals for $M_{(i)}$ according to the shifted exponential and the generalised Pareto distributions are

$$(\mu - \sigma \log(1 - v_{i,0.025}); \mu - \sigma \log(1 - v_{i,0.975})), \quad (4)$$

$$(\mu - \sigma \xi^{-1}((1 - v_{i,0.025})^{-\xi} - 1); \mu - \sigma \xi^{-1}((1 - v_{i,0.975})^{-\xi} - 1)), \quad (5)$$

respectively, where $v_{i,0.025}$ and $v_{i,0.975}$ are the 0.025 and 0.975 quantiles of $v_{(i)}$, respectively. Note that σ in the shifted exponential distribution is equal to the mean of $M - \mu$, and it can be presented in terms of b as $\sigma = b^{-1}(\log(10))^{-1}$.

The Gutenberg–Richter relationship for Southwest Iceland, presented according to the shifted exponential distribution as in (2), is shown in the top panel of Fig. 2, along with 95% prediction intervals for the ordered magnitudes given by (4). The points in Fig. 2 show the magnitudes of the observed earthquakes on the x-axis and the observed number of earthquakes above each magnitude on the y-axis. If the magnitudes of the earthquakes in this region follow the same exponential distribution, then the observed points should fall close to the straight line. We use the 95% prediction intervals for the ordered magnitudes to assess how close the observations fall to the line. Many of the observations with magnitudes between 4.49 and 5.2 fall above the prediction intervals, which indicates that the shifted exponential distribution with a constant scale parameter does not describe the earthquake magnitudes in the Southwest Iceland adequately well. Similar can be seen in the case of the generalised Pareto distribution with constant σ and ξ (see Fig. 2), where (3) is assumed, and the 95% prediction intervals for the ordered magnitudes are calculated according to (5). Although, a smaller number of observations with magnitudes between 4.49 and 5.2 fall above the prediction intervals compared to the exponential distribution, these results suggest that the generalised Pareto distribution, with constant parameters, does not describe the magnitude–frequency relationship of earthquake magnitudes in Southwest Iceland sufficiently well.

There are strong indications in the literature that suggest the seismogenic potential varies across the Southwest Iceland transform zone from west to east (down to $\sim 19.6^\circ\text{W}$), with earthquake maximum magnitudes thought to increase from ~ 5.5 to ~ 7 , respectively (see also, e.g., Bjarnason & Einarsson, 1991; Stefánsson et al., 1993; Árnadóttir et al., 2004; Clifton & Kattenhorn, 2006; Einarsson et al., 2020). In the region considered in this study, this physical trend explains most of the results, apart from those in the easternmost part as that seismicity is of volcanic origin, the maximum magnitudes of which are considered to be $\sim \text{M}4.5\text{--}5.5$. Thus, in the search for extensions of the models for earthquake magnitudes based on

the exponential and generalised Pareto distributions with constant parameters, for simplicity, we consider modelling using parameters that vary spatially along the east–west direction. These models are introduced in Sect. 3.4.

3.4 Spatial Modelling of Earthquake Magnitudes

Here, we propose two statistical models for the magnitude of earthquakes above $\mu = 4.5$ in Southwest Iceland using data from 1904 to 2019. The proposed models are motivated by the exploration of the data in Sect. 3.3 and the fact that the reference magnitude, $\mu = 4.5$, generally represents the minimum magnitude considered in PSHA. The first model assumes the generalised Pareto distribution for magnitude above μ with spatially varying σ and a constant ξ . In particular, it is assumed that σ varies only with longitude. The proposed statistical model is a Bayesian latent Gaussian model. Its data level, latent level, and hyperparameter level are specified below.

At the data level, the observed earthquake magnitudes above the threshold μ in Southwest Iceland are denoted by M_i , where $i \in \{1, \dots, n\}$ and n is the total number of earthquakes above μ in the Southwest Iceland region over the period from 1904 to 2019. The longitude coordinates of M_i are denoted by s_i . We assume that M_i follows the generalised Pareto distribution, that is,

$$M_i \sim \text{GP}(\xi, \sigma_i, \mu), \quad i \in \{1, \dots, n\}, \quad (6)$$

where ξ is the shape parameter, σ_i is the scale parameter, and μ is the threshold. Since the scale parameter varies spatially with longitude, s , then $\sigma_i = \sigma(s_i)$, and σ is considered as a continuous function of s . A simplified version of this model will also be considered here, namely, a model based on the generalised Pareto distribution with σ and ξ both equal to constants.

If $\xi < 0$, the probability density function of M_i is

$$\pi(M_i | \sigma_i, \xi, \mu) = \frac{1}{\sigma_i} \left(1 + \frac{\xi(M_i - \mu)}{\sigma_i} \right)^{-(1/\xi + 1)}, \quad i \in \{1, \dots, n\}, \quad (7)$$

for $\mu \leq M_i \leq \mu - \sigma_i/\xi$ and zero otherwise. On the other hand, if $\xi > 0$, the probability density in (7) is defined for $M_i \geq \mu$, but it is zero for $M_i < \mu$. If $\xi = 0$, the generalised Pareto distribution is equal to the shifted exponential distribution. Its probability density function is

$$\pi(M_i | \sigma_i, \mu) = \frac{1}{\sigma_i} \exp\left(-\frac{(M_i - \mu)}{\sigma_i}\right), \quad i \in \{1, \dots, n\}, \quad (8)$$

for $M_i \geq \mu$ and zero otherwise. The probability density in (8) is such that the scale parameter varies with longitude, that is, $\sigma_i = \sigma(s_i)$. The other proposed model for earthquake magnitudes is a Bayesian latent Gaussian model, which assumes the shifted exponential model (8) at the data level. Here, we will also consider a shifted exponential model with a constant σ .

The earthquake moment magnitude, M_w , is calculated from the seismic moment, an earthquake source property that is proportional to the stiffness of the rock, the fault area, and the amount of slip on the earthquake fault (Aki & Richards, 1980). In any given seismic region, the earthquake magnitudes are generally constrained by the available fault area, i.e., the thickness of the brittle (i.e., seismogenic) crust and the potential maximum fault length. Thus, it is reasonable to assume that the magnitude–frequency distribution has a natural upper limit, or maximum magnitude. Due to this fact, the shape parameter of the generalised Pareto distribution is assumed to be negative here. A lower bound for the shape parameter ξ is set equal to -0.75 to aid with the posterior inference while still supporting a plausible range of values of ξ . We assign a uniform prior density to ξ on the interval $[-0.75, 0]$ and define ξ as a hyperparameter. The upper bound for the earthquake magnitudes is a function of both the scale parameter and the shape parameter; thus, the upper bound will vary with the longitude. Since there is a limited amount of data here (the number of events is $n = 113$) to estimate the shape parameter, we opt for a shape parameter that is constant over the region of interest. Furthermore, since the spatial locations of the vast majority of earthquakes in Southwest Iceland appear to be concentrated along a relatively narrow lineament oriented from east to west, for simplicity therefore, we model σ only as a function of longitude as opposed to both longitude and latitude. We conducted a likelihood ratio test (Casella & Berger, 2002) to compare a null model based on the generalised Pareto distribution with a constant shape parameter and scale parameter modelled with a third-degree polynomial in the longitude to an alternative model based on the generalised Pareto distribution with a constant shape parameter and scale parameter modelled with a third-degree polynomial in both longitude and latitude that includes interaction terms. The null model was not rejected at a 0.01 significance level. Thus, we claim that it is sufficient to model these data with a generalised Pareto distribution with a scale parameter that varies with longitude.

The logarithm of the scale parameter is modelled at the latent level with cubic B-spline functions and a random walk process. This ensures that the scale parameter σ is positive, and its transformation, $\eta = \log(\sigma)$, is defined on the real line. The model for η corresponding to the i -th earthquake, with longitude s_i , is given by

$$\eta_i = \sum_{k=1}^K u_k B_k(s_i), \quad i \in \{1, \dots, n\},$$

where u_k is an unknown latent parameter, $B_k(\cdot)$ is a B-spline function, and $B_1(\cdot), \dots, B_K(\cdot)$ form a cubic B-spline basis (Wasserman, 2006) on an interval that includes the longitudes of the observed earthquakes. Let $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)^\top$ and

$\mathbf{u} = (u_1, \dots, u_K)^\top$ be such that

$$\boldsymbol{\eta} = A\mathbf{u},$$

where the elements of the matrix A are known and based on the cubic B-spline basis.

A priori, $\mathbf{u}$ is assumed to be governed by a random walk process (Rue & Held, 2005, Sect. 3.3.1, pp. 94–101). Its precision matrix is Q_u , defined as $Q_u = \sigma_u^{-2}Q$, where σ_u is an unknown hyperparameter, and Q is the following $K \times K$ precision matrix

$$Q = \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 & -1 \\ & & & & & -1 & 1 \end{bmatrix}; \quad (9)$$

see Rue and Held (2005, Sect. 3.3.1, p. 95). The joint distribution of $\mathbf{u}$ conditional on σ_u can be written as

$$\pi(\mathbf{u}|\sigma_u) = \pi(u_1) \prod_{k=2}^K \pi(u_k|u_{k-1}, \sigma_u),$$

where $\pi(u_k|u_{k-1}, \sigma_u)$ is a Gaussian density with mean u_{k-1} and variance σ_u^2 , and $\pi(u_1)$ is such that $\exp(u_1)$ follows an exponential distribution with mean λ_u^{-1} , that is, $\pi(u_1) = \lambda_u \exp(-\lambda_u \exp(u_1) + u_1)$. The parameter λ_u is chosen such that, a priori, the expected value of $\sigma(s_0)$ (where s_0 is the longitude furthest west in the interval for the B-spline basis) is equal to 0.5, which corresponds to $\lambda_u = 2$. In fact, $\sigma(s_0) = \exp(u_1)$, so $\sigma(s_0)$ follows, a priori, an exponential distribution with most of its probability mass over the interval $[0, 1.5]$. This interval contains plausible values for $\sigma(s_0)$ in the case of the data. The logarithm of the prior density of $\mathbf{u}$ conditional on σ_u is

$$\log \pi(\mathbf{u}|\sigma_u) = c_0 - \lambda_u \exp(u_1) + u_1 - \frac{(K-1)}{2} \log(\sigma_u^2) - \frac{\sigma_u^{-2}}{2} \mathbf{u}^\top Q \mathbf{u},$$

where c_0 is a constant independent of $\mathbf{u}$ and σ_u , see Rue and Held (2005, Sect. 3.1).

The parameters $u_2, \dots, u_K$ are transformed to improve the posterior sampling. The new parameters, $z_2, \dots, z_K$, are defined such that $u_k = u_1 + \sum_{j=2}^k \sigma_u z_j$, for $k \in \{2, \dots, K\}$, and their conditional prior density is $\pi(z_2, \dots, z_K|u_1, \sigma_u) = \prod_{j=2}^K \mathcal{N}(z_j|0, 1)$, that is, the z parameters are a priori independent and follow a Gaussian distribution with mean zero and variance one.

Table 1 Four statistical models for earthquake magnitudes

Distribution	Description of parameters
Exponential	Constant σ
Generalised Pareto	Constant σ and constant ξ
Exponential	Spatially varying σ
Generalised Pareto	Spatially varying σ and constant ξ

The parameter σ_u is a hyperparameter that governs the step size between each pair of u_k and u_{k+1} . We assign a PC prior density (Simpson et al., 2017) to σ_u , which is an exponential density in this case. To cover a reasonably wide range of values for the step size, we opt for the expected value of σ_u to be such that the ratio of $\sigma(s_{\text{end}})$ over $\sigma(s_0)$ (where s_{end} is the longitude furthest east in the interval for the B-spline basis) falls in the interval $[0.5, 2]$ with probability 0.95. This gives σ_u a prior mean equal to $\log(2)z_{0.975}^{-1}(K-1)^{-1/2}$, where $z_{0.975} = 1.96$, i.e., the 0.975 quantile of the standard Gaussian density.

The model, which assumes that the earthquake magnitudes over the threshold $\mu = 4.5$ follow an exponential distribution with a spatial varying scale parameter, uses the same B-spline basis for the scale parameter and the same prior densities for the unknown parameters related to $\sigma(s)$, as in the model above. The two simpler models, namely, the exponential model and the generalised Pareto model with unknown scale parameters that are constant over the spatial region of interest, are such that their scale parameters are assigned exponential prior densities with a mean equal to 0.5. The shape parameter of the latter model is unknown and constant over the spatial region. It is assigned a uniform prior density on the interval $[-0.75, 0]$. These four models are listed in Table 1. They will be used in Sect. 4 to analyse the earthquake magnitude data presented in Sect. 2.

3.5 Posterior Inference

In this chapter, we implement a Metropolis algorithm (e.g., Gelman et al., 2013) to sample from the posterior density of the unknown parameters. Metropolis algorithms use a symmetric proposal density, which simplifies the calculations. They work well when the sampled parameters are defined on the real line. The hyperparameters ξ and σ_u are transformed to the real line to make them better suited for the Metropolis algorithm. The shape parameter, ξ , which is assumed to be in the interval $[-0.75, 0]$, is transformed to $\psi_1 = \log(\xi + 0.75) - \log(-\xi)$. The prior density of ψ_1 is $\pi(\psi_1) = \exp(-\psi_1)(1 + \exp(-\psi_1))^{-2}$. The hyperparameter σ_u is transformed to $\psi_2 = \log(\sigma_u)$. Its prior density is $\pi(\psi_2) = \lambda_u \exp(-\lambda_u \exp(\psi_2) + \psi_2)$. We define u_1 as a hyperparameter and let $\theta = (\psi_1, \psi_2, u_1)^T$. The posterior

density of the unknown parameters is

$$\pi(\mathbf{z}, \boldsymbol{\theta} | \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \pi(\mathbf{z}) \pi(\mathbf{y} | \mathbf{z}, \boldsymbol{\theta}), \quad (10)$$

where $\mathbf{z} = (z_2, \dots, z_K)^\top$, $\mathbf{y} = (M_1, \dots, M_n)^\top$, and $\pi(\boldsymbol{\theta}) = \pi(\psi_1) \pi(\psi_2) \pi(u_1)$.

The Metropolis algorithm requires a proposal density. Here, the proposal density is based on a Gaussian approximation to the posterior density. The mean of this Gaussian approximation is the mode of the posterior density. Its precision matrix is the negative Hessian matrix of the logarithm of the posterior density, evaluated at the mode. Denote the parameter vector $(\mathbf{z}^\top, \boldsymbol{\theta}^\top)^\top$ by $\boldsymbol{\phi}$. The Gaussian approximation for $\pi(\boldsymbol{\phi} | \mathbf{y})$ is given by

$$\tilde{\pi}(\boldsymbol{\phi} | \mathbf{y}) = \mathcal{N}(\boldsymbol{\phi} | \hat{\boldsymbol{\phi}}, Q_{\boldsymbol{\phi} | \mathbf{y}}^{-1}),$$

where $\hat{\boldsymbol{\phi}}$ is the posterior mode, and the precision matrix, $Q_{\boldsymbol{\phi} | \mathbf{y}}$, is

$$Q_{\boldsymbol{\phi} | \mathbf{y}} = \left(- \frac{d^2 \log(\pi(\boldsymbol{\phi} | \mathbf{y}))}{d\phi_i d\phi_j} \right)_{i,j} \Big|_{\boldsymbol{\phi} = \hat{\boldsymbol{\phi}}}.$$

A new proposal, $\boldsymbol{\phi}^*$, is drawn from the following proposal density:

$$J_{\boldsymbol{\phi}, t}(\boldsymbol{\phi}^* | \boldsymbol{\phi}^{t-1}) = \mathcal{N}(\boldsymbol{\phi}^* | \boldsymbol{\phi}^{t-1}, c Q_{\boldsymbol{\phi} | \mathbf{y}}^{-1}),$$

where $c = 2.38^2 / \dim(\boldsymbol{\phi})$ is a scaling factor. This sampling scheme is applied in Sect. 4 to the model presented here, i.e., the generalised Pareto model with a spatial scale parameter. Modified versions of this sampling scheme are applied to the other three models that are specified in Sect. 3.4. This sampling scheme was proposed by Roberts et al. (1997). It works well when the posterior density is close to being Gaussian.

4 Results

The magnitudes of the earthquakes that took place in Southwest Iceland from 1904 to 2019 (see Sect. 2) were analysed by applying the two Bayesian latent Gaussian models and the two simple models presented in Sect. 3. The model parameters of the four models were inferred separately using the posterior sampling scheme in Sect. 3.5. For each model, four chains were used, each chain consisting of 105,000 iterations, thereof 5,000 for burn-in. Convergence in the posterior simulations was assessed with the Gelman–Rubin statistic (Gelman & Rubin, 1992; Gelman et al., 2013) and visually through the trace plots, the histograms, and the sample autocorrelation functions of the parameters. Convergence in the posterior simulations was reached according to these plots and the Gelman–Rubin statistic.

Table 2 shows results based on four models presented in Table 1, in particular, the Deviance Information Criterion (DIC) and the effective number of parameters (p_D) (Spiegelhalter et al., 2002, see also chapter “Bayesian Modeling in Engineering Seismology: Ground-Motion Models”). Table 2 also shows the posterior medians and the 95% posterior intervals for the shape parameters in the two models that are based on the generalised Pareto distribution. DIC is used to compare the predictive performance of two or more models that are applied to the same dataset. DIC takes into account both the fit of a model to the data and its complexity. The fit is measured with the data density, and the model complexity is measured with the effective number of parameters. Overfitting is avoided since a too complex model that provides a good fit to the data will have a too large effective number of parameters. A model that is too simple will not be selected since the small effective number of parameters will not compensate for the model’s poor fit. The best model, according to DIC, strikes a balance between the fit to the data and the model complexity.

The model, which assumes the generalised Pareto distribution with scale parameter that varies with longitude, performs best out of these four models with respect to DIC; see Table 2. The model that assumes the generalised Pareto distribution with a constant scale parameter over the region outperforms the model that assumes the exponential distribution with a scale parameter that varies with longitude. The model that assumes the exponential distribution with a constant scale parameter over the region is the least favourable of these four models. Figure 3 shows several quantiles of earthquake magnitude as a function of longitude for each of the four models; we refer to them here as quantile curves. The match between the estimated quantiles and the magnitudes of the earthquakes in the ICEL-NMAR catalogue is most convincing for the model that assumes the generalised Pareto distribution with a scale parameter that varies with longitude. The plotted quantile curves correspond to the 0.05, 0.25, 0.50, 0.75, and 0.95 quantiles. Therefore, one can expect about 5% of the data points to be below the 0.05 quantile curve; 5% above the 0.95 quantile curve; and 20%, 25%, 25%, and 20% between the four strips defined by the five quantile curves, the first being the 0.05 quantile curve and the last one being the 0.95 quantile curve. The total number of earthquakes is $n = 113$, and the model that assumes the generalised Pareto distribution with the scale parameter varying with

Table 2 Comparison of the four models: (i) exponential distribution with constant σ ; (ii) generalised Pareto distribution with constant σ ; (iii) exponential distribution with spatially varying σ ; (iv) generalised Pareto distribution with spatially varying σ . The table shows the Deviance Information Criterion (DIC), the effective number of parameters (p_D), the posterior estimate ($\hat{\xi}$), and the 95% posterior interval for ξ in the generalised Pareto models

Distribution	Description	$\hat{\xi}$	95% post. interval	DIC	p_D
Exponential	Constant σ			136.07	0.99
Generalised Pareto	Constant σ	−0.27	(−0.38; −0.11)	126.63	1.85
Exponential	Spatial σ			135.19	1.39
Generalised Pareto	Spatial σ	−0.60	(−0.74; −0.40)	97.41	4.65

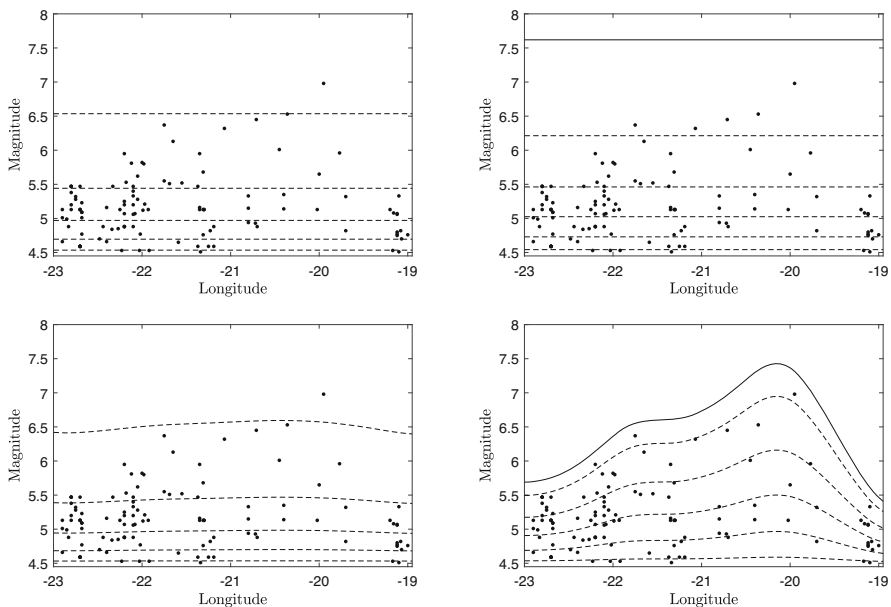


Fig. 3 The dashed lines show the 0.05, 0.25, 0.50, 0.75, and 0.95 quantiles of the earthquake magnitudes as a function of longitude according to the four models. The plots in the upper row correspond to models that assume the magnitudes are constant with respect to longitude, while in the lower row plots, it is assumed that the magnitudes vary with longitude. In the left column, the observed earthquake magnitudes above 4.49 are assumed to follow an exponential distribution, while in the right column, the magnitudes above 4.49 are assumed to follow a generalised Pareto distribution. The solid lines in the right column show the upper bounds of the two generalised Pareto models as a function of longitude

longitude is the one that gives the closest match between the fraction of the data falling within the strips and the above percentages.

Figure 4 shows the two main features of the model that assumes the generalised Pareto distribution with the spatially varying scale parameter, namely, the scale parameter and the upper bound as functions of longitude. The width and shape of the 95% posterior intervals support the claim that the shape parameter should vary with the longitude. The posterior median of the scale parameter is between 0.72 and 1.12 in the western part of the study area. The scale parameter stretches up to 1.77 in the eastern part of the Southwest Iceland but goes down to 0.56 in the easternmost part, which is from different origins (volcanic). The largest value of the upper bound is 7.42 according to the posterior median; however, the 95% posterior intervals suggest that the largest upper bound is in the interval between 7.04 and 8.44. The value 7.04 is essentially identical to the largest reported earthquake magnitude in the transform zone, while the value 8.44 is completely unrealistic given the physical tectonic constraints of the crust in the region. The 50% posterior interval suggests more plausible values for the upper bound, namely, values between 7.24 and 7.67.

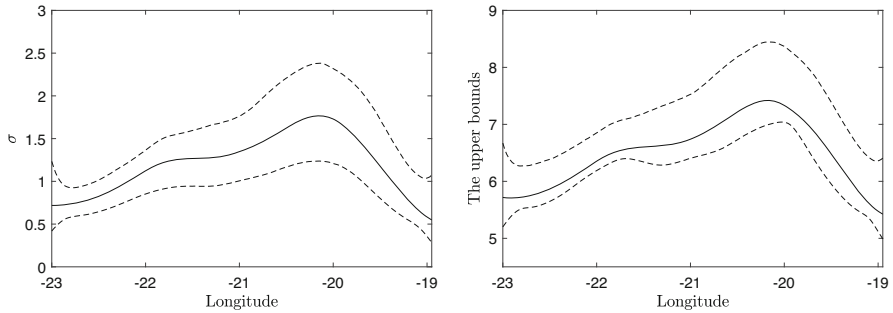


Fig. 4 The left and right plots show the scale parameter, σ , and the upper bound of the earthquake magnitude, respectively, as a function of longitude when assuming the generalised Pareto distribution with a scale parameter that varies with longitude. The solid lines show the posterior median, while the dashed lines show the 95% posterior intervals as a function of longitude

However, no reports or geological evidence of such large earthquake ruptures are found in the literature.

5 Conclusions

We proposed a Bayesian latent Gaussian model for spatial earthquake magnitude data that assumes the generalised Pareto distribution at the data level and a spatial model based on B-splines for the scale parameter at the latent level. This model was motivated by an exploration of earthquake magnitudes from the ICCEL-NMAR catalogue distributed across Southwest Iceland over the period ranging from 1904 to 2019. An analysis of these data based on the proposed model revealed that the distribution of earthquake magnitude varies with longitude.

The exponential distribution is often used to model earthquake magnitudes. In the case of the dataset analysed here, it is clear that the exponential distribution is an inadequate model. The exponential distribution does not have a natural upper bound. Therefore, it is not an adequate model for earthquake magnitudes in general, since the maximum seismogenic potential is in part limited by the thickness of the seismogenic crust, and earthquake fault rupture along the length may be hampered by the geometry of the fault system and the tectonic forces that drive the seismicity. Thus, it is appropriate to assume that the earthquake magnitude has an upper bound. For simplicity, due to the spatial configuration of the seismicity, we model its variation with longitude. An upper bound is induced within the generalised Pareto distribution by assuming that its shape parameter is negative.

The results here show that the conventional exponential model does not provide as good fit as the generalised Pareto distribution with a spatially varying scale parameter. We suggest that models for earthquake magnitudes include a spatial term;

however, to test the validity of the spatial term, these models should be compared to a model based on a generalized Pareto distribution with its shape and scale parameters assumed constant across the region of interest.

The proposed model could be extended to handle two spatial dimensions for earthquake magnitude datasets with large sample sizes and the events' locations spreading over both of the horizontal dimensions. Furthermore, a spatial model for the shape parameter could be considered. However, such a model would only be suited for datasets with a large number of observations within the region of interest, which is not the case for the ICCEL-NMAR dataset.

The frequency of earthquakes above the threshold needs to be modelled spatially across the region of interest to make the proposed model useful in PSHA. For example, in the case of the Southwest Iceland, the largest earthquakes are located in its eastern part. However, the frequency of earthquakes above the threshold magnitude $\mu = 4.5$ in the western part of the transform zone is higher than that in the eastern part, per unit of area. Future research involves setting forth a Poisson process that describes earthquake occurrence above a plausible reference magnitude, with an intensity that is based on a spatial model for earthquake magnitudes (similar to the one proposed here) and a spatial model for the frequency of earthquakes above the reference magnitude.

The proposed model assumes that the magnitude of each earthquake is exact, meaning it does not consider any potential measurement errors. Nevertheless, in reality, the earthquake magnitudes are estimated quantities, and therefore, there is uncertainty about their exact sizes (e.g., Aki & Richards, 1980). Thus, future research should aim to extend the proposed model to incorporate the uncertainty surrounding the earthquake magnitude.

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Improving Numerical Weather Forecasts by Bayesian Hierarchical Modelling



Joshua Lovegrove and Stefan Siebert

1 Introduction

Reliable modelling and prediction of complex interconnected systems such as the human body, galaxies, or the Earth's climate system has been a pipe dream of generations of scientists. However, the dream of a reality that is mathematically consistent, and thus fully predictable, and ultimately computable and controllable has been shattered by advances in mathematics and computer science (e.g., Chaitin, 1995; Moore, 1990). Fundamental uncertainties in predictions about the future of a complex system arise from imperfect knowledge of initial conditions, boundary conditions, and the detailed physical mechanisms that govern the system. But on a practical level, mathematical simplifications, missing system components, mis-specifications of the model, as well as numerical approximations also contribute to imperfect predictions. It is now accepted, and even expected, that predictions about complex real-world phenomena have to include information about these inherent uncertainties and limitations of predictions (Stainforth et al., 2007; Slingo & Palmer, 2011). Probabilistic forecasting, i.e., communicating probability distributions over possible future events, has therefore become common practice in many domains such as finance (Ohlson, 1980), epidemiology (Held et al., 2017), hydrology (Tartakovsky, 2007), and, particularly relevant for this chapter, weather and climate prediction (Buizza, 2008; Gneiting & Katzfuss, 2014)

With the explosive growth of computing power in the second half of the twentieth century, detailed simulations of large numerical models of the atmosphere have become feasible. The horizontal resolution of global coupled atmosphere-ocean models has been increased from 500 to 20 km since the 1970s, leading to a several

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orders of magnitudes increase in the number of numerical grid cells, all the whilst including ever more key physical processes such as aerosol chemistry and vegetation (Le Treut et al., 2007). Limited area regional climate models are now being run at sub-kilometre resolutions that allow representation of individual convective clouds (Kendon et al., 2017). Short-term weather prediction models are now detailed enough to represent turbulent flow on temporal and spatial scales small enough so as to be useful for wind farm operation and management (Gilbert et al., 2020). Exponential growth in computer model detailedness and resolution also happened in other scientific fields such as galaxy simulations (Kim et al., 2013) and agent-based modelling (Abar et al., 2017).

Problems with computer simulation models become apparent when the model is confronted with real-world observations to which the model was not explicitly tuned (such as the future). Predicting future, yet unseen, values makes shortcoming and failures of computer models especially apparent. Some researchers have even warned against using computer simulation for prediction altogether and limit their usage to analysis and exploration only (Oreskes, 2000).

In weather and climate prediction, despite careful parameter tuning and integration of data from multiple sources of information, numerical dynamical models usually do not provide accurate and reliable predictions. Typically, observed errors include unconditional biases (e.g., the model predicts too warm or too wet), conditional biases (e.g., the model predicts too warm in summer and too cold in winter), scaling errors (e.g., a one degree increase in model temperature does not correspond to a one degree increase in real-world temperature), drifts (e.g., the total energy content of the model state increases with run time), ensemble dispersion errors (e.g., multiple model runs started from perturbed initial conditions do not diverge at the same rate as the mean prediction errors increases), or lack of skill (e.g., vanishing correlation between predicted and observed state). The field of *forecast verification* (Jolliffe & Stephenson, 2012) is concerned with developing methodology to quantify these different errors, and *forecast recalibration* or *forecast postprocessing* (Siegert & Stephenson, 2019; Li et al., 2017) refers to using statistical methods to adjust numerical model forecast to better fit the observations by learning from the past prediction errors.

Forecast biases can often not be easily reduced or corrected by more careful model parameters tuning, using more and better data, or increasing the level of detail of the model. Statistical postprocessing is then a last resort to correct biases after the forecast model run has finished. In Sect. 2, we analyse seasonal temperature forecast data to motivate and illustrate the methods and ideas presented in this chapter. Section 3 focuses on forecast postprocessing methods, which fit numerical model output to observations with the aim to improve forecasts in the future. Output fields from numerical climate models have a rich correlation structure in space and time and across variables, which should be exploited by forecast postprocessing. Section 4 therefore presents an application of Bayesian hierarchical modelling with spatial priors for forecast postprocessing, based on recent developments by Hrafnkelsson et al. (2021) and Jóhannesson et al. (2022). Section 5 concludes the chapter with a discussion and directions for further research.

2 The Data

Forecasts of the atmosphere–ocean system can come in numerous flavours and can be distinguished along different dimensions. For example, deterministic point forecasts provide only a single trajectory of the future behaviour, whereas ensemble forecasts provide several possible solutions to quantify the prediction uncertainty, and probabilistic forecasts assign a probability distribution over all possible outcomes. Continuous forecast targets can be real-valued (such as temperature), whereas categorical forecasts only have a finite number of possible outcomes (such as precipitation type). All these different forecast types have to be evaluated by different forecast metrics. Jolliffe and Stephenson (2012) provide a complete and detailed summary of forecast types and strengths and weaknesses of the associated verification metrics.

To illustrate some of the shortcomings of numerical prediction models, we look at a specific example of temperature forecasts produced by a global ensemble forecasting system.

Key information about the model and observations used in this study are summarised in Table 1. The ERA-Interim reanalysis (Dee et al., 2011) supplied the “observations.” Reanalysis is used to solve the problem of incomplete historical records by combining past observations with weather model output to fill in missing data in a physically reasonable way. Retrospective forecasts were downloaded from the S2S (subseasonal-to-seasonal) database (Vitart et al., 2017) via the ECMWF data portal (ECMWF, 2021a). The study region for the data analysed in Sect. 4 was defined over Europe (34.5°N–69°N, 24°W–60° E aggregated to 1.5°) with 1368 grid points in total. Ensemble mean forecasts were calculated from ensembles with 10 members initialised on 29 June in each year and run up to 46 days into the future. That is, temperature on 30 June is forecast at 1 day lead time, 1 July at 2 days lead time, etc. Since models are continuously updated and since every reforecast data set is produced for one particular real-time forecast and model version, there is only one reforecast run per year with the same model version, and hence our sample size

Table 1 Summary of historical forecasts and verifying observation data used in this study

Variable	Daily mean 2-m temperature (T2M)
Forecast model	ECMWF IFS CY43R1 (ECMWF, 2021b)
Observation data	ERA-Interim reanalysis (Dee et al., 2011)
Initialisation date	29 June
Time period	2000–2019
Forecast lead time	1, 2, . . . , 46 days
Horizontal resolution	16 km up to day 15, 32 km after day 15
Vertical resolution	91 vertical levels
Temporal resolution	12-minute integration time step
Spatial domain	Global
Domain of interest	34.5°N–69°N, 24°W–60°E

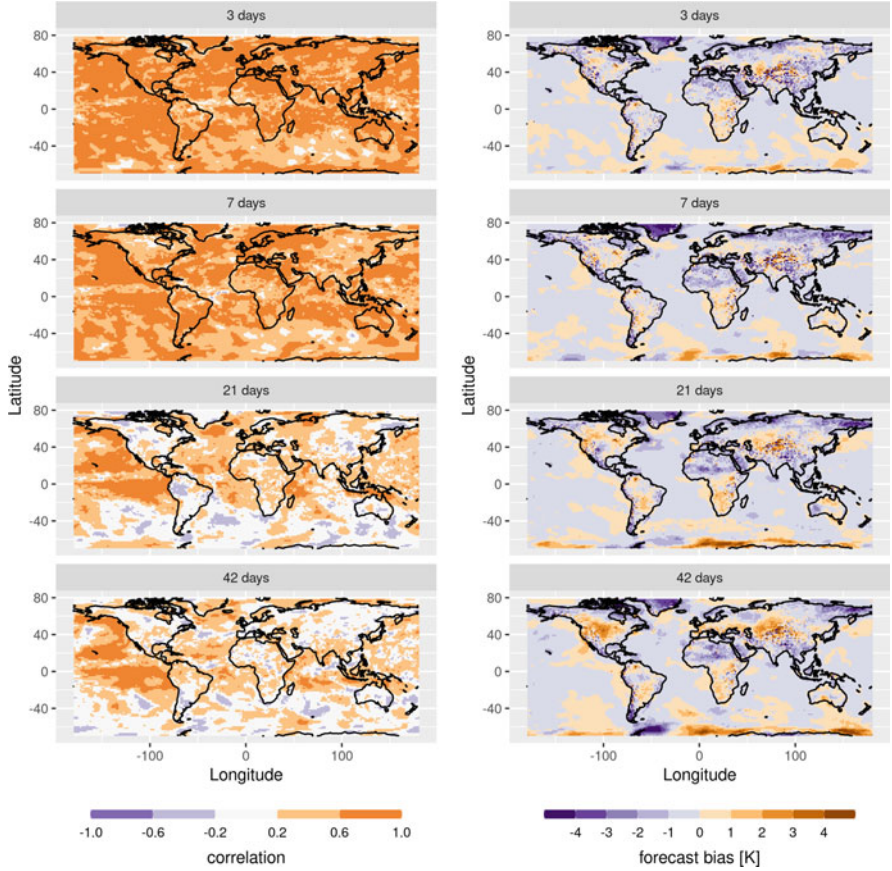


Fig. 1 Gridded maps of correlation (left) and bias (right) of daily average temperature forecasts versus reanalysis, at forecast lead times 3, 7, 21, and 42 days ahead (top to bottom). All forecasts were initialised on 29 June each year from 2000 to 2019 ($n = 20$) (Coarse-grainings of colour scales were applied to improve representation)

is $n = 20$ (see Manoussakis & Vitart, 2022 for details). Such a data set is a typical hindcast data set used in retrospective studies of weather forecast skill in terms of its spatial/temporal resolution and time period covered, although a long forecast range of 46 days is a special feature of the S2S database.

Figure 1 illustrates correlation and forecast bias, two key summary measures of the performance of the forecasting system. The correlation between forecast and observations is generally high at short lead times, indicating high predictability of temperature compared to the climatological mean. Largely due to sensitivity to initial conditions, correlation decays as a function of forecast lead time, leading to a decrease in forecast accuracy, ultimately rendering the predictions “useless” at 3 weeks lead time. There are pockets of long-range predictability, with high

correlation even at 42 days, in known regions of large scale ocean–atmosphere interactions such as the El-Niño Southern Oscillation (ENSO) in the tropical Pacific.

Forecast biases, defined as the time-averaged difference between forecast and observation, are a common and easily identifiable indicator of model discrepancy. The biases shown in Fig. 1 change as a function of location and lead time and appear to be spatially structured. Some biases are persistent for all lead times, such as the cold bias over Greenland, and some bias patterns are amplified over time, such as the warm biases over North America and the Southern Ocean. The next section will review methods that reduce forecast shortcoming through statistical modelling.

3 Statistical Methods for Quantifying and Correcting Forecast Errors

Managing forecast quality through statistical methods can be thought of in terms of two (related) fields, namely *forecast verification*, which is the identification and quantification of forecast errors through suitable performance metrics, and *statistical postprocessing*, which refers to the correction of forecast errors by statistical modelling.

Forecast verification is a mature field in the meteorological sciences (Jolliffe & Stephenson, 2012). Some of the earliest work on suitable reward functions for the assessment of forecasters came from the Earth Sciences (Brier, 1950). Scoring rules are mathematical functions that quantify how closely a forecast matches its verifying observation. A *proper* scoring rule is a scoring rule that, by virtue of its mathematical properties, forces an expectation maximising forecaster to state their honest belief, rather than adjusting the forecast to obtain a better score (Gneiting & Raftery, 2007). Today, the early developments in the assessment of weather forecasts echo across various fields such as medical research, hydrology, and economics (e.g., Linnet, 1989; Weijs et al., 2010; Armantier & Treich, 2013). Much research effort is spent on the development of new tools to identify and quantify different forecast errors for specific applications, such as extreme values (Lerch et al., 2017), spatial fields (Gilleland et al., 2009), and point forecasts (Ehm et al., 2016). There are specific tools to quantify the frequentist properties of probabilistic forecasts, such as the reliability diagram, and the reliability–resolution–uncertainty decomposition of verification scores (Siebert, 2017). In addition to identifying forecast shortcomings, an important application of forecast verification is the comparison of forecast skill between different forecasters or forecast models (Diebold & Mariano, 2002; Siebert et al., 2017).

Evidently, the identification and quantification of forecast errors is only the first step. Knowledge about weaknesses of a forecast system, gained through forecast verification, has to be communicated to users honestly, to manage expectations and inform choices. Ideally, though, knowledge about forecast errors should be used

to correct them through statistical postprocessing and thereby make forecasts more valuable to end users.

In statistical postprocessing, a training set of past forecast and observations is used to learn a suitable transformation that can be applied to future forecasts to correct their errors and improve their predictive accuracy and reliability. The removal of a constant forecast bias is the most obvious and straightforward statistical postprocessing method: if a temperature forecast is known to have been 2° too cold historically, it is rational to add 2° to future forecasts to correct for the cold bias.

Model output statistics (MOS, Glahn & Lowry, 1972; Glahn et al., 2009) is one of the earliest forecast postprocessing techniques. Glahn and Lowry (1972) defined MOS as “an objective weather forecasting technique which consists of determining a statistical relationship between a predictand and variables forecast by a numerical model at some projection time(s).” That is, a statistical model is fitted to explain a predictand in the real world by one or several outputs from the numerical model which act as covariates. This is done using a multiple linear regression in which predictands, such as observed values of 2-m air temperature, wind speed, or probability of precipitation, are related to predictors, which are generally raw forecasts from a numerical weather prediction model, but might also include latitude, longitude, altitude, a land–sea indicator, or various other external factors.

In the simplest case, MOS for post processing is a simple linear regression of the observation y_t at time t on the forecasts f_t issued by the numerical model for time instance t and the same forecast variable, i.e.,

$$y_t = \beta_0 + \beta_1(f_t - \bar{f}) + \sigma\epsilon_t,$$

where $\bar{f} = \frac{1}{n} \sum_t f_t$ and $\epsilon_t \sim \mathcal{N}(0, 1)$. Centering the forecasts by subtracting their time averages $\bar{f}$ simplifies some of the parameter inference.

Using maximum likelihood estimation yields the regression parameter estimators $\hat{\beta}_0 = \bar{y} = \frac{1}{n} \sum_t y_t$, $\hat{\beta}_1 = \frac{s_{fy}}{s_{ff}} = \frac{\sum_t (f_t - \bar{f})(y_t - \bar{y})}{\sum_t (f_t - \bar{f})^2}$, and $\hat{\sigma}^2 = \frac{1}{n} \sum_t (y_t - \hat{y}_t)^2$, where $\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1(f_t - \bar{f})$ is the t -th fitted value. (Bayesian inference based on reference priors results in the same estimators.) The predictive distribution for a new observation y^* based on a new forecast f^* is a Student's t -distribution with $n - 2$ degrees of freedom, centred on the predicted mean $\hat{\beta}_0 + \hat{\beta}_1(f^* - \bar{f})$ and scaled by the predictive standard deviation $\hat{\sigma} \left(1 + \frac{1}{n} + \frac{(f^* - \bar{f})^2}{\sum_t (f_t - \bar{f})^2}\right)^{1/2}$. More discussion of the probabilistic skill of the MOS predictive distribution can be found in Siebert et al. (2016).

Figure 2 shows the maximum likelihood estimates of the linear regression (MOS) parameters when fitted separately for each grid point and lead time of the ECMWF temperature forecasts. The fitted intercept $\hat{\beta}_0$ varies considerably in space but not much over lead time. The slope coefficient $\hat{\beta}_1$ shows more small scale variations in space than $\hat{\beta}_0$ and also changes visibly over lead time. The scale parameter $\hat{\sigma}$ is similarly variable in space and generally increases with lead time. In Sect. 4, we will

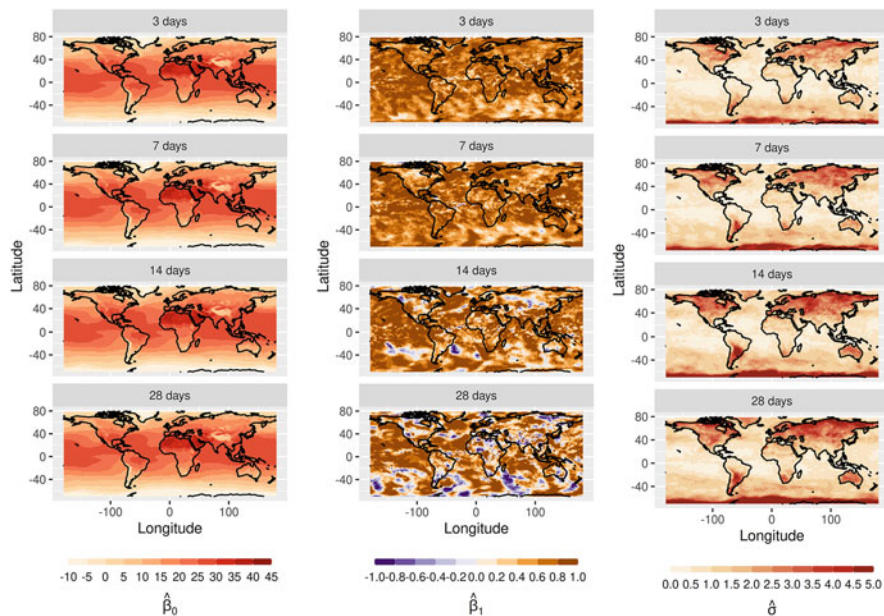


Fig. 2 Grid pointwise MOS parameter estimates $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\sigma}$ (from left to right) for 2-m temperature forecasts at lead times 3, 7, 14, and 28 days (from top to bottom)

motivate and apply a spatial smoothing model to these estimates and show that this leads to slight improvements of out-of-sample performance of the postprocessed forecasts. In the remainder of this section, we discuss a number of extensions of MOS for postprocessing ensemble forecasts and spatially extended forecasts, but the main developments in Sect. 4 will focus on simple MOS.

Various attempts have been made to extend MOS so that it could be used for the postprocessing of different types of forecasts. For ensemble forecasts, where several forecasts started from perturbed initial conditions are available for the same observation, an intuitive approach is to apply MOS to each ensemble member individually. However, this method leads to results that converge towards the climatological mean forecast (Wilks, 2006). A further problem is that the classical regression involved in MOS assumes that the predictand follows a Gaussian distribution with constant variance. In a non-linear dynamical system, one should expect to see varying levels of predictability (e.g., weather in the middle of a stable heat wave is better predictable than during rapidly changing weather in the middle of a cyclone). One would then expect that these different levels of predictability are represented by the ensemble spread, where, e.g., a larger ensemble variance implies more uncertainty in the forecast and, hence, greater predictive error. The spread–skill relationship of ensemble forecasts has been discussed in detail by Hopson (2014).

To make use of the ensemble spread information, Gneiting et al. (2005) proposed ensemble model output statistics (EMOS), also called nonhomogeneous Gaussian regression (NGR). The formulation of this method is similar to that of MOS; the observations y_t are written as linear functions of the ensemble members $f_{t,i}$, where t is time and i is the ensemble member index. Furthermore, the variance of the residuals is modelled as a linear function of the ensemble variance, and therefore the predictive variance is no longer constant. Mathematically, NGR can be written as Wilks (2018)

$$\begin{aligned} y_t &\sim \mathcal{N}(\mu_t, \sigma_t^2), \\ \mu_t &= \beta_0 + \beta_1 m_t, \text{ and} \\ \sigma_t^2 &= \gamma + \delta s_t^2, \end{aligned}$$

where m_t is the ensemble average forecast at time t , and s_t^2 is the ensemble variance at time t . The variance parameters γ and δ must be constrained to ensure that σ_t^2 is non-negative. Note that in the absence of a spread–skill relationship, we expect $\delta = 0$, which would recover the MOS forecast distributions with constant variance.

Gneiting et al. (2005) used NGR to produce probability forecasts of sea level pressure and found that the NGR forecasts were better calibrated than the raw NWP predictions. For example, the coverage of prediction intervals was more accurate. However, it was also reported that the estimate for the parameter δ was negligibly small in many instances, suggesting no systematic relationship between forecast error and ensemble spread, and hence the simple MOS model may be sufficient. Despite this, if a relationship between the predictive error and the ensemble spread does exist, then it may reasonably be expected that NGR forecasts will be more skillful than those given by MOS.

A further limitation of MOS lies in the assumption that the forecasting variable must have a Gaussian distribution (conditional on the ensemble forecasts). Extensions for the non-Gaussian case have been proposed but will not be explored here (see Li et al., 2017; Wilks, 2018).

Spatial NGR (Feldmann et al., 2015) combines the ideas of univariate NGR with the “geostatistical output perturbation” (GOP) approach (Gel et al., 2004) to account for the spatial correlations in numerical forecast fields. The GOP approach was introduced as an inexpensive substitute of a dynamical ensemble based on a single numerical weather prediction. The method dresses the deterministic forecast with a simulated forecast error field according to a spatial random process. The result of combining GOP with univariate NGR is a multivariate predictive distribution that is able to generate spatially coherent forecasts whilst retaining the univariate NGR marginals. An alternative approach to generating spatially coherent multivariate postprocessed forecasts is the Schaake shuffle (Clark et al., 2004; Schefzik et al., 2013), which considers only the inputs and outputs of a univariate postprocessing method. Univariate MOS involves postprocessing the raw forecasts at each grid point individually, such that any existing spatial structure is lost. The Schaake shuf-

file retains spatial structure by reconstructing Spearman's rank correlation structure between locations based on the rank ordering of the raw forecasts. Jóhannesson et al. (2022) illustrate the benefit of rank reordering in an application to flood frequencies using a Bayesian hierarchical modelling.

4 Spatial Statistical Modelling for Forecast Postprocessing

4.1 A Bayesian Hierarchical Modelling Framework

Figure 1 shows that correlations and biases between forecasts and observations can differ substantially between different spatial locations and different lead times. In statistical postprocessing, we should thus allow regression parameters (and other details of the postprocessing model) to be different for different locations and lead times, rather than using a one-size-fits-all approach to postprocessing for all forecasts. However, when fitting the postprocessing models individually for all locations and lead times, as was done in Fig. 2, each set of regression parameters is only estimated from the 20 forecast–observation pairs from that location and lead time. Hence, despite the large number of available forecast–observation pairs, we find ourselves in a small sample size situation, where estimation uncertainty in the model parameters is likely to have an effect on the quality of the postprocessed forecast. We should look for ways to use the abundance of data better to reduce estimation uncertainty in postprocessing parameters.

Also, in Fig. 1, we see that correlations and biases at nearby locations, whilst different, tend to be similar to each other. Considering the strong spatial dependency of the forecast and observation data, spatial correlations of fitted parameters are not surprising. In the light of such strong spatial similarities, fitting postprocessing models independently at each location and lead time seems like a poor use of the available information. In this section, we propose a postprocessing model that is flexible enough to allow parameters at a different location to be different whilst also exploiting the spatial similarities by *borrowing strength* from data at nearby locations. In essence, we increase the effective sample size by making the regression parameter estimates at a grid point not only dependent on the hindcast data at that grid point but also dependent on data at neighbouring grid points. In this chapter, we will not account for correlations across the forecast lead time dimension, but the method could well be extended to that effect.

A practical approach for borrowing strength across space for parameter estimation is by applying a Bayesian hierarchical model consisting of three levels: response level, latent level, and hyperparameter level.

At the *response level*, we assume statistical models for the observed data that are conditionally independent between individual grid points, given the model parameters. Denote by $\mathbf{y}_s = (y_{s,1}, \dots, y_{s,N_t})'$ and $\mathbf{f}_s = (f_{s,1}, \dots, f_{s,N_t})'$ the vectors of observations and numerical model forecasts, respectively, at grid point

$s = (i, j)$ with longitude index i and latitude index j . Furthermore, denote by the vector $\mathbf{x}_s = (\alpha_s, \beta_s, \tau_s)$ the 3-vector of regression coefficients at location s . The likelihood function for the regression parameters at grid point s is then given by the joint conditional probability of the data:

$$\begin{aligned} L(\mathbf{x}_s; \mathbf{y}_s, \mathbf{f}_s) &= \prod_{t=1}^{N_t} p(y_{s,t} | \mathbf{x}_s) \\ &= (2\pi e^{\tau_s})^{-N_t/2} \exp \left\{ -\frac{e^{-\tau_s}}{2} \sum_{t=1}^{N_t} [y_{s,t} - (\alpha_s + \beta_s(f_{s,t} - \bar{f}_s))]^2 \right\}. \end{aligned} \quad (1)$$

Centering the model predictions by subtracting their mean $\bar{f}_s = N_t^{-1} \sum_t f_{s,t}$ is not strictly necessary, but it eliminates the asymptotic correlation of the maximum likelihood estimators for α_s and β_s , which will be convenient later on. Maximising (1) with respect to α_s , β_s , and τ_s yields the usual OLS estimators of the regression parameters at location s .

At the *latent level*, the regression parameters collected in the vector $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_{N_s})'$ are modelled as spatially correlated random variables. Their spatial correlation is encoded in their joint prior distribution $p(\mathbf{x} | \boldsymbol{\theta})$ which can depend on hyperparameters $\boldsymbol{\theta}$. Here, we assume that the vectors $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_{N_s})'$, $\boldsymbol{\beta} = (\beta_1, \dots, \beta_{N_s})'$, and $\boldsymbol{\tau} = (\tau_1, \dots, \tau_{N_s})'$ are independent, so that their joint prior distribution is

$$p(\mathbf{x} | \boldsymbol{\theta}) = p(\boldsymbol{\alpha} | \boldsymbol{\theta}_\alpha) p(\boldsymbol{\beta} | \boldsymbol{\theta}_\beta) p(\boldsymbol{\tau} | \boldsymbol{\theta}_\tau),$$

where the hyperparameters are $\boldsymbol{\theta} = \{\boldsymbol{\theta}_\alpha, \boldsymbol{\theta}_\beta, \boldsymbol{\theta}_\tau\}$.

Here, we model $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\tau}$ by 2-dimensional random walk (RW2D) models. The RW2D for the field $\boldsymbol{\alpha}$ is defined through a conditional Gaussian model for each $\alpha_{i,j}$. When (i, j) is an interior point of the grid, the distribution of $\alpha_{i,j}$, conditional on other elements in $\boldsymbol{\alpha}$, is defined as

$$\alpha_{i,j} | \boldsymbol{\alpha}_{-(i,j)} \sim \mathcal{N} \left(\frac{1}{4} (\alpha_{i+1,j} + \alpha_{i-1,j} + \alpha_{i,j+1} + \alpha_{i,j-1}), \frac{1}{4} \theta_\alpha^{-1} \right), \quad (2)$$

where θ_α is a precision parameter. The fields $\boldsymbol{\beta}$ and $\boldsymbol{\tau}$ are defined similarly. The RW2D model assumes that the value at grid point (i, j) is equal to the average over its four nearest neighbours plus an independent Gaussian distributed “innovation.” Large values of the precision parameter θ_α produce a 2D spatial field $\boldsymbol{\alpha}$ that is smooth, whereas small values of θ_α lead to a rough spatial field with weak spatial dependency. The RW2D model in (2) implies that the vector $\boldsymbol{\alpha}$ has a multivariate Gaussian distribution with a sparse and rank deficient precision matrix $\mathbf{Q}_\alpha$, making it an *intrinsic Gaussian Markov Random Field* (IGMRF) (see Rue & Held, 2005 for details). The joint density of the vector $\boldsymbol{\alpha}$ under an RW2D prior is proportional to

$$p(\boldsymbol{\alpha}|\theta_\alpha) \propto \theta_\alpha^{(N_s-1)/2} \exp \left\{ -\frac{1}{2} \boldsymbol{\alpha}' Q_\alpha \boldsymbol{\alpha} \right\}.$$

Sparseness of the precision matrix enables use of sparse methods for solving linear systems, which allows for efficient computational inference. In this chapter, we use independent RW2D priors for each $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\tau}$, with smoothness hyperparameters θ_α , θ_β , and θ_τ .

To complete the Bayesian hierarchical model, we specify the hyperparameter level in the form of a prior distribution $p(\boldsymbol{\theta}) = p(\theta_\alpha, \theta_\beta, \theta_\tau)$. We assume that the precision hyperparameters are a priori independent and have Exponential prior distributions with rate parameter fixed at $\lambda = 5 \times 10^{-5}$:

$$p(\boldsymbol{\theta}) = p(\theta_\alpha)p(\theta_\beta)p(\theta_\tau) \propto e^{-\lambda(\theta_\alpha+\theta_\beta+\theta_\tau)}.$$

We have specified the likelihood function $p(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})$, the prior distribution at the latent level $p(\mathbf{x}|\boldsymbol{\theta})$, and the prior for the hyperparameters $p(\boldsymbol{\theta})$. The joint distribution of all random quantities in our model is given by the product

$$p(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta}) = p(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta}) p(\mathbf{x}|\boldsymbol{\theta}) p(\boldsymbol{\theta}).$$

We want to learn about the latent variables $\mathbf{x}$ (the postprocessing parameters) by inferring their marginal posterior distribution, given by

$$p(\mathbf{x}|\mathbf{y}) = \int p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y}) p(\boldsymbol{\theta}|\mathbf{y}) d\boldsymbol{\theta}. \quad (3)$$

The posterior distribution of the latent variables $\mathbf{x}$ conditional on hyperparameters is given by

$$p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y}) = \frac{p(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta})}{\int p(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta}) d\mathbf{x}}, \quad (4)$$

and the marginal posterior distribution of the hyperparameters is given by

$$p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta})}{p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})p(\mathbf{y})}. \quad (5)$$

Computationally, our challenge will be to calculate the integrals in (3) and (4) as well as the normalisation constant $p(\mathbf{y})$ in (5).

In the following, we sketch the derivation of the “Max-and-Smooth” method for our particular application; see Hrafnkelsson et al. (2021) for the derivation of the method for general latent Gaussian models with a multivariate link function. We will first focus on the posterior distribution $p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$. To make the inference tractable, we apply a Laplace approximation to the likelihood function $p(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})$. That is, we approximate $p(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta})$ by a Gaussian distribution in $\mathbf{x}$. To this end, we first Taylor-expand the local log-likelihood $f(\mathbf{x}_s) = \log p(\mathbf{y}_s|\mathbf{x}_s, \boldsymbol{\theta})$ around the

mode $\hat{\mathbf{x}}_s = \underset{\mathbf{x}_s}{\operatorname{argmax}} f(\mathbf{x}_s)$ to second order:

$$f(\mathbf{x}_s) = \log p(\mathbf{y}_s | \mathbf{x}_s, \boldsymbol{\theta}) \approx \text{const} + \frac{1}{2}(\mathbf{x}_s - \hat{\mathbf{x}}_s)' H_s(\hat{\mathbf{x}}_s)(\mathbf{x}_s - \hat{\mathbf{x}}_s),$$

where the mode $\hat{\mathbf{x}}_s = (\hat{\alpha}_s, \hat{\beta}_s, \hat{\tau}_s)'$ is the vector of local MLEs with elements

$$\begin{aligned}\hat{\alpha}_s &= \bar{y}_s, \\ \hat{\beta}_s &= \frac{\sum_t (y_{s,t} - \bar{y}_s)(f_{s,t} - \bar{f}_s)}{\sum_t (f_{s,t} - \bar{f}_s)^2}, \\ \hat{\tau}_s &= \log \frac{1}{N_t} \sum_t (y_{s,t} - \hat{f}_{s,t})^2,\end{aligned}$$

and where $\hat{f}_{s,t} = \hat{\alpha}_s + \hat{\beta}_s(f_{s,t} - \bar{f}_s)$. The Hessian matrix (the matrix of mixed second derivatives) of $f(\mathbf{x}_s)$, evaluated at the mode, is given by

$$H_s(\hat{\mathbf{x}}_s) = \begin{pmatrix} -N_t e^{-\hat{\tau}_s} & 0 & 0 \\ 0 & -e^{-\hat{\tau}_s} \sum_t (f_{s,t} - \bar{f}_s)^2 & 0 \\ 0 & 0 & -\frac{N_t}{2} \end{pmatrix}. \quad (6)$$

Due to conditional independence at the response level, the full log-likelihood function $f(\mathbf{x}) = \log p(\mathbf{y} | \mathbf{x}, \boldsymbol{\theta})$ is thus approximated by

$$f(\mathbf{x}) \approx \text{const} - \frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}})' \mathbf{Q}_{y|x}(\mathbf{x} - \hat{\mathbf{x}}), \quad (7)$$

where $\hat{\mathbf{x}} = (\hat{\mathbf{x}}'_1, \dots, \hat{\mathbf{x}}'_{N_s})'$ and the block-diagonal matrix

$$\mathbf{Q}_{y|x} = \text{bdiag}(-H(\hat{\mathbf{x}}_1), -H(\hat{\mathbf{x}}_2), \dots).$$

Using the Laplace approximation of the likelihood, the posterior distribution of $\mathbf{x}$ is given by

$$\begin{aligned}p(\mathbf{x} | \boldsymbol{\theta}, \mathbf{y}) &= C_1 p(\mathbf{y} | \mathbf{x}) p(\mathbf{x} | \boldsymbol{\theta}) \\ &= C_2 \exp \left\{ -\frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}})' \mathbf{Q}_{y|x}(\mathbf{x} - \hat{\mathbf{x}}) - \frac{1}{2} \mathbf{x}' \mathbf{Q}_x \mathbf{x} \right\} \\ &= C_3 \exp \left\{ -\frac{1}{2} \mathbf{x}' (\mathbf{Q}_x + \mathbf{Q}_{y|x}) \mathbf{x} + (\mathbf{Q}_{y|x} \hat{\mathbf{x}})' \mathbf{x} \right\} \\ &= C_4 \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_{x|y})' \mathbf{Q}_{x|y}(\mathbf{x} - \boldsymbol{\mu}_{x|y}) \right\}, \quad (8)\end{aligned}$$

where $C_1, \dots, C_4$ are constants that do not depend on $\mathbf{x}$ and (8) is proportional to the density of a multivariate Gaussian distribution with expectation

$$\boldsymbol{\mu}_{\mathbf{x}|\mathbf{y}} = (\mathcal{Q}_{\mathbf{x}} + \mathcal{Q}_{\mathbf{y}|\mathbf{x}})^{-1} \mathcal{Q}_{\mathbf{y}|\mathbf{x}} \hat{\mathbf{x}}$$

and precision matrix

$$\mathcal{Q}_{\mathbf{x}|\mathbf{y}} = \mathcal{Q}_{\mathbf{x}} + \mathcal{Q}_{\mathbf{y}|\mathbf{x}}.$$

We note that the approximated posterior distribution $p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$ is a multivariate Gaussian distribution, from whose expectation vector we can extract point estimates of the regression parameters and from whose covariance matrix we can extract uncertainty estimates. By setting $\mathcal{Q}_{\mathbf{x}} = 0$, we would have $\boldsymbol{\mu}_{\mathbf{x}|\mathbf{y}} = \hat{\mathbf{x}}$, i.e., the posterior expectation is then equal to the maximum likelihood estimates we would get without specifying a spatial prior for $\mathbf{x}$. In general, since both $\mathcal{Q}_{\mathbf{x}}$ and $\mathcal{Q}_{\mathbf{y}|\mathbf{x}}$ are sparse matrices, the posterior precision $\mathcal{Q}_{\mathbf{x}|\mathbf{y}}$ is also sparse, which allows for an efficient computation of $\boldsymbol{\mu}_{\mathbf{x}|\mathbf{y}}$ even if the spatial domain is large. Lastly, it should be noted that the quality of the posterior inference depends on the quality of the Gaussian approximation of the likelihood. If the likelihood function has a strongly skewed or even multi-modal shape, a Gaussian approximation will be poor.

Hrafnkelsson et al. (2021) gave the following alternative interpretation of the approximated posterior derived in (8). Consider a model where the local MLEs $\hat{\mathbf{x}}_s = (\hat{\alpha}_s, \hat{\beta}_s, \hat{\tau}_s)'$ are interpreted as independent “noisy measurements” of the “true” parameter values $\mathbf{x}_s = (\alpha_s, \beta_s, \tau_s)'$. The measurement error model for the MLEs is derived from asymptotic likelihood theory, where the MLEs are asymptotically unbiased, $E[\hat{\mathbf{x}}_s] \approx \mathbf{x}_s$, and their variance is asymptotically equal to the expected information matrix (inverse of the negative Hessian matrix of the log-likelihood, evaluated at the mode). Replacing the expected information by the observed information (Efron & Hinkley, 1978), the MLEs $\hat{\mathbf{x}}$ have a $3N_s$ -dimensional Gaussian distribution, with mean vector equal to the “true” $\mathbf{x}$ and precision matrix equal to $\mathcal{Q}_{\mathbf{y}|\mathbf{x}}$ defined in (7):

$$\hat{\mathbf{x}} \sim \mathcal{N}(\mathbf{x}, \mathcal{Q}_{\mathbf{y}|\mathbf{x}}^{-1}). \quad (9)$$

Analogous to the derivation of (8) the “true” parameter values $\mathbf{x}$ have a joint Gaussian prior distribution with mean zero and precision matrix $\mathcal{Q}_{\mathbf{x}}$

$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathcal{Q}_{\mathbf{x}}^{-1}). \quad (10)$$

Under this measurement error model, the posterior of $\mathbf{x}$ given $\hat{\mathbf{x}}$ is proportional to $\exp \left\{ -\frac{1}{2} \mathbf{x}' (\mathcal{Q}_{\mathbf{y}|\mathbf{x}} + \mathcal{Q}_{\mathbf{x}}) \mathbf{x} + (\mathcal{Q}_{\mathbf{y}|\mathbf{x}} \hat{\mathbf{x}})' \mathbf{x} \right\}$, which is the same posterior we get under approximation of the posterior distribution using a Laplace approximation. This interpretation of the Bayesian hierarchical model suggests that we can infer the posterior distribution of the latent field $\mathbf{x}$ in a two-step procedure:

1. Calculate the local MLEs $\hat{\mathbf{x}}_s$ and the Hessian matrices $H_s(\hat{\mathbf{x}}_s)$ at each location s .
2. Calculate the posterior $p(\mathbf{x}|\hat{\mathbf{x}})$ by combining the surrogate-likelihood $p(\hat{\mathbf{x}}|\mathbf{x})$ from the measurement error model in (9) with the spatial prior $p(\mathbf{x}|\boldsymbol{\theta})$ in (10).

Since the first step is a maximisation step and the inference of the posterior with a spatial prior in the second step usually has a smoothing effect, the two-step method introduced by Hrafnkelsson et al. (2021) and further studied by Jóhannesson et al. (2022) has been dubbed Max-and-Smooth.

In our particular regression model, a further simplification presents itself: since the Hessian $H_s(\hat{\mathbf{x}}_s)$ is diagonal, and our priors for $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\tau}$ are independent, the joint posterior factors into the product of three individual posteriors:

$$p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y}) = p(\boldsymbol{\alpha}|\hat{\boldsymbol{\alpha}}, \boldsymbol{\theta}_\alpha) p(\boldsymbol{\beta}|\hat{\boldsymbol{\beta}}, \boldsymbol{\theta}_\beta) p(\boldsymbol{\tau}|\hat{\boldsymbol{\tau}}, \boldsymbol{\theta}_\tau),$$

and hence Max-and-Smooth can be applied to the three regression parameters individually. That is, the factorisation of the joint posterior means that we can calculate MLEs at each location and subsequently infer $\boldsymbol{\alpha}$ from the MLEs of $\boldsymbol{\alpha}$ and the spatial prior $p(\boldsymbol{\alpha}|\boldsymbol{\theta}_\alpha)$ separately from the posterior distributions of $\boldsymbol{\beta}$ and $\boldsymbol{\tau}$.

The regression parameters can be inferred by posterior inference of the latent variables $\mathbf{x}$ at fixed values of the hyperparameters $\boldsymbol{\theta}$. If the hyperparameters can be fixed at reasonable values or if it can be shown that the effect of changing hyperparameters is negligible for a given application, then inference at fixed hyperparameters is sufficient. If we want to also infer the hyperparameters and propagate their uncertainty into the uncertainty of $\mathbf{x}$, we have to infer the hyperparameters by calculating $p(\boldsymbol{\theta}|\mathbf{y})$ via (5) and propagate their posterior uncertainty to inference about $\mathbf{x}$ by calculating $p(\mathbf{x}|\mathbf{y})$ via (3). To infer $p(\boldsymbol{\theta}|\mathbf{y})$, the normalisation constant $p(\mathbf{y}) = \int [p(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta})/p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})] d\boldsymbol{\theta}$ can be calculated by numerical quadrature because the dimensionality of the hyperparameter space is small. Using the Gaussian approximation of $p(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$, the integral in (3) can be approximated by a finite mixture of Gaussians, with weights derived from $p(\boldsymbol{\theta}|\mathbf{y})$. The details on this procedure, dimensionality considerations, as well as numerical refinements can be found in Rue et al. (2009).

We have shown that the spatial fields of regression parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\tau}$ can be inferred independently and that the inferential framework can be separated into the two-step Max-and-Smooth process. As a result of these simplifications, we can calculate MLEs of the regression parameters and then infer their posterior distributions under a spatial prior using the R-INLA software (Lindgren & Rue, 2015). R-INLA performs fully Bayesian inference in latent Gaussian models using integrated nested Laplace approximations (Rue et al., 2017; Bakka et al., 2018). In the next section, we will use Max-and-Smooth via R-INLA to infer MOS parameters under a spatial prior and show that the resulting parameter estimates improve out-of-sample predictions of European summer surface temperatures.

4.2 *Application of Max-and-Smooth to Statistical Forecast Postprocessing*

The model proposed in Sect. 4.1 constitutes an approximate statistical framework to improve parameter estimates in statistical postprocessing by borrowing strength from data at neighbouring grid points via a spatial prior in a Bayesian hierarchical model. In this section, we will apply the methodology to a specific region of the global data set and evaluate its practical merit through cross validation.

We focus our analysis on the gridded surface temperature forecasts in a region that covers most of Europe and parts of Asia and Africa. We estimate postprocessing parameters by Max-and-Smooth and evaluate their performance, at each forecast lead time individually, i.e., we do not apply any smoothing across the lead time dimension.

For the smoothing step, we use the R-INLA software to approximate the posterior distributions of the regression parameters α , β , and τ under a latent Gaussian model with spatially correlated priors. We first calculate the local MLEs $\hat{\alpha}_s$, $\hat{\beta}_s$, and $\hat{\tau}_s$ with the usual least squares estimators. These are passed to R-INLA and smoothed under a 2D random walk prior. R-INLA calculates approximations of the posterior densities of the latent spatial fields α , β , and τ , and we use the estimated posterior means and marginal posterior variances in the following analyses.

Figure 3 compares the local MLEs $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\tau}$ with the posterior means of α , β , and τ under a 2D random walk prior. A certain amount of smoothing is visible in all parameters, indicating that information is shared between spatially close locations in the desired way. The degree of smoothing differs between the three parameters, with the least degree of smoothing visible in α and most smoothing visible in β and τ . This can be explained by differences in the asymptotic variances of the MLEs which enter the precision matrix $Q_{y|x}$ through the Hessian (cf. (6)). The interpretation of these differences in the context of the fitted LGM is that the “measurements” of α are relatively more precise than β and τ and therefore better constrain the posterior estimates.

Figure 4 illustrates differences between the regression lines before and after spatial smoothing of regression parameters. Depending on the differences of MLEs at neighbouring grid points, the smoothed regression slope can be either larger or smaller than the unsmoothed slope. This shows that the effect of spatial smoothing of parameter is not the same as shrinkage towards zero as in, e.g., ridge regression or lasso regression. Furthermore, the effect of smoothing on the resulting forecast can be substantial. For a given forecast value, the linear response predicted by the regression model can differ by several degrees between the unsmoothed and smoothed parameters.

Figure 5 compares the posterior densities of hyperparameters. The lower posterior mean precision for the field α confirms that the intercept experiences less smoothing than the other parameters as discussed above. The prior for all three hyperparameters is almost uniform, and the posteriors are relatively sharp, which suggests that the hyperparameters are well constrained by the data.

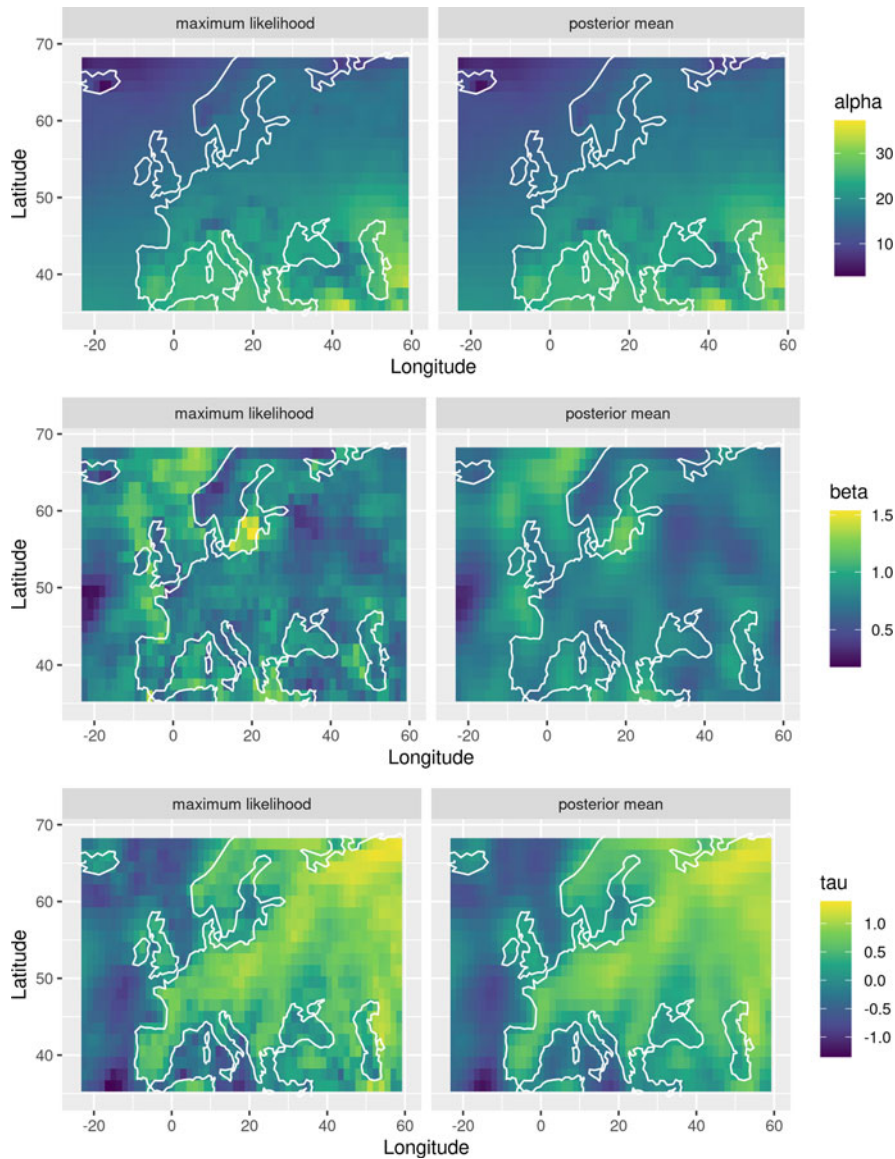


Fig. 3 Comparison of local maximum likelihood estimates of regression parameters (left panels) with posterior means under a spatial RW2D prior (right panels)

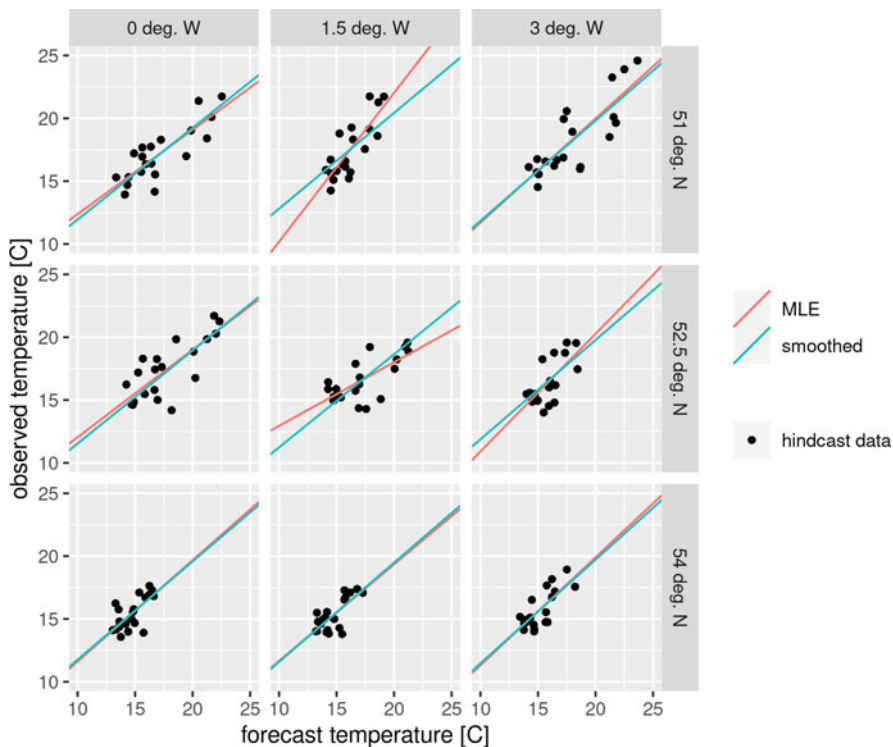


Fig. 4 Illustration of the effect of spatial smoothing of intercept and regression slopes on a 3 by 3 grid point domain

We apply leave-one-out cross validation to assess out-of-sample performance and avoid overfitting, i.e., we fit the MLEs on 19 years, smooth them with R-INLA, and make a prediction for the left-out year by postprocessing the left-out forecast using the smoothed parameters, and compare that prediction to the left-out observation. The procedure is repeated 20 times, leaving out each year in turn.

The goal of the method is to borrow strength from data at neighbouring grid points to improve parameter estimates and ultimately make better postprocessed forecasts. We quantify the effect of spatial smoothing by comparing mean squared errors (MSEs) of the postprocessed predictions before and after smoothing the regression parameters.

$$\text{MSE} = \frac{1}{N_t N_s} \sum_{t=1}^{N_t} \sum_{s=1}^{N_s} \left[y_{s,t} - \tilde{f}_{s,t} \right]^2$$

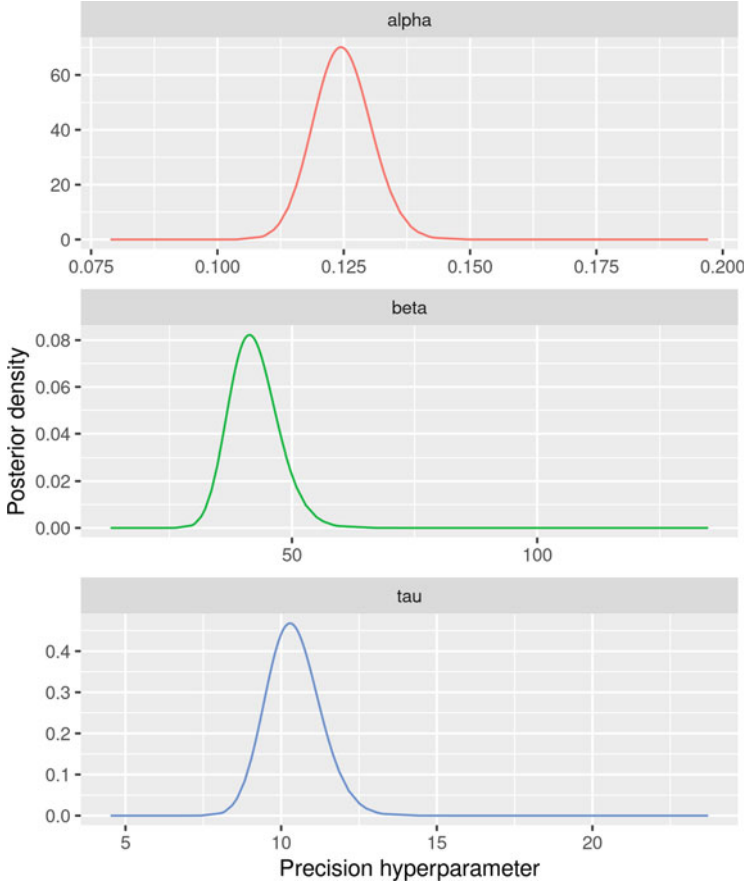


Fig. 5 Marginal posterior distributions of the three hyperparameters that control the amount of spatial smoothing

where

$$\tilde{f}_{s,t} = \tilde{\alpha}_{s,-t} + \tilde{\beta}_{s,-t}(f_{s,t} - \bar{f}_{s,t})$$

where $\tilde{\alpha}_{s,-t}$ denotes the posterior mean of α_s calculated on the data set from which the year t was excluded. The MSE is calculated independently for each forecast lead time. To have a benchmark for the overall usefulness of the postprocessed predictions, we also calculate the leave-one-out MSE of the climatological mean forecast at each location.

Figure 6 shows leave-one-out MSEs of the postprocessed forecasts with unsmoothed (MLE) parameter estimates, smoothed parameter estimates, and also the MSE of the climatological forecasts. The postprocessed forecasts have an initially low MSE, about half of the climatological MSE at lead time 1 day. The



Fig. 6 Cross-validated mean squared errors (averaged over time and the study domain) of the climatological forecast and the regression-based forecasts with unsmoothed and smoothed parameters

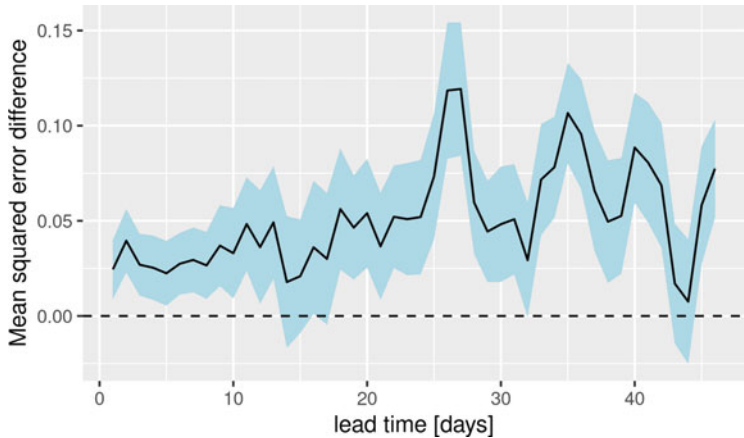


Fig. 7 Difference in cross-validated mean squared errors between regression based forecasts with unsmoothed and smoothed parameters. The ribbon has a half width of 1.96 standard errors. Positive MSE differences indicate that smoothed parameters improve the forecasts

MSE grows exponentially with lead time, as would be expected in a non-linear dynamical system with sensitivity to initial conditions. The MSE saturates at values around the climatological MSE after about 2 weeks. The difference in MSE between smoothed and unsmoothed forecasts is relatively small and difficult to assess based on Fig. 6.

To compare the MSE of smoothed and unsmoothed forecasts, we plot the MSE difference between forecasts with unsmoothed and smoothed regression parameters in Fig. 7. Positive MSE differences in this plot indicate that smoothing has decreased

the MSE and thus improved the postprocessed forecasts. We observe a small but consistent and statistically significant improvement of smoothing at all lead times. The MSE improvement of around 0.05 is relatively small when compared to the absolute value of MSE between 3 and 7 observed in Fig. 6. However, for practical purposes, the improvement is important at lead times between 10 and 14 days where the postprocessed forecasts offer only a small improvement over the climatological forecast. For example at lead time 13 days, the climatological MSE is 6.58, the MSE with unsmoothed parameters is 6.42, and the MSE with smoothed parameters is 6.37. That means that compared to the improvement of the postprocessed forecast versus climatology, parameter smoothing offers an additional *relative* improvement of $(6.42 - 6.37)/(6.58 - 6.42) \approx 31\%$. Note that the improvements reported here are in-sample. Applying the method out of sample, at locations where no training data is available to estimate the regression parameters, larger improvements can be expected.

The statistical postprocessing model does not only provide the best guess forecast $\tilde{f}_{s,t}$ but also an entire predictive distribution function. In our particular case, at fixed values of the postprocessing parameters α , β , and τ and a new numerical forecast value f^* , the forecast distribution is Gaussian, with expectation $\tilde{f} = \alpha_s + \beta_s(f^* - \tilde{f})$ and variance e^{τ_s} . If the parameters are estimated and have an estimation variance associated with them, the predictive distribution in the standard regression framework is a Student's t-distribution with $n - 2$ degrees of freedom, expectation $\tilde{f}$, and variance $\text{var}(\hat{\alpha}) + (f^* - \tilde{f})^2 \text{var}(\hat{\beta}) + e^{\hat{\tau}}$ where the variances of $\hat{\alpha}$ and $\hat{\beta}$ are the asymptotic variances of the MLEs. For the purpose of conciseness, we assume here that predictive distributions can be constructed in the same way when $\hat{\alpha}$ and $\hat{\beta}$ are derived in a Bayesian hierarchical model with a spatial prior, and the variances are the marginal posterior variances.

One method to evaluate the quality of predictive distribution functions is to analyse their coverage properties. We construct 95% prediction intervals from the parameter estimates and their variances from the unsmoothed and smoothed parameter estimates and calculate their frequency of covering the verifying observation. Well-calibrated 95% prediction intervals should cover the observations 95% of the time, and large deviations of the observed coverage from this nominal coverage are an indicator of poor model fit or a potential flaw in the methodology, e.g., due to simplifications and approximations.

Figure 8 shows the average coverage frequencies of 95% prediction intervals for all lead times. The coverage frequency of the prediction intervals derived from the unsmoothed parameters falls short of the nominal value, up to 2% which seems acceptable in practice. The smoothed parameters produce prediction intervals with even lower coverage, approximately 0.5% lower than the unsmoothed parameters. The reduction is consistent at all lead times. The coverage results suggest that parameter smoothing leads to a slight deterioration of the reliability of prediction intervals.

We have conflicting results about the benefit of spatial smoothing of post-processing parameters. The mean squared error, a measure of forecast accuracy, improves, but coverage of the prediction intervals, a measure of reliability of

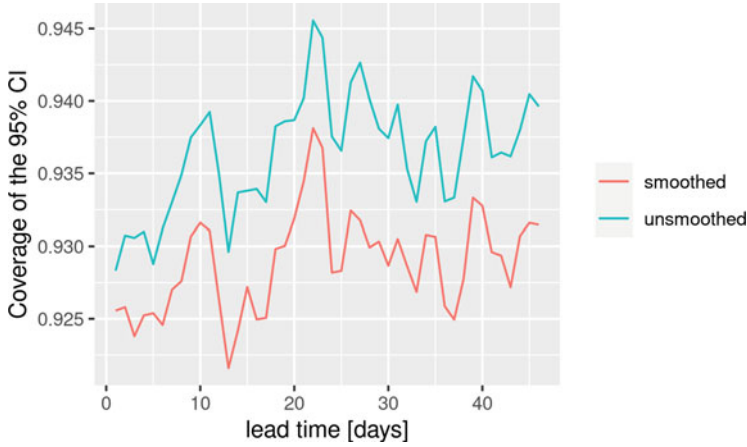


Fig. 8 Coverage frequencies of prediction intervals of the regression-based forecasts with unsmoothed and smoothed parameters. The nominal coverage is 95%

probabilistic forecasts, becomes slightly worse. The logarithmic score (also log-score) is a scoring rule to evaluate density forecasts (Good, 1952; Bernardo, 1979). The logarithmic score is sensitive to both accuracy and reliability (Bröcker, 2009). The log-score is defined as the logarithm of the predictive density assigned to the verifying observation. To establish an ordering between competing forecast schemes for the same observations, the average log-score difference can be used. Hence, let $p_{1,1}(\cdot), \dots, p_{N_s, N_t}(\cdot)$ be the predictive densities generated by the postprocessing with unsmoothed parameters and $q_{1,1}(\cdot), \dots, q_{N_s, N_t}(\cdot)$ the respective densities generated with smoothed parameters, both for verifying observations $y_{1,1}, \dots, y_{N_s, N_t}$. Then, the log-score difference is given by

$$\frac{1}{N_s N_t} \sum_{s=1}^{N_s} \sum_{t=1}^{N_t} [\log q_{s,t}(y_{s,t}) - \log p_{s,t}(y_{s,t})].$$

Positive log-score differences indicate here that densities derived from spatially smoothed parameters assign on average higher densities to the observation than those derived from unsmoothed parameters. In particular, a positive log-score difference of Δ implies that the better forecast assigns (in the geometric mean) e^Δ times as much density to the verifying observations.

We see in Fig. 9 that log-score differences between the two postprocessing schemes are small throughout, but positive and statistically significant. Smoothing regression parameters has an overall positive effect on the performance on probabilistic temperature forecasts, despite the slight deterioration of the coverage properties of the prediction intervals. Another notable feature is the apparent trend in log-score difference between lead times 1 and 10. The larger increase at short lead

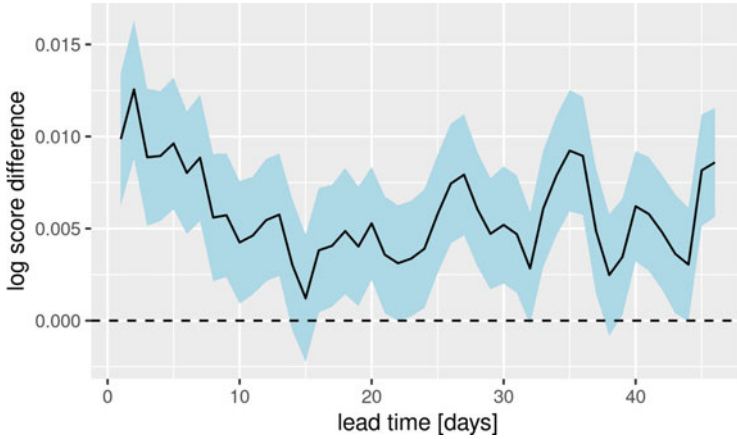


Fig. 9 Difference in the logarithmic score of regression-based forecasts with unsmoothed and smoothed parameters. The ribbon has a half-width of 1.96 standard errors of the mean. Positive values indicate the smoothed parameters improve the forecasts

times might imply that more useful information can be borrowed from neighbouring grid points for short-term than for long-term forecasts.

An R script for making weather forecasts with Bayesian hierarchical models is available at <https://github.com/sieste/lgmbook-chapter-weather-forecasts>

5 Discussion and Conclusion

A detailed understanding of the physical processes of the Earth's atmosphere ocean system allows us to formulate highly complex models and implement them numerically for forecasting. However, a retrospective comparison with observations reveals various shortcomings of numerical model forecasts. This chapter reviewed some common failure modes of numerical weather prediction models, as well as statistical modelling approaches to correct them. Fitting a simple linear regression model to observations, with the numerical weather model output used as a covariate, is known to be a robust and effective method for statistical postprocessing. We have illustrated the improvements that it offers over the direct model output in Sect. 4.2. A novel application of spatial statistical modelling in a latent Gaussian modelling framework was shown to consistently improve the postprocessed forecasts further. The verification results suggest that especially for weather forecasts on long time scales (weeks), the forecast improvement due to spatially smoothing regression parameters is not only statistically significant but also of practical importance.

The methodology presented here uses the recently proposed Max-and-Smooth method (Hrafnkelsson et al., 2021; Jóhannesson et al., 2022) for inference in latent Gaussian models. Max-and-Smooth offers an attractive simplification of a full

Bayesian inference in models where observed data can be modelled as conditionally independent given a spatial-temporally correlated latent field. The method is appealing for its simplicity. Since the full likelihood is replaced by an approximation that only depends on summary statistics (the maximum likelihood estimates and the observed information), Max-and-Smooth reduces the computational complexity of the likelihood calculation. However, the accuracy of the inferred posterior depends on how well the likelihood is approximated by a Gaussian, which can be poor especially if the sample size is small.

Future studies should consider different forecast targets than temperature, such as wind speed and precipitation, to check whether the improvements observed in this chapter are robust. Furthermore, smoothing parameters only in space could be extended by smoothing parameters also across time in a dynamic linear regression framework and also across forecast lead time. The simple 2D random walk model might be improved by a more careful model checking, and additional covariates such as latitude or land/sea mask should be considered for inclusion in the model at the latent level. These modifications might lead to further improvements and also eliminate the deterioration of coverage properties of prediction intervals observed in this chapter. Lastly, the assumption of conditional independence at the response level is questionable because observations and forecasts should be assumed to be “directly” spatially and temporally correlated. Using a spatio-temporal statistical model not only at the latent level but also at the response level might further improve the postprocessed forecasts, although such an extension makes the inference framework more complicated. In summary, this study offers evidence that a simple smoothing step applied to postprocessing parameter estimates can improve statistical weather forecasts and provides an encouraging direction for research and application of latent Gaussian modelling.

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Bayesian Latent Gaussian Models for High-Dimensional Spatial Extremes



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1 Introduction

Extreme-value theory (Coles, 2001; Beirlant et al., 2004; Davison & Huser, 2015) has become the standard probabilistic tool to build statistical models and make inference for high-impact extreme events occurring in a wide range of geo-environmental applications (e.g., Davison & Gholamrezaee, 2012; Reich & Shaby, 2012; Huser & Davison, 2014; Jonathan et al., 2014; Asadi et al., 2015; Jalbert et al., 2017; Vettori et al., 2019; Engelke & Hitz, 2020). Classical extreme-value models rely on asymptotic arguments for block maxima or high threshold exceedances when the block size or the threshold, respectively, increases arbitrarily. On the one hand, the celebrated Fisher–Tippett theorem states that, in the univariate case, the only possible non-degenerate limit distribution for renormalized block maxima is the generalized extreme-value (GEV) distribution, and this motivates its use in practice for modeling maxima with large but *finite* block sizes—typically, yearly blocks in environmental applications. On the other hand, the Pickands–Balkema–de Haan theorem states that the only possible non-degenerate limit distribution for high threshold exceedances is the generalized Pareto (GP) distribution, and this motivates its use in practice for modeling peaks over high but *finite* thresholds—often taken as the empirical 95%-threshold. These two seemingly different modeling techniques

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are in fact intrinsically related to each other through a Poisson point process representation that provides a unified description of the asymptotic upper tail; see Coles (2001) for more details. While these approaches are theoretically equivalent, they all have their pros and cons in practice. The block maximum approach avoids complications with intra-block seasonality and temporal dependence, but it has been criticized for being wasteful of data given that it only uses one value per block and ignores other smaller but potentially large events. By contrast, the threshold exceedance approach uses all extreme observations and often requires a detailed modeling of temporal dependence, which may either be a desideratum or a nuisance. While the “horse-racing” between the block maximum and threshold exceedance approaches is still debated (Bücher & Zhou, 2021), it is often believed that the threshold exceedance approach offers a richer, more convenient, and easily interpretable modeling framework, especially in the spatial context. In this work, we build Bayesian hierarchical models for high-resolution spatial precipitation extremes over Saudi Arabia, by exploiting the Poisson point process representation based on peaks over threshold.

Several approaches have been proposed for modeling spatial extremes; see Davison et al. (2012), Davison et al. (2019), and Huser and Wadsworth (2022) for comprehensive reviews on this topic. One possibility is to generalize univariate asymptotic models to the spatial setting, in such a way to not only model the marginal distribution of extremes accurately but also to capture their potentially strong dependencies using models justified by extreme-value theory. This leads to the class of max-stable processes for spatially indexed block maxima (Padoan et al., 2010), and to generalized Pareto processes for spatial threshold exceedances defined in terms of a certain risk functional (Thibaud & Opitz, 2015). While these asymptotic models are backed up by strong theoretical arguments, their complicated probabilistic structure leads to awkward likelihood functions, which are computationally demanding to evaluate, thus limiting likelihood-based inference to relatively small dimensions (Huser & Davison, 2013; Castruccio et al., 2016; de Fondeville & Davison, 2018; Huser et al., 2019). Various types of “sub-asymptotic” extreme models that circumvent some of the issues of asymptotic models have also been proposed (Wadsworth & Tawn, 2012, 2022; Huser et al., 2017, 2021; Huser & Wadsworth, 2019; Zhong et al., 2022), though they usually still remain difficult to fit in relatively high dimensions. Alternatively, when the main focus is to obtain accurate *marginal* return level estimates at observed and unobserved locations, while capturing response level dependence is of secondary importance, it is convenient to rely on latent Gaussian models (LGMs). These models often assume conditional independence among observations and can thus be fitted more efficiently in high dimensions, while performing well at spatial smoothing and prediction by borrowing strength across locations when spatial effects are embedded within model parameters. LGMs have been successfully applied in a wide range of applications (see, e.g., Rue et al., 2009, and the references therein) and belong to the broader class of Bayesian hierarchical models (Banerjee et al., 2003; Diggle & Ribeiro, 2007), whose specification is described at three levels: (i) the response level, specifying a parametric distribution for the observed

data, (ii) the latent level, modeling unknown parameters, potentially transformed with a link function, using fixed and random effects, and (iii) the hyperparameter level, specifying prior distributions for hyperparameters. In the case of LGMs, all latent variables are specified with a joint Gaussian distribution. When a multivariate link function is used to transform parameters jointly such that fixed/random effects are embedded within multiple linear predictors at the latent level, then we usually refer to these models as *extended* LGMs (Geirsson et al., 2020; Hrafnkelsson et al., 2021; Jóhannesson et al., 2022). While the response level is designed to capture the marginal stochastic variability using an appropriate probability distribution family but often assumes conditional independence (of the data given the parameters) for computational convenience, the latent level is designed to capture non-stationary spatio-temporal variation, trends, and dependencies through random and covariate effects. When the goal is to accurately predict the probability of extremes at unobserved locations and to smooth return level estimates across space, it is key to incorporate spatial dependence at the latent level to borrow strength across locations. However, the exact form of dependence assumed at the latent level is not crucial for spatial prediction and so the multivariate Gaussianity assumption of LGMs is not a major limitation. The hyperparameter level is used to sufficiently, but not overly, constrain model parameters with prior distributions, in order to stabilize estimation of all parameters and latent variables, and/or incorporate expert knowledge if desired.

Several types of extended LGMs have already been used in the extreme literature. Key differences between the proposed models are mainly with respect to the actual data likelihood being used at the response level (e.g., based either on block maxima or on peaks over threshold), the detailed latent level specification (e.g., whether or not random effects are specified as Gaussian Markov random fields with a sparse precision matrix), and the actual method of inference (e.g., “exact” Markov chain Monte Carlo algorithms, or approximate Bayesian inference methods). While all these differences may appear to be relatively minor at first sight, they have in fact important implications in terms of the methodology’s scalability to high dimensions and the accuracy of the final return level estimates. The first attempt to model extremes with an LGM in the literature is the paper by Coles and Casson (1998), who analyzed data generated from a climatological model at 55 equally spaced locations to characterize hurricane risk along the eastern coastlines of the United States. A Poisson point process likelihood for high threshold exceedances was used, whereby marginal location, scale, and shape parameters were further modeled using a relatively simple representation in terms of mutually independent Gaussian spatial processes, and the inference was performed using a basic Metropolis–Hastings MCMC algorithm with random walk proposals. Later, Cooley et al. (2007) built an LGM designed for the spatial interpolation of extreme return levels, using the GP distribution for high threshold exceedances, and the inference was performed by MCMC based on a precipitation dataset available at 56 weather stations. Huerta and Sansó (2007) proposed a similar model for spatio-temporal extremes, fitted using a customized MCMC algorithm, based on the GEV distribution for block maxima (with replicated observations at 19 stations). Other similar LGMs were

also proposed by Davison et al. (2012), Hrafnkelsson et al. (2012), Geirsson et al. (2015), and Dyrddal et al. (2015), but these models were all applied to block maxima data in relatively small dimensions, due to the computational burden and difficulties related to the convergence of Markov chains in MCMC algorithms. Rue et al. (2009) developed the integrated nested Laplace approximation (INLA), a fast and accurate approximate Bayesian solution for estimating generic LGMs, and it was later exploited by Opitz et al. (2018) and Castro-Camilo et al. (2019) to fit relatively simple spatio-temporal extreme-value models based on the GP distribution, but unfortunately, the R-INLA software currently does not support extended LGMs, where distinct random effects control the behavior of multiple parameters at the latent level. This is a major limitation when the data likelihood function has several parameters (e.g., the location and scale) that display a complex spatially varying behavior. Geirsson et al. (2020) later developed the LGM split sampler, which provides significant improvements in the mixing of Markov chains for extended LGMs and thus reduces the overall computational burden, but they illustrated their algorithm on a relatively small extreme example. With gridded or areal data, it is also possible to model random effects using Gaussian Markov random fields (GMRFs), which have a sparse precision matrix. This offers major computational gains by improving the efficiency of random effects' updates in large dimensions (see, e.g., Sang & Gelfand, 2009, 2010; Cooley & Sain, 2010; Jalbert et al., 2017, for extreme-value data examples). More recently, Jóhannesson et al. (2022) modeled a complex spatio-temporal dataset of yearly river flow maxima available at several hundreds of irregularly spaced stations over the United Kingdom using an extended LGM based on the GEV distribution, which embeds multiple latent spatial random effects defined in terms of stochastic partial differential equations (SPDEs). The SPDE approach is intrinsically linked to GMRFs (Lindgren et al., 2011), and this yields fast inference thanks to the sparsity of precision matrices. Moreover, instead of using an “exact” MCMC algorithm, Jóhannesson et al. (2022) leveraged *Max-and-Smooth*, a two-step approximate Bayesian MCMC-based inference scheme recently proposed by Hrafnkelsson et al. (2021), which shares some similarities with INLA to some degree and achieves exceptional speed and accuracy for fitting extended LGMs with replicates. In spatial applications, this method essentially consists of the two following consecutive steps: first, in the “Max” step, maximum likelihood estimates of model parameters and their observed information matrices are computed at each site separately, and then, in the “Smooth” step, these parameter estimates are smoothed jointly using an approximate LGM where the likelihood function has been approximated with a Gaussian density function, while properly accounting for the parameter uncertainty from the first step. Unlike other MCMC schemes for extremes, the Gaussian–Gaussian conjugacy of this approximate LGM can be exploited to improve the mixing of Markov chains and reduce the computational burden drastically.

In this book chapter, we showcase the modeling and inference approach proposed by Jóhannesson et al. (2022) on a new high-resolution precipitation dataset, but we make a crucial modification: instead of modeling block maxima with the GEV distribution, we here exploit the more convenient and informative framework of

threshold exceedances. This allows us to exploit all information available from the upper tail for inference, thus reducing the overall estimation uncertainty. Moreover, similarly to Coles and Casson (1998) but unlike Cooley et al. (2007), we here use the Poisson point process likelihood instead of the GP likelihood, which allows us to model the three marginal parameters jointly (rather than having to fit two separate models to estimate the GP parameters and the threshold exceedance probability) and to avoid an overly strong influence of the threshold choice on the results. We also investigate Max-and-Smooth and the accuracy of the Gaussian likelihood approximation in this context and demonstrate its usefulness in practice.

For illustration, we consider daily precipitation (mm) data, obtained from the Tropical Rainfall Measuring Mission (TRMM, Version 7) available over the period 2000–2019 without missing values, at a spatial resolution of $0.25^\circ \times 0.25^\circ$ over Saudi Arabia; the dataset is available at <https://gpm.nasa.gov/data-access/downloads/trmm>. Saudi Arabia has a diverse geography (Arabian desert, steppes, mountain ranges, volcanic lava fields, etc.) and has, for the most part, a hot desert climate with very high daytime temperatures during the summer and a sharp temperature drop at night, with the exception of the southwestern region, which features a semi-arid climate. Although the annual precipitation is very low overall, certain regions of Saudi Arabia have been regularly affected over the last two decades by short but intense convective storms, causing devastating flash floods, extensive damage, and fatalities (Deng et al., 2015; Yesubabu et al., 2016). Considering the whole of Saudi Arabia, our dataset comprises 2738 grid cells, resulting in approximately 20 million spatio-temporal observations in total. Figure 1 shows spatial maps of the observations for two extreme days: the day with the highest spatial average precipitation and the day with the highest daily precipitation localized on a single grid cell. In both cases, these days fall within the month of April. In the left panel, the grid cells with higher daily precipitation amounts are mainly stretched between the latitudes 18°N and 26°N . In the right panel, we observe that the grid cells with high precipitation amounts are localized near the Asir mountains along the southwestern coastline of the Red Sea, although the precipitation amount drops quickly away from the “wettest” pixel, indicating that the tail dependence is relatively short range. To explore the marginal precipitation distribution, Fig. 2 displays histograms of the positive precipitation distribution, as well as high threshold exceedances, for three representative grid cells (sites 208, 1716, and 1905) shown in Fig. 1. For the three chosen grid cells, the percentages of rainy days are 3.70%, 3.16%, and 4.53%, respectively. Here, the threshold is chosen as the 75% empirical quantile of positive precipitation intensities, and it therefore varies across space. Despite the majority of the data being zeros (corresponding to dry days), the histograms illustrate that there are still a relatively large number of high positive precipitation amounts, such that the corresponding density is highly skewed and heavy-tailed. Further exploratory analysis (not shown) reveals that there is no discernible long-term trend across years, but there is a clear intra-year seasonal pattern that varies across space. While most of the grid cells receive the highest precipitation amounts in April, a large fraction of the northern part of the country and the eastern coastline near the Persian Gulf receive the highest

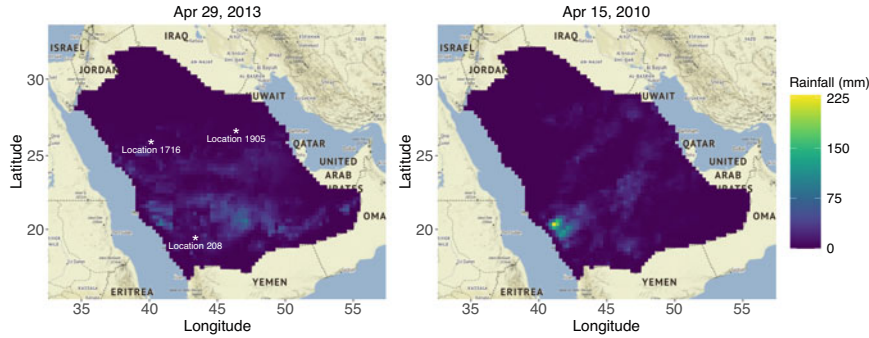


Fig. 1 Spatial maps of daily precipitation (mm) for the day with the highest spatial average (left), and the day with the highest daily precipitation at a single grid cell (right). Three representative locations (sites 208, 1716, and 1905), used for exploratory data analysis and model checking, are highlighted in the left panel

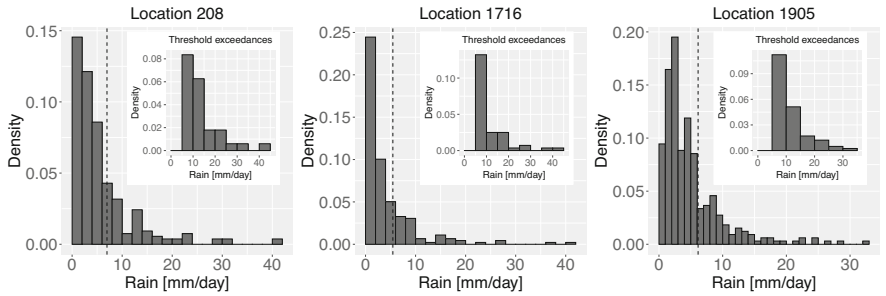


Fig. 2 Histograms of observed positive precipitation amounts at three representative grid cells (sites 208, 1716, and 1905, from left to right) shown in Fig. 1. The smaller histograms in each sub-panel are based on high threshold exceedances, where the threshold is taken as the 75% empirical quantile of the positive precipitation amounts observed at the corresponding site. For the three chosen grid cells, the percentages of rainy days are 3.70%, 3.16%, and 4.53%, respectively

precipitation amounts in November. For simplicity, in this work, we ignore any temporal trend and dependence and assume that the observations at each grid cell are independent and identically distributed across time. Modeling spatially varying seasonality patterns and time dependence is out of the scope of this chapter but is of practical relevance and deserves further investigation in future research. Our main goal here is to illustrate the usefulness of extended LGMs for extreme-value analysis in a high-resolution peaks-over-threshold context when combined with the powerful Max-and-Smooth inference method.

The rest of this chapter is organized as follows. In Sect. 2, we provide some background information on univariate extreme-value theory and the Poisson point process formulation of extremes. In Sect. 3, we specify our proposed spatial extended LGM in detail and describe the SPDE approach for modeling spatial random effects. In Sect. 4, we describe the Max-and-Smooth inference method and

demonstrate its accuracy in the peaks-over-threshold context. In Sect. 5, we present the results from our Saudi Arabian precipitation application. We finally conclude in Sect. 6 with some discussion and perspectives on future research.

2 Univariate Extreme-Value Theory Background

Assume that $Y_1, Y_2, \dots, \overset{\text{iid}}{\sim} F_Y$ is a sequence of independent and identically distributed (iid) random variables with distribution F_Y , and let $M_n = \max\{Y_1, \dots, Y_n\}$. The variables Y_i can be thought of as daily precipitation measurements observed over time at a single site, while M_n may represent the annual maximum precipitation amount. According to Fisher–Tippett theorem, if there exist sequences of constants $a_n > 0$ and $b_n \in \mathbb{R}$ such that, as $n \rightarrow \infty$,

$$\Pr\{(M_n - b_n)/a_n \leq z\} \rightarrow G(z), \quad (1)$$

for some non-degenerate cumulative distribution function G , then G is necessarily a generalized extreme-value (GEV) distribution, which may be expressed as

$$G(z) = \begin{cases} \exp\left[-\{1 + \xi(z - \mu)/\sigma\}_+^{-1/\xi}\right], & \text{if } \xi \neq 0, \\ \exp\left[-\exp\{-(z - \mu)/\sigma\}\right], & \text{if } \xi = 0, \end{cases} \quad (2)$$

defined on $\{z \in \mathbb{R} : 1 + \xi(z - \mu)/\sigma > 0\}$, where $\mu \in \mathbb{R}$, $\sigma > 0$, and $\xi \in \mathbb{R}$ are location, scale, and shape parameters, respectively, and $a_+ = \max\{a, 0\}$. Return levels are then defined as high quantiles of G , i.e.,

$$G^{-1}(1 - p) = \begin{cases} \mu - \sigma[1 - \{-\log(1 - p)\}^{-\xi}]/\xi, & \text{if } \xi \neq 0, \\ \mu - \sigma \log\{-\log(1 - p)\}, & \text{if } \xi = 0, \end{cases} \quad (3)$$

for small probabilities p . If the GEV distribution is used as a model for yearly block maxima, then the M -year return level z_M is obtained by setting $p = 1/M$ in (3), i.e., $z_M = G^{-1}(1 - 1/M)$. In other words, the M -year return level is the value that is expected to be exceeded once every M years on average, under temporal stationarity. The most important model parameter in (2), as far as the estimation of return levels is concerned, is thus the shape parameter ξ . When $\xi < 0$, G is the reverse Weibull distribution, where the support has a finite upper bound. When $\xi = 0$, G is the Gumbel distribution with a light tail (i.e., exponentially decaying tail or thinner). When $\xi > 0$, G is the Fréchet distribution with a heavy tail (i.e., tail heavier than that of an exponential distribution). Therefore, it is crucial to accurately estimate ξ when extreme quantiles need to be computed, and in the spatial context, it is important to borrow strength across neighboring locations to reduce the estimation uncertainty and improve estimates of ξ and high quantiles.

The convergence in (1) is the theoretical justification for using the GEV distribution to model block maxima in practice. However, this approach may be wasteful of data since other extreme observations may be disregarded if they are smaller than the block maximum. In geo-environmental applications, the block size n is often naturally considered to be one year to avoid the intricate modeling of seasonality. Choosing n to be one month may also be an option, though seasonality then needs to be modeled and this block size may not be large enough to justify using an asymptotic extreme-value model. A convenient alternative approach is to rely on asymptotic models for high threshold exceedances.

Under the assumption that (1) holds, the Pickands–Balkema–de Haan theorem implies that the distribution of $Y - u \mid Y > u$ may be approximated, for large thresholds u , by the generalized Pareto (GP) distribution $H(y)$, defined as

$$H(y) = \begin{cases} 1 - (1 + \xi y / \kappa_u)_+^{-1/\xi} & \text{if } \xi \neq 0, \\ 1 - \exp(-y / \kappa_u) & \text{if } \xi = 0, \end{cases} \quad (4)$$

where $\kappa_u > 0$ is a threshold-dependent scale parameter and $\xi \in \mathbb{R}$ is the same shape parameter as above. In practice, this result justifies fitting the GP distribution (4) to extreme observations that exceed a high pre-determined threshold u . However, to get a complete description of the tail behavior, this model needs to be supplemented by a Bernoulli model for the threshold exceedance indicators, i.e., $\mathbb{I}(Y_i > u)$, to estimate the threshold exceedance probability $\zeta_u = \Pr(Y > u)$. Estimating the model parameters κ_u , ξ , and ζ_u with two separate models for $Y_i - u \mid Y_i > u$ and $\mathbb{I}(Y_i > u)$ may sometimes be inconvenient. Moreover, it can be shown that the choice of threshold u has an effect on the value of the scale parameter κ_u in (4), which makes it less easily interpretable, especially in the presence of covariates.

To circumvent these issues, it is convenient to rely on the Poisson point process representation of extremes (Davison & Smith, 1990), which naturally connects the GEV and GP asymptotic distributions. A point process defined on a specific set is a probabilistic rule for the occurrence and position of point events. Consider the set of random points

$$\left\{ \left(\frac{i}{n+1}, \frac{Y_i - b_n}{a_n} \right)^\top ; i = 1, \dots, n \right\}, \quad (5)$$

where a_n and b_n may be chosen as in (2). Denoting the points in (5) by N_n , let $N_n(A_0)$ denote, by abuse of notation, the number of these points in some region of the form $A_0 = [0, 1] \times (u, \infty)$ for some large enough u . By the independence of Y_i , $i = 1, \dots, n$, $N_n(A_0)$ follows a binomial distribution with a non-trivial success probability, and thus, N_n is a valid point process. Furthermore, using the standard convergence of a binomial distribution to a Poisson limit, the limiting distribution of $N_n(A_0)$ can be shown to be Poisson; thus, assuming (1) holds, the point process (5)

converges to a Poisson point process with mean measure

$$\Lambda([t_1, t_2] \times (y, \infty)) = (t_2 - t_1)\{1 + \xi(y - \mu)/\sigma\}_+^{-1/\xi}, \quad (6)$$

for suitable regions of the form $A = [t_1, t_2] \times (y, \infty)$ with $0 \leq t_1 < t_2 \leq 1$ and $y > u_L$ for some limiting lower bound u_L and where μ , σ , and ξ are as in the GEV parametrization (2). It can indeed be shown that the GEV representation (2) and GP representation (4) can both be seen as a consequence of the Poisson point process convergence result. The intensity function corresponding to the mean measure (6) can be obtained by differentiation, i.e., $\lambda(t, y) = -\frac{d^2}{dt dy} \Lambda([t_0, t] \times [y, \infty)) = \sigma^{-1}\{1 + \xi(y - \mu)/\sigma\}_+^{-1/\xi-1}$, for $0 \leq t_0 < t \leq 1$. In practice, this limiting Poisson process can be fitted to the original, non-normalized points $\{(i/(n+1), Y_i); i = 1, \dots, n\}$ since the constants a_n and b_n may be absorbed into the location and scale parameters. The corresponding likelihood function for fitting this process to data observed in the region $A_u = [0, 1] \times (u, \infty)$, for some high threshold u , is

$$\begin{aligned} L(\mu, \sigma, \xi; A_u) &= \exp\{-\Lambda(A_u)\} \prod_{i=1}^{N_u} \lambda\{i/(n+1), Y_{(n-i+1)}\} \\ &\propto \exp\left\{-n_{\text{block}} \left(1 + \xi \frac{u - \mu}{\sigma}\right)_+^{-1/\xi}\right\} \prod_{i=1}^{N_u} \frac{1}{\sigma} \left(1 + \xi \frac{Y_{(n-i+1)} - \mu}{\sigma}\right)_+^{-1/\xi-1}, \end{aligned} \quad (7)$$

where $Y_{(1)} < \dots < Y_{(n)}$ are the order statistics and N_u is the (random) number of observations Y_i exceeding the threshold u , the parameters μ , σ and ξ are as before, and n_{block} may be chosen to rescale the intensity function in order that the interpretation of the parameters $\{\mu, \sigma, \xi\}$ estimated from (7) matches that obtained from block maxima in (2) for some desired block size; see Coles (2001). For example, if n_{block} is equal to the number of years of observations, then the parameters $\{\mu, \sigma, \xi\}$ estimated from (7) will theoretically correspond to those that would be obtained by fitting the GEV distribution to yearly maxima. The benefits of the likelihood inference approach based on (7) are that, unlike the likelihood based on the GP distribution (4), all three marginal parameters $\{\mu, \sigma, \xi\}$ are modeled at once, thus facilitating uncertainty assessment of return levels (which are a function of all three parameters), and that parameter estimates are (relatively) invariant to the chosen threshold u . Moreover, there is a direct correspondence between the GEV and Poisson point process parameterizations, which eases interpretation, and return levels can readily be obtained using (3).

In the frequentist framework, we can estimate the model parameters from an iid dataset $Y_1, \dots, Y_n$, by directly maximizing the log-likelihood corresponding to (7) numerically, thus obtaining maximum likelihood estimates (MLEs) of μ , σ , and ξ . Alternatively, to stabilize the estimation of the shape parameter ξ , which is often subject to high uncertainty, and thus to avoid unrealistic estimates, we may penalize

certain values of ξ through an additional prior $\pi(\xi)$. In the context of flood data modeling, Martins and Stedinger (2000) proposed using a beta density shifted to the interval $(-0.5, 0.5)$, with mean 0.10 and standard deviation 0.122. Jóhannesson et al. (2022) instead used a symmetric beta prior and also shifted to the interval $(-0.5, 0.5)$, but with mean zero and standard deviation 0.167. Once the prior $\pi(\xi)$ is specified, we can maximize the “generalized likelihood function,” defined as the product of the actual likelihood function (7) and the prior $\pi(\xi)$, thus providing robust parameter estimates. In our Saudi Arabian precipitation application, we consider the same beta prior for ξ as in Jóhannesson et al. (2022), i.e., we take it to be a symmetric Beta(4, 4) density over $(-0.5, 0.5)$. The main reason for choosing this prior is that it avoids excessively small or large estimates of ξ (thus preventing overly short tails when $\xi < 0.5$, as well as infinite-variance models when $\xi > 0.5$), even when there are only a limited number of threshold exceedances. Henceforth, estimates obtained by maximizing the generalized likelihood function will simply be referred to as MLEs. As explained in Sect. 4, obtaining MLEs and a reliable estimate of their observed information matrix, at each site separately, is required to perform fully Bayesian inference with Max-and-Smooth. The left column of Fig. 3 shows MLEs (using the additional prior $\pi(\xi)$) obtained by fitting, separately at each location, the limiting Poisson point process model to extreme precipitation peaks exceeding the site-specific 75% empirical quantile of positive precipitation intensities (i.e., excluding the zeros). When accounting for zero precipitation values, this threshold varies spatially and corresponds to the 90.16–99.74% marginal quantile level depending on the location, which gives an average of 66 threshold exceedances (out of 7245 days) per location (minimum 19, interquartile range 40–70, maximum 713 threshold exceedances). As expected, values of the location parameter μ and the scale σ are higher in the southwestern part of Saudi Arabia. Estimates of ξ tend to be slightly positive overall, but the spatial pattern is much more chaotic and noisy, with values of ξ ranging from -0.2 to 0.4 . The jittery pattern of ξ is less realistic in the sense that the values at two nearby pixels are expected to be similar to each other due to relatively homogeneous geographical and environmental conditions, especially in the middle of the country. Thus, a spatial model appropriately smoothing values of ξ (and μ and σ , as well) would give more sensible results and improve return level estimation. In the next section, we describe our spatial LGM framework to achieve this.

3 Latent Gaussian Modeling Framework

We now explain the general framework of latent Gaussian models (LGMs) and illustrate it here in the spatial setting of our Saudi Arabian precipitation application, although LGMs can be applied much more generally in other contexts (see, e.g., Rue et al., 2009; Hrafnkelsson et al., 2021). As mentioned in the introduction, an LGM consists of three hierarchical levels (the response level, the latent level, and the hyperparameter level), which are detailed below.

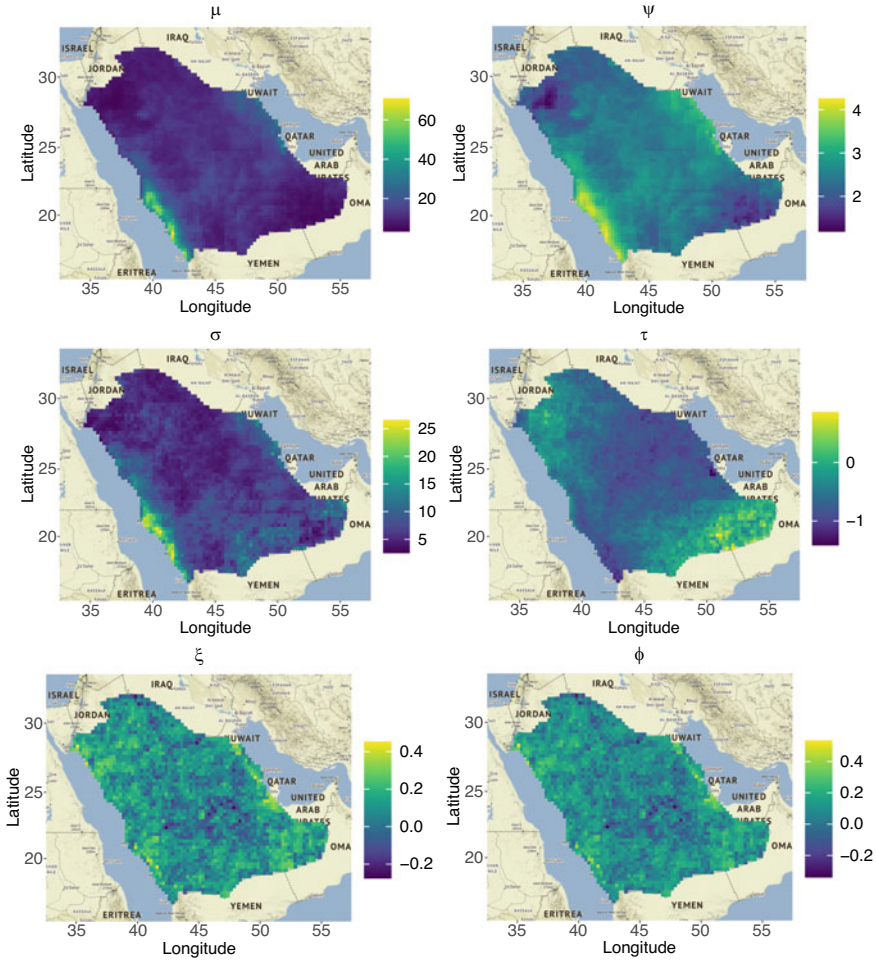


Fig. 3 Maximum likelihood estimates of the spatial Poisson point process parameters μ_i , σ_i , and ξ_i (left) and their transformations ψ_i , τ_i , and ϕ_i (right), respectively

3.1 Response Level Specification

For simplicity, we assume that there are no missing values and that the number of observations is the same at each location. Let $Y_t(s)$ denote the daily precipitation process over Saudi Arabia, and we denote by y_{it} the observed amount on day $t \in \{1, \dots, T\}$ at spatial location $s_i \in \mathcal{D} = \{s_1, \dots, s_N\}$, where T and N denote the total number of temporal replicates per location and the number of locations, respectively. Let $\mathbf{y}_i = (y_{i1}, \dots, y_{iT})^\top$ be the vector containing all observations at location s_i , and let $\mathbf{y} = (\mathbf{y}_1^\top, \dots, \mathbf{y}_N^\top)^\top$ be the combined vector of all observations.

At the response level of the Bayesian hierarchical model, we describe stochastic fluctuations using a parametric family $\pi(\mathbf{y} \mid \varphi_1, \dots, \varphi_K)$ that depends on K parameters. Assuming that the observations y_{it} have density $\pi(\cdot \mid \varphi_{1i}, \dots, \varphi_{Ki})$ and are conditionally independent (across both space and time) given spatially varying parameters $\boldsymbol{\varphi}_1 = (\varphi_{11}, \dots, \varphi_{1N})^\top, \dots, \boldsymbol{\varphi}_K = (\varphi_{K1}, \dots, \varphi_{KN})^\top$, we can then write the probability density function of $\mathbf{y}$, conditional on $\boldsymbol{\varphi}_1, \dots, \boldsymbol{\varphi}_K$, as

$$\pi(\mathbf{y} \mid \boldsymbol{\varphi}_1, \dots, \boldsymbol{\varphi}_K) = \prod_{i=1}^N \pi(\mathbf{y}_i \mid \varphi_{1i}, \dots, \varphi_{Ki}) = \prod_{i=1}^N \prod_{t=1}^T \pi(y_{it} \mid \varphi_{1i}, \dots, \varphi_{Ki}). \quad (8)$$

Hereafter, we refer to the density $\pi(\mathbf{y} \mid \boldsymbol{\varphi}_1, \dots, \boldsymbol{\varphi}_K)$ as the *likelihood* function, when viewed as a function of the parameters $\boldsymbol{\varphi}_1, \dots, \boldsymbol{\varphi}_K$. In our spatial extreme context based on peaks over threshold, the likelihood in (8) is constructed by multiplying sitewise Poisson point process likelihoods of the form (7), with $K = 3$ and $\varphi_1 \equiv \mu$, $\varphi_2 \equiv \sigma$, and $\varphi_3 \equiv \xi$. Thus, we assume that spatial variability among marginal extremes may be captured through the underlying spatially varying Poisson point process parameters $\boldsymbol{\varphi}_1 \equiv \boldsymbol{\mu} = (\mu_1, \dots, \mu_N)^\top$, $\boldsymbol{\varphi}_2 \equiv \boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_N)^\top$, and $\boldsymbol{\varphi}_3 \equiv \boldsymbol{\xi} = (\xi_1, \dots, \xi_N)^\top$, though in practice some of these parameters may be assumed to have a more parsimonious formulation. The spatial structure of parameters is specified at the latent level.

3.2 Latent Level Specification and Multivariate Link Function

At the latent level, we first suitably transform parameters and then model them through spatially structured and/or unstructured Gaussian model components. To be more precise, let $g : \Omega \rightarrow \mathbb{R}^K$ be a K -variate bijective link function that transforms the original parameters (with support in $\Omega \subset \mathbb{R}^K$) as $g(\varphi_1, \dots, \varphi_K) = (\eta_1, \dots, \eta_K)^\top$, in such a way that the domain of each transformed parameter η_j , $j = 1, \dots, K$, is the whole real line. At each location, the original parameters $(\varphi_{1i}, \dots, \varphi_{Ki})^\top$ are thus transformed through g as $(\eta_{1i}, \dots, \eta_{Ki})^\top = g(\varphi_{1i}, \dots, \varphi_{Ki})$, $i = 1, \dots, N$, and we can then combine them across locations like above as $\boldsymbol{\eta}_1 = (\eta_{11}, \dots, \eta_{1N})^\top, \dots, \boldsymbol{\eta}_K = (\eta_{K1}, \dots, \eta_{KN})^\top$. The link function can be chosen to “Gaussianize” the behavior of parameters (i.e., to get a bell-shaped posterior distribution for the transformed parameters, with low skewness, stable posterior variance across space, thin tails, etc.), or to ensure they reflect certain desired properties (e.g., having their support on the whole real line), or to reduce confounding issues between latent parameters if their estimates appear to be strongly correlated. Once the link function is specified, the general formulation of the latent model, which consists in modeling the transformed parameters using fixed

and random effects, may be expressed as

$$\begin{aligned}
 \eta_1 &= X_1 \beta_1 + w_1 + \varepsilon_1, \\
 \eta_2 &= X_2 \beta_2 + w_2 + \varepsilon_2, \\
 &\vdots \\
 \eta_K &= X_K \beta_K + w_K + \varepsilon_K,
 \end{aligned} \tag{9}$$

where $\beta_1, \dots, \beta_K$ are fixed (covariate) effects specified with independent Gaussian priors (see further details below), $X_1, \dots, X_K$ are the corresponding design matrices comprising observed covariates, $w_1, \dots, w_K$ are zero-mean spatially structured Gaussian random effects, and $\varepsilon_1, \dots, \varepsilon_K$ are zero-mean unstructured Gaussian “noise” (i.e., everywhere-independent) effects capturing small-scale variability. All the fixed and random effects in (9) are assumed to be mutually independent. Moreover, although there are other possibilities, we shall here define the spatially structured random effects w_j , $j = 1, \dots, K$, by discretizing a stochastic partial differential equation (SPDE) approximating a Matérn Gaussian field, which yields sparse precision matrices and thus fast inference.

In our spatial extreme context, we transform the parameters $(\varphi_1, \varphi_2, \varphi_3)^\top \equiv (\mu, \sigma, \xi)^\top$ jointly using a multivariate link function g . We therefore obtain three jointly transformed parameter vectors at the latent level, namely $\psi = (\psi_1, \dots, \psi_N)^\top$, $\tau = (\tau_1, \dots, \tau_N)^\top$, and $\phi = (\phi_1, \dots, \phi_N)^\top$, where $(\eta_{1i}, \eta_{2i}, \eta_{3i})^\top \equiv (\psi_i, \tau_i, \phi_i)^\top = g(\mu_i, \sigma_i, \xi_i)$. Similar to Jóhannesson et al. (2022), our choice of link function is justified by the following arguments. Observing that the estimated location parameters $\hat{\mu}_i$ are all positive and right-skewed (see the top left panel of Fig. 3) and that the location and scale parameters, $\hat{\mu}_i$ and $\hat{\sigma}_i$, are strongly linearly correlated, we transform them jointly as $\psi_i = \log(\mu_i)$, and $\tau_i = \log(\sigma_i/\mu_i)$, $i = 1, \dots, N$. Moreover, as the estimation of the shape parameter is generally subject to high uncertainty, we follow Jóhannesson et al. (2022) and consider the transformation $\phi_i = h(\xi_i)$, where

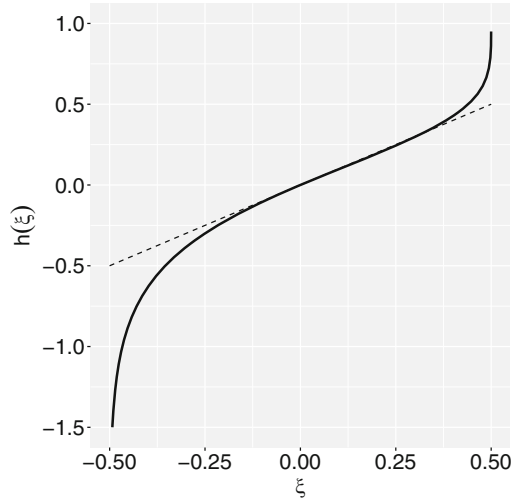
$$\phi = h(\xi) = a_\phi + b_\phi \log[-\log\{1 - (\xi + 0.5)^{c_\phi}\}], \tag{10}$$

with $c_\phi = 0.8$, $b_\phi = -c_\phi^{-1} \log\{1 - 0.5^{c_\phi}\}\{1 - 0.5^{c_\phi}\}2^{c_\phi-1} = 0.39563$, and $a_\phi = -b_\phi \log[-\log\{1 - 0.5^{c_\phi}\}] = 0.062376$. The corresponding inverse transformation can be easily computed as

$$\xi = h^{-1}(\phi) = (1 - \exp[-\exp\{(\phi - a_\phi)/b_\phi\}])^{1/c_\phi} - 0.5. \tag{11}$$

This specific transformation, displayed in Fig. 4, prevents overly small and large values of ξ by restricting its domain to the interval $(-0.5, 0.5)$; it conveniently ensures that $h(\xi) \approx \xi$ for $\xi \approx 0$, which implies that the transformed shape parameter ϕ can be interpreted similarly to ξ in case of light or moderately heavy

Fig. 4 The transformation $h : (-0.5, 0.5) \rightarrow \mathbb{R}$ (continuous black curve). The reference line with intercept zero and slope one is shown as a dashed line



tails, and it also makes sure that the asymptotic variance of the MLE behaves reasonably; see Jóhannesson et al. (2022) for more details. Therefore, overall, we can write the chosen multivariate link function as $g : (0, \infty)^2 \times (-0.5, 0.5) \rightarrow \mathbb{R}^3$ with $g(\mu, \sigma, \xi) = (\log(\mu), \log(\sigma/\mu), h(\xi))^T$. MLEs for the transformed parameters $(\psi_i, \tau_i, \phi_i)^T = g(\mu_i, \sigma_i, \xi_i)$, $i = 1, \dots, N$, are shown in the right column of Fig. 3. By inspecting and comparing the color balance and color scale of the different maps in Fig. 3, we can deduce that the distribution of transformed parameters looks more symmetric, spatially well-behaved, and that the very strong correlation between μ and σ has been comparatively somewhat reduced in the transformed parameters. We can also see that there seems to be a relatively smooth underlying signal in the transformed parameters, but that the sitewise estimates are somewhat noisy, especially as far as the shape parameter is concerned. This justifies modeling transformed parameters using a spatial latent model of the form (9).

We now detail more specifically each of the model components involved in the latent structure (9), in the context of our Saudi Arabian precipitation application. More general structures might be considered in other applications. In case relevant covariate information is available, it can be easily incorporated into the modeling of the transformed parameters ψ , τ , and ϕ , but we here do not have any meaningful covariates to use (but see, e.g., Jóhannesson et al., 2022, who use catchment descriptors as covariates in their flood frequency analysis), and thus, we only consider an intercept with some additional spatial effect terms. Moreover, while the spatially structured effects provide great flexibility in modeling latent parameters, we also need to be careful not to allow unnecessary additional computational burden by allowing spatial effects for some parameters that do not exhibit any spatial structured variability. Hence, we first investigate whether or not the spatial effects w_j in (9) are truly required to model spatial variability in each of the transformed parameters ψ , τ , and ϕ . To assess this, we compute binned empirical variograms

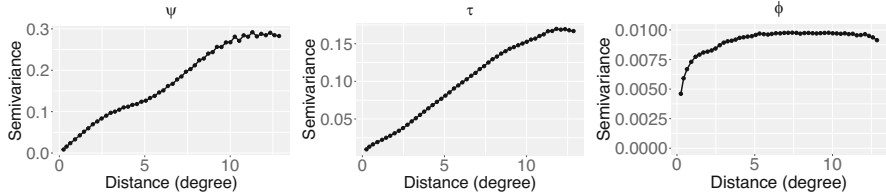


Fig. 5 Binned empirical variograms based on the MLEs of the spatially varying transformed model parameters ψ_i , τ_i , and ϕ_i (left to right), $i = 1, \dots, N$

for each of the spatial fields of the MLEs. Figure 5 shows the results for each transformed parameter. We can see that the variograms of ψ and τ indicate long-range spatial dependence, while the variogram of ϕ does not give as much evidence of a strong spatial dependence, but rather seems to indicate that about 50–75% of the marginal variability is due to unstructured noise. This corroborates the initial impression given by Fig. 3. Thus, here, we construct a relatively parsimonious spatial model, where both ψ and τ are modeled using an intercept, a spatially structured term, and a spatially unstructured noise term, while the transformed shape parameter ϕ is modeled solely using an intercept and a spatially unstructured noise term but does not involve any spatially structured term. This is rather an extreme case of smoothing of the values ϕ_i , $i = 1, \dots, N$, namely, assuming its spatially structured term is the same across the spatial domain.

To model spatially structured effects, we here consider the class of Gaussian processes driven by a Matérn correlation structure, which may be approximated based on finite-dimensional Gaussian Markov random fields (GMRFs). GMRFs are structured in terms of conditional independence relationships, which lead to sparse precision matrices that can be conveniently summarized with a graphical representation, and this can be exploited in practice to speed up computations; see Rue and Held (2005) for more details on GMRFs. A direct link between continuous-space Gaussian processes with a dense Matérn correlation structure, and GMRFs with a sparse precision matrix, may indeed be established theoretically through the stochastic partial differential equation (SPDE) approach (Lindgren et al., 2011), and this is the model structure that we shall use here.

Therefore, in our application, we can rewrite the latent level (9) more specifically as

$$\begin{aligned}\psi &= \beta_\psi \mathbf{1}_N + \mathbf{A} \mathbf{w}_\psi^* + \varepsilon_\psi, \\ \tau &= \beta_\tau \mathbf{1}_N + \mathbf{A} \mathbf{w}_\tau^* + \varepsilon_\tau, \\ \phi &= \beta_\phi \mathbf{1}_N + \varepsilon_\phi,\end{aligned}\tag{12}$$

where $\mathbf{1}_N$ is an N -dimensional vector of ones representing the intercept, and β_ψ , β_τ , and β_ϕ are the corresponding coefficients, the vectors $\mathbf{w}_\psi^*$ and $\mathbf{w}_\tau^*$ are independent spatially structured random effects defined on a triangulated mesh $\mathcal{D}^*$ covering the region of interest, the matrix $\mathbf{A}$ in (12) is a projection matrix from $\mathcal{D}^*$ (mesh nodes)

to $\mathcal{D}$ (data locations), and the vectors $\mathbf{e}_\psi$, $\mathbf{e}_\tau$, and $\mathbf{e}_\phi$ are spatially unstructured noise terms, i.e., nugget effects. To be more precise, the mesh $\mathcal{D}^*$ should be a relatively fine spatial discretization of the domain of interest, which we can construct using the function `inla.mesh.2d` from the package INLA (www.r-inla.org). Note that the number of mesh nodes and the mesh itself may be defined independently of the data locations, and its resolution should mainly depend on the effective correlation range of the process of interest. Typically, longer range processes can be approximated using coarser meshes. Figure 6 displays the mesh that we used in our Saudi Arabian precipitation application. It contains 1196 mesh nodes in total, which is much less than the total number of grid cells (2738), thus saving computational time and memory, and it was found to yield satisfactory results. Having constructed the mesh, we can then define the spatially structured random effects in (12) as $\mathbf{w}_\psi^* \sim \mathcal{N}_{|\mathcal{D}^*|}(\mathbf{0}, s_\psi^2 \mathbf{Q}_\psi^{-1})$ and $\mathbf{w}_\tau^* \sim \mathcal{N}_{|\mathcal{D}^*|}(\mathbf{0}, s_\tau^2 \mathbf{Q}_\tau^{-1})$, where $|\mathcal{D}^*|$ denotes the number of mesh nodes in $\mathcal{D}^*$, $\mathcal{N}_{|\mathcal{D}^*|}$ denotes the $|\mathcal{D}^*|$ -variate Gaussian distribution, s_ψ^2 and s_τ^2 denote the marginal variances of $\mathbf{w}_\psi^*$ and $\mathbf{w}_\tau^*$, respectively,

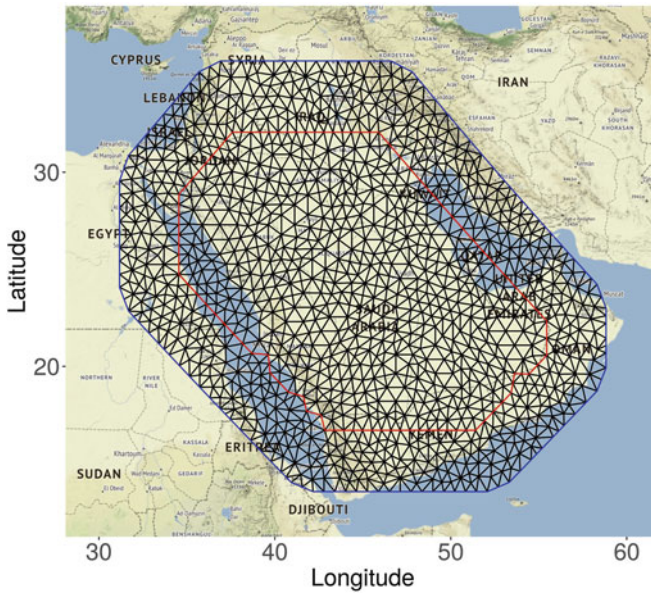


Fig. 6 Triangulated mesh $\mathcal{D}^*$ over Saudi Arabia, which is used to construct the spatial random effects for the parameter surfaces, based on the Matérn stochastic partial differential equation (SPDE) model. In the R function `inla.mesh.2d` used to create this spatial mesh discretizing the spatial domain, `cutoff` (the minimum allowed distance between points) is set to 0.8, `offset` (the size of the inner and outer extensions around the data locations) is set to 0.08, and `max.edge` (a vector of the maximum allowed triangle edge lengths in the inner domain and in the outer extension) is set to $(0.1, 1)^\top$. The inner domain boundary and the outer extension boundary are presented in red and blue, respectively. The outer extension avoids boundary effects in the SPDE approximation; see Lindgren et al. (2011) for more details

and $\mathbf{Q}_{\rho_\psi}$ and $\mathbf{Q}_{\rho_\tau}$ are sparse precision matrices defined through the Matérn SPDE–GMRF relationship in terms of some range parameters ρ_ψ and ρ_τ , respectively; see Lindgren et al. (2011) for mathematical details. Thus, the final covariance matrices of $\mathbf{w}_\psi = \mathbf{A}\mathbf{w}_\psi^*$ and $\mathbf{w}_\tau = \mathbf{A}\mathbf{w}_\tau^*$ are $s_\psi^2 \mathbf{A}^\top \mathbf{Q}_{\rho_\psi}^{-1} \mathbf{A}$ and $s_\tau^2 \mathbf{A}^\top \mathbf{Q}_{\rho_\tau}^{-1} \mathbf{A}$, respectively, and they are approximate Matérn covariance matrices with range parameters ρ_ψ and ρ_τ , respectively. Throughout this chapter, we fix the Matérn smoothness parameter to one, which produces reasonably smooth realizations justified by the variograms in Fig. 5. As for the nugget effects $\boldsymbol{\varepsilon}_\psi$, $\boldsymbol{\varepsilon}_\tau$, and $\boldsymbol{\varepsilon}_\phi$ in (12), they are defined as zero-mean white Gaussian noises with marginal variance σ_ψ^2 , σ_τ^2 , and σ_ϕ^2 , respectively, i.e.,

$$\begin{aligned}\boldsymbol{\varepsilon}_\psi &\sim \mathcal{N}_N(\mathbf{0}, \sigma_\psi^2 \mathbf{I}_N), \\ \boldsymbol{\varepsilon}_\tau &\sim \mathcal{N}_N(\mathbf{0}, \sigma_\tau^2 \mathbf{I}_N), \\ \boldsymbol{\varepsilon}_\phi &\sim \mathcal{N}_N(\mathbf{0}, \sigma_\phi^2 \mathbf{I}_N),\end{aligned}$$

where $\mathbf{I}_N$ is the identity matrix.

We can then rewrite the three equations in (12) more compactly in matrix form as

$$\boldsymbol{\eta} = \mathbf{Z}\mathbf{v} + \boldsymbol{\varepsilon}, \quad (13)$$

where

$$\begin{aligned}\boldsymbol{\eta} &= (\boldsymbol{\psi}^\top, \boldsymbol{\tau}^\top, \boldsymbol{\phi}^\top)^\top, \quad \mathbf{v} = (\beta_\psi, \mathbf{w}_\psi^{*\top}, \beta_\tau, \mathbf{w}_\tau^{*\top}, \beta_\phi)^\top, \quad \boldsymbol{\varepsilon} = (\boldsymbol{\varepsilon}_\psi^\top, \boldsymbol{\varepsilon}_\tau^\top, \boldsymbol{\varepsilon}_\phi^\top)^\top, \\ \mathbf{Z} &= \begin{pmatrix} \mathbf{1}_N & \mathbf{A} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \mathbf{1}_N & \mathbf{A} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \mathbf{1}_N \end{pmatrix},\end{aligned}$$

where the dots denote null vectors/matrices of appropriate dimension. Using the latent model specification (13), we write $\pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta})$ for the multivariate Gaussian density of the transformed model parameters $\boldsymbol{\eta}$ given the fixed and random effects $\mathbf{v}$ and some hyperparameter vector $\boldsymbol{\theta}$ (here, comprising the marginal standard deviations and range parameters of random effects). It follows that $\pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta})$ is the density of an $3N$ -dimensional Gaussian distribution, i.e., $\mathcal{N}_{3N}(\mathbf{0}, \text{bdiag}(\sigma_\psi^2 \mathbf{I}_N, \sigma_\tau^2 \mathbf{I}_N, \sigma_\phi^2 \mathbf{I}_N))$, where “bdiag” refers to a block diagonal matrix. Moreover, we write $\pi(\mathbf{v} \mid \boldsymbol{\theta})$ for the multivariate Gaussian density of $\mathbf{v}$ given $\boldsymbol{\theta}$. Since the fixed and random effects are assumed to have independent zero-mean Gaussian priors, $\pi(\mathbf{v} \mid \boldsymbol{\theta})$ is the density of a $\mathcal{N}_{3+2|\mathcal{D}^*|}(\mathbf{0}, \text{bdiag}(\sigma_{\beta_\psi}^2, s_\psi^2 \mathbf{Q}_{\rho_\psi}^{-1}, \sigma_{\beta_\tau}^2, s_\tau^2 \mathbf{Q}_{\rho_\tau}^{-1}, \sigma_{\beta_\phi}^2))$ distribution, where $\sigma_{\beta_\psi}^2$, and $\sigma_{\beta_\tau}^2, \sigma_{\beta_\phi}^2$ are user-specified prior variances for intercept coefficients. Specifically, we here choose independent vague Gaussian priors in our application, i.e., $\beta_l \sim \mathcal{N}(0, \sigma_{\beta_l}^2)$ with $\sigma_{\beta_l}^2 = 100^2$, for each $l \in \{\psi, \tau, \phi\}$. The hyperparameter

vector θ is crucial in controlling the behavior of latent effects, in order to stabilize estimation while avoiding overly restrictive behaviors. This is controlled at the hyperparameter level by assigning a suitable prior density $\pi(\theta)$ to θ . The details are included in the next section.

3.3 Hyperparameter Level Specification

At the hyperparameter level, we finally assign prior distributions to all hyperparameters (i.e., here, the vector $\theta = (\sigma_\psi, s_\psi, \rho_\psi, \sigma_\tau, s_\tau, \rho_\tau, \sigma_\phi)^\top$). Additionally, we set a prior on the shape parameter ξ to stabilize its estimation, similar to Martins and Stedinger (2000) and Jóhannesson et al. (2022). In our precipitation application, we use the following specification:

- For the nugget standard deviations σ_ψ , σ_τ , and σ_ϕ , we use penalized complexity (PC) priors (Simpson et al., 2017) that are parametrization-invariant prior distributions designed to prevent overfitting, by shrinking hyperparameters toward a basic reference model—here, a model without nugget effects. We can show that, in this case, the PC prior for σ_l , $l \in \{\psi, \tau, \phi\}$, is an exponential distribution with some user-defined rate parameter controlling how concentrated the prior for σ_l (and, thus, also the prior for the ϵ_l noise term) is around zero; see Theorem 2.1 of Fuglstad et al. (2019). In other words, we set $\pi(\sigma_l) = \lambda_{\sigma_l} \exp(-\lambda_{\sigma_l} \sigma_l)$, $\sigma_l > 0$, for some rate parameter $\lambda_{\sigma_l} > 0$, $l \in \{\psi, \tau, \phi\}$.
- For the parameter vectors $(s_\psi, \rho_\psi)'$ and $(s_\tau, \rho_\tau)'$ of the spatially structured Matérn random effects, we use the same PC prior as in Fuglstad et al. (2019), defined as

$$\pi(s_l, \rho_l) = \lambda_{s_l} \lambda_{\rho_l} \rho_l^{-2} \exp(-\lambda_{s_l} s_l - \lambda_{\rho_l} \rho_l^{-1}), \quad s_l > 0, \rho_l > 0, \quad l \in \{\psi, \tau\},$$

where $\lambda_{\rho_l}, \lambda_{s_l} > 0$ are user-defined rate parameters, selected according to Theorem 2.6 of Fuglstad et al. (2019).

- For the shape parameter $\xi = (\xi_1, \dots, \xi_N)^\top$, we consider independent Beta(4, 4) priors shifted to the interval $(-0.5, 0.5)$, which then induces a prior on the transformed shape parameter $\phi = (\phi_1, \dots, \phi_N)^\top$. Given the transformation $\phi = h(\xi)$ in (10) (with inverse transformation $\xi = h^{-1}(\phi)$ in (11)), the prior

density for each ϕ_i may thus be expressed as

$$\pi(\phi_i) = \frac{1}{B(4, 4)b_\phi c_\phi} \left\{ h^{-1}(\phi_i) + \frac{1}{2} \right\}^{4-c_\phi} \left\{ \frac{1}{2} - h^{-1}(\phi_i) \right\}^3 \\ \times \exp \left\{ \frac{\phi_i - a_\phi}{b_\phi} - \exp \left(\frac{\phi_i - a_\phi}{b_\phi} \right) \right\},$$

where $B(\cdot, \cdot)$ is the beta function, and a_ϕ , b_ϕ , and c_ϕ are specified in (10). Hence, we have $\pi(\boldsymbol{\phi}) = \prod_{i=1}^N \pi(\phi_i)$, with $\pi(\phi_i)$ defined as above, i.e., a transformed Beta(4, 4) density.

We then assume that the priors for $(s_\psi, \rho_\psi)^\top$, $(s_\tau, \rho_\tau)^\top$, σ_ψ , σ_τ , σ_ϕ , and $\boldsymbol{\phi}$ are mutually independent. In particular, the prior for hyperparameters can be written as $\pi(\boldsymbol{\theta}) = \pi(\sigma_\psi) \times \pi(s_\psi, \rho_\psi) \times \pi(\sigma_\tau) \times \pi(s_\tau, \rho_\tau) \times \pi(\sigma_\phi)$.

3.4 Summarized Full Model Specification

In summary, the model structure is specified at three levels. The response level is specified by the density $\pi(\mathbf{y} \mid \boldsymbol{\eta})$, which is the same as the likelihood function (8), but now expressed in terms of the transformed parameters $\boldsymbol{\eta} = (\boldsymbol{\psi}^\top, \boldsymbol{\tau}^\top, \boldsymbol{\phi}^\top)^\top$. Taking the additional transformed beta prior $\pi(\boldsymbol{\phi})$ into consideration, we obtain the generalized likelihood function

$$L(\mathbf{y} \mid \boldsymbol{\eta}) = \pi(\mathbf{y} \mid \boldsymbol{\eta}) \times \pi(\boldsymbol{\phi}). \quad (14)$$

The latent level is specified through (13) by the multivariate Gaussian density $\pi(\boldsymbol{\eta}, \mathbf{v} \mid \boldsymbol{\theta}) = \pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta}) \times \pi(\mathbf{v} \mid \boldsymbol{\theta})$, where each term $\pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta})$ and $\pi(\mathbf{v} \mid \boldsymbol{\theta})$ is a multivariate Gaussian density itself as made precise in Sect. 3.2. Finally, the hyperparameter level is specified by the prior density $\pi(\boldsymbol{\theta})$ detailed in Sect. 3.3.

Thus, by exploiting this hierarchical representation, the posterior density is

$$\pi(\boldsymbol{\eta}, \mathbf{v}, \boldsymbol{\theta} \mid \mathbf{y}) \propto L(\boldsymbol{\eta} \mid \mathbf{y}) \pi(\boldsymbol{\eta}, \mathbf{v} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) = L(\boldsymbol{\eta} \mid \mathbf{y}) \pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta}) \pi(\mathbf{v} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta}). \quad (15)$$

In order to perform Bayesian inference, sampling or numerical approximation of the posterior density (15) is required. Usually, Markov chain Monte Carlo algorithms can be used to generate approximate samples from (15). However, the complicated form of the generalized likelihood function, expressed in terms of the Poisson point process likelihood (7), prevents Gibbs sampling to update model parameters, and the presence of multiple high-dimensional latent spatial random effects makes it computationally impractical. Alternative computational solutions need to be found. The integrated nested Laplace approximation (INLA) is a highly popular

approximate Bayesian inference technique for LGMs, but unfortunately, it does not apply, at least in its current implementation in R, to *extended* LGMs with a multivariate link function. In the next section, we describe Max-and-Smooth, which consists in approximating the generalized likelihood function $L(\mathbf{y} \mid \boldsymbol{\eta})$ in (14) with a Gaussian density, to exploit the conjugacy of Gaussian–Gaussian LGMs for fast MCMC-based inference.

4 Approximate Bayesian Inference with Max-and-Smooth

Max-and-Smooth is a fully Bayesian inference scheme designed to fit extended LGMs and relies on two successive steps: (1) “Max” step: in the spatial setting, we first obtain parameter estimates independently at each site, as well as a robust estimate of their observed information matrix, and use them to suitably approximate the generalized likelihood function $L(\boldsymbol{\eta} \mid \mathbf{y})$ in (15) with a Gaussian density, and (2) “Smooth” step: exploiting this likelihood approximation, we then smooth parameter estimates by fitting an alternative LGM with Gaussian likelihood (later also called Gaussian–Gaussian model), which is a surrogate-model for the exact extended LGM. This surrogate-model is fitted using a straightforward MCMC algorithm, which treats the parameter estimates as “noisy data” with a known covariance structure (obtained from the “Max” step). This two-step inference approach can be shown to be equivalent (up to the likelihood approximation) to fitting the original LGM in one single step. The quality of the Gaussian approximation will thus determine the method’s accuracy, and this implies that Max-and-Smooth requires to have enough temporal replicates at each site. Significant computational gains can be obtained with Max-and-Smooth thanks to the conjugacy of the Gaussian–Gaussian surrogate-model, and also because the computational burden of the “Smooth” step is insensitive to the number of temporal replicates, unlike other conventional MCMC or INLA methods. We now describe each step in our peaks-over-threshold extreme-value setting and study the approximation’s accuracy in this context.

4.1 “Max” Step: Computing MLEs and Likelihood Approximation

The first step of Max-and-Smooth is to obtain MLEs and to approximate the generalized likelihood function with a (rescaled) Gaussian likelihood. From (8) and (14), the generalized likelihood function can be written, thanks to the conditional independence assumption, as the product $L(\boldsymbol{\eta} \mid \mathbf{y}) = \prod_{i=1}^N L(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i)$, where $L(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i) = \pi(\mathbf{y}_i \mid \psi_i, \tau_i, \phi_i)\pi(\phi_i)$ are sitewise generalized likelihoods and $\mathbf{y} = (\mathbf{y}_1^\top, \dots, \mathbf{y}_N^\top)^\top$, $\mathbf{y}_i = (y_{i1}, \dots, y_{iT})^\top$, $i = 1, \dots, N$. For each site $i = 1, \dots, N$, we compute the MLE as $(\hat{\psi}_i, \hat{\tau}_i, \hat{\phi}_i)^\top = \arg \max_{(\psi, \tau, \phi)} L(\psi, \tau, \phi \mid \mathbf{y}_i)$

and then approximate the i -th generalized likelihood function $L(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i)$ by a possibly rescaled Gaussian density with mean $(\hat{\psi}_i, \hat{\tau}_i, \hat{\phi}_i)^\top$ and covariance matrix $\Sigma_{\eta, \mathbf{y}_i} = \mathbf{Q}_{\eta, \mathbf{y}_i}^{-1}$, where $\mathbf{Q}_{\eta, \mathbf{y}_i}$ denotes the observed information matrix, i.e., the negative Hessian matrix of $\log L(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i)$ evaluated at $(\hat{\psi}_i, \hat{\tau}_i, \hat{\phi}_i)^\top$. Therefore, $L(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i) \approx c_i \hat{L}(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i)$, where $\hat{L}(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i)$ denotes the density of $\mathcal{N}_3((\hat{\psi}_i, \hat{\tau}_i, \hat{\phi}_i)^\top, \Sigma_{\eta, \mathbf{y}_i})$ and $c_i \geq 0$ is some nonnegative constant independent of $(\psi_i, \tau_i, \phi_i)^\top$. Combining all locations together, we can thus approximate the overall generalized likelihood function as $L(\boldsymbol{\eta} \mid \mathbf{y}) = \prod_{i=1}^N L(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i) \approx \prod_{i=1}^N c_i \hat{L}(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i) \propto \prod_{i=1}^N \hat{L}(\psi_i, \tau_i, \phi_i \mid \mathbf{y}_i) := \hat{L}(\boldsymbol{\eta} \mid \mathbf{y})$.

To clarify this approximation with a toy example, let us consider the setting where $W_1, \dots, W_m \stackrel{\text{iid}}{\sim} \mathcal{N}(\gamma, 1)$. The likelihood function for γ is $L(\gamma \mid W_1, \dots, W_m) = \prod_{k=1}^m [(2\pi)^{-1/2} \exp\{-\frac{1}{2}(W_k - \gamma)^2\}]$, which may be rewritten as a product of two terms, namely the constant $c = m^{-1/2}(2\pi)^{-(m-1)/2} \exp\{-\frac{1}{2}(m^{-1} \sum_{k=1}^m W_k^2 - \bar{W}^2)\}$ and $m^{1/2}(2\pi)^{-1/2} \exp\{-\frac{1}{2}m(\gamma - \bar{W})^2\}$, where $\bar{W} = m^{-1} \sum_{k=1}^m W_k$. Here, the first term c is indeed constant with respect to γ , whereas the second term is the Gaussian density with mean $\bar{W}$, the MLE for γ , and variance m^{-1} ; we can easily show that the negative Hessian of $\log L(\gamma \mid W_1, \dots, W_m)$ is indeed m . In this toy example, we consider the data distribution to be Gaussian, and hence, the Gaussian approximation holds as an equality. In more general settings with alternative non-Gaussian likelihoods, the Gaussian approximation is justified thanks to the large-sample properties of the MLE; see Schervish (1995) for example. Given that the Gaussian density is the *asymptotic* form of the likelihood under mild regularity conditions, this ensures that this likelihood approximation will be accurate, provided the number of temporal replicates is large enough.

In our extreme-value context, $\hat{L}(\boldsymbol{\eta} \mid \mathbf{y})$ is the density of a $3N$ -dimensional Gaussian distribution with mean $\hat{\boldsymbol{\eta}} = (\hat{\boldsymbol{\psi}}^\top, \hat{\boldsymbol{\tau}}^\top, \hat{\boldsymbol{\phi}}^\top)^\top$, where $\hat{\boldsymbol{\psi}} = (\hat{\psi}_1, \dots, \hat{\psi}_N)^\top$, $\hat{\boldsymbol{\tau}} = (\hat{\tau}_1, \dots, \hat{\tau}_N)^\top$, and $\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \dots, \hat{\phi}_N)^\top$ and covariance matrix $\Sigma_{\eta, \mathbf{y}} = \mathbf{Q}_{\eta, \mathbf{y}}^{-1}$ constructed by properly stacking the entries of $\Sigma_{\eta, \mathbf{y}_1}, \dots, \Sigma_{\eta, \mathbf{y}_N}$. The full posterior density (15) may then be approximated based on $\hat{L}(\boldsymbol{\eta} \mid \mathbf{y})$ as

$$\pi(\boldsymbol{\eta}, \mathbf{v}, \boldsymbol{\theta} \mid \mathbf{y}) \approx \hat{\pi}(\boldsymbol{\eta}, \mathbf{v}, \boldsymbol{\theta} \mid \mathbf{y}) \propto \hat{L}(\boldsymbol{\eta} \mid \mathbf{y}) \pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta}) \pi(\mathbf{v} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta}). \quad (16)$$

Now, consider a surrogate-LGM where the sitewise MLEs $\hat{\boldsymbol{\eta}}$ are treated as the data, with distribution $\hat{\boldsymbol{\eta}} \sim \mathcal{N}_{3N}(\boldsymbol{\eta}, \mathbf{Q}_{\eta, \mathbf{y}}^{-1})$, where $\mathbf{Q}_{\eta, \mathbf{y}}$ is defined as above and assumed to be known. Through the lenses of this surrogate-model, we can interpret the vector $\hat{\boldsymbol{\eta}}$ as a noisy measurement of the true unknown parameter vector $\boldsymbol{\eta}$. While the likelihood of the surrogate-model is different from the likelihood of the original model, the latent and hyperparameter levels are kept the same. Writing the likelihood function of the surrogate-model as $L^*(\boldsymbol{\eta} \mid \hat{\boldsymbol{\eta}}) = \pi(\hat{\boldsymbol{\eta}} \mid \boldsymbol{\eta})$, the

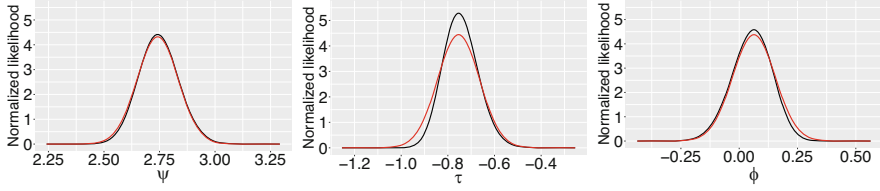


Fig. 7 Normalized likelihood functions (black) of the parameters ψ_i , τ_i , and ϕ_i (left to right), with the corresponding approximate densities obtained from the Gaussian likelihood approximation (red), for a representative grid cell (site 208) with 67 threshold exceedances

corresponding posterior density may be written as

$$\begin{aligned} \pi(\boldsymbol{\eta}, \mathbf{v}, \boldsymbol{\theta} \mid \hat{\boldsymbol{\eta}}) &\propto \pi(\hat{\boldsymbol{\eta}} \mid \boldsymbol{\eta})\pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta})\pi(\mathbf{v} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta}) \\ &= L^*(\boldsymbol{\eta} \mid \hat{\boldsymbol{\eta}})\pi(\boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta})\pi(\mathbf{v} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta}), \end{aligned} \quad (17)$$

where $L^*(\boldsymbol{\eta} \mid \hat{\boldsymbol{\eta}}) = \hat{L}(\boldsymbol{\eta} \mid \mathbf{y})$. Hence, the posterior density for the surrogate-model in (17) is the same as the approximated posterior density in (16) that provides a good approximation to the exact posterior density in (15). This justifies using the surrogate-model to make inference, instead of relying on the exact but more complicated posterior density.

To verify the accuracy of the Gaussian likelihood approximation in our peaks-over-threshold extreme-value context, Fig. 7 displays the true normalized likelihood together with the Gaussian approximation at a representative grid cell (site 208) with 67 threshold exceedances (out of 268 positive precipitation intensities, for a total of 7245 temporal replicates); see Fig. 1 for the exact location. The likelihood approximation is very accurate for ψ_i , while it is comparatively less accurate for ϕ_i and τ_i . The results are expected to improve quickly when a larger number of threshold exceedances become available, similarly to the findings of Jóhannesson et al. (2022) in the case of the GEV distribution, which is in line with asymptotic results for the posterior density (Schervish, 1995). While the asymptotic Gaussianity of the MLEs involves the expected information, here we use the observed information, and the justification follows from Efron and Hinkley (1978).

4.2 “Smooth” Step: Fitting the Gaussian–Gaussian Surrogate-model

The second step of Max-and-Smooth is to fit the approximate surrogate-model with (17) as its posterior, based on the sitewise MLEs $\hat{\boldsymbol{\eta}}$ (and their precision matrices $\mathbf{Q}_{\boldsymbol{\eta}, \mathbf{y}}^{-1}$) pre-computed in the “Max” step. We here briefly describe some key aspects of MCMC-based inference for the surrogate-model. The hierarchical representation

of the surrogate-model is

$$\begin{aligned}
 \hat{\boldsymbol{\eta}} \mid \boldsymbol{\eta} &\sim \mathcal{N}_{3N}(\boldsymbol{\eta}, \mathbf{Q}_{\boldsymbol{\eta}, \mathbf{y}}^{-1}), \\
 \boldsymbol{\eta} \mid \mathbf{v}, \boldsymbol{\theta} &\sim \mathcal{N}_{3N}(\mathbf{Z}\mathbf{v}, \text{bdiag}(\sigma_{\psi}^2 \mathbf{I}_N, \sigma_{\tau}^2 \mathbf{I}_N, \sigma_{\phi}^2 \mathbf{I}_N)), \\
 \mathbf{v} \mid \boldsymbol{\theta} &\sim \pi(\mathbf{v} \mid \boldsymbol{\theta}) = \pi(\beta_{\psi}) \times \pi(\beta_{\tau}) \times \pi(\beta_{\phi}) \times \pi(\mathbf{w}_{\psi} \mid \boldsymbol{\theta}) \times \pi(\mathbf{w}_{\tau} \mid \boldsymbol{\theta}), \\
 \boldsymbol{\theta} &\sim \pi(\boldsymbol{\theta}) = \pi(\sigma_{\psi}) \times \pi(s_{\psi}, \rho_{\psi}) \times \pi(\sigma_{\tau}) \times \pi(s_{\tau}, \rho_{\tau}) \times \pi(\sigma_{\phi}),
 \end{aligned}$$

where the terms involved at the latent and hyperparameter levels are detailed in Sects. 3.2 and 3.3, respectively, and the notation is the same as earlier.

Thanks to the conjugate structure of the Gaussian–Gaussian surrogate-LGM, it is easy to verify that the full conditional density of $(\boldsymbol{\eta}^{\top}, \mathbf{v}^{\top})^{\top}$, i.e., $\pi(\boldsymbol{\eta}, \mathbf{v} \mid \hat{\boldsymbol{\eta}}, \boldsymbol{\theta})$, is multivariate Gaussian. While this density is $(5N + 3)$ -dimensional, we can exploit the sparsity of its precision matrix (defined in terms of $\mathbf{Q}_{\boldsymbol{\eta}, \mathbf{y}}$, $\mathbf{Q}_{\rho_{\psi}}$, and $\mathbf{Q}_{\rho_{\tau}}$) for fast sampling. Notice that Gibbs sampling updates for latent parameters would not be possible for the original LGM based on the exact Poisson point process likelihood and that Metropolis–Hastings updates (or variants thereof) would be required for all of the $5N + 3$ variables. The great benefit of the surrogate-model is that latent variables can be updated simultaneously by Gibbs sampling, which drastically reduces the computational burden and greatly improves the mixing and convergence of Markov chains. As for the hyperparameter vector $\boldsymbol{\theta}$, its (approximate) marginal posterior density, i.e., $\pi(\boldsymbol{\theta} \mid \hat{\boldsymbol{\eta}})$, is known up to a constant. Although $\pi(\boldsymbol{\theta} \mid \hat{\boldsymbol{\eta}})$ does not have a closed form, posterior samples of $\boldsymbol{\theta}$ can be easily obtained using the Metropolis–Hastings algorithm or grid sampling, for example. The fact that the exact marginal posterior density of $\boldsymbol{\theta}$, i.e., $\pi(\boldsymbol{\theta} \mid \mathbf{y})$, can be approximated with the marginal posterior of $\boldsymbol{\theta}$ under the surrogate-model is another important aspect of Max-and-Smooth that greatly improves posterior sampling.

A crucial point to note is that once sitewise MLEs $\hat{\boldsymbol{\eta}}$ are obtained, the rest of the computational time, required to fit the surrogate-model in the “Smooth” step, does not depend on the temporal dimension anymore. Moreover, obtaining MLEs in the “Max” step is often quite fast and can be performed in parallel across spatial locations. As an illustration, the “Max” step takes only a few seconds for our Saudi Arabian precipitation dataset with 2738 grid cells in total. Thus, Max-and-Smooth is doubly beneficial for spatio-temporal datasets with large temporal dimensions: first as the number of time replicates grows, the Gaussian approximation becomes very accurate, and second, the relative computational cost with respect to an “exact” MCMC inference scheme becomes negligible. We also note that although the inference is done in two steps, the uncertainty involved in the “Max” step is properly propagated into the “Smooth” step in a way that provides a valid approximation to the (exact) full posterior distribution.

5 Saudi Arabian Precipitation Extremes Application

In this section, we fit the latent Gaussian model of Sect. 3 to the Saudi Arabian TRMM daily precipitation dataset and draw approximate Bayesian inference using the Max-and-Smooth approach presented in Sect. 4.

The model is fitted to high threshold exceedances, based on the Poisson point process likelihood, using site-specific thresholds taken as the 75% empirical quantile of positive precipitation intensities. When taking zero precipitation values into account, the unconditional threshold probability level varies spatially (from about 90.16% at a few locations to about 99.74%), and it is often higher than the 99% quantile. On the scale of daily precipitation, the chosen thresholds range from about 1 to 15 mm, depending on the location. Fine-tuning the threshold at each site would be tedious with such a large spatial dimension and so we opted to choose the threshold in a pragmatic way to keep a reasonable number of threshold exceedances at each site (from 19 to 713 for an average of 66 threshold exceedances out of 7245 days, i.e., about 3.3 per year on average), while providing a suitable bias–variance trade-off and making sure that the MLEs and their covariance matrices obtained in the “Max” step are reliable.

In the “Smooth” step, we run the Gaussian–Gaussian surrogate-LGM for 10,000 MCMC iterations and remove 2000 burn-in iterations. Some posterior summary statistics for the estimated hyperparameters are presented in Table 1. All the intercept coefficients β_ψ , β_τ , and β_ϕ are significantly different from zero based on 95% credible intervals and are positive for ψ and ϕ but negative for τ . The posterior mean of β_ϕ is about 0.1, and since the standard deviation σ_ϕ of the corresponding nugget effect is about 0.06, the estimated transformed shape parameters ϕ_i are usually within the range $[-0.02, 0.22]$. This implies that the precipitation distribution is moderately heavy-tailed at most sites. The marginal standard deviations of the spatially structured SPDE effects, s_ψ and s_τ , are significantly larger than the respective standard deviations of the spatially unstructured nugget effects, σ_ψ and σ_τ . This indicates that these transformed location and scale parameters vary quite

Table 1 Posterior summary statistics for the estimated hyperparameters.

Hyperparameter	Mean	SD	2.5%	50%	97.5%
β_ψ	2.6545	0.3454	1.9755	2.6499	3.3576
σ_ψ	0.0442	0.0025	0.0392	0.0442	0.0491
s_ψ	0.5956	0.1347	0.4125	0.5690	0.9562
ρ_ψ	8.1123	2.0028	5.3706	7.7501	13.4077
β_τ	−0.5519	0.2340	−1.0185	−0.5543	−0.0745
σ_τ	0.0028	0.0025	0.0001	0.0020	0.0095
s_τ	0.3795	0.1007	0.2546	0.3587	0.6380
ρ_τ	8.3504	2.4316	5.2165	7.8593	14.3888
β_ϕ	0.0973	0.0020	0.0932	0.0973	0.1012
σ_ϕ	0.0593	0.0017	0.0559	0.0593	0.0626

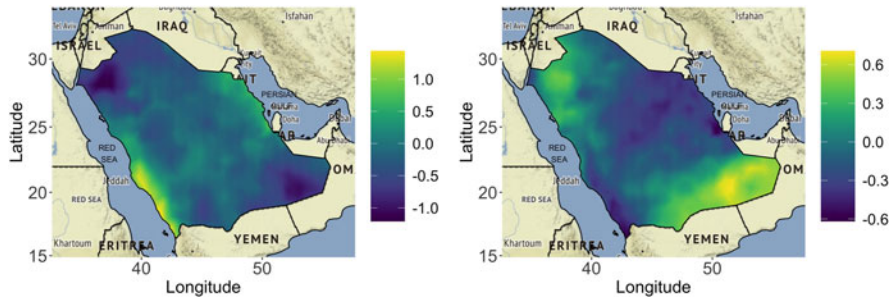


Fig. 8 Posterior means of the spatially structured SPDE effects, w_ψ^* (left) and w_τ^* (right), included in the latent structure for ψ and τ , respectively, projected to a fine grid covering Saudi Arabia in Fig. 6

smoothly over space and, thus, that it is important to include spatially structured effects in these parameters at the latent level. Both range parameters ρ_ψ and ρ_τ are fairly large and similar to each other, which indicates long-range spatial dependence and corroborates the empirical variograms in Fig. 5.

Figure 8 displays the posterior mean of the spatially structured SPDE random effects w_ψ^* and w_τ^* , included in the latent structure for ψ and τ , respectively, projected onto a fine grid covering Saudi Arabia shown in Fig. 6. The spatial patterns in the estimates of w_ψ^* and w_τ^* are similar to those observed in the maps of the MLEs, $\hat{\psi}_i$ and $\hat{\tau}_i$, plotted in Fig. 3. Higher values of the posterior means of w_ψ^* and w_τ^* are indeed observed near the southwestern and southeastern corners of the region of study, respectively. Since w_ψ^* is involved in the linear model specification for ψ_i and varies approximately between -1 and 1.5 , the latent spatially structured spatial effect scales the original location parameter $\mu_i = \exp(\psi_i)$ by a factor ranging from about 0.37 to 4.48 . Similarly, the posterior mean of w_τ varies approximately between -0.6 and 0.6 , which translates into a multiplicative factor for the original scale parameter σ_i that ranges from about 0.55 to 1.82 , after taking μ_i into account. We also stress that our model is believed to be accurate only within the spatial domain where the data are observed (Saudi Arabia), despite the fact that a large number of mesh nodes are outside the data domain, which are necessary only for edge effect correction in approximating the Matérn correlation using SPDE.

Figure 9 shows scatterplots of model-based estimates (i.e., posterior means) of the original and transformed model parameters plotted against their preliminary sitewise MLEs. We can see that the estimated location parameters $\hat{\mu}_i$ vary highly across space, with values ranging from about 5 to 60 depending on the location. Despite this high spatial variation, the model-based posterior estimates appear to be consistent with preliminary sitewise estimates of μ_i throughout the domain. This shows that the flexible LGM structure that we fitted is able to accurately capture complex spatial patterns in the location parameter, thanks to the latent spatially structured and unstructured effects, while reducing the posterior uncertainty by pooling information across locations. Posterior estimates of the scale parameter

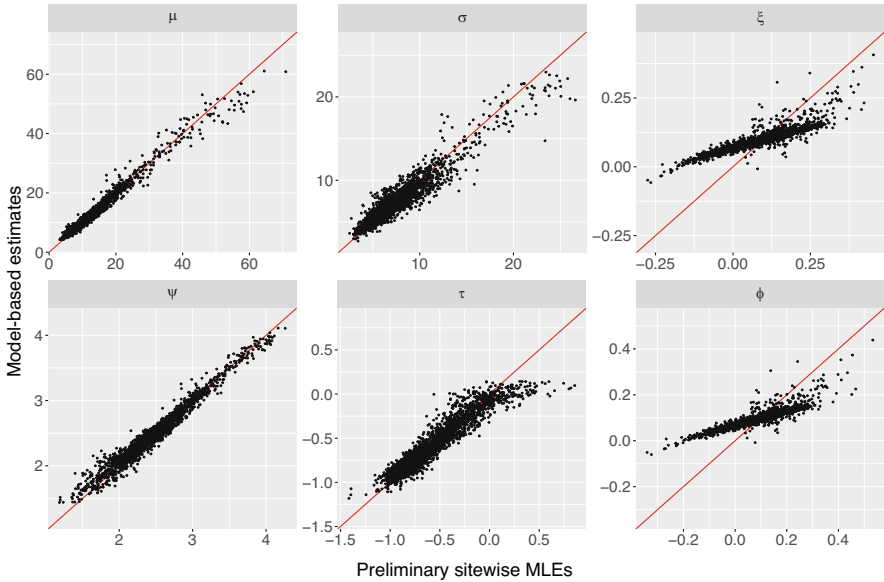


Fig. 9 Scatterplots of model-based posterior estimates of the original Poisson point process parameters (top), μ_i , σ_i , and ξ_i (left to right), and transformed parameters (bottom), ψ_i , τ_i , and ϕ_i (left to right), plotted with respect to their corresponding preliminary sitewise MLEs. The main diagonal (with intercept zero and unit slope) is shown in red

σ_i also appear to be generally consistent with preliminary sitewise estimates, although the points in the scatterplots are a bit more spread out around the main diagonal. This slightly larger variability is not an indication of a lack of fit, but it reflects the fact that sitewise estimates are quite noisy (mostly due to maximum likelihood estimation uncertainty), while the model-based posterior estimates are smoother over space (thanks to the shrinkage induced by the spatial structure of the LGM). As for the shape parameter ξ_i , there are significant differences between the model-based posterior estimates (which vary roughly between 0 and 0.25) and the preliminary sitewise MLEs (which vary roughly between -0.25 and 0.40). These huge differences are due to two main reasons: first, the latent structure of ϕ in (12) does not involve a spatially structured random effect, which makes it a bit less flexible than the models for ψ and τ , and second, the use of a PC prior for the standard deviation parameter σ_ϕ of the latent nugget effect induces relatively strong shrinkage toward a spatially constant shape parameter. Nevertheless, we believe that the posterior estimates of ξ_i are much more reasonable than the preliminary sitewise MLEs, which have large uncertainties and yield unrealistic tail behaviors, from bounded upper tails when $\xi_i < 0$ to very heavy-tailed when $\xi_i \approx 0.4$. By contrast, our Bayesian LGM framework succeeds in reducing posterior uncertainty by shrinkage and spatial pooling of information, in order to obtain satisfactory results.

We then illustrate the practical benefit of the proposed LGM framework by computing return levels and posterior predictive densities. Return levels can be estimated by plugging model-based posterior estimates $\hat{\mu}_i$, $\hat{\sigma}_i$, and $\hat{\xi}_i$ into (3), whereas the posterior predictive density at a spatial location s_i may be approximated as

$$\pi_i^*(\tilde{y} \mid \mathbf{y}) \approx \pi_i^*(\tilde{y} \mid \hat{\boldsymbol{\eta}}) = \iiint \pi_{u_i}(\tilde{y} \mid \boldsymbol{\eta}_i) \pi(\boldsymbol{\eta}, \mathbf{v}, \boldsymbol{\theta} \mid \hat{\boldsymbol{\eta}}) d\boldsymbol{\eta} d\mathbf{v} d\boldsymbol{\theta}, \quad (18)$$

where $\pi_{u_i}(\tilde{y} \mid \boldsymbol{\eta}_i)$ is the conditional density of $\tilde{Y}$ given $\tilde{Y} > u_i$, with $\tilde{Y} \sim G_{\text{GEV}(\mu_i, \sigma_i, \xi_i)}^{1/B}(y)$ (where $G_{\text{GEV}(\mu_i, \sigma_i, \xi_i)}^{1/B}(y) = \{G_{\text{GEV}(\mu_i, \sigma_i, \xi_i)}(y)\}^{1/B}$ by convention), and $G_{\text{GEV}(\mu_i, \sigma_i, \xi_i)}$ is the GEV distribution function with parameters μ_i , σ_i , and ξ_i , B is the block size (here, taken as $B = 365.25$ to represent yearly blocks), and u_i is the threshold chosen for the i -th location. Figure 10 shows within-sample posterior predictive densities (top panels) and estimated return level plots (bottom panels) for three representative sites (sites 208, 1716, and 1905) taken from three spatially distant parts of the domain (recall Fig. 1). The top panels display kernel density estimates of posterior predictive samples for each selected site, obtained by sampling from (18). We observe that the densities are right-skewed for all three cases and are relatively well calibrated with the observations. The bottom

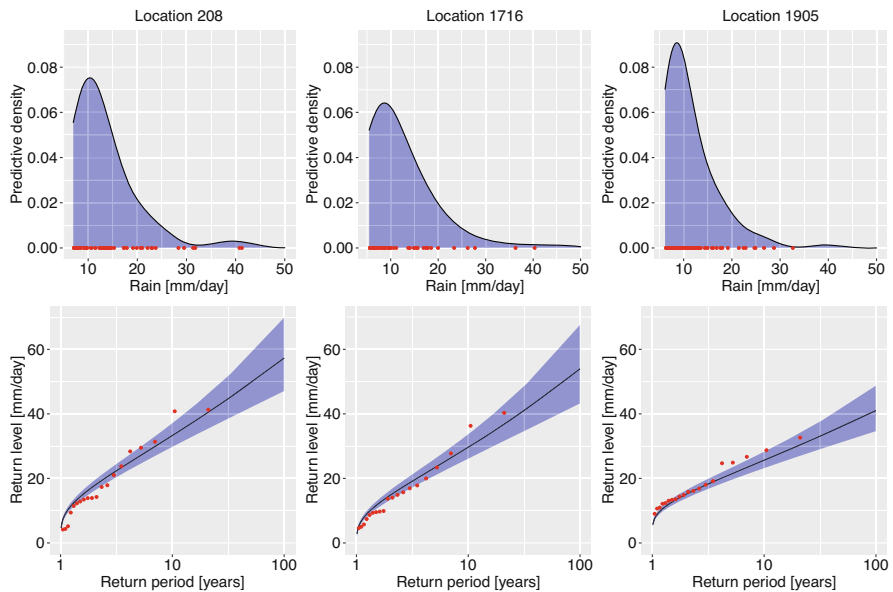


Fig. 10 Within-sample posterior predictive densities at three representative grid cells (top), and return level estimates plotted as a function of the return period on a logarithmic scale (bottom). In the return level plots, black lines are the posterior means, blue bands are 95% pointwise credible intervals, and red dots are the ordered observations at each site

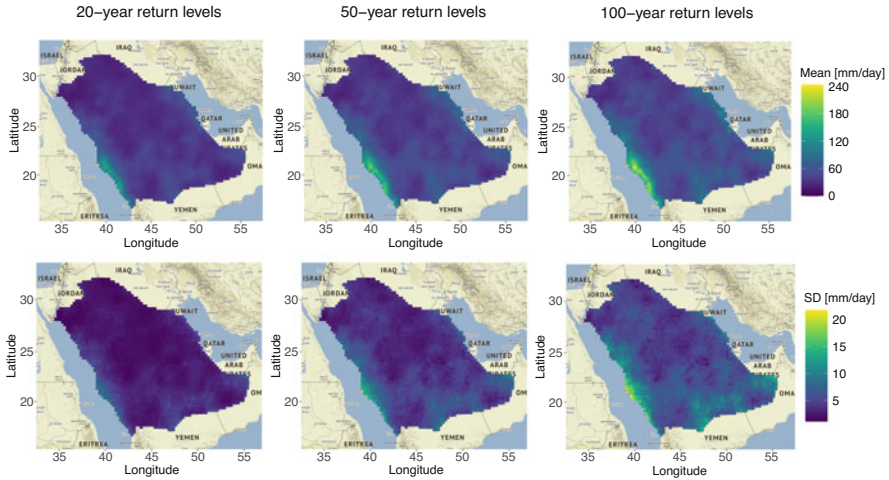


Fig. 11 Spatial maps of M -year return level estimates (top), with $M = 20, 50, 100$ (left to right), and their corresponding posterior standard deviations (bottom)

panels display estimated return levels plotted as a function of the return period, with associated 95% pointwise credible intervals, as well as the order statistics at each site. We can see that the observations are generally contained within, or just slightly outside, the credible bands, indicating that the proposed LGM fits the data reasonably well.

Finally, Fig. 11 displays maps of M -year return level estimates corresponding to the return periods of $M = 20, 50$, and 100 years, as well as their respective sitewise posterior standard deviations. As expected, return level estimates are realistically higher near the coasts of the Red Sea and the Persian Gulf. They range from 16.7 to 140.7 mm for $M = 20$ years, from 21.5 to 192.9 mm for $M = 50$ years, and from 25.2 to 242.9 mm for $M = 100$ years, with the highest precipitation amount expected at the grid cell with coordinate $(39.875^\circ\text{E}, 20.875^\circ\text{N})$, close to the major cities of Jeddah and Makkah. Furthermore, while the standard deviations are quite high near the coast of the Red Sea due to the large spread of the precipitation distribution and the high variability of threshold exceedances in this region, they are also quite high near the arid southeastern region characterized by a drier climate, which leads to a smaller number of threshold exceedances available to fit the model in this area. Comparable results were obtained by Davison et al. (2019) after adjustment (as, in that paper, the authors mistakenly forgot a multiplication factor 3 when aggregating three-hourly precipitation rates (mm/h) to daily precipitation (mm/day)) for a smaller region around Jeddah and Makkah, based on max-stable processes fitted to annual precipitation maxima.

6 Discussion and Conclusion

In this chapter, we have shown how complex latent Gaussian models (LGMs) for extremes, based on the convenient and informative Poisson point process representation for peaks-over-threshold events, can be suitably constructed and efficiently fitted using Max-and-Smooth to massive spatio-temporal datasets. Our proposed modeling framework assumes that Gaussian fixed and spatial random effects are embedded within model parameters at the latent level, in order to flexibly capture spatial variability, non-stationary patterns, and dependencies. Our model relies on the stochastic partial differential equation (SPDE) representation of Matérn random fields, whose discretization leads to Gaussian Markov random fields (GMRFs) with sparse precision matrices. In case of strongly non-stationary spatial trends with gridded or areal data, it would also be possible to replace the latent stationary SPDE effects with intrinsic conditional autoregressive (CAR) priors, for example, but we have not explored this here. Exploiting GMRFs at the latent level, combined with an efficient Bayesian inference scheme, makes it possible to tackle complicated problems in very high dimensions. The Saudi Arabian precipitation dataset that we analyzed in this chapter is the proof of this, since it comprises 2738 grid cells and numerous temporal replicates at each site, and we did not reach our computational limits by any measure. Our proposed methodology indeed scales up to even bigger and higher resolution datasets. Importantly, using latent SPDE random effects combined with Max-and-Smooth implies that the computational burden is only moderately impacted by the spatial and temporal dimensions. Computationally speaking, it is indeed crucial to realize that what matters the most is the number of mesh nodes used to discretize the SPDE, which should depend on the spatial effective correlation range and the size of the domain, rather than the actual number of observation locations. Another important performance-limiting factor is the number of hyperparameters, which should be kept relatively small. Moreover, the great benefit of Max-and-Smooth is to perform inference in two separate steps. The first step is typically very fast and consists in computing sitewise maximum likelihood estimates (MLEs), which can be done in parallel. The second step consists in fitting an approximate Gaussian–Gaussian surrogate-LGM to the MLEs, so that the number of temporal replicates only *indirectly* affects the model fit through the variability of MLEs but is irrelevant in terms of the computational time. Therefore, leveraging both Max-and-Smooth and SPDE random effects provides a powerful toolbox for analyzing massive and complex datasets using extended LGMs. We also stress that although we studied extreme precipitation data in this work, this methodology applies more generally and, if suitably adapted, could potentially be used in other contexts and a variety of statistical applications (see Hrafnkelsson et al., 2021).

The use of high threshold exceedances, which contrasts with the block maximum approach advocated by Jóhannesson et al. (2022) and many other authors, allows us to exploit information from all extreme peaks-over-threshold events (on a daily scale), potentially leading to a drastic uncertainty reduction, and opens the door

to more detailed modeling of intra-year seasonal variability, as well as extremal clusters that are formed due to temporal dependence. In this work, we have decided to ignore these important issues for the sake of simplicity, by assuming that the data were iid in time, but it would be interesting in future research to extend our spatial LGM framework by including temporally structured latent random effects, which capture temporal non-stationary patterns and long-term time trends. We also stress here that, unlike the model of Cooley et al. (2007), our approach relies on the Poisson point process representation of extremes, which allows us to conveniently describe the data's tail behavior using a single LGM (rather than two separate models for threshold exceedances and threshold exceedance occurrences), which gives a unified treatment of uncertainty. Moreover, model parameters are relatively insensitive to the threshold choice (unlike LGMs based on the generalized Pareto distribution) and have a one-to-one correspondence with the block maximum approach based on the GEV distribution, which facilitates interpretation.

Beyond the distributional assumptions at the response level and the specific latent model structure, an important modeling aspect with extended LGMs is the specification of a suitable multivariate link function. In our extreme-value context, we transformed the location and scale parameters jointly to avoid overly strong correlations between latent variables, and we used a rather peculiar transformation for the shape parameter that facilitates interpretation, while avoiding pathological behaviors. To prevent estimating models with overly short or heavy tails, we made the a priori choice of restricting the shape parameter to the interval $(-0.5, 0.5)$, thus ensuring finite variance. Our approach has, thus, some links with the recently introduced concept of property-preserving penalized complexity (P^3C) priors (Castro-Camilo et al., 2021).

Overall, this chapter demonstrates the effectiveness of LGMs in estimating spatial return level surfaces with a real, large-scale, geo-environmental application (i.e., precisely, the modeling of precipitation extremes over the whole Saudi Arabian territory). However, the conditional independence assumption at the response level is clearly a limitation when the data themselves exhibit strong spatial dependence. Ignoring this issue might lead to underestimating the uncertainty of estimated parameters and return levels for univariate or spatial quantities (Jóhannesson et al., 2022). To circumvent this issue, a possibility involves post-processing posterior predictive samples simulated from the model at the observed sites by modifying their ranks in such a way to match the data's empirical copula, while keeping the same marginal distributions (Jóhannesson et al., 2022). Another possibility might be to explicitly model response level dependence similar to Sang and Gelfand (2010), who proposed an LGM characterized by a Gaussian copula at the response level. However, while this approach corrects the unrealistic conditional independence assumption to some extent, it leads to heavier computations. Furthermore, the Gaussian copula is quite rigid in its joint tail and does not comply with classical extreme-value theory. In the same vein, Hazra and Huser (2021) recently fitted a Bayesian hierarchical model, constructed from a Dirichlet process mixture of low-rank Student's t processes, to model sea surface temperature data in high dimensions (all the way from low to high quantiles). The fast Bayesian inference was made

possible thanks to the availability of Gibbs updates and to the low-rank structure stemming from the sparse set of suitably chosen spatial basis functions. However, it is not clear how to adapt their methodology to the case where interest lies in making inference for extremes only, modeled with a Poisson point process or GEV likelihood function, thus preventing observations in the bulk of the distribution from affecting the estimation of the tail structure. In the future research, it would be interesting to extend Max-and-Smooth to the case of extended LGMs with response level dependence characterized by various types of copula models.

For reproducibility reasons and to make our methodology easily accessible to the whole statistics community and beyond, our R implementation of Max-and-Smooth with a simple example can be found in the Supplementary Material. Furthermore, it can also be downloaded from the following GitHub repository: https://github.com/arnabstatswithR/max_and_smooth.

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